A UNIFORM BOUND FOR LOG CALABI–YAU ITERATES OF PLANE ENDOMORPHISMS

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Abstract. We prove that for the $17280 \cdot 2^{200}$-th iterate of any polarized endomorphism of the projective plane, the pair consisting of the plane and the normalized ramification divisor is log canonical. Hence Question 6.3 of Luo–Meng has an affirmative answer, with a bound on the iterate that is explicit and independent of the degree of the endomorphism.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let X be a normal projective variety and $q \geq 2$ an integer. A surjective endomorphism $f : X \to X$ is *q-polarized* if there exists an ample divisor H on X such that

$$f^*H \sim qH.$$

Such an f is finite, and we let $R(f)$ be the ramification divisor of f, that is, the divisor on X with

$$K_X = f^*K_X + R(f).$$

On $\mathbb{P}^2$, since $\mathrm{Pic}(\mathbb{P}^2)$ is generated by $\mathcal{O}_{\mathbb{P}^2}(1)$, an endomorphism f is q-polarized if and only if $f^*\mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^2}(q)$. For such f and any integer $k \geq 1$, $\deg R(f^k) = 3(q^k - 1)$, so $K_{\mathbb{P}^2} + \frac{1}{q^k-1}R(f^k) \equiv 0$, and thus $\left(\mathbb{P}^2, \frac{1}{q^k-1}R(f^k)\right)$ is log Calabi–Yau if and only if it is lc.

Gongyo proposed the following conjecture, see [Men23].

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Conjecture 1.1 ([Men23, Conjecture 1.3]). *Let $q \geq 2$ be an integer, X a smooth projective variety, and $f : X \to X$ a q-polarized endomorphism. Then there exists an integer $k \geq 1$ such that $\left(X, \frac{1}{q^k-1}R(f^k)\right)$ is lc.*

Let X be a normal projective variety and $f : X \to X$ a q-polarized endomorphism. When $\dim X = 2$, an irreducible curve $C \subset X$ is *totally periodic* if $f^{-k}(C) = C$ for some integer $k \geq 1$. The conclusion of Conjecture 1.1, namely that

$$\left(X, \tfrac{1}{q^k-1}R(f^k)\right) \text{ is lc for some integer } k \geq 1,$$

was proved in the following cases:

(1) When $X = \mathbb{P}^n$ for some integer $n \geq 1$ and f is a Galois cover, with $k = 1$ [LZ26].
(2) When X is a smooth projective surface, except when $X \cong \mathbb{P}^2$ and there does not exist a totally periodic curve on X [Men23].
(3) When X is a normal projective surface with $\rho(X) \geq 2$ or with a totally periodic curve [CZ25].
(4) When $X = \mathbb{P}^2$ [LM26, Theorem 1.3].
(5) When X is $\mathbb{Q}$-Gorenstein [Yos26, Theorem 1.4].
(6) When X is any normal projective variety [Liu26, Theorem 1.5].

Case (4) reads as follows.

Theorem 1.2 ([LM26, Theorem 1.3]). *Let $q \geq 2$ be an integer and $f : \mathbb{P}^2 \to \mathbb{P}^2$ a q-polarized endomorphism. Then there exists an integer $k \geq 1$ such that $\left(\mathbb{P}^2, \frac{1}{q^k-1}R(f^k)\right)$ is lc.*

In October 2026, Yoshikawa proved case (5) in the following form.

Theorem 1.3 ([Yos26, Theorem 1.4]). *Let $q \geq 2$ be an integer, X a normal $\mathbb{Q}$-Gorenstein projective variety, and $f : X \to X$ a q-polarized endomorphism. Then there exists an integer $k \geq 1$ such that $\left(X, \frac{1}{q^{km}-1}R(f^{km})\right)$ is lc for any integer $m \geq 1$.*

Case (6) is contained in [Liu26, Theorem 1.5]: for any integer $q \geq 2$, any normal projective variety X, and any q-polarized endomorphism $f : X \to X$, there exists an integer $k \geq 1$ such that for any integer $m \geq 1$, the pair $\left(X, \frac{1}{q^{km}-1}R(f^{km})\right)$ is lc and

$$K_X + \tfrac{1}{q^{km}-1}R(f^{km}) \sim_{\mathbb{Q}} 0,$$

which is the case $\Delta = 0$ of [CZ25, Conjecture 1.5] on pairs (X, Δ). By [Liu26, Theorem 1.3 and Corollary 1.4], the conjecture of Broustet and Gongyo [BG17, Conjecture 1.2] holds: any normal projective variety X admitting a q-polarized endomorphism for some integer $q \geq 2$ is of Calabi–Yau type, that is, there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that (X, B) is lc and $K_X + B \sim_{\mathbb{Q}} 0$.

In Theorems 1.2 and 1.3 and in [Liu26, Theorem 1.5] the integer k depends on f, and none of [LM26, Theorem 1.3], [Yos26, Theorem 1.4], and [Liu26, Theorem 1.5] contains a bound on k in terms of f. Since a smooth projective variety is normal and $\mathbb{Q}$-Gorenstein, Theorem 1.3 with $m = 1$ implies Conjecture 1.1. What remains open is a bound on k that is uniform in f.

Already on $\mathbb{P}^2$, the integer $k = 1$ does not suffice: for the 3-polarized endomorphism

$$f : \mathbb{P}^2 \to \mathbb{P}^2, \quad [x : y : z] \to \left[x^3 + y^3 : x(y^2 + z^2) : z(3y^2 + 2z^2)\right],$$

the pair $\left(\mathbb{P}^2, \frac{1}{2}R(f)\right)$ is not lc [LZ26, Example 1.5]. Luo and Meng asked the following question, where an effective bound means a bound that can be computed.

Question 1.4 ([LM26, Question 6.3]). Let $q \geq 2$ be an integer. Does there exist an effective bound $N(q)$ such that for any q-polarized endomorphism $f : \mathbb{P}^2 \to \mathbb{P}^2$, the pair $\left(\mathbb{P}^2, \frac{1}{q^k-1}R(f^k)\right)$ is lc for some integer $1 \leq k \leq N(q)$? Can $N(q)$ be taken independent of q?

In this paper we answer Question 1.4 affirmatively, with a bound independent of both q and f. The main result of this paper is the following.

Theorem 1.5. *Let $q \geq 2$ be an integer and $f : \mathbb{P}^2 \to \mathbb{P}^2$ a q-polarized endomorphism. Then*

$$\left(\mathbb{P}^2, \frac{1}{q^{17280n}-1} R(f^{17280n})\right)$$

is lc for any integer $n \geq 2^{200}$. In particular, for $N := 17280 \cdot 2^{200}$, the pair $\left(\mathbb{P}^2, \frac{1}{q^N-1} R(f^N)\right)$ is lc.

We do not know the smallest integer N_0 such that $\left(\mathbb{P}^2, \frac{1}{q^{N_0}-1} R(f^{N_0})\right)$ is lc for any integer $q \geq 2$ and any q-polarized endomorphism f of $\mathbb{P}^2$, and $N_0 \geq 2$ by [LZ26, Example 1.5].

Sketch of the proof. Let $q \geq 2$ be an integer and f a q-polarized endomorphism of $\mathbb{P}^2$. For any integer $k \geq 1$, let $B_k := \frac{1}{q^k-1} R(f^k)$, and let

$$g := f^{17280} \quad \text{and} \quad Q := q^{17280},$$

so that $B_{17280n} = \frac{1}{Q^n-1} R(g^n)$ for any integer $n \geq 1$. For a divisorial valuation u over $\mathbb{P}^2$, let $A(u)$ be the log discrepancy of u with respect to $(\mathbb{P}^2, 0)$, and for a finite morphism $h : \mathbb{P}^2 \to \mathbb{P}^2$, let $h_* u$ be the divisorial valuation defined by $(h_* u)(\zeta) := u(h^* \zeta)$ for rational functions ζ on $\mathbb{P}^2$. Since $A(h_* u) = A(u) + u(R(h))$ (Lemma 2.4), for any integer $n \geq 1$ the pair $(\mathbb{P}^2, B_{17280n})$ is lc if and only if

$$A((g^n)_* u) \leq Q^n A(u) \quad \text{for any divisorial valuation } u \text{ over } \mathbb{P}^2.$$

Let $n \geq 2^{200}$ be an integer, assume that $(\mathbb{P}^2, B_{17280n})$ is not lc, and let C be the reduced curve whose irreducible components are the irreducible curves P with $\mathrm{mult}_P B_{288} > \frac{3}{5}$. The proof derives a contradiction in four steps.

(1) $\mathrm{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ for a closed point p with $g(p) = p$, and exactly one irreducible component L of C contains p. Moreover, L is a line with $g(L) = L$ (Proposition 5.2, Theorem 9.1, and Construction-Theorem 10.1).

(2) There exist affine linear coordinates x, y centered at p with $L = \{x = 0\}$, a polynomial $\phi \in \mathbb{C}[x]$, and a rational number $1 < t < 11$ such that the monomial valuation

$$v := \mathrm{mon}_{x, y-\phi(x)}(1, t),$$

with weights 1 and t on x and $y - \phi(x)$, satisfies $v(B_{17280n}) > A(v) = 1 + t$ (Construction-Theorem 10.4, which starts from a weighted blow-up with negative log discrepancy [Kaw17, Theorem 1(ii)]).

(3) Let $\lambda := c_\infty(g, p)$ be the asymptotic attraction rate of g at p (Definition 2.11) and $\sigma := 1 - \lambda/Q$. Let $e := e_L(g)$ be the ramification index of g along L (Definition 2.2). For any integer $b \geq 0$, let $w_b := (g^b)_* v$, $m_b := w_b(\mathfrak{m}_p)$, and $\nu_b := w_b / m_b$. Then

$$0 < \sigma < \frac{\log 32}{n}, \quad 0 < 1 - \frac{e}{Q} \leq \frac{10^{12}}{n}, \quad \text{and} \quad 2 < t < 11$$

(Theorems 11.1 and 12.1). Since $n \geq 2^{200}$, the set $I := \{10^6 + 4i \mid i \in \mathbb{Z},\ 0 \leq i \leq 600\}$ satisfies $I \subset [10^6, n/10^{20}]$, the range of b in Theorem 12.1(3), and for any $b \in I$,

$$\nu_b(L) = 1, \quad \left|\frac{m_b}{Q^b} - 1\right| \leq 10^{-7}, \quad \frac{13}{10} < \alpha(\nu_b) < 12, \quad \text{and} \quad A(\nu_b) < 13$$

(Theorem 12.1(3) and Proposition 12.3), where α is the skewness of normalized valuations at p (Definition 2.8).

(4) For any $b \in I$ with $b + 4 \in I$, $\alpha(\nu_{b+4}) < \alpha(\nu_b) - \frac{1}{50}$ (Section 13). Thus

$$\alpha(\nu_{10^6+2400}) < 12 - \frac{600}{50} = 0,$$

which contradicts $\alpha(\nu_{10^6+2400}) \geq 1$.

The proof of (1) uses Theorem 4.1 and Propositions 5.4, 5.5, and 6.1, and the proof of Theorem 9.1 uses Theorem 8.2 and Propositions 8.8 and 9.5. The bound $\sigma > 0$ in (3) uses Proposition 7.5.

The estimates in (1), (3), and (4) use the inner product defined by

$$\langle \delta_\mu, \delta_{\mu'} \rangle := \alpha(\mu \wedge \mu')$$

on the real vector space with basis $\{\delta_\mu\}_\mu$, where μ ranges over the normalized valuations at p with $\alpha(\mu) < +\infty$ (Definition 8.1, [FJ04, Section 7.12]), and $\mu \wedge \mu'$ is the greatest lower bound for the order with $\mu \leq \mu'$ if $\mu(\zeta) \leq \mu'(\zeta)$ for any $\zeta \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p}$. For a finite morphism of germs of smooth surfaces $h : (X, p) \to (X', p')$ of local degree d and normalized valuations $\nu_1, \dots, \nu_N$ at p with finite skewness and $h_*\nu_i = m_i\nu_i'$, where ν_i' is normalized,

$$\Big(d\alpha(\nu_i \wedge \nu_j) - m_i m_j \alpha(\nu_i' \wedge \nu_j')\Big)_{1 \leq i,j \leq N} \text{ is positive semidefinite}$$

(Lemma 8.3, whose input is [FJ04, Theorem 8.13]).

The proof of (4) compares ν_b with a monomial valuation below it. Let β_b be the largest value of ν_b on a line through p, attained on a line M, and let x_b, y_b be affine linear coordinates centered at p with $L = \{x_b = 0\}$ and $M = \{y_b = 0\}$. Then $\mu_b := \mathrm{mon}_{x_b,y_b}(1, \beta_b)$ satisfies

$$\mu_b \leq \nu_b, \quad \alpha(\mu_b) = \beta_b, \quad \text{and} \quad \beta_b - 1 > \frac{1}{40}$$

(Lemma 8.5, using $\alpha(\nu_b) > \frac{13}{10}$ and $A(\nu_b) < 13$). Write $(g^4)_*\nu_b = aQ^4\nu_{b+4}$ and $(g^4)_*\mu_b = a'Q^4\mu_b'$ with μ_b' normalized, so that $|a - 1| < 10^{-6}$ by (3). Since

$$\deg_p(g^4) \leq Q^8 \quad \text{and} \quad \alpha(\nu_b \wedge \mu_b) = \alpha(\mu_b) = \beta_b,$$

where $\deg_p(g^4)$ is the local degree of g^4 at p (Definition 2.2), we obtain

$$a^2\alpha(\nu_{b+4}) - 2aa'\alpha(\nu_{b+4} \wedge \mu_b') + a'^2\alpha(\mu_b') \leq \alpha(\nu_b) - \beta_b$$

(Lemma 8.3 for g^4, the valuations ν_b, μ_b, and the coefficients $1, -1$). The meet $\nu_{b+4} \wedge \mu_b'$ is close to the multiplicity valuation mult_p at p:

$$\alpha(\nu_{b+4} \wedge \mu_b') = 1 \ \text{ if } \mu_b'(L) > 1, \quad \text{and} \quad \alpha(\nu_{b+4} \wedge \mu_b') \leq \alpha(\mu_b') < 1 + 2 \cdot 10^{-6} \ \text{ if } \mu_b'(L) = 1.$$

The first equality holds since $\nu_{b+4}(L) = 1 < \mu_b'(L)$, and the second bound uses

$$\mu_b((g^4)^*H) < Q^4(1 + 11 \cdot 10^{-8})$$

for any line $H \neq L$ through p (Lemma 8.6 and Definition-Lemma 11.2). Thus the skewness β_b of $\nu_b \wedge \mu_b$ drops to less than $1 + 2 \cdot 10^{-6}$ for the images, and since $|a - 1| < 10^{-6}$,

$$\alpha(\nu_{b+4}) < \alpha(\nu_b) - (\beta_b - 1) + 38 \cdot 10^{-6} < \alpha(\nu_b) - \frac{1}{50}.$$

The iterates in the proofs of [LM26, Theorem 1.3] for f without totally periodic curves and of [Yos26, Theorem 1.4] depend on f through a first iterate fixing the points with asymptotic attraction rate q, resp. making the components of a closed subset of $\mathbb{P}^2$ totally invariant [Yos26, Theorem 4.8], and through constants depending on f [LM26, Proposition 2.14(iii)], [Yos26, Proposition 8.3]. Here they are replaced by $g = f^{17280}$ in (1), the exponent 12 of Proposition 7.5, and the bounds on σ and $1 - \frac{e}{Q}$ in (3), which follow from the failure of lc.

Structure of the paper. Section 2 recalls multiplier ideals, valuative trees, and eigenvaluations, and contains the formula for log discrepancies under finite morphisms and a contact inequality along smooth curves. Section 3 studies the ramification of iterates of f along invariant curves of degree at most 4. Section 4 studies the iterates f^k with $6 \mid k$. Section 5 proves that a non-lc point is fixed by an iterate of f whose order does not depend on f. Section 6 classifies the curves along which the ramification divisor has large coefficients, and proves that the non-lc points on such curves are contained in a single invariant line. Section 7 estimates the asymptotic attraction rate at a non-lc point. Section 8 bounds the norms and inner products of the tree measures of

plane divisors (Definition 8.1), in particular of the ramification divisor and the pullbacks of lines, compares the inner products of normalized valuations with those of their pushforwards under finite germs, and bounds the skewness of a normalized valuation τ in terms of the thinness of τ and the values of τ on lines. Section 9 proves log canonicity at the points outside the curves of Section 6. Section 10 reduces the problem to a monomial valuation transverse to the invariant line. Section 11 estimates the eigenvaluations at a fixed point of the invariant line. Section 12 studies the orbit of the monomial valuation. Section 13 proves Theorem 1.5.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs, valuations and ramification.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let $h : X \to Y$ be a finite surjective morphism between smooth varieties. The *ramification divisor* of h is the divisor $R(h) \geq 0$ with $K_X = h^*K_Y + R(h)$. For a prime divisor P on X, the *ramification index* of h along P is $e_P(h) := \operatorname{mult}_P h^*h(P)$, so that $\operatorname{mult}_P R(h) = e_P(h) - 1$, and $e_P(h' \circ h) = e_P(h)e_{h(P)}(h')$ for any finite surjective morphism $h' : Y \to Z$ to a smooth variety Z. For a closed point $x \in X$, the *local degree* of h at x is

$$\deg_x h := \dim_{\mathbb{C}} \mathcal{O}_{X,x}/(h^*\mathfrak{m}_{h(x)})\mathcal{O}_{X,x},$$

where $\mathfrak{m}_y$ denotes the ideal sheaf of a closed point y of a variety. For an integer $Q \geq 1$, an *endomorphism of* $\mathbb{P}^2$ *of degree* Q is a morphism $g : \mathbb{P}^2 \to \mathbb{P}^2$ with

$$g^*\mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^2}(Q).$$

For $Q \geq 2$, the endomorphisms of $\mathbb{P}^2$ of degree Q are exactly the Q-polarized endomorphisms of $\mathbb{P}^2$. Such a g is finite and flat, and $\sum_{x \in g^{-1}(y)} \deg_x g = Q^2$ for any closed point $y \in \mathbb{P}^2$. Thus the topological degree of g is Q^2, while "degree Q" refers to the last display.

Definition 2.3. For a smooth surface X and a divisorial valuation v over X, let

$$A_X(v) := a(v, X, 0) \quad \text{and} \quad A(v) := A_{\mathbb{P}^2}(v).$$

For a pair (X, B) and a closed point $x \in X$, we say that (X, B) is *lc at* x (resp. *klt at* x) if $a(E, X, B) \geq 0$ (resp. $a(E, X, B) > 0$) for any prime divisor E over X whose center on X contains x. We denote by $\operatorname{Nlc}(X, B)$ the set of closed points $x \in X$ at which (X, B) is not lc.

For a divisorial valuation $v = c\,\mathrm{ord}_E$ over X, with E a prime divisor on a normal variety X' and $\pi : X' \to X$ a projective birational morphism, an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X and an ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$, let $v(D) := c\,\mathrm{mult}_E\,\pi^*D$ and $v(\mathfrak{a}) := \min\{v(\phi) \mid \phi \in \mathfrak{a}_\xi\}$, where ξ is the generic point of $\mathrm{center}_X\,v$. If X is smooth, let $K_{X'/X} := K_{X'} - \pi^*K_X$. Then $A_X(v) = c(1 + \mathrm{mult}_E\,K_{X'/X})$ when X is a smooth surface, and since $K_{X'} + (\pi^*B - K_{X'/X}) = \pi^*(K_X + B)$ for any $\mathbb{R}$-divisor B on X,

$$a(v, X, B) = A_X(v) - v(B).$$

Lemma 2.4. *Let $h : X \to Y$ be a finite surjective morphism between smooth surfaces and v a divisorial valuation over X. Then h_*v, defined by $(h_*v)(\phi) := v(h^*\phi)$ for any rational function ϕ on Y, is a divisorial valuation over Y with $\mathrm{center}_Y(h_*v) = h(\mathrm{center}_X\,v)$, and*

$$A_Y(h_*v) = A_X(v) + v(R(h)).$$

In particular, for an endomorphism g of $\mathbb{P}^2$ of degree $Q \geq 2$ and a closed point $x \in \mathbb{P}^2$, the pair $\left(\mathbb{P}^2, \frac{1}{Q-1}R(g)\right)$ is lc at x if and only if $A(g_*v) \leq QA(v)$ for any divisorial valuation v over $\mathbb{P}^2$ whose center contains x.

Proof. Write $v = c\,\mathrm{ord}_E$, where $c > 0$ is a real number and E is a prime divisor over X. Since h is finite, $h(\mathrm{center}_X\,v)$ is closed. If a rational function ϕ on Y is regular at the generic point of $h(\mathrm{center}_X\,v)$, then $h^*\phi$ is regular at the generic point of $\mathrm{center}_X\,v$, and $h^*\phi$ vanishes along $\mathrm{center}_X\,v$ if ϕ vanishes along $h(\mathrm{center}_X\,v)$. Hence $\mathrm{center}_Y(h_*v) = h(\mathrm{center}_X\,v)$. Since h_*v is a divisorial valuation over Y [ZS60, Chapter VI, §6, Lemma 2, Corollary 1, and §14, Theorem 31], there exist projective birational morphisms $\pi : X' \to X$ and $\pi' : Y' \to Y$ from normal varieties, prime divisors $E \subset X'$ and $E' \subset Y'$, and an integer $r \geq 1$ such that $h' := \pi'^{-1} \circ h \circ \pi$ is a morphism near the generic point of E, $h'^*E' = rE$ there, and $(h_*v)(\phi) = cr\,\mathrm{mult}_{E'}\,\mathrm{div}(\pi'^*\phi)$ for any rational function ϕ on Y. Near the generic point of E, the Hurwitz formula gives $K_{X'} = h'^*K_{Y'} + (r-1)E$, while $K_{X'} = \pi^*h^*K_Y + \pi^*R(h) + K_{X'/X}$ and $K_{Y'} = \pi'^*K_Y + K_{Y'/Y}$. Hence $1 + \mathrm{mult}_E\,K_{X'/X} + \mathrm{mult}_E\,\pi^*R(h) = r(1 + \mathrm{mult}_{E'}\,K_{Y'/Y})$. Multiplying by c, we obtain $A_X(v) + v(R(h)) = A_Y(h_*v)$.

For the second sentence, let $B := \frac{1}{Q-1}R(g)$. For any divisorial valuation v over $\mathbb{P}^2$,

$$QA(v) - A(g_*v) = (Q-1)A(v) - v(R(g)) = (Q-1)a(v, \mathbb{P}^2, B)$$

by the first sentence. Since $a(v, \mathbb{P}^2, B) \geq 0$ for any divisorial valuation v whose center contains x if and only if $(\mathbb{P}^2, B)$ is lc at x, the lemma follows. □

2.2. **Inversion of adjunction and multiplier ideals.** For a divisorial valuation v over X centered at x, we also write v for the extension of v to $\widehat{\mathcal{O}}_{X,x}$, which is finite on $\widehat{\mathcal{O}}_{X,x}\backslash\{0\}$ [FJ04, Section 1.5.3 and Proposition 2.10]. For a germ $C = \{t = 0\}$ we set $v(C) := v(t)$, and for $D = \sum_i d_iD_i$ with $D_i = \{\phi_i = 0\}$ near x we set

$$(D \cdot C)_x := \sum_i d_i \dim_{\mathbb{C}} \widehat{\mathcal{O}}_{X,x}/(\phi_i, t).$$

Lemma 2.5. *Let X be a smooth surface, $x \in X$ a closed point, C a germ at x of a smooth curve, algebraic or analytic, and $D \geq 0$ an $\mathbb{R}$-divisor on X with $C \not\subset \mathrm{Supp}\,D$. Then*

$$v(D) \leq (D \cdot C)_x\,(A_X(v) - v(C))$$

for any divisorial valuation v over X centered at x. In particular,

$$v(D) \leq (\mathrm{mult}_x\,D)\,(A_X(v) - v(\mathfrak{m}_x))$$

for any such v.

Proof. Write $D = \sum_i d_iD_i$ with $d_i > 0$ and $D_i = \{\phi_i = 0\}$ near x, and let $r := (D \cdot C)_x$. If $r = 0$, then $x \notin D_i$ for any i, so $v(D) = 0$ and the inequality holds.

Assume that $r > 0$. Possibly replacing t, where $C = \{t = 0\}$, by the truncation of t in degree $< N$ with respect to local parameters of X at x for $N \gg 0$, which changes neither $v(t)$ nor the ideals (ϕ_i, t), we may assume that C is algebraic. Possibly shrinking X to a neighborhood of x, we may assume that C is a smooth irreducible curve on X and $C \cap \operatorname{Supp} D \subset \{x\}$. Then $D|_C = rx$, since the coefficient of x in $D|_C$ is $(D \cdot C)_x = r$. For any real number $0 < \epsilon < 1$, $(C, \frac{1-\epsilon}{r} D|_C)$ is klt, so $(X, C + \frac{1-\epsilon}{r} D)$ is plt near x by inversion of adjunction. By $a(v, X, C + \frac{1-\epsilon}{r} D) \geq 0$,

$$A_X(v) - v(C) - \frac{1-\epsilon}{r} v(D) \geq 0.$$

Letting ϵ tend to 0, we obtain $v(D) \leq r(A_X(v) - v(C))$.

For the last sentence, let y_1, y_2 be local parameters of X at x and $C_1 := \{\lambda_1 y_1 + \lambda_2 y_2 = 0\}$, where $(\lambda_1, \lambda_2) \in \mathbb{C}^2$ is general, so that the tangent line of C_1 at x is not contained in the tangent cone of $\operatorname{Supp} D$ at x. Then C_1 is smooth, $C_1 \not\subset \operatorname{Supp} D$, $(D \cdot C_1)_x = \operatorname{mult}_x D$, and $v(C_1) \geq v(\mathfrak{m}_x)$. Since $\operatorname{mult}_x D \geq 0$, the first inequality for C_1 implies $v(D) \leq (\operatorname{mult}_x D)(A_X(v) - v(\mathfrak{m}_x))$. □

Definition 2.6 ([Laz10, Definition 1.3])**.** Let X be a smooth projective variety, $D \geq 0$ a $\mathbb{Q}$-divisor on X, and $h : X' \to X$ a log resolution of (X, D). The *multiplier ideal* of D is

$$\mathcal{J}(X, D) := h_* \mathcal{O}_{X'} \left(K_{X'/X} - \lfloor h^* D \rfloor \right).$$

Lemma 2.7. *Let X be a smooth projective variety and $D \geq 0$ a $\mathbb{Q}$-divisor on X. Then:*

(1) For a closed point $x \in X$, $\mathcal{J}(X, D)_x = \mathcal{O}_{X,x}$ if and only if (X, D) is klt at x.

(2) For any rational number $0 < \epsilon \ll 1$, $\mathcal{J}(X, (1-\epsilon)D)$ does not depend on ϵ, and

$$\{x \in X \text{ closed} \mid \mathcal{J}(X, (1-\epsilon)D)_x \neq \mathcal{O}_{X,x}\} = \operatorname{Nlc}(X, D).$$

(3) For any prime divisor P on X, $\mathcal{J}(X, D) = \mathcal{O}_X(-\lfloor \operatorname{mult}_P D \rfloor P)$ at the generic point of P.

(4) For any divisorial valuation v over X and any non-zero local section ϕ of $\mathcal{J}(X, D)$ at the generic point of $\operatorname{center}_X v$,

$$v(\phi) + a(v, X, 0) > v(D).$$

In particular, $v(\phi) + A_X(v) > v(D)$ if $\dim X = 2$.

Proof. Let $h : X' \to X$ be a log resolution of (X, D), and write

$$h^* D = \sum_i r_i E_i \quad \text{and} \quad K_{X'/X} = \sum_i b_i E_i,$$

where E_i runs over the prime divisors on X', and $r_i = b_i = 0$ for all but finitely many i. Then $a(E_i, X, D) = 1 + b_i - r_i$, and by [Laz10, Definition 1.3 and (1.1)], a local section ϕ of $\mathcal{O}_X$ belongs to $\mathcal{J}(X, D)$ if and only if

$$\operatorname{mult}_{E_i} \operatorname{div}(h^* \phi) \geq \lfloor r_i \rfloor - b_i \text{ for any } i.$$

(1) and (3) follow from the criterion, since $a(E_i, X, D) = 1 + b_i - r_i$ and h is an isomorphism over the generic point of P.

(2) For $0 < \epsilon \ll 1$ any $\lfloor (1-\epsilon) r_i \rfloor$ does not depend on ϵ, and $\lfloor (1-\epsilon) r_i \rfloor \leq b_i$ if and only if $r_i \leq 1 + b_i$, that is, $a(E_i, X, D) \geq 0$. Thus, applying the criterion to $\phi = 1$, $\mathcal{J}(X, (1-\epsilon)D)_x = \mathcal{O}_{X,x}$ if and only if $a(E_i, X, D) \geq 0$ for any i with $x \in h(E_i)$, that is, (X, D) is lc at x.

(4) Write $v = c \operatorname{ord}_E$. Since $\mathcal{J}(X, D)$ does not depend on the log resolution [Laz10, Proposition 1.6], we may assume that $E = E_i$ for some i. Then

$$\operatorname{mult}_E \operatorname{div}(h^* \phi) \geq \lfloor r_i \rfloor - b_i > r_i - 1 - b_i,$$

and multiplying by c implies the inequality since $a(v, X, 0) = c(1 + b_i)$ and $v(D) = c r_i$. □

2.3. **The valuative tree.**

Definition 2.8 ([FJ04, Sections 1.2, 1.5, 3.4 and 3.6, Definitions 2.23 and 3.23])**.** Let X be a smooth surface and $p \in X$ a closed point, and let $\mathfrak{m}_p$ also denote the maximal ideal of $\widehat{\mathcal{O}}_{X,p}$. For a function $\nu : \widehat{\mathcal{O}}_{X,p} \to [0, +\infty]$ and an ideal $I \subset \widehat{\mathcal{O}}_{X,p}$, let $\nu(I) := \inf\{\nu(\phi) \mid \phi \in I\}$. A *normalized valuation at* p is a function $\nu : \widehat{\mathcal{O}}_{X,p} \to [0, +\infty]$ such that

$$\nu(\phi\psi) = \nu(\phi) + \nu(\psi), \quad \nu(\phi + \psi) \geq \min\{\nu(\phi), \nu(\psi)\}, \quad \nu(c) = 0, \quad \text{and} \quad \nu(\mathfrak{m}_p) = 1$$

for any $\phi, \psi \in \widehat{\mathcal{O}}_{X,p}$ and any $c \in \mathbb{C}^*$.

We write $\nu \leq \mu$ if $\nu(\phi) \leq \mu(\phi)$ for any $\phi \in \widehat{\mathcal{O}}_{X,p}$, and $\nu < \mu$ if $\nu \leq \mu$ and $\nu \neq \mu$. For normalized valuations $\mu \leq \nu$ at p, let

$$[\mu, \nu] := \{\nu' \mid \nu' \text{ is a normalized valuation at } p \text{ and } \mu \leq \nu' \leq \nu\} \quad \text{and} \quad [\mu, \nu) := [\mu, \nu] \backslash \{\nu\}.$$

The *multiplicity valuation* mult_p is the normalized valuation at p defined by

$$\mathrm{mult}_p(\phi) := \sup\{k \in \mathbb{N} \mid \phi \in \mathfrak{m}_p^k\}.$$

An *irreducible formal curve germ at* p is an irreducible element $\phi \in \mathfrak{m}_p$, up to multiplication by units of $\widehat{\mathcal{O}}_{X,p}$. For $\phi, \psi \in \widehat{\mathcal{O}}_{X,p}$, let

$$(\phi \cdot \psi)_p := \dim_{\mathbb{C}} \widehat{\mathcal{O}}_{X,p}/(\phi, \psi) \in \mathbb{N} \cup \{+\infty\}.$$

For an irreducible formal curve germ ϕ at p, the *curve valuation* ν_ϕ is the normalized valuation at p defined by

$$\nu_\phi(\psi) := \frac{(\phi \cdot \psi)_p}{\mathrm{mult}_p \phi} \quad \text{for } \psi \in \widehat{\mathcal{O}}_{X,p}.$$

For a curve $P \subset X$ through p that is smooth at p, or a germ at p of a smooth curve, we write $\nu_P := \nu_\phi$, where ϕ is a local equation of P at p. The *skewness* of a normalized valuation ν at p is

$$\alpha(\nu) := \sup\left\{ \frac{\nu(\phi)}{\mathrm{mult}_p \phi} \middle| \phi \in \mathfrak{m}_p \backslash \{0\} \right\}.$$

Let $D = \sum_i d_i D_i$ be an $\mathbb{R}$-divisor on X, where the D_i are distinct prime divisors, and let ν be a normalized valuation at p such that $D \geq 0$ or $\alpha(\nu) < +\infty$. We let

$$\nu(D) := \sum_i d_i \nu(\phi_i),$$

where ϕ_i is a local equation of D_i at p if $p \in D_i$, and $\phi_i = 1$ if $p \notin D_i$.

For regular parameters x, y at p, so that $\widehat{\mathcal{O}}_{X,p} = \mathbb{C}[[x, y]]$, and real numbers $a, b > 0$, the *monomial valuation* $\mathrm{mon}_{x,y}(a, b)$ is defined by

$$\mathrm{mon}_{x,y}(a, b)(\phi) := \min\{ai + bj \mid c_{ij} \neq 0\} \quad \text{for } \phi = \sum_{i,j \geq 0} c_{ij} x^i y^j \in \mathbb{C}[[x, y]],$$

which is a normalized valuation at p when $\min\{a, b\} = 1$. A normalized valuation at p is *divisorial* if it is a positive multiple of a divisorial valuation over X centered at p.

The *multiplicity* of a normalized valuation ν at p is

$$m(\nu) := \min\{\mathrm{mult}_p \phi \mid \phi \text{ is an irreducible formal curve germ at } p \text{ with } \nu \leq \nu_\phi\},$$

where $m(\nu) := +\infty$ if there does not exist such ϕ [FJ04, Section 3.4], and the *thinness* $A(\nu)$ of ν is that of [FJ04, Section 3.6], given by the formula in Theorem 2.9(3). The *quasimonomial* and the *infinitely singular* normalized valuations at p are those of [FJ04, Definition 2.23].

Theorem 2.9 ([FJ04, Theorem 3.14, Corollary 3.15, Proposition 3.25, Theorem 3.26, Sections 3.4 and 3.6, Proposition 3.44, Remark 3.50])**.** *Let X be a smooth surface and $p \in X$ a closed point. Then:*

(1) *The set of normalized valuations at p, ordered by $\leq$, is a rooted tree with root* mult_p. *In particular, for any normalized valuation ν at p,* $\mathrm{mult}_p \leq \nu$ *and* $[\mathrm{mult}_p, \nu]$ *is totally ordered by $\leq$. Any two normalized valuations ν, μ at p have a greatest lower bound $\nu \wedge \mu$. The skewness α takes values in $[1, +\infty]$, $\alpha(\mathrm{mult}_p) = 1$, and for any normalized valuation ν at p, α restricts to a strictly increasing bijection of $[\mathrm{mult}_p, \nu]$ onto $[1, \alpha(\nu)]$.*

(2) *For any normalized valuation ν at p and any irreducible formal curve germ ϕ at p,*

$$\nu(\phi) = \mathrm{mult}_p \phi \cdot \alpha(\nu \wedge \nu_\phi).$$

For any two distinct irreducible formal curve germs ϕ, ψ at p,

$$(\phi \cdot \psi)_p = \mathrm{mult}_p \phi \cdot \mathrm{mult}_p \psi \cdot \alpha(\nu_\phi \wedge \nu_\psi).$$

(3) *The multiplicity m takes values in $\mathbb{N}^+ \cup \{+\infty\}$. For any normalized valuation ν at p, m is increasing on $[\mathrm{mult}_p, \nu]$ and*

$$A(\nu) = 2 + \int_{\mathrm{mult}_p}^{\nu} m(\mu) d\alpha(\mu).$$

In particular, $A(\nu) \geq 1 + \alpha(\nu)$.

(4) *If ν is a divisorial normalized valuation at p, then $A(\nu) = A_X(\nu)$.*

(5) *For any irreducible formal curve germ ϕ at p, $\alpha(\nu_\phi) = +\infty$, and there exists a normalized valuation $\mu < \nu_\phi$ at p such that $m(\nu) = \mathrm{mult}_p \phi$ for any $\nu \in [\mu, \nu_\phi)$.*

Lemma 2.10. *Let $p \in \mathbb{P}^2$ be a closed point and ν a normalized valuation at p with $\alpha(\nu) < +\infty$. Then:*

(1) $\mathrm{mult}_p \leq \nu \leq \alpha(\nu)\,\mathrm{mult}_p$.

(2) *For any non-zero $\phi, \psi \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p}$ without common irreducible factor,*

$$(\phi \cdot \psi)_p \geq \frac{\nu(\phi)\nu(\psi)}{\alpha(\nu)}.$$

(3) *For any $\mathbb{R}$-divisors $D_1, D_2 \geq 0$ on $\mathbb{P}^2$ without common component,*

$$\nu(D_1)\nu(D_2) \leq \alpha(\nu) \deg D_1 \deg D_2.$$

Proof. (1) follows from Theorem 2.9(1) and the definition of α.

For (2), we first let ϕ, ψ be distinct irreducible formal curve germs at p, and let $s := \alpha(\nu \wedge \nu_\phi)$ and $t := \alpha(\nu \wedge \nu_\psi)$. Since $\nu \wedge \nu_\phi, \nu \wedge \nu_\psi \in [\mathrm{mult}_p, \nu]$ and $[\mathrm{mult}_p, \nu]$ is totally ordered, possibly switching ϕ and ψ, we may assume that $\nu \wedge \nu_\phi \leq \nu \wedge \nu_\psi$. Since $\nu \wedge \nu_\phi \leq \nu_\phi$ and $\nu \wedge \nu_\phi \leq \nu \wedge \nu_\psi \leq \nu_\psi$, we have $\nu \wedge \nu_\phi \leq \nu_\phi \wedge \nu_\psi$. Thus by Theorem 2.9(1), $s \leq \alpha(\nu_\phi \wedge \nu_\psi)$ and $s \leq t \leq \alpha(\nu)$. By Theorem 2.9(2),

$$(\phi \cdot \psi)_p = \mathrm{mult}_p \phi \cdot \mathrm{mult}_p \psi \cdot \alpha(\nu_\phi \wedge \nu_\psi) \geq \mathrm{mult}_p \phi \cdot \mathrm{mult}_p \psi \cdot \frac{st}{\alpha(\nu)} = \frac{\nu(\phi)\nu(\psi)}{\alpha(\nu)}.$$

For non-zero ϕ, ψ without common irreducible factor, we write $\phi = u \prod_{i=1}^{n_1} \phi_i$ and $\psi = u' \prod_{j=1}^{n_2} \psi_j$, where u, u' are units and ϕ_i, ψ_j are irreducible formal curve germs at p with $\phi_i \neq \psi_j$ for any i, j. Then

$$(\phi \cdot \psi)_p = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} (\phi_i \cdot \psi_j)_p \geq \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \frac{\nu(\phi_i)\nu(\psi_j)}{\alpha(\nu)} = \frac{\nu(\phi)\nu(\psi)}{\alpha(\nu)}.$$

For (3), we write $D_1 = \sum_i a_i P_i$ and $D_2 = \sum_j b_j P'_j$, where $a_i, b_j \geq 0$ and the P_i, P'_j are prime divisors with $P_i \neq P'_j$ for any i, j. Let ϕ_i and ψ_j be the local equations of P_i and P'_j at p as in Definition 2.8. Since $P_i \neq P'_j$, $(\phi_i \cdot \psi_j)_p = (P_i \cdot P'_j)_p < +\infty$, so ϕ_i and ψ_j have no common irreducible factor. By (2) and Bezout's theorem,

$$\nu(\phi_i)\nu(\psi_j) \leq \alpha(\nu)(P_i \cdot P'_j)_p \leq \alpha(\nu) \deg P_i \deg P'_j.$$

Multiplying by $a_i b_j$ and adding over i, j, we obtain (3). The lemma follows. □

2.4. Attraction rates and eigenvaluations.

Definition 2.11 ([FJ07, Definitions 2.2, 2.3, 4.1 and 4.3])**.** Let X be a smooth surface, $p \in X$ a closed point, and $h : (X, p) \to (X, p)$ a finite morphism of germs. For a normalized valuation ν at p, let $(h_*\nu)(\phi) := \nu(h^*\phi)$ for $\phi \in \widehat{\mathcal{O}}_{X,p}$, and let

$$c(h, \nu) := \nu(h^*\mathfrak{m}_p) \quad \text{and} \quad h_\bullet\nu := \frac{h_*\nu}{c(h, \nu)},$$

where $h^*\mathfrak{m}_p$ denotes the ideal of $\widehat{\mathcal{O}}_{X,p}$ generated by $\{h^*\phi \mid \phi \in \mathfrak{m}_p\}$. Since h is finite, $h^*\mathfrak{m}_p$ is $\mathfrak{m}_p$-primary, so $1 \le c(h, \nu) < +\infty$ and $h_\bullet\nu$ is a normalized valuation at p. The *attraction rate* of h is $c(h, \mathrm{mult}_p)$, and the *asymptotic attraction rate* of h is

$$c_\infty(h, p) := \lim_{n\to+\infty} c(h^n, \mathrm{mult}_p)^{1/n}.$$

An *eigenvaluation* of h is a normalized valuation ω at p such that $h_\bullet\omega = \omega$ and $c(h, \omega) = c_\infty(h, p)$.

For an endomorphism g of $\mathbb{P}^2$ and a closed point $p \in \mathbb{P}^2$ with $g(p) = p$, we write $c(g, \nu)$, $g_\bullet\nu$, $c_\infty(g, p)$, and eigenvaluations of g at p for those of the germ $g : (\mathbb{P}^2, p) \to (\mathbb{P}^2, p)$.

Theorem 2.12 ([FJ07, Theorem 4.2 and Lemma 4.7])**.** *Let X be a smooth surface, $p \in X$ a closed point, x, y regular parameters at p, and $h : (X, p) \to (X, p)$ a finite morphism of germs. Let*

$$\psi := \frac{\partial(h^*x)}{\partial x}\frac{\partial(h^*y)}{\partial y} - \frac{\partial(h^*x)}{\partial y}\frac{\partial(h^*y)}{\partial x} \in \widehat{\mathcal{O}}_{X,p}$$

be the Jacobian determinant of h. Then:

1. *If $c_\infty(h, p) > 1$, then there exists an eigenvaluation ω of h such that one of the following holds:*
   - (a) *ω is quasimonomial.*
   - (b) *ω is infinitely singular or the curve valuation of an irreducible analytic curve germ, and there exists a normalized valuation $\mu < \omega$ at p such that $c(h, \nu) = c_\infty(h, p)$ and $h_\bullet\nu > \nu$ for any $\nu \in [\mu, \omega)$.*
2. *For any normalized valuation ν at p,*

$$c(h, \nu)A(h_\bullet\nu) = A(\nu) + \nu(\psi).$$

Proof. In (1)(b), [FJ07, Theorem 4.2] states $c(h, \mu) = c_\infty(h, p)$. Since $\mu \le \nu \le \omega$ implies $\mu(h^*\mathfrak{m}_p) \le \nu(h^*\mathfrak{m}_p) \le \omega(h^*\mathfrak{m}_p)$, and $c(h, \omega) = c_\infty(h, p)$, we have $c(h, \nu) = c_\infty(h, p)$ for any $\nu \in [\mu, \omega)$. The other statements are [FJ07, Theorem 4.2 and Lemma 4.7]. □

3. Log canonicity along invariant curves

Theorem 3.1. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 4$ and $C \subset \mathbb{P}^2$ an irreducible curve of degree ≤ 4 with $g(C) = C$. Let $\pi : \overline{C} \to C$ be the normalization and $h : \overline{C} \to \overline{C}$ the morphism with $\pi \circ h = g \circ \pi$, and let $x \in C$ be a closed point with $h^2(z) \ne z$ for any $z \in \pi^{-1}(x)$. Then $\left(\mathbb{P}^2, \frac{1}{Q^n-1}R(g^n)\right)$ is lc at x for any integer $n \ge 8192$.*

We prove Theorem 3.1 in Subsection 3.2.

3.1. Restriction of the ramification divisor to an invariant curve.

Lemma 3.2. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 2$ and $C \subset \mathbb{P}^2$ an irreducible curve of degree d with $g(C) = C$. Let $e := e_C(g)$, write $g^*C = eC + D$, let $\pi : \overline{C} \to C$ be the normalization, let $h : \overline{C} \to \overline{C}$ be the morphism with $\pi \circ h = g \circ \pi$, and let $\Delta := \mathrm{Diff}_{\overline{C}}(0)$ be the different of $(\mathbb{P}^2, C)$ on $\overline{C}$* [Kol13, Definition 4.2]*. Then $\Delta \ge 0$* [Kol13, Proposition 4.5(1)]*, $\pi^*\omega_C = \omega_{\overline{C}}(\Delta)$, $\deg h = Q$, $1 \le e \le Q$, $\deg \pi^*(D|_C) = d^2(Q - e)$,*

$$\deg \Delta = (d - 1)(d - 2) - 2h^1(\overline{C}, \mathcal{O}_{\overline{C}}),$$

and the following hold.

(1) For any integer $n \geq 1$, let $D_n := \sum_{i=0}^{n-1} e^{n-1-i}(g^i)^* D$ and $R'_n := R(g^n) - (e^n - 1)C$. Then $(g^n)^* C = e^n C + D_n$, $e_C(g^n) = e^n$, $R'_n \geq 0$, $C \not\subset \operatorname{Supp} R'_n$, and on $\overline{C}$,

$$\pi^*(R'_n|_C) = \sum_{i=0}^{n-1} e^{n-1-i}(h^i)^* \pi^*(D|_C) + R(h^n) + \Delta - (h^n)^* \Delta.$$

(2) For any $z \in \overline{C}$,

$$d^2(Q - e) \geq (\operatorname{mult}_z \Delta - 1)Q + 1 - \deg \Delta.$$

(3) If $(\mathbb{P}^2, C)$ is not lc, then $d \geq 3$ and

$$e \leq \left(1 - \frac{1}{2d^2}\right) Q.$$

(4) If $d \geq 4$ and $(\mathbb{P}^2, C)$ is lc, then

$$e \leq \left(1 - \frac{1}{d^2}\right) Q.$$

Proof. The line bundle $\mathcal{L} := \pi^*(\mathcal{O}_{\mathbb{P}^2}(1)|_C)$ has degree d, and $h^*\mathcal{L} \cong \pi^*(\mathcal{O}_{\mathbb{P}^2}(Q)|_C)$ has degree Qd. Thus $(\deg h)d = Qd$, and $\deg h = Q$. Since C is a Cartier divisor of degree d on $\mathbb{P}^2$, $\omega_C \cong \mathcal{O}_C(d-3)$ by adjunction, and $\pi^*\omega_C = \omega_{\overline{C}}(\Delta)$ by the definition of Δ [Kol13, Definition 4.2], so

$$d(d-3) = \deg \pi^* \omega_C = 2h^1(\overline{C}, \mathcal{O}_{\overline{C}}) - 2 + \deg \Delta,$$

which is the formula for $\deg \Delta$. Since $K_{\mathbb{P}^2} + C$ is Cartier, Δ has integer coefficients [Kol13, Definition 4.2].

Since $g(C) = C$, we have $e \geq 1$, $D \geq 0$, and $C \not\subset \operatorname{Supp} D$. Since $\deg g^* C = Qd$, we have $\deg D = d(Q - e)$. Thus $1 \leq e \leq Q$ and

$$\deg \pi^*(D|_C) = D \cdot C = d^2(Q - e). \tag{3.1}$$

For any integer $i \geq 0$, $C \not\subset \operatorname{Supp}(g^i)^* D$, since $g^i(C) = C \not\subset \operatorname{Supp} D$. If $(g^n)^* C = e^n C + D_n$ for some $n \geq 1$, then

$$(g^{n+1})^* C = (g^n)^*(eC + D) = e^{n+1} C + e D_n + (g^n)^* D = e^{n+1} C + D_{n+1}.$$

Since $g^* C = eC + D_1$, induction on n gives $(g^n)^* C = e^n C + D_n$ and $C \not\subset \operatorname{Supp} D_n$ for any $n \geq 1$. Thus $e_C(g^n) = e^n$, so $\operatorname{mult}_C R(g^n) = e^n - 1$ by Definition 2.2, that is, $R'_n \geq 0$ and $C \not\subset \operatorname{Supp} R'_n$. In particular, $R'_1|_C$ and $D|_C$ are Cartier divisors on C with $R'_1|_C \geq 0$ and $D|_C \geq 0$.

(1) Fix $z \in \overline{C}$, and let $w := h(z)$, $x := \pi(z)$, and $y := \pi(w) = g(x)$. Let σ and τ be local equations of C at x and at y respectively, and let Ω and Ω' be generators of $\omega_{\mathbb{P}^2}$ near x and near y respectively. Since $g^* C = eC + D$ and $K_{\mathbb{P}^2} = g^* K_{\mathbb{P}^2} + (e-1)C + R'_1$, near x we may write

$$g^* \tau = \sigma^e a \quad \text{and} \quad g^* \Omega' = \sigma^{e-1} \kappa \Omega,$$

where a and κ are regular functions near x with $\operatorname{div}(a) = D$ and $\operatorname{div}(\kappa) = R'_1$. Let Res_C denote the Poincaré residue along C, and let

$$\eta := \pi^* \operatorname{Res}_C(\Omega/\sigma) \quad \text{and} \quad \eta' := \pi^* \operatorname{Res}_C(\Omega'/\tau),$$

rational differentials on $\overline{C}$. Since Ω/σ and Ω'/τ generate $\omega_{\mathbb{P}^2}(C)$ near x and near y respectively, the definition of Δ [Kol13, Definition 4.2] implies

$$\operatorname{mult}_z \operatorname{div}(\eta) = -\operatorname{mult}_z \Delta \quad \text{and} \quad \operatorname{mult}_w \operatorname{div}(\eta') = -\operatorname{mult}_w \Delta.$$

We claim that

$$\pi^* \left((\kappa/a)|_C\right) \eta = e h^* \eta'. \tag{3.2}$$

Both sides of (3.2) are rational differentials on $\overline{C}$ and $\pi \circ h = g \circ \pi$, so it suffices to prove

$$(\kappa/a)|_C \operatorname{Res}_C(\Omega/\sigma) = e(g|_C)^* \operatorname{Res}_C(\Omega'/\tau)$$

near a general point x' of C near x. Since x' is general, C is smooth at x' and at $g(x')$, and $a(x') \neq 0$. Near x' and near $g(x')$ we may write

$$\frac{\Omega}{\sigma} = \frac{d\sigma}{\sigma} \wedge \beta \quad \text{and} \quad \frac{\Omega'}{\tau} = \frac{d\tau}{\tau} \wedge \beta'$$

with regular 1-forms β and β', so that $\operatorname{Res}_C(\Omega/\sigma) = \beta|_C$ and $\operatorname{Res}_C(\Omega'/\tau) = \beta'|_C$. Then

$$\frac{\kappa}{a} \cdot \frac{\Omega}{\sigma} = g^* \frac{\Omega'}{\tau} = \left(e \frac{d\sigma}{\sigma} + \frac{da}{a} \right) \wedge g^* \beta'.$$

Since κ/a and da/a are regular near x', taking residues along C implies $(\kappa/a)|_C \beta|_C = e(g|_C)^*(\beta'|_C)$ near x'. This proves (3.2).

Let $\lambda := \deg_z h$. If ζ is a local parameter at w, then $h^*\zeta$ is a unit times the λ-th power of a local parameter at z, so

$$\operatorname{mult}_z \operatorname{div}(h^*\eta') = \lambda(\operatorname{mult}_w \operatorname{div}(\eta') + 1) - 1.$$

Since κ and a are local equations of R_1' and D near x, taking $\operatorname{mult}_z \operatorname{div}$ of the left side and of the right side of (3.2), we obtain

$$\operatorname{mult}_z \pi^*(R_1'|_C) = \operatorname{mult}_z \pi^*(D|_C) + (\lambda - 1) + \operatorname{mult}_z \Delta - \lambda \operatorname{mult}_{h(z)} \Delta. \tag{3.3}$$

Since $\operatorname{mult}_z R(h) = \lambda - 1$ and $\operatorname{mult}_z h^*\Delta = \lambda \operatorname{mult}_{h(z)} \Delta$, (3.3) is the coefficient at z of the formula of (1) for $n = 1$. As z is arbitrary, the formula of (1) holds for $n = 1$. For any $n \geq 1$, h^n satisfies $\pi \circ h^n = g^n \circ \pi$, and $(g^n)^* C = e^n C + D_n$, so the case $n = 1$ applied to g^n gives

$$\pi^*(R_n'|_C) = \pi^*(D_n|_C) + R(h^n) + \Delta - (h^n)^*\Delta.$$

Since $\pi \circ h^i = g^i \circ \pi$, we have $\pi^*(((g^i)^* D)|_C) = (h^i)^* \pi^*(D|_C)$ for any $i \geq 0$, and (1) follows.

(2) Let $m := \operatorname{mult}_z \Delta$. For any $z' \in h^{-1}(z)$, (3.3) at z' and $\pi^*(R_1'|_C) \geq 0$ imply

$$\operatorname{mult}_{z'} \pi^*(D|_C) \geq (m-1) \deg_{z'} h + 1 - \operatorname{mult}_{z'} \Delta.$$

Since $\sum_{z' \in h^{-1}(z)} \deg_{z'} h = Q$, $\#h^{-1}(z) \geq 1$, $\Delta \geq 0$, and $\pi^*(D|_C) \geq 0$, (3.1) implies

$$\begin{aligned} d^2(Q-e) &\geq \sum_{z' \in h^{-1}(z)} \operatorname{mult}_{z'} \pi^*(D|_C) \geq (m-1)Q + \#h^{-1}(z) - \sum_{z' \in h^{-1}(z)} \operatorname{mult}_{z'} \Delta \\ &\geq (m-1)Q + 1 - \deg \Delta. \end{aligned}$$

(3) Assume that $(\mathbb{P}^2, C)$ is not lc. Since $(\mathbb{P}^2, C)$ is lc outside C, it is not lc at some closed point of C, so $(\overline{C}, \Delta)$ is not lc by [Kaw07, Theorem]. Since Δ has integer coefficients, there exists $z \in \overline{C}$ with $\operatorname{mult}_z \Delta \geq 2$. Thus $2 \leq \deg \Delta \leq (d-1)(d-2)$, so $d \geq 3$.

Suppose that $e = Q$. Then $e_C(g^n) = Q^n$ for any $n \geq 1$ by (1), so (2) for g^n and $\operatorname{mult}_z \Delta \geq 2$ imply

$$0 \geq Q^n + 1 - \deg \Delta$$

for any $n \geq 1$, a contradiction. Thus $Q - e \geq 1$.

If $Q \geq 2(\deg \Delta - 1)$, then (2) and $\operatorname{mult}_z \Delta \geq 2$ imply $d^2(Q-e) \geq Q + 1 - \deg \Delta \geq \frac{Q}{2}$. If $Q < 2(\deg \Delta - 1)$, then $\deg \Delta \leq (d-1)(d-2) < d^2$ implies $\frac{Q}{2d^2} < 1 \leq Q - e$. In both cases

$$Q - e \geq \frac{Q}{2d^2},$$

which is (3).

(4) Assume that $d \geq 4$ and $(\mathbb{P}^2, C)$ is lc. Then $(\overline{C}, \Delta)$ is lc by [Kaw07, Theorem], so $\operatorname{mult}_z \Delta \in \{0, 1\}$ for any $z \in \overline{C}$. Let $\Gamma := \operatorname{Supp} \Delta$ and $\Gamma' := h^{-1}(\Gamma)$. Thus $\#\Gamma = \deg \Delta$. By the Riemann–Hurwitz formula,

$$\deg R(h) = (Q-1)\left(2 - 2h^1(\overline{C}, \mathcal{O}_{\overline{C}})\right) \geq 0,$$

so $h^1(\overline{C}, \mathcal{O}_{\overline{C}}) \le 1$, $\deg R(h) \le 2(Q-1)$, and $\deg \Delta \ge (d-1)(d-2) - 2 \ge 4$. Since

$$\sum_{z \in \Gamma'} \deg_z h = Q\#\Gamma \quad \text{and} \quad \sum_{z \in \Gamma'} (\deg_z h - 1) \le \deg R(h),$$

we have

$$\#\Gamma' \ge Q \deg \Delta - 2(Q-1).$$

For any $z \in \Gamma'$, $\operatorname{mult}_{h(z)} \Delta = 1$, so (3.3) and $\pi^*(R_1'|_C) \ge 0$ imply $\operatorname{mult}_z \pi^*(D|_C) \ge 1 - \operatorname{mult}_z \Delta$. Thus by (3.1),

$$d^2(Q-e) \ge \sum_{z \in \Gamma'} (1 - \operatorname{mult}_z \Delta) \ge \#\Gamma' - \deg \Delta \ge (\deg \Delta - 2)(Q-1) \ge 2(Q-1) \ge Q,$$

which is (4). □

Lemma 3.3. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 2$ and $C \subset \mathbb{P}^2$ a smooth irreducible curve with $g(C) = C$. Let $e := e_C(g)$ and $h := g|_C$, and write*

$$g^*C = eC + D \quad \text{and} \quad R(g^n) = (e^n - 1)C + R_n' \quad \text{for } n \ge 1.$$

Let $x \in C$ be a closed point, and for any integer $i \ge 0$ let

$$c_i := \prod_{j=0}^{i-1} \frac{\deg_{h^j(x)} h}{Q} \quad \text{and} \quad \alpha_i := \frac{\operatorname{mult}_{h^i(x)}(D|_C)}{Q}.$$

Then:

(1) $\deg C \le 3$.

(2) For any integer $n \ge 1$,

$$\frac{1 + \operatorname{mult}_x(R_n'|_C)}{Q^n} = c_n + \sum_{i=0}^{n-1} \alpha_i \left(\frac{e}{Q}\right)^{n-1-i} c_i.$$

(3) For any integer $n \ge 1$ with $1 + \operatorname{mult}_x(R_n'|_C) \le Q^n$, the pair $\left(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n)\right)$ is lc at x.

Proof. (1) Let $d := \deg C$. Since C is smooth, $\pi = \mathrm{id}_C$ and $\Delta = 0$, so Lemma 3.2 implies $\deg h = Q$ and $(d-1)(d-2) = 2h^1(C, \mathcal{O}_C)$. By the Riemann–Hurwitz formula,

$$\deg R(h) = (Q-1)\left(2 - 2h^1(C, \mathcal{O}_C)\right) \ge 0,$$

so $h^1(C, \mathcal{O}_C) \le 1$. Thus $(d-1)(d-2) \le 2$, and $d \le 3$.

(2) By Lemma 3.2(1), with $\pi = \mathrm{id}_C$ and $\Delta = 0$,

$$R_n'|_C = R(h^n) + \sum_{i=0}^{n-1} e^{n-1-i} (h^i)^*(D|_C).$$

Local degrees on the smooth curve C multiply under composition, so

$$\deg_x(h^i) = \prod_{j=0}^{i-1} \deg_{h^j(x)} h = Q^i c_i$$

for any $i \ge 0$. Thus

$$\operatorname{mult}_x R(h^n) = \deg_x(h^n) - 1 = Q^n c_n - 1,$$
$$\operatorname{mult}_x (h^i)^*(D|_C) = \deg_x(h^i) \operatorname{mult}_{h^i(x)}(D|_C) = Q^{i+1} \alpha_i c_i.$$

Substituting the formulas for $\operatorname{mult}_x R(h^n)$ and $\operatorname{mult}_x (h^i)^*(D|_C)$ into the formula for $R_n'|_C$ above and dividing by Q^n, we obtain (2).

(3) Assume that $1 + \operatorname{mult}_x(R_n'|_C) \le Q^n$. Possibly shrinking $\mathbb{P}^2$ to a neighborhood of x, we may assume that $\operatorname{Supp}(R_n'|_C) \subset \{x\}$. Then $\operatorname{mult}_x(R_n'|_C) \le Q^n - 1$, so $\left(C, \frac{1}{Q^n - 1} R_n'|_C\right)$ is lc. Since

C is smooth, the different of $\frac{1}{Q^n-1}R'_n$ on C is $\frac{1}{Q^n-1}R'_n|_C$, so $\left(\mathbb{P}^2, C+\frac{1}{Q^n-1}R'_n\right)$ is lc near C by inversion of adjunction. Since $e \le Q$ by Lemma 3.2, $0 \le e^n-1 \le Q^n-1$ and

$$\frac{1}{Q^n-1}R(g^n) = \frac{e^n-1}{Q^n-1}C + \frac{1}{Q^n-1}R'_n \le C + \frac{1}{Q^n-1}R'_n.$$

Thus $\left(\mathbb{P}^2, \frac{1}{Q^n-1}R(g^n)\right)$ is lc at x. □

Lemma 3.4. *Let $\overline{C}$ be a smooth projective curve, let $h : \overline{C} \to \overline{C}$ be a finite morphism of degree $Q \ge 4$, and let $z \in \overline{C}$ be a closed point such that $h^2(z) \ne z$. For any integer $i \ge 0$, let*

$$c_i := \prod_{j=0}^{i-1} \frac{\deg_{h^j(z)} h}{Q}.$$

Then:

(1) $\sum_{i\ge 0} c_i \le 8$, and $c_i \le \frac{8}{i+1}$ for any integer $i \ge 0$.

(2) For any integer $n \ge 2$, any real numbers $t \in (0,1]$ and $\beta \ge 0$, and any real numbers $\alpha_0, \dots, \alpha_{n-1} \in [0, \beta(1-t)]$,

$$c_n + \sum_{i=0}^{n-1} \alpha_i t^{n-1-i} c_i < \frac{8+32\beta}{n}.$$

Proof. For any integer $i \ge 0$, let $z_i := h^i(z)$ and $u_i := \deg_{z_i} h/Q$. For any $w \in \overline{C}$,

$$\sum_{y \in h^{-1}(w)} \deg_y h = Q. \tag{3.4}$$

Thus $u_i \in (0,1]$ and $c_{i+1} = c_i u_i \le c_i$ for any $i \ge 0$.

Since $\operatorname{mult}_y R(h) = \deg_y h - 1$ for any $y \in \overline{C}$, the Riemann–Hurwitz formula implies

$$\sum_{y\in\overline{C}} (\deg_y h - 1) = \deg R(h) = (Q-1)\left(2 - 2h^1(\overline{C}, \mathcal{O}_{\overline{C}})\right) \le 2Q-2.$$

Hence $\sum_{j=0}^{2} \deg_{y_j} h \le 2Q+1$ for any three distinct points $y_0, y_1, y_2 \in \overline{C}$, and the inequality of arithmetic and geometric means implies

$$\prod_{j=0}^{2} \frac{\deg_{y_j} h}{Q} \le \left(\frac{2Q+1}{3Q}\right)^3 \le \left(\frac{3}{4}\right)^3 = \frac{27}{64}, \tag{3.5}$$

where the second inequality holds since $Q \ge 4$.

(1) Let $m := \min\{i \in \mathbb{N} \mid h^2(z_i) = z_i\}$, where $\min \emptyset := +\infty$. Since $h^2(z) \ne z$, $m \ge 1$. For any integer $i \ge 0$ with $i+2 \le m$, the points z_i, z_{i+1}, z_{i+2} are distinct, since $h^2(z_i) \ne z_i$ and $h^2(z_{i+1}) \ne z_{i+1}$ (as $i+1 < m$), so $h(z_i) \ne z_i$ and $h(z_{i+1}) \ne z_{i+1}$. Let i be an integer with $0 \le i < m$, and write $i = 3k+j$ with integers $k \ge 0$ and $0 \le j \le 2$. For any integer $0 \le k' \le k-1$, $3k'+2 < i < m$, so (3.5) for the points $z_{3k'}, z_{3k'+1}, z_{3k'+2}$ implies

$$c_i \le c_{3k} = \prod_{k'=0}^{k-1} u_{3k'} u_{3k'+1} u_{3k'+2} \le \left(\frac{27}{64}\right)^k.$$

Therefore

$$\sum_{0\le i<m} c_i \le 3 \sum_{k\ge 0} \left(\frac{27}{64}\right)^k = \frac{192}{37} < 6. \tag{3.6}$$

Assume that $m < +\infty$. Let $r := \min\{r' \in \{1,2\} \mid h^{r'}(z_m) = z_m\}$. Then $z_{i+r} = z_i$ and $h^2(z_i) = z_i$ for any $i \ge m$. Let $y := z_{m+r-1}$. Then $h(y) = z_m = h(z_{m-1})$, and $y \ne z_{m-1}$

since $h^2(y) = y$ and $h^2(z_{m-1}) \neq z_{m-1}$. By (3.4) for $w = z_m$, $\deg_y h + \deg_{z_{m-1}} h \leq Q$, that is, $u_{m-1} \leq 1 - u_{m+r-1}$. Let $\gamma := \prod_{j=0}^{r-1} u_{m+j}$. Since $\gamma \leq u_{m+r-1}$,

$$u_{m-1} \leq 1 - \gamma.$$

In particular, $\gamma < 1$ as $u_{m-1} > 0$. Since $u_{i+r} = u_i$ for any $i \geq m$, for any integers $k \geq 0$ and $0 \leq j \leq r-1$,

$$c_{m+kr+j} = c_{m-1} u_{m-1} \gamma^k \prod_{j'=0}^{j-1} u_{m+j'} \leq c_{m-1}(1-\gamma)\gamma^k.$$

Thus

$$\sum_{i \geq m} c_i \leq r c_{m-1}(1-\gamma) \sum_{k \geq 0} \gamma^k = r c_{m-1} \leq 2.$$

Together with (3.6), $\sum_{i \geq 0} c_i < 8$ whether m is finite or not. Since $c_{i+1} \leq c_i$ for any $i \geq 0$,

$$(i+1)c_i \leq \sum_{i'=0}^{i} c_{i'} \leq 8$$

for any $i \geq 0$. (1) follows.

(2) For any integer $N \geq 0$ and any $t \in (0, 1]$,

$$(1-t)\sum_{j=0}^{N} t^j = 1 - t^{N+1} \leq 1, \quad \text{hence} \quad (1-t)t^N \leq \frac{1}{N+1}. \tag{3.7}$$

Let $k := \lfloor n/2 \rfloor$. Since $n \geq 2$,

$$k \leq n-1, \quad n-k \geq \frac{n}{2}, \quad \text{and} \quad k+2 > \frac{n}{2}.$$

Since $t \leq 1$ and $\alpha_i \leq \beta(1-t)$ for any i, (1) and (3.7) for $N = n-1-k$ imply

$$\sum_{i=0}^{k} \alpha_i t^{n-1-i} c_i \leq \beta(1-t)t^{n-1-k} \sum_{i=0}^{k} c_i \leq 8\beta(1-t)t^{n-1-k} \leq \frac{8\beta}{n-k} \leq \frac{16\beta}{n}.$$

By (1), $c_i \leq \frac{8}{k+2}$ for any $i \geq k+1$. Thus (3.7) for $N = n-k-2$ implies

$$\sum_{i=k+1}^{n-1} \alpha_i t^{n-1-i} c_i \leq \frac{8\beta}{k+2}(1-t) \sum_{j=0}^{n-k-2} t^j \leq \frac{8\beta}{k+2} \leq \frac{16\beta}{n},$$

where the sum over i and the sum over j are 0 if $k = n-1$. Since $c_n \leq \frac{8}{n+1} < \frac{8}{n}$ by (1), we obtain

$$c_n + \sum_{i=0}^{n-1} \alpha_i t^{n-1-i} c_i < \frac{8}{n} + \frac{16\beta}{n} + \frac{16\beta}{n} = \frac{8+32\beta}{n}.$$

The lemma follows. □

Lemma 3.5. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 4$, and let $C \subset \mathbb{P}^2$ be an irreducible curve of degree $d \leq 4$ with $g(C) = C$. Let $\pi : \overline{C} \to C$ be the normalization, let $h : \overline{C} \to \overline{C}$ be the morphism with $\pi \circ h = g \circ \pi$, let $e := e_C(g)$, and for any integer $n \geq 1$ write*

$$R(g^n) = (e^n - 1)C + R'_n.$$

Let $x \in C$ be a closed point such that $h^2(z) \neq z$ for any $z \in \pi^{-1}(x)$. Then for any integer $n \geq 2$ and any divisorial valuation v over $\mathbb{P}^2$ centered at x,

$$v(R'_n) < \left(\frac{8+32d^2}{n} Q^n + 6\right) A(v).$$

Proof. Write $g^*C = eC + D$, fix an integer $n \geq 2$ and $z \in \pi^{-1}(x)$, and let $\Delta := \mathrm{Diff}_{\overline{C}}(0)$. By Lemma 3.2, $\deg h = Q$, $\frac{e}{Q} \in (0,1]$, $\deg \pi^*(D|_C) = d^2(Q-e)$, $\deg \Delta \leq (d-1)(d-2) \leq 6$, $R'_n \geq 0$, and $C \not\subset \operatorname{Supp} R'_n$. For $i \geq 0$ let $c_i := \prod_{j=0}^{i-1} \frac{\deg_{h^j(z)} h}{Q}$ and $a_i := \operatorname{mult}_{h^i(z)} \pi^*(D|_C)$, and let $r_z := \operatorname{mult}_z \pi^*(R'_n|_C)$. Since $\deg_z(h^i) = Q^i c_i$, $\Delta \geq 0$, and $\operatorname{mult}_z \Delta \leq 6$, the formula of Lemma 3.2(1) at z implies

$$\frac{r_z}{Q^n} \leq c_n + \sum_{i=0}^{n-1} \frac{a_i}{Q} \left(\frac{e}{Q}\right)^{n-1-i} c_i + \frac{6}{Q^n}.$$

Since $0 \leq a_i \leq d^2(Q-e)$, Lemma 3.4(2) with $t = \frac{e}{Q}$, $\beta = d^2$, and $\alpha_i = \frac{a_i}{Q}$ implies

$$r_z < \frac{8 + 32d^2}{n} Q^n + 6. \tag{3.8}$$

Since $R'_n \geq 0$, $C \not\subset \operatorname{Supp} R'_n$ and $\#\pi^{-1}(x) \leq \operatorname{mult}_x C$, we have

$$\operatorname{mult}_x R'_n \cdot \operatorname{mult}_x C \leq (R'_n \cdot C)_x = \sum_{z \in \pi^{-1}(x)} r_z \leq \operatorname{mult}_x C \cdot \max_{z \in \pi^{-1}(x)} r_z,$$

where the equality is [Ful08, Section 7.5, Proposition 2]. Hence $\operatorname{mult}_x R'_n \leq \max_{z \in \pi^{-1}(x)} r_z$. By the last sentence of Lemma 2.5 and (3.8),

$$v(R'_n) \leq (\operatorname{mult}_x R'_n)\,(A(v) - v(\mathfrak{m}_x)) \leq (\operatorname{mult}_x R'_n) A(v) < \left(\frac{8 + 32d^2}{n} Q^n + 6\right) A(v),$$

where the last inequality holds since $A(v) > 0$. The lemma follows. □

3.2. **Points away from 2-periodic points of the normalization.**

Proof of Theorem 3.1. Let $d := \deg C \leq 4$ and $e := e_C(g)$, write $g^*C = eC + D$, and for $n \geq 1$ let $R'_n := R(g^n) - (e^n - 1)C$, the divisor of Lemma 3.2(1). By Lemma 3.2, $t := e/Q \in (0,1]$ and $\deg \pi^*(D|_C) = d^2(Q-e)$.

If C is singular, then $d \geq 3$. A singular irreducible plane cubic has a unique singular point, which is a node or a cusp, and $(\mathbb{P}^2, C)$ is not lc at a cusp. Thus one of the following holds:

(1) C is smooth (**Step 1**).
(2) C is singular, and $d = 4$ or $(\mathbb{P}^2, C)$ is not lc (**Step 2**).
(3) $d = 3$ and C has a node (**Steps 3–7**).

Step 1. In this step we prove the theorem for any $n \geq 300$ when C is smooth.

Here $\pi = \mathrm{id}_C$ and $h = g|_C$, and $h^2(x) \neq x$. By Lemma 3.3(1), $d \leq 3$. For any $i \geq 0$ let

$$c_i := \prod_{j=0}^{i-1} \frac{\deg_{h^j(x)} h}{Q} \quad \text{and} \quad \alpha_i := \frac{\operatorname{mult}_{h^i(x)}(D|_C)}{Q}.$$

The c_i and α_i are the numbers of Lemma 3.3, and the c_i are those of Lemma 3.4 for $z = x$. Since $D|_C \geq 0$ and $\deg(D|_C) = d^2(Q-e)$, $0 \leq \alpha_i \leq d^2(1-t) \leq 9(1-t)$ for any i. By Lemma 3.2, $\deg h = Q \geq 4$. Thus Lemma 3.3(2) and Lemma 3.4(2) for $z = x$ and $\beta = 9$ imply, for any $n \geq 300$,

$$\frac{1 + \operatorname{mult}_x(R'_n|_C)}{Q^n} < \frac{8 + 32 \cdot 9}{n} = \frac{296}{n} < 1.$$

By Lemma 3.3(3), $(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n))$ is lc at x for any $n \geq 300$.

Step 2. In this step we prove the theorem for any $n \geq 128d^3$ when C is singular, and $d = 4$ or $(\mathbb{P}^2, C)$ is not lc.

If $(\mathbb{P}^2, C)$ is not lc, then Lemma 3.2(3) implies $d \geq 3$ and $e \leq (1 - \frac{1}{2d^2})Q$. If $(\mathbb{P}^2, C)$ is lc, then $d = 4$ and Lemma 3.2(4) implies $e \leq (1 - \frac{1}{d^2})Q$. Thus in both cases $d \in \{3, 4\}$ and $t \leq 1 - \frac{1}{2d^2}$. Since $e \leq Q$, and since $(1-s)^n \leq (1+s)^{-n} \leq \frac{1}{1+ns}$ for $0 < s < 1$,

$$\frac{e^n - 1}{Q^n - 1} \leq t^n \leq \left(1 - \frac{1}{2d^2}\right)^n < \frac{2d^2}{n}.$$

Let v be a divisorial valuation centered at x. Since $\operatorname{mult}_x C \leq d$, the last sentence of Lemma 2.5 implies

$$v(C) \leq (\operatorname{mult}_x C)\,(A(v) - v(\mathfrak{m}_x)) \leq dA(v).$$

By Lemma 3.5, for any $n \geq 2$,

$$v(R'_n) < \left(\frac{8 + 32d^2}{n} Q^n + 6\right) A(v).$$

Since $\frac{Q^n}{Q^n - 1} \leq 2$ and $Q^n - 1 \geq n$, for any $n \geq 128d^3$ we obtain

$$\begin{aligned}
\frac{v(R(g^n))}{(Q^n - 1)A(v)} &= \frac{(e^n - 1)v(C) + v(R'_n)}{(Q^n - 1)A(v)} \\
&< \frac{2d^3}{n} + \frac{16 + 64d^2}{n} + \frac{6}{n} \\
&\leq \frac{1}{64} + \frac{1}{2d} + \frac{11}{64d^3} \\
&\leq \frac{1}{64} + \frac{1}{6} + \frac{11}{1728} < 1,
\end{aligned}$$

where the last line uses $d \geq 3$. Thus for any $n \geq 128d^3$,

$$v(R(g^n)) \leq (Q^n - 1)A(v) \quad \text{for any divisorial valuation } v \text{ centered at } x. \tag{3.9}$$

We show that, for any $n \geq 1$, (3.9) implies that $(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n))$ is lc at x. Let E be a prime divisor over $\mathbb{P}^2$ with $x \in \operatorname{center}_{\mathbb{P}^2} E$. If $\operatorname{center}_{\mathbb{P}^2} E = \{x\}$, then (3.9) for the valuation v of E is the inequality $a(E, \mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n)) \geq 0$. Otherwise E is a curve $P \subset \mathbb{P}^2$ with $x \in P$, and we need $\operatorname{mult}_P R(g^n) \leq Q^n - 1$. Fix a branch of P at x. Let $\mu_1 : Y_1 \to \mathbb{P}^2$ be the blow-up of x, and let E_1 be the exceptional prime divisor of μ_1. For $j \geq 1$, let $p_j \in E_j$ be the point on the strict transform of the branch, let $\mu_{j+1} : Y_{j+1} \to \mathbb{P}^2$ be the composition of the blow-up of p_j with μ_j, and let E_{j+1} be the exceptional prime divisor of the blow-up of p_j. For any $j \gg 0$, p_j is not contained in E_i for $i < j$, and $(\mu_j)_*^{-1} P$ is smooth at p_j, so

$$a(E_{j+1}, \mathbb{P}^2, 0) = a(E_j, \mathbb{P}^2, 0) + 1 \quad \text{and} \quad \operatorname{mult}_{E_{j+1}} \mu_{j+1}^* P = \operatorname{mult}_{E_j} \mu_j^* P + 1.$$

Thus $a(E_j, \mathbb{P}^2, 0)/\operatorname{mult}_{E_j} \mu_j^* P \to 1$ as $j \to +\infty$. Since $R(g^n) \geq (\operatorname{mult}_P R(g^n))P$, (3.9) for the valuation v of E_j implies

$$(\operatorname{mult}_{E_j} \mu_j^* P) \cdot \operatorname{mult}_P R(g^n) \leq (Q^n - 1)a(E_j, \mathbb{P}^2, 0)$$

for any j, so $\operatorname{mult}_P R(g^n) \leq Q^n - 1$. Thus $(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n))$ is lc at x for any $n \geq 128d^3$.

Step 3. In this step we assume that $d = 3$ and C has a node, and we compute the coefficients of $\pi^*(R'_n|_C)$.

Let o be the node of C and $\pi^{-1}(o) = \{z_1, z_2\}$. Then $\overline{C} \cong \mathbb{P}^1$. By Lemma 3.2, $\deg \Delta = (d-1)(d-2) - 2h^1(\overline{C}, \mathcal{O}_{\overline{C}}) = 2$. Since $K_{\mathbb{P}^2} + C$ is Cartier, Δ is a $\mathbb{Z}$-divisor [Kol13, Definition 4.2], and since C is singular at o and smooth outside o, $\operatorname{Supp} \Delta = \{z_1, z_2\}$ [Kol13, Proposition 4.5(1)]. Thus $\Delta = z_1 + z_2$. By Lemma 3.2, $\deg h = Q$. Thus $\deg_y h \leq Q$ for any $y \in \overline{C}$. For $z \in \overline{C}$ and $i \geq 0$ let

$$c_i := \prod_{j=0}^{i-1} \frac{\deg_{h^j(z)} h}{Q} \quad \text{and} \quad \alpha_i := \frac{\operatorname{mult}_{h^i(z)} \pi^*(D|_C)}{Q}.$$

They depend on the choice of $z \in \overline{C}$. The sequence $(c_i)_{i\geq 0}$ is decreasing, and Lemma 3.2 implies $0 \leq \alpha_i \leq 9(1-t)$ for any i.

Since $\deg_z(h^i) = Q^i c_i$ for any $i \geq 0$, taking coefficients at z in Lemma 3.2(1) implies, for any $n \geq 1$,

$$\operatorname{mult}_z \pi^*(R'_n|_C) = Q^n \left(c_n + \sum_{i=0}^{n-1} \alpha_i t^{n-1-i} c_i \right) - 1 + \operatorname{mult}_z \Delta - Q^n c_n \operatorname{mult}_{h^n(z)} \Delta. \tag{3.10}$$

Step 4. In **Steps 4–6** we assume that $d = 3$, C has a node and $x = o$. In this step we show

$$\deg_{z_1} h + \deg_{z_2} h \leq 9(Q-e). \tag{3.11}$$

By assumption, $h^2(z_1) \neq z_1$ and $h^2(z_2) \neq z_2$. Thus $g(o) \neq o$: if $g(o) = o$, then $h(\{z_1, z_2\})$ is contained in $\pi^{-1}(g(o)) = \{z_1, z_2\}$, and a map from $\{z_1, z_2\}$ to itself has a point $z \in \{z_1, z_2\}$ with $h^2(z) = z$. Since $\pi(h(z_1)) = \pi(h(z_2)) = g(o) \neq o$, we have $h(z_1), h(z_2) \notin \{z_1, z_2\}$. Let $V := h^{-1}(\{z_1, z_2\})$. Then $V \cap \{z_1, z_2\} = \emptyset$, and for any $z \in V$, $\operatorname{mult}_z \Delta = 0$ and $\operatorname{mult}_{h(z)} \Delta = 1$. By Lemma 3.2(1), $R'_1 \geq 0$ and $C \not\subset \operatorname{Supp} R'_1$, so (3.10) for $n = 1$ implies

$$0 \leq \operatorname{mult}_z \pi^*(R'_1|_C) = \operatorname{mult}_z \pi^*(D|_C) - 1 \quad \text{for any } z \in V.$$

Thus, by Lemma 3.2,

$$\#V \leq \sum_{z\in V} \operatorname{mult}_z \pi^*(D|_C) \leq \deg \pi^*(D|_C) = 9(Q-e).$$

Since $h^{-1}(z_1) \cap h^{-1}(z_2) = \emptyset$ and $\deg h = Q$, we have $\sum_{z\in V} \deg_z h = 2Q$. Since $\overline{C} \cong \mathbb{P}^1$, the Riemann–Hurwitz formula implies $\deg R(h) = 2Q - 2$, and the coefficient of $R(h)$ at $y \in \overline{C}$ is $\deg_y h - 1$. Since $z_1, z_2 \notin V$,

$$2Q - \#V + (\deg_{z_1} h - 1) + (\deg_{z_2} h - 1) = \sum_{y\in V\cup\{z_1,z_2\}} (\deg_y h - 1) \leq 2Q - 2.$$

Thus $\deg_{z_1} h + \deg_{z_2} h \leq \#V \leq 9(Q-e)$, which is (3.11). In particular, $e < Q$, so $t < 1$. Moreover, for $z \in \{z_1, z_2\}$ and any $i \geq 1$, (3.11) implies

$$c_i \leq c_1 = \frac{\deg_z h}{Q} \leq 9(1-t). \tag{3.12}$$

Step 5. In this step we let $z := z_1$ and show that, for any $n \geq 1$,

$$\operatorname{mult}_z \pi^*(R'_n|_C) \leq \frac{90}{n}(Q^n - e^n). \tag{3.13}$$

Since $\operatorname{mult}_z \Delta = 1$ and $Q^n c_n \operatorname{mult}_{h^n(z)} \Delta \geq 0$, (3.10) and $\alpha_i \leq 9(1-t)$ imply

$$\frac{\operatorname{mult}_z \pi^*(R'_n|_C)}{Q^n} \leq c_n + 9(1-t) \sum_{i=0}^{n-1} t^{n-1-i} c_i.$$

Since $h^2(z) \neq z$, Lemma 3.4(1) implies

$$\sum_{i\geq 0} c_i \leq 8 \quad \text{and} \quad c_i \leq \frac{8}{i+1} \text{ for any } i \geq 0.$$

Since $(c_i)_i$ is decreasing and $(t^{n-1-i})_i$ is increasing, $\sum_{i,j=0}^{n-1} (c_i - c_j)(t^{n-1-i} - t^{n-1-j}) \leq 0$, that is,

$$n \sum_{i=0}^{n-1} t^{n-1-i} c_i \leq \left(\sum_{i=0}^{n-1} c_i \right) \left(\sum_{i=0}^{n-1} t^{n-1-i} \right) \leq \frac{8(1-t^n)}{1-t}.$$

Thus

$$9(1-t) \sum_{i=0}^{n-1} t^{n-1-i} c_i \leq \frac{72}{n}(1-t^n).$$

Since $\frac{1}{t} \geq 2-t$, we have $t^{-n} \geq (1+(1-t))^n \geq 1+n(1-t)$, so

$$1-t^n \geq \frac{n(1-t)}{1+n(1-t)}.$$

If $n(1-t) \leq 1$, then $1-t^n \geq \frac{n(1-t)}{2}$, and (3.12) implies $c_n \leq 9(1-t) \leq \frac{18}{n}(1-t^n)$. If $n(1-t) > 1$, then $1-t^n > \frac{1}{2}$, so

$$c_n \leq \frac{8}{n+1} < \frac{16}{n+1}(1-t^n) < \frac{18}{n}(1-t^n).$$

Thus $c_n \leq \frac{18}{n}(1-t^n)$, so

$$\operatorname{mult}_z \pi^*(R'_n|_C) \leq Q^n(1-t^n)\left(\frac{18}{n}+\frac{72}{n}\right) = \frac{90}{n}(Q^n-e^n).$$

Step 6. In this step we prove the theorem for any $n \geq 90$ when $d=3$, C has a node and $x=o$.

Let C_1 and C_2 be the branches of C at o, where π maps the germ of $\overline{C}$ at z_1 isomorphically onto C_1 and the germ at z_2 isomorphically onto C_2. Then C_1 and C_2 are smooth germs meeting transversally at o, and $(R'_n \cdot C_1)_o = \operatorname{mult}_{z_1} \pi^*(R'_n|_C)$. Let v be a divisorial valuation centered at o. Lemma 2.5 for C_1 and (3.13) for $z=z_1$ imply

$$v(R'_n) \leq (R'_n \cdot C_1)_o\,(A(v)-v(C_1)) \leq \frac{90}{n}(Q^n-e^n)A(v).$$

Near o, $C=C_1+C_2$ is a simple normal crossing divisor, so $(\mathbb{P}^2, C)$ is lc at o, that is, $v(C) \leq A(v)$. Thus for any $n \geq 90$,

$$v(R(g^n)) = (e^n-1)v(C)+v(R'_n) \leq \left((e^n-1)+\frac{90}{n}(Q^n-e^n)\right)A(v) \leq (Q^n-1)A(v).$$

By **Step 2**, $(\mathbb{P}^2, \frac{1}{Q^n-1}R(g^n))$ is lc at o for any $n \geq 90$.

Step 7. In this step we prove the theorem for any $n \geq 300$ when $d=3$, C has a node and $x \neq o$.

Let $\pi^{-1}(x) = \{z\}$. Then $z \notin \{z_1, z_2\}$, $h^2(z) \neq z$, and π is an isomorphism near z, so $\operatorname{mult}_x(R'_n|_C) = \operatorname{mult}_z \pi^*(R'_n|_C)$. Since $\operatorname{mult}_z \Delta = 0$ and $Q^n c_n \operatorname{mult}_{h^n(z)} \Delta \geq 0$, (3.10) implies

$$1+\operatorname{mult}_x(R'_n|_C) \leq Q^n\left(c_n + \sum_{i=0}^{n-1} \alpha_i t^{n-1-i} c_i\right).$$

Since $0 < t \leq 1$ and $0 \leq \alpha_i \leq 9(1-t)$, Lemma 3.4(2) for $\beta = 9$ implies, for any $n \geq 300$,

$$\frac{1+\operatorname{mult}_x(R'_n|_C)}{Q^n} < \frac{8+32\cdot 9}{n} = \frac{296}{n} < 1.$$

Thus $\operatorname{mult}_x(R'_n|_C) < Q^n-1$. Possibly shrinking $\mathbb{P}^2$ to a neighborhood of x, we may assume that C is smooth and $C \cap \operatorname{Supp} R'_n \subset \{x\}$. Then $(C, \frac{1}{Q^n-1}R'_n|_C)$ is lc, and by inversion of adjunction, $(\mathbb{P}^2, C+\frac{1}{Q^n-1}R'_n)$ is lc at x. Since $\frac{e^n-1}{Q^n-1} \leq 1$,

$$\frac{1}{Q^n-1}R(g^n) = \frac{e^n-1}{Q^n-1}C + \frac{1}{Q^n-1}R'_n \leq C + \frac{1}{Q^n-1}R'_n,$$

so $(\mathbb{P}^2, \frac{1}{Q^n-1}R(g^n))$ is lc at x for any $n \geq 300$.

Since $d \leq 4$, we have $8192 \geq \max\{300, 128d^3, 90\}$. By **Steps 1, 2, 6 and 7**, the theorem follows. □

4. Iterates of order divisible by six

Theorem 4.1. *Let f be an endomorphism of $\mathbb{P}^2$ of degree $q \geq 2$, $m \geq 1$ an integer, and $B := \frac{1}{q^{6m}-1} R(f^{6m})$. Then:*

(1) For any irreducible curve $C \subset \mathbb{P}^2$, $\operatorname{mult}_C B \leq 1$.
(2) Either $(\mathbb{P}^2, B)$ is lc, or $\operatorname{Nlc}(\mathbb{P}^2, B) = \{p\}$ for a closed point p and $v(B) \leq A(v) + v(\mathfrak{m}_p)$ for any divisorial valuation v over $\mathbb{P}^2$.
(3) For any divisorial valuation v over $\mathbb{P}^2$, $v(B) \leq \frac{3}{2} A(v)$, equivalently

$$A((f^{6m})_* v) \leq \frac{3q^{6m}-1}{2} A(v).$$

(4) If v is a divisorial valuation centered at a closed point p with $v(B) > A(v)$, then $A(v) < 100 v(\mathfrak{m}_p)$.

Compare (1) with [Men23, Theorem 3.1], where the iterate depends on f.
We prove Theorem 4.1 in Subsection 4.2.

4.1. Coefficients of iterates.

Lemma 4.2. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$ and $C \subset \mathbb{P}^2$ an irreducible curve. Then:*

(1) $e_C(g^6) \leq Q^6$.
(2) If $g^6(C) \neq C$, then $\operatorname{mult}_C \frac{1}{Q^6-1} R(g^6) \leq \frac{8}{9}$, and $\operatorname{mult}_C \frac{1}{Q^6-1} R(g^6) < \frac{2}{3}$ if $\deg C \geq 2$.

Proof. Since $\deg R(g) = 3(Q-1)$ and $\operatorname{mult}_D R(g) = e_D(g) - 1$ for any irreducible curve D, for any finite set $\mathcal{D}$ of irreducible curves we have

$$\sum_{D \in \mathcal{D}} (e_D(g) - 1) \deg D \leq 3(Q-1). \tag{4.1}$$

Let D' be an irreducible curve. Then $\deg g^* D' = Q \deg D'$, and for any irreducible curve D with $g(D) = D'$, the coefficient of D in $g^* D'$ is $e_D(g)$. Thus for any finite set $\mathcal{D}$ of irreducible curves with $g(D) = D'$ for any $D \in \mathcal{D}$,

$$\sum_{D \in \mathcal{D}} e_D(g) \deg D \leq Q \deg D'. \tag{4.2}$$

In particular, $e_D(g) \deg D \leq Q \deg g(D)$ for any irreducible curve D. Moreover, by (4.1) with $\mathcal{D} = \{D\}$ and since $e_D(g) \leq 2(e_D(g) - 1)$ when $e_D(g) \geq 2$, for any irreducible curve D,

$$e_D(g) \leq 3Q - 2 \leq Q^2, \quad \text{and} \quad e_D(g) \deg D \leq 6(Q-1) < 6Q \ \text{ if } e_D(g) \geq 2. \tag{4.3}$$

For any integer $i \geq 0$, let $C_i := g^i(C)$ and $e_i := e_{C_i}(g)$. Thus $C_0 = C$. Since ramification indices of extensions of discrete valuation rings are multiplicative in towers,

$$e_C(g^6) = \prod_{i=0}^{5} e_i \quad \text{and} \quad \operatorname{mult}_C R(g^6) = \prod_{i=0}^{5} e_i - 1.$$

Proof of (2). Assume that $g^6(C) \neq C$. We show

$$\frac{e_C(g^6)}{Q^6} \leq \frac{8}{9}, \quad \text{and} \quad \frac{e_C(g^6)}{Q^6} < \frac{2}{3} \ \text{ if } \deg C \geq 2. \tag{4.4}$$

This implies (2): by (4.4), $e_C(g^6) \leq Q^6$, so

$$\operatorname{mult}_C \frac{1}{Q^6-1} R(g^6) = \frac{e_C(g^6) - 1}{Q^6 - 1} \leq \frac{e_C(g^6)}{Q^6}.$$

Case 1. No curve occurs more than twice in the sequence $C_0, \dots, C_5$. Let $N := \#\{C_0, \dots, C_5\}$, let $i_1 < \cdots < i_N$ be the indices $i \in \{0, \dots, 5\}$ with $C_i \notin \{C_0, \dots, C_{i-1}\}$, and for any $j \in \{1, \dots, N\}$ let

$$N_j := \#\{i \in \{0, \dots, 5\} \mid C_i = C_{i_j}\} \in \{1, 2\}.$$

Then $\sum_{j=1}^{N} N_j = 6$, so $N \geq 3$. We claim that $N \geq 4$. Suppose that $N = 3$. Then $N_1 = N_2 = N_3 = 2$ since $\sum_j N_j = 6$ and $N_j \leq 2$, so $C_0 = C_i$ for some $i \in \{1, \dots, 5\}$. Let $n := \min\{k \in \{1, \dots, 5\} \mid C_k = C_0\}$. Then $C_{i+n} = C_i$ for any $i \geq 0$ and $C_0, \dots, C_{n-1}$ are pairwise distinct (if $C_i = C_j$ with $0 \leq i < j < n$, then $C_{i+n-j} = C_n = C_0$ with $1 \leq i + n - j < n$), so $n = \#\{C_0, \dots, C_5\} = N = 3$. Thus $g^6(C) = C_6 = C_0 = C$, a contradiction.

By (4.1) with $\mathcal{D} = \{C_{i_1}, \dots, C_{i_N}\}$ and since $\deg C_{i_j} \geq 1$,

$$\sum_{j=1}^{N} e_{i_j} \leq 3Q - 3 + N.$$

By the weighted arithmetic-geometric mean inequality for the numbers e_{i_j}/N_j with weights $N_j/6$,

$$e_C(g^6) = \prod_{j=1}^{N} e_{i_j}^{N_j} \leq \prod_{j=1}^{N} N_j^{N_j} \cdot \left(\frac{3Q - 3 + N}{6}\right)^6.$$

For $N = 4, 5, 6$, the numbers $N_1, \dots, N_N$ are $2, 2, 1, 1$, resp. $2, 1, 1, 1, 1$, resp. $1, \dots, 1$ up to order, so $\prod_{j=1}^{N} N_j^{N_j} = 4^{6-N}$. Since $Q \geq 2$ and $N \geq 3$,

$$\frac{3Q - 3 + N}{6Q} = \frac{1}{2} + \frac{N - 3}{6Q} \leq \frac{N + 3}{12}.$$

Thus

$$\frac{e_C(g^6)}{Q^6} \leq 4^{6-N} \left(\frac{N + 3}{12}\right)^6 \in \left\{\left(\frac{343}{432}\right)^2, \frac{256}{729}, \frac{729}{4096}\right\}.$$

Since $3 \cdot 343^2 = 352947 < 373248 = 2 \cdot 432^2$, $3 \cdot 256 < 2 \cdot 729$ and $3 \cdot 729 < 2 \cdot 4096$, we obtain $\frac{e_C(g^6)}{Q^6} < \frac{2}{3}$.

Case 2. Some curve occurs at least three times in the sequence $C_0, \dots, C_5$. Let

$$n_0 := \min\{k \geq 0 \mid C_k = C_{k+j} \text{ for some integer } j \geq 1\} \quad \text{and} \quad n := \min\{j \geq 1 \mid C_{n_0} = C_{n_0+j}\}.$$

Then $C_{i+n} = C_i$ for any $i \geq n_0$, and $C_0, \dots, C_{n_0+n-1}$ are pairwise distinct. Each of $C_0, \dots, C_{n_0-1}$ occurs once in the sequence $(C_i)_{i \geq 0}$, so the curve occurring three times is C_i for some i with $n_0 \leq i \leq n_0 + n - 1$, and the indices i' with $C_{i'} = C_i$ are $i, i + n, i + 2n, \dots$. Thus

$$n_0 + 2n \leq i + 2n \leq 5, \quad \text{so} \quad (n, n_0) \in \{(1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 1)\}.$$

If $n_0 = 0$, then $C_{i+n} = C_i$ for any $i \geq 0$ and n divides 6, so $g^6(C) = C$, a contradiction. It remains to consider the following four cases.

Case 2.1. $(n, n_0) = (1, 1)$. Then $C_0 \neq C_1 = g(C_1)$, $e_1 \leq Q$ by (4.2) with $\mathcal{D} = \{C_1\}$ and $D' = C_1$, and $e_C(g^6) = e_0 e_1^5$. If $e_1 = 1$, then $e_C(g^6) = e_0 \leq Q^2 \leq \frac{1}{16} Q^6$ by (4.3). Assume that $e_1 \geq 2$, and let $y := e_1/Q \in (0, 1]$. Since $g(C_0) = g(C_1) = C_1$, (4.2) with $\mathcal{D} = \{C_0, C_1\}$ implies $e_0 \deg C + e_1 \deg C_1 \leq Q \deg C_1$, that is,

$$e_0 \deg C \leq Q(1 - y) \deg C_1.$$

Multiplying by $\frac{e_1}{Q^2 \deg C}$ and applying (4.3) to C_1, we obtain

$$\frac{e_0 e_1}{Q^2} \leq (1 - y) \frac{e_1 \deg C_1}{Q \deg C} \leq \frac{6(1 - y)}{\deg C}.$$

Since the maximum of $(1-y)y^4$ on $[0,1]$ is $\frac{4^4}{5^5}$,

$$\frac{e_C(g^6)}{Q^6} = \frac{e_0e_1}{Q^2}\cdot y^4 \le \frac{6(1-y)y^4}{\deg C} \le \frac{1536}{3125\deg C} < \frac{2}{3}.$$

Case 2.2. $(n,n_0)=(1,2)$. Then C_0, C_1, C_2 are pairwise distinct, $g(C_2)=C_2$, $e_2 \le Q$ by (4.2) with $\mathcal{D}=\{C_2\}$ and $D'=C_2$, and $e_C(g^6)=e_0e_1e_2^4$. If $e_2=1$, then $e_C(g^6)=e_0e_1 \le Q^4 \le \frac{1}{4}Q^6$ by (4.3). Assume that $e_2 \ge 2$, and let $y := e_2/Q \in (0,1]$. By (4.2) with $\mathcal{D}=\{C_0\}$, and with $\mathcal{D}=\{C_1,C_2\}$ and $D'=C_2$,

$$e_0\deg C \le Q\deg C_1 \quad\text{and}\quad e_1\deg C_1 \le Q(1-y)\deg C_2,$$

so $e_0e_1\deg C \le Q^2(1-y)\deg C_2$. Multiplying by $\frac{e_2}{Q^3\deg C}$ and applying (4.3) to C_2, we obtain

$$\frac{e_0e_1e_2}{Q^3} \le (1-y)\frac{e_2\deg C_2}{Q\deg C} \le \frac{6(1-y)}{\deg C}.$$

Since the maximum of $(1-y)y^3$ on $[0,1]$ is $\frac{27}{256}$,

$$\frac{e_C(g^6)}{Q^6} = \frac{e_0e_1e_2}{Q^3}\cdot y^3 \le \frac{6(1-y)y^3}{\deg C} \le \frac{81}{128\deg C} < \frac{2}{3}.$$

Case 2.3. $(n,n_0)=(1,3)$. Then C_0, C_1, C_2, C_3 are pairwise distinct, $g(C_3)=C_3$, $e_3 \le Q$ by (4.2) with $\mathcal{D}=\{C_3\}$ and $D'=C_3$, and $e_C(g^6)=e_0e_1e_2e_3^3$. Let $y := e_3/Q \in (0,1]$. By (4.1) with $\mathcal{D}=\{C_0,C_1,C_2,C_3\}$, $e_0+e_1+e_2+e_3 \le 3Q+1$. By the arithmetic-geometric mean inequality, $e_0e_1e_2 \le \left(\frac{3Q+1-e_3}{3}\right)^3$. Since the function $y(3+\frac{1}{Q}-y)$ of y is increasing on $[0,1]$ and $Q \ge 2$,

$$\frac{e_C(g^6)}{Q^6} \le \left(\frac{y(3+\frac{1}{Q}-y)}{3}\right)^3 \le \left(\frac{2+\frac{1}{Q}}{3}\right)^3 \le \left(\frac{5}{6}\right)^3 < \frac{2}{3}.$$

Case 2.4. $(n,n_0)=(2,1)$. Then C_0, C_1, C_2 are pairwise distinct, $g(C_0)=C_1$, $g(C_1)=C_2$, $g(C_2)=C_1$, and $e_C(g^6)=e_0e_1^3e_2^2$. By (4.2) with $\mathcal{D}=\{C_1\}$ and with $\mathcal{D}=\{C_2\}$,

$$e_1\deg C_1 \le Q\deg C_2 \quad\text{and}\quad e_2\deg C_2 \le Q\deg C_1,$$

so $y := e_1e_2/Q^2 \in (0,1]$.

Suppose that $e_1=1$. By (4.1) with $\mathcal{D}=\{C_0,C_2\}$, $e_0+e_2 \le 3Q-1$. By the arithmetic-geometric mean inequality for $e_0, \frac{e_2}{2}, \frac{e_2}{2}$,

$$\frac{e_C(g^6)}{Q^6} = \frac{4e_0\cdot\frac{e_2}{2}\cdot\frac{e_2}{2}}{Q^6} \le \frac{4}{Q^6}\left(\frac{e_0+e_2}{3}\right)^3 < \frac{4}{Q^3} \le \frac{1}{2}.$$

Suppose that $e_1 \ge 2$. Since $g(C_0)=g(C_2)=C_1$, (4.2) with $\mathcal{D}=\{C_0,C_2\}$ implies

$$e_0\deg C + e_2\deg C_2 \le Q\deg C_1.$$

Since $e_1\deg C_1 \le Q\deg C_2$, we have $e_2\deg C_2 \ge \frac{e_2}{Q}e_1\deg C_1 = Qy\deg C_1$, so

$$e_0\deg C \le Q(1-y)\deg C_1.$$

Multiplying by $\frac{e_1}{Q^2\deg C}$ and applying (4.3) to C_1, we obtain

$$\frac{e_0e_1}{Q^2} \le (1-y)\frac{e_1\deg C_1}{Q\deg C} \le \frac{6(1-y)}{\deg C}.$$

Since the maximum of $(1-y)y^2$ on $[0,1]$ is $\frac{4}{27}$,

$$\frac{e_C(g^6)}{Q^6} = \frac{e_0e_1}{Q^2}\cdot y^2 \le \frac{6(1-y)y^2}{\deg C} \le \frac{8}{9\deg C}.$$

In each of Cases 1 and 2.1–2.4, the bound obtained for $\frac{e_C(g^6)}{Q^6}$ is $\le \frac{8}{9}$, and $< \frac{2}{3}$ if $\deg C \ge 2$. This proves (4.4), hence (2).

Proof of (1). If $g^6(C) \neq C$, then (1) follows from (4.4). Assume that $g^6(C) = C$, and let $n := \min\{k \geq 1 \mid C_k = C\}$. Then n divides 6 and $e_{i+n} = e_i$ for any $i \geq 0$. Since $e_i \deg C_i \leq Q \deg C_{i+1}$ for any $i \in \{0, \dots, n-1\}$ by (4.2) and $C_n = C_0$,

$$\prod_{i=0}^{n-1} e_i \cdot \prod_{i=0}^{n-1} \deg C_i \leq Q^n \prod_{i=1}^{n} \deg C_i = Q^n \prod_{i=0}^{n-1} \deg C_i,$$

so $\prod_{i=0}^{n-1} e_i \leq Q^n$. Thus

$$e_C(g^6) = \prod_{i=0}^{5} e_i = \left(\prod_{i=0}^{n-1} e_i\right)^{6/n} \leq Q^6.$$

The lemma follows. □

4.2. **The non-lc point.**

Lemma 4.3. *Let $k \geq 1$ be an integer and $D \geq 0$ a $\mathbb{Q}$-divisor on $\mathbb{P}^2$ of degree k with $\operatorname{mult}_P D < 2$ for any prime divisor P on $\mathbb{P}^2$. Let C be the reduced (possibly empty) curve whose irreducible components are the prime divisors P with $\operatorname{mult}_P D > 1$. Then there exist a rational number $\epsilon_0 \in (0,1)$ and a (possibly empty) 0-dimensional closed subscheme $Z \subset \mathbb{P}^2$ such that*

$$\mathcal{J}\left(\mathbb{P}^2, (1-\epsilon)D\right) = \mathcal{O}_{\mathbb{P}^2}(-C) \otimes \mathcal{I}_Z$$

for any rational number $\epsilon \in (0, \epsilon_0)$,

$$\operatorname{length} Z \leq h^0\left(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(k-3-\deg C)\right),$$

and $\operatorname{Nlc}(\mathbb{P}^2, D) = C \cup \operatorname{Supp} Z$.

Proof. By Lemma 2.7 (2), there exists a rational number $\epsilon_0 \in (0,1)$ such that $\mathcal{J} := \mathcal{J}(\mathbb{P}^2, (1-\epsilon)D)$ does not depend on $\epsilon \in (0, \epsilon_0)$ and $\lfloor (1-\epsilon)\operatorname{mult}_P D \rfloor = \lceil \operatorname{mult}_P D \rceil - 1$ for any $\epsilon \in (0, \epsilon_0)$ and any prime divisor $P \subset \operatorname{Supp} D$. Since $\operatorname{mult}_P D < 2$, $\lceil \operatorname{mult}_P D \rceil - 1$ is 1 if $P \subset C$ and 0 otherwise. Thus, by Lemma 2.7 (3), at the generic point η_P of any prime divisor P on $\mathbb{P}^2$, $\mathcal{J}$ is the maximal ideal $\mathfrak{m}_{\eta_P}$ if $P \subset C$ and trivial otherwise. Since the local rings of $\mathbb{P}^2$ are unique factorization domains, $\mathcal{J} = \mathcal{O}_{\mathbb{P}^2}(-C) \otimes \mathcal{I}_Z$ for a (possibly empty) 0-dimensional closed subscheme $Z \subset \mathbb{P}^2$. By Lemma 2.7 (2), $\operatorname{Nlc}(\mathbb{P}^2, D)$ is the set of closed points at which $\mathcal{J}$ is non-trivial, that is, $C \cup \operatorname{Supp} Z$.

Fix a rational number $\epsilon \in (0, \epsilon_0)$, and let L be a divisor linearly equivalent to k times a line. Since $L - (1-\epsilon)D$ is a $\mathbb{Q}$-divisor of degree $k\epsilon > 0$, Nadel vanishing [Laz10, Theorem 2.4] implies $H^1(\mathbb{P}^2, \mathcal{I}_Z \otimes \mathcal{O}_{\mathbb{P}^2}(k-3-\deg C)) = 0$. Thus the exact sequence

$$0 \to \mathcal{I}_Z \otimes \mathcal{O}_{\mathbb{P}^2}(k-3-\deg C) \to \mathcal{O}_{\mathbb{P}^2}(k-3-\deg C) \to \mathcal{O}_Z \to 0$$

implies

$$\operatorname{length} Z = h^0(Z, \mathcal{O}_Z) \leq h^0\left(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(k-3-\deg C)\right).$$

The lemma follows. □

Lemma 4.4. *Let $D \geq 0$ be a $\mathbb{Q}$-divisor on $\mathbb{P}^2$ of degree 3 such that $\operatorname{mult}_C D \leq 1$ for any irreducible curve $C \subset \mathbb{P}^2$. Then either $(\mathbb{P}^2, D)$ is lc, or there exists a closed point $p \in \mathbb{P}^2$ such that $\operatorname{Nlc}(\mathbb{P}^2, D) = \{p\}$ and*

$$v(D) \leq A(v) + v(\mathfrak{m}_p)$$

for any divisorial valuation v over $\mathbb{P}^2$.

Proof. By Lemma 4.3 with $k = 3$ (since $\operatorname{mult}_C D \leq 1$ for any irreducible curve C, the reduced curve of Lemma 4.3 is empty and the 0-dimensional subscheme Z given by Lemma 4.3 satisfies $\operatorname{length} Z \leq h^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) = 1$), there exists a rational number $\epsilon_0 \in (0,1)$ such that, for any rational number $\epsilon \in (0, \epsilon_0)$, $\mathcal{J}(\mathbb{P}^2, (1-\epsilon)D)$ is $\mathcal{O}_{\mathbb{P}^2}$ or $\mathfrak{m}_p$ for a closed point $p \in \mathbb{P}^2$ independent of ϵ, and $\operatorname{Nlc}(\mathbb{P}^2, D) = \operatorname{Supp} Z$. Assume that $(\mathbb{P}^2, D)$ is not lc. Then

$$\operatorname{Nlc}(\mathbb{P}^2, D) = \{p\} \quad \text{and} \quad \mathcal{J}(\mathbb{P}^2, (1-\epsilon)D) = \mathfrak{m}_p \text{ for any rational number } \epsilon \in (0, \epsilon_0).$$

Let v be a divisorial valuation over $\mathbb{P}^2$, let η be the generic point of $\mathrm{center}_{\mathbb{P}^2} v$, fix a rational number $\epsilon \in (0, \epsilon_0)$, and let $\phi \in (\mathfrak{m}_p)_\eta$ be an element with $v(\phi) = v(\mathfrak{m}_p)$. Since $\mathcal{J}(\mathbb{P}^2, (1-\epsilon)D) = \mathfrak{m}_p$, Lemma 2.7 (4) for $(1-\epsilon)D$ implies $v(\mathfrak{m}_p) + A(v) > (1-\epsilon)v(D)$. Letting ϵ tend to 0, we obtain $v(D) \leq A(v) + v(\mathfrak{m}_p)$. The lemma follows. □

Lemma 4.5. *Let $p \in \mathbb{P}^2$ be a closed point, ν a normalized divisorial valuation at p, and $C \subset \mathbb{P}^2$ an irreducible curve of degree d. Assume that $\nu(C) > A(\nu) - 1$. Then*

$$(\nu(C) - A(\nu) + 1)^2 \leq (d-1)^2 \alpha(\nu).$$

In particular,

$$\nu(C) \leq A(\nu) - 1 + (d-1)\sqrt{A(\nu) - 1}.$$

Proof. Since $1 \leq \alpha(\nu) \leq A(\nu) - 1 < +\infty$ by Theorem 2.9 (1), (3), (4), we have $\nu(C) > A(\nu) - 1 \geq 1$, so $p \in C$. We let x, y be *affine linear coordinates centered at p*, that is, affine coordinates on an affine chart $\mathbb{A}^2 \subset \mathbb{P}^2$ containing p with $x(p) = y(p) = 0$, and let $\phi \in \mathbb{C}[x, y]$ be an irreducible polynomial of degree at most d with $C \cap \mathbb{A}^2 = \{\phi = 0\}$. Then $\nu(C) = \nu(\phi)$. We write $\phi_x := \partial\phi/\partial x$ and $\phi_y := \partial\phi/\partial y$.

We show that

$$\nu(\phi_x) \geq \nu(\phi) - A(\nu) + 1 \quad \text{and} \quad \nu(\phi_y) \geq \nu(\phi) - A(\nu) + 1. \tag{4.5}$$

By Definition 2.8, there exist a prime divisor E over $\mathbb{P}^2$ with center p and a real number $c > 0$ such that $\nu(\psi) = c \operatorname{mult}_E \operatorname{div}(h^*\psi)$ for any non-zero $\psi \in \mathbb{C}[x, y]$, where $h : X' \to \mathbb{P}^2$ is a projective birational morphism from a smooth surface X' on which E is a prime divisor. Near a general closed point of E, we let t, z be local coordinates on X' with $E = \{t = 0\}$, and write $h^*\phi = t^a u$ and $h^*x = t^b w$, where $a := \operatorname{mult}_E \operatorname{div}(h^*\phi)$, $b := \operatorname{mult}_E \operatorname{div}(h^*x)$, and u, w are regular. Then

$$h^*(\mathrm{d}\phi \wedge \mathrm{d}x) = t^{a+b-1} \left(au\mathrm{d}t \wedge \mathrm{d}w + bw\mathrm{d}u \wedge \mathrm{d}t + t\mathrm{d}u \wedge \mathrm{d}w\right),$$

so the coefficient of $h^*(\mathrm{d}\phi \wedge \mathrm{d}x)$ with respect to $\mathrm{d}t \wedge \mathrm{d}z$ has order at least $a + b - 1$ along E. Since

$$\mathrm{d}\phi \wedge \mathrm{d}x = -\phi_y \mathrm{d}x \wedge \mathrm{d}y$$

and the coefficient of $h^*(\mathrm{d}x \wedge \mathrm{d}y)$ has order $\operatorname{mult}_E(K_{X'} - h^*K_{\mathbb{P}^2})$ along E, we obtain

$$\operatorname{mult}_E \operatorname{div}(h^*\phi_y) + \operatorname{mult}_E(K_{X'} - h^*K_{\mathbb{P}^2}) \geq a + b - 1.$$

Since $A(\nu) = c\,(\operatorname{mult}_E(K_{X'} - h^*K_{\mathbb{P}^2}) + 1)$ by Definition 2.1 and Theorem 2.9 (4), and $\nu(x) \geq \nu(\mathfrak{m}_p) = 1$, multiplying by c implies the second inequality of (4.5). Exchanging x and y implies the first one.

Since $\nu(\phi) > A(\nu) - 1$, (4.5) implies $\nu(\phi_x) > 0$ and $\nu(\phi_y) > 0$, that is, each of ϕ_x, ϕ_y is 0 or vanishes at p. If $\phi_y = 0$, then $\phi \in \mathbb{C}[x]$ is irreducible, so $\deg \phi = 1$ and ϕ_x is a non-zero constant, a contradiction. Thus $\phi_y \neq 0$. Exchanging x and y, we have $\phi_x \neq 0$. Both ϕ_x and ϕ_y have degree at most $d - 1$.

Let $\psi := \gcd(\phi_x, \phi_y) \in \mathbb{C}[x, y]$. We show that $\psi(p) \neq 0$. Suppose that $\psi(p) = 0$, and let $\Gamma \subset \mathbb{A}^2$ be an irreducible curve with $p \in \Gamma \subset \{\psi = 0\}$. Since ϕ_x and ϕ_y vanish on Γ, the restriction of ϕ to the non-singular locus of Γ has zero differential. Since Γ is irreducible and the characteristic is 0, ϕ is constant on Γ, and the constant is $\phi(p) = 0$. Thus $\Gamma = C \cap \mathbb{A}^2$, and ϕ_x, ϕ_y vanish on $C \cap \mathbb{A}^2$, so any point of $C \cap \mathbb{A}^2$ is a singular point of C, which contradicts the finiteness of the singular locus of the reduced curve C.

We write $\phi_x = \psi\phi_1$ and $\phi_y = \psi\phi_2$. Then $\phi_1, \phi_2 \in \mathbb{C}[x, y]$ are coprime, have degree at most $d - 1$, and satisfy $\nu(\phi_1) = \nu(\phi_x)$ and $\nu(\phi_2) = \nu(\phi_y)$ since $\psi(p) \neq 0$. By Bezout's theorem for the closures in $\mathbb{P}^2$ of $\{\phi_1 = 0\}$ and $\{\phi_2 = 0\}$, which have no common component,

$$(\phi_1 \cdot \phi_2)_p \leq (d-1)^2.$$

In particular $(\phi_1 \cdot \phi_2)_p < +\infty$, so ϕ_1 and ϕ_2 have no common irreducible factor in $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$. By Lemma 2.10 (2) and (4.5),

$$(d-1)^2 \geq (\phi_1 \cdot \phi_2)_p \geq \frac{\nu(\phi_1)\nu(\phi_2)}{\alpha(\nu)} \geq \frac{(\nu(C) - A(\nu) + 1)^2}{\alpha(\nu)}.$$

Since $\alpha(\nu) \leq A(\nu) - 1$ and $\nu(C) - A(\nu) + 1 > 0$, taking square roots, we obtain $\nu(C) \leq A(\nu) - 1 + (d-1)\sqrt{A(\nu) - 1}$. The lemma follows. □

Lemma 4.6. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$ and $C \subset \mathbb{P}^2$ a smooth irreducible curve of degree d with $g(C) = C$. Then $v(R(g)) \leq (Q-1)A(v)$ for any divisorial valuation v centered at a closed point of C with $v(C) \geq d^2\,(A(v) - v(C))$.*

Proof. Let x be the center of v, let $e := e_C(g)$, write $g^*C = eC + D$, let $h := g|_C$, and let $R_1' := R(g) - (e-1)C$. By Lemma 3.2 (1) with $n = 1$, $\pi = \mathrm{id}_C$, and $\Delta = 0$ (since C is smooth), $R_1'|_C = D|_C + R(h)$. Thus, by Lemma 3.2,

$$(R_1' \cdot C)_x = \mathrm{mult}_x(R_1'|_C) \leq \deg(D|_C) + \deg_x h - 1 \leq d^2(Q-e) + Q - 1.$$

Since C is smooth, $(\mathbb{P}^2, C)$ is lc, so $A(v) - v(C) \geq 0$. Thus Lemma 2.5 for R_1' implies $v(R_1') \leq \big(Q - 1 + d^2(Q-e)\big)\,(A(v) - v(C))$. Since $v(R(g)) = (e-1)v(C) + v(R_1')$ and $e \leq Q$,

$$(Q-1)A(v) - v(R(g)) \geq (Q-e)\,\big(v(C) - d^2\,(A(v) - v(C))\big) \geq 0.$$

The lemma follows. □

Lemma 4.7. *Let f be an endomorphism of $\mathbb{P}^2$ of degree $q \geq 2$, $m \geq 1$ an integer, $B := \frac{1}{q^{6m}-1}R(f^{6m})$, $p \in \mathbb{P}^2$ a closed point, and ν a normalized divisorial valuation at p with $A(\nu) \geq 12$ and $\nu(B) > A(\nu)$. Then $\nu(C) \leq \frac{\deg C}{3}A(\nu)$ for any line or smooth conic $C \subset \mathbb{P}^2$.*

Proof. Let $d := \deg C$, and suppose that $\nu(C) > \frac{d}{3}A(\nu)$. Then $\nu(C) > 0$, so $p \in C$. By Theorem 2.9 (1), (3), (4), $1 \leq \alpha(\nu) \leq A(\nu) - 1 < +\infty$. By Lemma 4.2 (1) for the endomorphism f^m of degree q^m, $e_C(f^{6m}) \leq q^{6m}$. Since $\mathrm{mult}_C R(f^{6m}) = e_C(f^{6m}) - 1$ (Definition 2.2), we have $\mathrm{mult}_C B \leq 1$. Since $\deg R(f^{6m}) = 3(q^{6m} - 1)$, the $\mathbb{Q}$-divisor $B - (\mathrm{mult}_C B)C \geq 0$ has degree $3 - d\,\mathrm{mult}_C B$ and no common component with C. For any real number $t > 0$, we let

$$\gamma(t) := \mathrm{mult}_C B \cdot t + \frac{(A(\nu) - 1)d(3 - d\,\mathrm{mult}_C B)}{t}.$$

Since $\nu(B) = \mathrm{mult}_C B \cdot \nu(C) + \nu(B - (\mathrm{mult}_C B)C)$, Lemma 2.10 (3) for C and $B - (\mathrm{mult}_C B)C$, together with $\alpha(\nu) \leq A(\nu) - 1$, implies $\nu(B) \leq \gamma(\nu(C))$.

Let $H \subset \mathbb{P}^2$ be a line through p that is not tangent to C at p. Then $C + H$ is simple normal crossing near p, so $(\mathbb{P}^2, C + H)$ is lc at p, and

$$A(\nu) - \nu(C) - \nu(H) = a(\nu, \mathbb{P}^2, C + H) \geq 0.$$

Since $\nu(H) \geq \nu(\mathfrak{m}_p) = 1$, we have

$$\nu(C) \leq A(\nu) - 1. \tag{4.6}$$

Since $d\,\mathrm{mult}_C B \leq 2$, γ is convex on $(0, +\infty)$. By the assumption, $\nu(B) \leq \gamma(\nu(C))$, and (4.6),

$$A(\nu) < \nu(B) \leq \gamma(\nu(C)) \quad \text{and} \quad \frac{d}{3}A(\nu) < \nu(C) \leq A(\nu) - 1.$$

On the other hand,

$$\frac{1}{A(\nu)}\gamma\left(\frac{d}{3}A(\nu)\right) = \frac{d\,\mathrm{mult}_C B}{3} + \frac{9(A(\nu) - 1)}{A(\nu)^2}\left(1 - \frac{d\,\mathrm{mult}_C B}{3}\right) < 1,$$

since $\frac{d\,\mathrm{mult}_C B}{3} \leq \frac{2}{3}$ and $9(A(\nu) - 1) < A(\nu)^2$. Since γ is convex,

$$\gamma(\nu(C)) \leq \max\left\{\gamma\left(\frac{d}{3}A(\nu)\right), \gamma(A(\nu) - 1)\right\},$$

so $\gamma(A(\nu)-1) > A(\nu)$, that is,

$$\operatorname{mult}_C B \cdot (A(\nu)-1) + d(3 - d \operatorname{mult}_C B) > A(\nu).$$

Since $3d-d^2-1 = 1$ for $d \in \{1,2\}$ and $A(\nu)-1-d^2 \geq 11-d^2$, we obtain $\operatorname{mult}_C B > 1 - \frac{1}{A(\nu)-1-d^2}$, which is $\geq \frac{9}{10}$ if $d=1$ and $\geq \frac{6}{7}$ if $d=2$. Since $\frac{9}{10} > \frac{8}{9}$ and $\frac{6}{7} > \frac{2}{3}$, Lemma 4.2 (2) for the endomorphism f^m implies $f^{6m}(C) = C$.

By the assumption, $\nu(C)^2 > \frac{d^2}{9}A(\nu)^2 > d^2(A(\nu)-1)$, so $\gamma(\nu(C))$, which equals

$$\operatorname{mult}_C B \cdot \left(\nu(C) - \frac{(A(\nu)-1)d^2}{\nu(C)}\right) + \frac{3d(A(\nu)-1)}{\nu(C)},$$

is increasing in $\operatorname{mult}_C B$. Since $\operatorname{mult}_C B \leq 1$ and $d(3-d) = 2$, we obtain $A(\nu) < \nu(C) + \frac{2(A(\nu)-1)}{\nu(C)}$, that is,

$$\nu(C)^2 - A(\nu)\nu(C) + 2(A(\nu)-1) > 0.$$

Since $A(\nu) \geq 12$, we have $16A(\nu)^2 - 200A(\nu) + 200 > 0$, that is, $\sqrt{A(\nu)^2 - 8A(\nu) + 8} > \frac{3}{5}A(\nu)$. Thus the roots

$$t_\pm := \frac{A(\nu) \pm \sqrt{A(\nu)^2 - 8A(\nu) + 8}}{2}$$

of $t^2 - A(\nu)t + 2(A(\nu)-1)$ satisfy $t_- < \frac{1}{5}A(\nu)$ and $t_+ > \frac{4}{5}A(\nu)$. Since $\nu(C) > \frac{1}{3}A(\nu) > t_-$ by the assumption and $\nu(C)^2 - A(\nu)\nu(C) + 2(A(\nu)-1) > 0$, we have $\nu(C) > t_+$, so

$$\nu(C) > \frac{4}{5}A(\nu) \geq \frac{d^2}{d^2+1}A(\nu).$$

Thus

$$\nu(C) \geq d^2\left(A(\nu) - \nu(C)\right),$$

and Lemma 4.6 for the endomorphism f^{6m} and the divisorial valuation ν implies $\nu(B) \leq A(\nu)$, a contradiction. □

Proof of Theorem 4.1. (1) Let $C \subset \mathbb{P}^2$ be an irreducible curve. By Lemma 4.2 (1) for the endomorphism f^m of degree q^m, $e_C(f^{6m}) \leq q^{6m}$. Since $\operatorname{mult}_C R(f^{6m}) = e_C(f^{6m}) - 1$ (Definition 2.2), we have $\operatorname{mult}_C B \leq 1$.

(2) Since $\deg R(f^{6m}) = 3(q^{6m}-1)$, $B \geq 0$ is a $\mathbb{Q}$-divisor of degree 3. By (1), Lemma 4.4 with $D = B$ implies (2).

(3) We first show that

$$2v(\mathfrak{m}_x) \leq A(v) \tag{4.7}$$

for any closed point $x \in \mathbb{P}^2$ and any divisorial valuation v over $\mathbb{P}^2$. If the center of v is not x, then $v(\mathfrak{m}_x) = 0 < A(v)$. If the center of v is x, then $\nu := v/v(\mathfrak{m}_x)$ is a divisorial normalized valuation at x by Definition 2.8, and $A(\nu) \geq 1 + \alpha(\nu) \geq 2$ by Theorem 2.9 (1), (3), (4). Since $A(\nu) = A(v)/v(\mathfrak{m}_x)$ by Definition 2.1, this is (4.7).

Let v be a divisorial valuation over $\mathbb{P}^2$. If $(\mathbb{P}^2, B)$ is lc, then $v(B) \leq A(v)$. Otherwise, by (2) and (4.7),

$$v(B) \leq A(v) + v(\mathfrak{m}_p) \leq \frac{3}{2}A(v).$$

By Lemma 2.4 for f^{6m},

$$A((f^{6m})_* v) = A(v) + v(R(f^{6m})) = A(v) + (q^{6m}-1)v(B).$$

Thus $v(B) \leq \frac{3}{2}A(v)$ if and only if

$$A((f^{6m})_* v) \leq A(v) + \frac{3}{2}(q^{6m}-1)A(v) = \frac{3q^{6m}-1}{2}A(v).$$

(4) Let v be a divisorial valuation over $\mathbb{P}^2$ centered at a closed point p such that $v(B) > A(v)$, and suppose that $A(v) \geq 100v(\mathfrak{m}_p)$. Since the center of v is p, $v(\mathfrak{m}_p) > 0$. By Definition 2.1 and the definitions of $v(D)$ and $v(\mathfrak{a})$ after Definition 2.3, $A(tv) = tA(v)$, $(tv)(B) = tv(B)$, and

$(tv)(\mathfrak{m}_p) = tv(\mathfrak{m}_p)$ for any real number $t > 0$. By Definition 2.8, $\nu := v/v(\mathfrak{m}_p)$ is a normalized divisorial valuation at p. Thus

$$\nu(B) > A(\nu) \geq 100.$$

By Theorem 2.9 (1), (3), (4), $1 \leq \alpha(\nu) \leq A(\nu) - 1 < +\infty$.

If $\nu(P) \leq \frac{\deg P}{3} A(\nu)$ for any prime divisor $P \subset \operatorname{Supp} B$, then

$$\nu(B) = \sum_{P \subset \operatorname{Supp} B} \operatorname{mult}_P B \cdot \nu(P) \leq \frac{A(\nu)}{3} \sum_{P \subset \operatorname{Supp} B} \operatorname{mult}_P B \cdot \deg P = A(\nu),$$

a contradiction. Thus there exists an irreducible component C of B, of degree d, such that

$$\nu(C) > \frac{d}{3} A(\nu). \tag{4.8}$$

In particular $\nu(C) > 0$, so $p \in C$. Let $D_2 := B - (\operatorname{mult}_C B)C$. Since C and $D_2 \geq 0$ have no common component and $\deg D_2 = 3 - d \operatorname{mult}_C B \geq 0$, Lemma 2.10 (3) implies

$$\nu(C)\nu(D_2) \leq \alpha(\nu) d (3 - d \operatorname{mult}_C B).$$

Since $\alpha(\nu) \leq A(\nu) - 1$, we have

$$\nu(B) \leq \operatorname{mult}_C B \cdot \nu(C) + \frac{(A(\nu) - 1) d (3 - d \operatorname{mult}_C B)}{\nu(C)}. \tag{4.9}$$

We distinguish three cases.

Case 1. $d \geq 4$. By (4.8), $\nu(C) - A(\nu) + 1 > \frac{d-3}{3} A(\nu) + 1 > 0$. In particular, $\nu(C) > A(\nu) - 1$. By Lemma 4.5 and $\alpha(\nu) \leq A(\nu) - 1$,

$$\frac{(d-3)^2}{9} A(\nu)^2 < (\nu(C) - A(\nu) + 1)^2 \leq (d-1)^2 \alpha(\nu) < (d-1)^2 A(\nu).$$

Since $d - 1 \leq 3(d - 3)$ for $d \geq 4$, we obtain

$$A(\nu) < 9 \left(\frac{d-1}{d-3} \right)^2 \leq 81,$$

which contradicts $A(\nu) \geq 100$.

Case 2. $d \in \{1, 2\}$. Then C is a line or a smooth conic, and Lemma 4.7 gives $\nu(C) \leq \frac{d}{3} A(\nu)$, which contradicts (4.8).

Case 3. $d = 3$. By (4.8), $\nu(C) > A(\nu)$, so $a(\nu, \mathbb{P}^2, C) = A(\nu) - \nu(C) < 0$ and $(\mathbb{P}^2, C)$ is not lc at p. We show that

$$\operatorname{mult}_C B \leq \frac{17}{18}. \tag{4.10}$$

If $f^{6m}(C) \neq C$, then $\operatorname{mult}_C B < \frac{2}{3}$ by Lemma 4.2 (2) for the endomorphism f^m. If $f^{6m}(C) = C$, then Lemma 3.2 (3) for the endomorphism f^{6m} implies $e := e_C(f^{6m}) \leq \frac{17}{18} q^{6m}$, so

$$\operatorname{mult}_C B = \frac{e-1}{q^{6m}-1} \leq \frac{e}{q^{6m}} \leq \frac{17}{18}.$$

Since C is a divisor of degree 3 with coefficient 1 and $(\mathbb{P}^2, C)$ is not lc at p, Lemma 4.4 with $D = C$ implies $\nu(C) \leq A(\nu) + 1$. By (4.9) with $d = 3$, $A(\nu) < \nu(C) \leq A(\nu) + 1$, and (4.10),

$$\begin{aligned}
\nu(B) &\leq \operatorname{mult}_C B \cdot (A(\nu) + 1) + \frac{9(1 - \operatorname{mult}_C B)(A(\nu) - 1)}{A(\nu)} \\
&\leq \operatorname{mult}_C B \cdot (A(\nu) + 1) + 9(1 - \operatorname{mult}_C B) \\
&= A(\nu) + 1 - (1 - \operatorname{mult}_C B)(A(\nu) - 8) \leq A(\nu) + 1 - \frac{A(\nu) - 8}{18} < A(\nu),
\end{aligned}$$

where the last inequality holds since $A(\nu) \geq 100$. The last display contradicts $\nu(B) > A(\nu)$.

Therefore $A(v) < 100 v(\mathfrak{m}_p)$, and the theorem follows. □

4.3. **A non-lc monomial valuation.**

Lemma 4.8. *Let $p \in \mathbb{P}^2$ be a closed point, let x, y be regular parameters of $\mathcal{O}_{\mathbb{P}^2,p}$, let a, b be real numbers such that $0 < 10a \le b$, and let $v := \mathrm{mon}_{x,y}(a,b)$. Then for any irreducible curve $C \subset \mathbb{P}^2$ of degree $d \ge 3$,*

$$v(C) \le \frac{d}{3}(a+b).$$

Proof. Let $\psi \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p} = \mathbb{C}[[x,y]]$ be a local equation of C at p, and let $k := v(C) = v(\psi)$. If $p \notin C$, then $k = 0$. Thus we may assume that $p \in C$. Since C is reduced, ψ is reduced.

Step 1. In this step we prove that for any non-zero $\xi, \zeta \in \mathbb{C}[[x,y]]$ without common irreducible factor,

$$(\xi \cdot \zeta)_p \ge \frac{v(\xi)v(\zeta)}{ab}. \tag{4.11}$$

Let $t := b/a$ and $\nu := \mathrm{mon}_{x,y}(1,t) = v/a$. Since $t \ge 1$, $i + tj \le t(i+j)$ for any monomial $x^i y^j$ and $\nu(y) = t$, the normalized valuation ν satisfies $\alpha(\nu) = t$. By Lemma 2.10(2),

$$(\xi \cdot \zeta)_p \ge \frac{\nu(\xi)\nu(\zeta)}{\alpha(\nu)} = \frac{v(\xi)v(\zeta)}{a^2 t} = \frac{v(\xi)v(\zeta)}{ab}.$$

Step 2. In this step we prove the lemma when $d \ge 4$. Suppose that $k > \frac{d}{3}(a+b)$. Let ψ_x, ψ_y be the partial derivatives of ψ with respect to x, y. Since $\partial(x^i y^j)/\partial x = i x^{i-1} y^j$ and $\partial(x^i y^j)/\partial y = j x^i y^{j-1}$,

$$v(\psi_x) \ge k - a \quad \text{and} \quad v(\psi_y) \ge k - b,$$

where $k - b > k - (a+b) > 0$. We let

$$\mu_p(C) := \dim_{\mathbb{C}} \mathbb{C}[[x,y]]/(\psi_x, \psi_y) \tag{4.12}$$

be the *Milnor number* of C at p, let

$$\delta_p(C) := \dim_{\mathbb{C}} \overline{\mathcal{O}}_{C,p}/\mathcal{O}_{C,p},$$

where $\overline{\mathcal{O}}_{C,p}$ is the integral closure of $\mathcal{O}_{C,p}$, and let $r_p(C)$ be the number of irreducible factors of ψ. Since ψ is reduced, $\mu_p(C) < +\infty$ and

$$\mu_p(C) = 2\delta_p(C) - r_p(C) + 1$$

by [Mil68, Theorem 10.5], see [GP18, formula (M) and Proposition 4.2] for $\mu_p(C)$ defined by (4.12). Since $\mu_p(C) < +\infty$, ψ_x and ψ_y are non-zero without common irreducible factor, and $\mu_p(C) = (\psi_x \cdot \psi_y)_p$. By (4.11),

$$\mu_p(C) \ge \frac{(k-a)(k-b)}{ab} = \frac{(a+b)^2}{ab} \cdot \frac{k}{a+b}\left(\frac{k}{a+b} - 1\right) + 1.$$

Since $b/a \ge 10$, $(a+b)^2/(ab) = b/a + a/b + 2 \ge \frac{121}{10}$. Since $k/(a+b) > d/3 \ge 4/3$ and $z(z-1)$ is strictly increasing in $z \ge 1/2$,

$$\mu_p(C) > \frac{121}{10} \cdot \frac{d}{3}\left(\frac{d}{3} - 1\right) + 1 = \frac{121}{90} d(d-3) + 1.$$

On the other hand, let $\overline{C}$ be the normalization of C. Since

$$\sum_{p' \in C} \delta_{p'}(C) = p_a(C) - h^1(\overline{C}, \mathcal{O}_{\overline{C}}) \le p_a(C)$$

and $r_p(C) \ge 1$,

$$\mu_p(C) \le 2\delta_p(C) \le 2p_a(C) = (d-1)(d-2) = d(d-3) + 2.$$

Since $d \geq 4$, $d(d-3) \geq 4$, so $\frac{31}{90}d(d-3) \geq \frac{124}{90} > 1$ and

$$\frac{121}{90}d(d-3) + 1 > d(d-3) + 2,$$

a contradiction. Thus $k \leq \frac{d}{3}(a+b)$.

Step 3. In this step we prove the lemma when $d = 3$, that is, we prove that $k \leq a + b$. Write $\psi = \sum_{i,j\geq 0} c_{ij}x^i y^j$. Since $k \leq ia + jb$ for any (i, j) with $c_{ij} \neq 0$ and $b \geq 10a$, we have

$$k \leq \max\{a, b, 2a, a + b, 3a\} = a + b$$

as soon as one of $c_{10}, c_{01}, c_{20}, c_{11}, c_{30}$ is non-zero. Suppose that $c_{10} = c_{01} = c_{20} = c_{11} = 0$. Since C is an irreducible cubic, $\operatorname{mult}_p C = 2$, so $c_{02} \neq 0$.

Let S be the curve $\{y = 0\}$ in a neighborhood of p, which is smooth at p. Since $c_{00} = c_{10} = c_{20} = 0$, $\psi(x, 0) = \sum_{i\geq 3} c_{i0}x^i$, so

$$(C \cdot S)_p = \dim_{\mathbb{C}} \mathbb{C}[[x]]/(\psi(x, 0)) \geq 3,$$

with equality if and only if $c_{30} \neq 0$. Let $\pi : X' \to \mathbb{P}^2$ be the blow-up of p, let E_1 be the exceptional prime divisor, let $C' := \pi_*^{-1}C$ and $S' := \pi_*^{-1}S$, and let $z_0 \in E_1$ be the point corresponding to the tangent direction of S at p. Then S' intersects E_1 transversally at z_0. Since the quadratic part of ψ is $c_{02}y^2$, the tangent cone of C at p is the double line in the tangent direction of S, so $C' \cap E_1 = \{z_0\}$. Since $p_a(C) = 1$ and $\operatorname{mult}_p C = 2$, $p_a(C') = p_a(C) - 1 = 0$, so C' is smooth because $\sum_{p'\in C'} \delta_{p'}(C') \leq p_a(C')$. Since $(C' \cdot E_1) = \operatorname{mult}_p C = 2$ and $C' \cap E_1 = \{z_0\}$, C' is tangent to E_1 at z_0. Thus C' and S' are smooth at z_0 with distinct tangent lines. Since $\pi^*C = C' + 2E_1$,

$$(C \cdot S)_p = (\pi^*C \cdot S')_{z_0} = (C' \cdot S')_{z_0} + 2(E_1 \cdot S')_{z_0} = 1 + 2 = 3.$$

Thus $c_{30} \neq 0$, so $k \leq 3a \leq a + b$.

The lemma follows. □

Proposition 4.9. *Let f be an endomorphism of $\mathbb{P}^2$ of degree $q \geq 2$, let $m \geq 1$ be an integer, let $B := \frac{1}{q^{6m}-1}R(f^{6m})$, and let v be a divisorial valuation centered at a closed point $p \in \mathbb{P}^2$. Assume that $v(B) > A(v)$. Then there exist affine linear coordinates x, y centered at p, a polynomial $\phi \in \mathbb{C}[x]$ with $\phi(0) = 0$, and positive integers a, b satisfying the following.*

(1) $a \leq b < 11a$.
(2) $a \deg \phi \leq b$.
(3) $w := \operatorname{mon}_{x,y-\phi(x)}(a, b)$ satisfies $w(B) > A(w)$.

Proof. **Step 1.** In this step we find regular parameters x_1, x_2 of $\mathcal{O}_{\mathbb{P}^2,p}$ and positive integers $a \leq b$ such that $u_1 := \operatorname{mon}_{x_1,x_2}(a, b)$ satisfies

$$u_1(B) > A(u_1) = a + b. \tag{4.13}$$

Since the center of v is p and $A(v) - v(B) < 0$, $(\mathbb{P}^2, B)$ is not lc at p. Write $B = \sum_i c_i B_i$, where the B_i are distinct prime divisors and $c_i > 0$. By [Kaw17, Theorem 1(ii)] for the germ of $\mathbb{P}^2$ at p and the $\mathbb{R}$-ideal $\prod_i \mathcal{O}_{\mathbb{P}^2}(-B_i)^{c_i}$ (a formal product of ideal sheaves with positive real exponents, whose log discrepancy at a prime divisor E on Y for a projective birational morphism $h : Y \to \mathbb{P}^2$ is $1 + \operatorname{mult}_E K_{Y/\mathbb{P}^2} - \sum_i c_i \operatorname{mult}_E h^*B_i$ [Kaw17, p. 188], which equals $a(E, \mathbb{P}^2, B)$), there exist regular parameters x_1, x_2 of $\mathcal{O}_{\mathbb{P}^2,p}$ and positive integers a, b such that the exceptional divisor F of the weighted blow-up of $\mathbb{P}^2$ with $\operatorname{wt}(x_1, x_2) = (a, b)$ satisfies $\operatorname{center}_{\mathbb{P}^2} F = p$ and $a(F, \mathbb{P}^2, B) < 0$. Possibly swapping x_1 and x_2, we may assume that $a \leq b$. Since u_1 is a positive real multiple of the divisorial valuation given by F, $a(u_1, \mathbb{P}^2, B) < 0$ by Definition 2.1, that is, $u_1(B) > A(u_1)$. Since $(\mathbb{P}^2, \{x_1x_2 = 0\})$ is log smooth near p and the weighted blow-up is toric in the coordinates x_1, x_2, F is an lc place of $(\mathbb{P}^2, \{x_1x_2 = 0\})$ near p, so $A(u_1) = u_1(x_1x_2) = a + b$.

Step 2. In this step we prove that $b < 11a$. Suppose that $b \geq 11a$. Since $\deg R(f^{6m}) = 3(q^{6m}-1)$ by the ramification formula,

$$\deg B = \sum_i c_i \deg B_i = 3.$$

Since $b \geq 10a$, by Lemma 4.8,

$$u_1(B_i) \leq \frac{\deg B_i}{3}(a+b) \tag{4.14}$$

for any B_i with $\deg B_i \geq 3$. If (4.14) held for all B_i with $\deg B_i \leq 2$, then $u_1(B) \leq \frac{\deg B}{3}(a+b) = a+b$, contradicting (4.13). Thus there exists a component C of B with $d := \deg C \in \{1,2\}$ and $u_1(C) > \frac{d}{3}(a+b)$, so C is a line or a smooth conic. Let $\nu := u_1/a = \mathrm{mon}_{x_1,x_2}(1, b/a)$. Then ν is a normalized divisorial valuation at p with $A(\nu) = 1 + \frac{b}{a} \geq 12$, and $\nu(B) > A(\nu)$ by (4.13). By Lemma 4.7, $\nu(C) \leq \frac{d}{3}A(\nu)$, that is, $u_1(C) \leq \frac{d}{3}(a+b)$, a contradiction. Thus $b < 11a$.

Step 3. In this step we construct x, y, ϕ. Let x', y' be regular parameters of $\mathcal{O}_{\mathbb{P}^2,p}$, and let ν' be a valuation on $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$, non-negative on $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$, with $\nu'(x') \geq a$ and $\nu'(y') \geq b$. We first prove that

$$\nu' \geq \mathrm{mon}_{x',y'}(a,b). \tag{4.15}$$

Let $\xi = \sum_{i,j} \gamma_{ij} x'^i y'^j \in \mathbb{C}[[x', y']]$ be non-zero, and let $N > \mathrm{mon}_{x',y'}(a,b)(\xi)$ be a real number. Then ξ is the sum of the finitely many terms $\gamma_{ij}x'^i y'^j$ with $ia + jb < N$ and an element of the ideal generated by the monomials $x'^i y'^j$ with $ia + jb \geq N$. Since $\nu'(x'^i y'^j) \geq ia + jb$ for any (i,j), $\nu'(\xi) \geq \mathrm{mon}_{x',y'}(a,b)(\xi)$.

Fix an affine chart $\mathbb{A}^2 \subset \mathbb{P}^2$ containing p. Since the affine linear functions on $\mathbb{A}^2$ vanishing at p span $\mathfrak{m}_p/\mathfrak{m}_p^2$, there exists an affine linear function x vanishing at p such that x, x_2 are regular parameters of $\mathcal{O}_{\mathbb{P}^2,p}$. Let y be an affine linear function vanishing at p such that x, y are affine linear coordinates centered at p, and let $u_2 := \mathrm{mon}_{x,x_2}(a,b)$. Since $a \leq b$, $u_1(\mathfrak{m}_p) = u_2(\mathfrak{m}_p) = a$. Thus $u_1(x) \geq a$, $u_2(x_1) \geq a$, and $u_1(x_2) = u_2(x_2) = b$, so $u_1 = u_2$ by (4.15) for $(x', y') = (x, x_2)$ and for $(x', y') = (x_1, x_2)$.

In $\widehat{\mathcal{O}}_{\mathbb{P}^2,p} = \mathbb{C}[[x, x_2]]$, write

$$y = \psi(x) + x_2\chi, \qquad \text{where } \psi(x) := y|_{x_2=0} \in \mathbb{C}[[x]] \text{ and } \chi \in \mathbb{C}[[x, x_2]].$$

Then $\psi(0) = 0$, and $\chi(p) \neq 0$ since x, y are regular parameters. Let $n := \lfloor b/a \rfloor$, and let $\phi \in \mathbb{C}[x]$ be the sum of the terms of ψ of degree at most n. Then $\phi(0) = 0$, $a \deg \phi \leq na \leq b$, and

$$\psi - \phi \in x^{n+1}\mathbb{C}[[x]], \qquad (n+1)a > b.$$

Since $\phi(x) - \phi'(0)x \in \mathfrak{m}_p^2$, x and $y - \phi(x)$ are regular parameters of $\mathcal{O}_{\mathbb{P}^2,p}$. Let

$$w := \mathrm{mon}_{x,y-\phi(x)}(a,b).$$

Since

$$y - \phi(x) = (\psi - \phi)(x) + x_2\chi \quad \text{and} \quad x_2 = \chi^{-1}\big(y - \phi(x) - (\psi - \phi)(x)\big),$$

we have

$$u_2(y - \phi(x)) \geq \min\{(n+1)a, b\} = b \quad \text{and} \quad w(x_2) \geq \min\{b, (n+1)a\} = b.$$

Since moreover $u_2(x) = w(x) = a$, (4.15) for $(x', y') = (x, y - \phi(x))$ and for $(x', y') = (x, x_2)$ implies $w = u_2 = u_1$. By (4.13) and **Step 2**, $w(B) > A(w)$ and $a \leq b < 11a$.

The proposition follows. □

5. Return of non-lc points to a fixed point

Set-up 5.1. Let q, f, B_n, C, g, and Q be as follows.

(1) f is an endomorphism of $\mathbb{P}^2$ of degree q, where $q \geq 2$ is an integer.
(2) $B_n := \frac{1}{q^n-1}R(f^n)$ for any integer $n \geq 1$.
(3) C is the reduced (possibly empty) curve on $\mathbb{P}^2$ whose irreducible components are the irreducible curves P with $\operatorname{mult}_P B_{288} > \frac{3}{5}$.
(4) $g := f^{17280}$ and $Q := q^{17280}$.

Proposition 5.2. *Notation and conditions as in Set-up 5.1. Let $n \geq 4 \cdot 10^8$ be an integer such that $(\mathbb{P}^2, B_{17280n})$ is not lc. Then $\operatorname{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ for a closed point p with $g(p) = p$.*

Proposition 5.2 is proved in Subsection 5.2, starting from Construction-Proposition 5.3(3).

5.1. The curve C.

Construction-Proposition 5.3. Notation and conditions as in Set-up 5.1. Then $\deg C \leq 4$, and there exists a (possibly empty) 0-dimensional closed subscheme $Z \subset \mathbb{P}^2$ such that:

(1) For any sufficiently small rational number $\epsilon > 0$,
$$\mathcal{J}\left(\mathbb{P}^2, \frac{5}{3}(1-\epsilon)B_{288}\right) = \mathcal{O}_{\mathbb{P}^2}(-C) \otimes \mathcal{I}_Z.$$
(2) The length of Z satisfies
$$\operatorname{length} Z \leq h^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2 - \deg C)).$$
(3) For any integer $n \geq 1$, $\operatorname{Nlc}(\mathbb{P}^2, B_{288n}) \subset C \cup \operatorname{Supp} Z$.

Proof. Let $D := \frac{5}{3}B_{288}$. Since $R(f^{288}) \geq 0$, $D \geq B_{288} \geq 0$. By Theorem 4.1(1) for $m = 48$, $\operatorname{mult}_P B_{288} \leq 1$ for any irreducible curve $P \subset \mathbb{P}^2$, so

$$\operatorname{mult}_P D \leq \frac{5}{3} \quad \text{for any irreducible curve } P \subset \mathbb{P}^2. \tag{5.1}$$

Since $\deg R(f^{288}) = 3(q^{288} - 1)$, $\deg D = 5$. We show that

$$\operatorname{Nlc}(\mathbb{P}^2, B_{288n}) \subset \operatorname{Nlc}(\mathbb{P}^2, D) \quad \text{for any integer } n \geq 1. \tag{5.2}$$

Since $D \geq B_{288}$, (5.2) holds for $n = 1$. Let $n \geq 2$ be an integer and $x \in \operatorname{Nlc}(\mathbb{P}^2, B_{288n})$. By Lemma 2.4 for the endomorphism f^{288n} of degree q^{288n}, there exists a divisorial valuation v over $\mathbb{P}^2$ whose center contains x such that
$$A\left((f^{288n})_* v\right) > q^{288n} A(v).$$
Let $w := (f^{288})_* v$. By Lemma 2.4, w is a divisorial valuation over $\mathbb{P}^2$, and since $f^{288n} = f^{288(n-1)} \circ f^{288}$,
$$(f^{288n})_* v = (f^{288(n-1)})_* w.$$
By Theorem 4.1(3) for $m = 48(n-1)$ and the valuation w,
$$A\left((f^{288n})_* v\right) \leq \frac{3q^{288(n-1)} - 1}{2} A(w).$$
Since $A(v) > 0$, we obtain
$$A(w) > \frac{2q^{288n}}{3q^{288(n-1)} - 1} A(v) > \frac{2q^{288}}{3} A(v).$$
By Lemma 2.4 for f^{288}, $A(w) = A(v) + v(R(f^{288})) = A(v) + (q^{288} - 1)v(B_{288})$. Thus
$$v(D) = \frac{5}{3} \cdot \frac{A(w) - A(v)}{q^{288} - 1} > \frac{5(2q^{288} - 3)}{9(q^{288} - 1)} A(v) > A(v),$$

where the last inequality holds because $5(2q^{288}-3)-9(q^{288}-1)=q^{288}-6>0$. Since the center of v contains x, $(\mathbb{P}^2, D)$ is not lc at x. This proves (5.2).

By (5.1), $\operatorname{mult}_P D \le \frac{5}{3} < 2$ for any prime divisor P on $\mathbb{P}^2$, and $\operatorname{mult}_P D > 1$ if and only if $\operatorname{mult}_P B_{288} > \frac{3}{5}$, that is, $P \subset C$. Since $\deg D = 5$, Lemma 4.3 for $k=5$ gives a (possibly empty) 0-dimensional closed subscheme $Z \subset \mathbb{P}^2$ with (1), (2), and $\operatorname{Nlc}(\mathbb{P}^2, D) = C \cup \operatorname{Supp} Z$, so (3) follows from (5.2). Since $D \ge \sum_P (\operatorname{mult}_P D)P$, where the sum is over the irreducible components P of C, and $\operatorname{mult}_P D > 1$ for any such P, $\deg C < \deg D = 5$, so $\deg C \le 4$. □

Proposition 5.4. *Notation and conditions as in Set-up 5.1, and let Z be as in Construction-Proposition 5.3. Then:*

(1) If $P \subset \mathbb{P}^2$ is an irreducible curve with $f^{6k}(P) \ne P$ for $k = 1, 2, 3$, then $\operatorname{mult}_P B_{6n} < \frac{26}{n+1}$ for any integer $n \ge 1$.

(2) For any irreducible component P of C, $f^{288}(P) = P$.

(3) For any integer $n \ge 8192$,

$$\operatorname{Nlc}(\mathbb{P}^2, B_{288n}) \subset (\operatorname{Supp} Z \backslash C) \cup \bigcup_P \pi_P\left(\left\{z \in \overline{P} \mid h_P^2(z) = z\right\}\right),$$

where the union is over the irreducible components P of C, $\pi_P : \overline{P} \to P$ is the normalization, and $h_P : \overline{P} \to \overline{P}$ is the morphism with $\pi_P \circ h_P = f^{288} \circ \pi_P$.

Proof. (1) We first prove three inequalities. Let $U_1, \dots, U_N \subset \mathbb{P}^2$ be pairwise distinct irreducible curves. By Definition 2.2, $\operatorname{mult}_{U_j} R(f^6) = e_{U_j}(f^6) - 1$ for any j, and $\deg R(f^6) = 3(q^6-1)$. Since $R(f^6) \ge 0$,

$$\sum_{j=1}^{N} \left(e_{U_j}(f^6) - 1\right) \deg U_j \le 3(q^6-1). \tag{5.3}$$

If moreover there exists an irreducible curve $V \subset \mathbb{P}^2$ with $f^6(U_j) = V$ for any j, then $\operatorname{mult}_{U_j}(f^6)^*V = e_{U_j}(f^6)$ for any j and $\deg(f^6)^*V = q^6 \deg V$, so

$$\sum_{j=1}^{N} e_{U_j}(f^6) \deg U_j \le q^6 \deg V. \tag{5.4}$$

If $N = 4$, then since $\deg U_j \ge 1$ for any j, (5.3) implies $\sum_{j=1}^{4} e_{U_j}(f^6) \le 3q^6 + 1$, so by the inequality of arithmetic and geometric means and since $q^6 \ge 64$,

$$\prod_{j=1}^{4} \frac{e_{U_j}(f^6)}{q^6} \le \left(\frac{3q^6+1}{4q^6}\right)^4 \le \left(\frac{25}{32}\right)^4 = \frac{390625}{1048576} < \frac{1}{2}. \tag{5.5}$$

Let P be as in (1). For any integer $i \ge 0$, let $P_i := f^{6i}(P)$ and $u_i := e_{P_i}(f^6)/q^6$, and for any integer $n \ge 0$, let

$$a_n := \prod_{i=0}^{n-1} u_i,$$

so $a_0 = 1$. By Lemma 4.2(1) for the endomorphism f of degree q, $0 < u_i \le 1$ for any i, so a_n is decreasing in n and $a_n \le 1$. Since $f^{6n} = f^{6(n-1)} \circ f^6$ and ramification indices of extensions of discrete valuation rings are multiplicative, induction on n implies

$$e_P(f^{6n}) = \prod_{i=0}^{n-1} e_{P_i}(f^6) = q^{6n} a_n.$$

Thus for any integer $n \ge 1$,

$$\operatorname{mult}_P B_{6n} = \frac{e_P(f^{6n}) - 1}{q^{6n} - 1} = \frac{q^{6n} a_n - 1}{q^{6n} - 1} \le a_n,$$

where the inequality holds because $a_n \le 1$. Since a_n is decreasing, $(n+1)a_n \le \sum_{i=0}^{n} a_i$. Thus it suffices to show that

$$\sum_{n=0}^{+\infty} a_n < 26. \tag{5.6}$$

Let $\rho := (25/32)^4 < \frac{1}{2}$. Let $n_0 \in \mathbb{N}^+ \cup \{+\infty\}$, and assume that $P_i, P_{i+1}, P_{i+2}, P_{i+3}$ are pairwise distinct for any integer $i \ge 0$ with $i+3 < n_0$. For any integers $j \ge 0$ and $0 \le i \le 3$ with $4j+i < n_0$, $a_{4j+i} \le a_{4j} \le \rho^j$ by (5.5) applied to the blocks $P_{4j'}, \dots, P_{4j'+3}$ for $0 \le j' < j$. Thus

$$\sum_{0 \le n < n_0} a_n \le 4 \sum_{j=0}^{+\infty} \rho^j = \frac{4}{1-\rho} < 8. \tag{5.7}$$

Suppose that there do not exist integers $i \ge 0$ and $k \in \{1,2,3\}$ with $P_{i+k} = P_i$. If $P_{i+i_1} = P_{i+i_2}$ for integers $i \ge 0$ and $0 \le i_1 < i_2 \le 3$, then $P_{i+i_1+(i_2-i_1)} = P_{i+i_1}$, which is impossible. Thus (5.7) holds for $n_0 = +\infty$, and (5.6) follows.

Suppose that there exist integers $i \ge 0$ and $k \in \{1,2,3\}$ with $P_{i+k} = P_i$. Let

$$m := \min\{i \in \mathbb{N} \mid P_{i+k} = P_i \text{ for some } k \in \{1,2,3\}\}.$$

Since $f^{6k}(P) \ne P$ for $k = 1,2,3$, $m \ge 1$. Let k be the smallest positive integer with $P_{m+k} = P_m$, so $k \le 3$ and $P_{i+k} = P_i$ for any $i \ge m$. The curves $P_0, \dots, P_{m-1}$ are pairwise distinct: if $P_{i_1} = P_{i_2}$ with $0 \le i_1 < i_2 < m$, then $P_{i_1} = P_{i_1+j(i_2-i_1)}$ for any $j \in \mathbb{N}$. Choosing j with $i_3 := i_1 + j(i_2 - i_1) \ge m$, we get $P_{i_1+k} = f^{6k}(P_{i_3}) = P_{i_3+k} = P_{i_3} = P_{i_1}$, contradicting the minimality of m. By (5.7) for $n_0 = m$,

$$\sum_{n=0}^{m-1} a_n < 8.$$

Let $\gamma := \prod_{i=m}^{m+k-1} u_i \le 1$. Since $u_{i+k} = u_i$ for any $i \ge m$ and $u_i \le 1$ for any i, $a_{m+jk+i} \le a_m \gamma^j$ for any integers $j \ge 0$ and $0 \le i < k$. Thus if $\gamma < 1$, then

$$\sum_{n=m}^{+\infty} a_n \le \frac{k a_m}{1-\gamma}. \tag{5.8}$$

It remains to show that $\sum_{n=m}^{+\infty} a_n \le 18$. We consider three cases.

Case 1: $u_i \le \frac{1}{2}$ for some $i \in \{m, \dots, m+k-1\}$. Then $\gamma \le \frac{1}{2}$, and by (5.8), $\sum_{n=m}^{+\infty} a_n \le 2k a_m \le 6$.

Case 2: $u_i > \frac{1}{2}$ for any $i \in \{m, \dots, m+k-1\}$, and the degrees $\deg P_m, \dots, \deg P_{m+k-1}$ are not all equal. For any $i \ge m$, $\deg P_i \le 6$: otherwise, since $e_{P_i}(f^6) > \frac{q^6}{2}$ and $q^6 > 8$,

$$\left(e_{P_i}(f^6) - 1\right) \deg P_i > 7\left(\frac{q^6}{2} - 1\right) > 3(q^6 - 1),$$

contradicting (5.3) for $N = 1$. Since $P_{m+k} = P_m$ and the degrees are not all equal, there exists $i \in \{m, \dots, m+k-1\}$ with $\deg P_i > \deg P_{i+1}$. By (5.4) for $N = 1$, $U_1 = P_i$, and $V = P_{i+1}$,

$$u_i \le \frac{\deg P_{i+1}}{\deg P_i} \le \frac{\deg P_i - 1}{\deg P_i} \le \frac{5}{6}.$$

Thus $\gamma \le \frac{5}{6}$, and by (5.8), $\sum_{n=m}^{+\infty} a_n \le 6k \le 18$.

Case 3: $u_i > \frac{1}{2}$ for any $i \in \{m, \dots, m+k-1\}$, and $\deg P_i = \deg P_m$ for any $i \ge m$. As in Case 2, $\deg P_m \le 6$. If $P_{m-1} = P_{m+k-1}$, then $P_{m-1+k} = f^{6k}(P_{m+k-1}) = P_{m+2k-1} = P_{m+k-1} = P_{m-1}$, contradicting the minimality of m. Thus $P_{m-1} \ne P_{m+k-1}$, and

$$f^6(P_{m-1}) = P_m = P_{m+k} = f^6(P_{m+k-1}).$$

By (5.4) for $N = 2$, $(U_1, U_2) = (P_{m-1}, P_{m+k-1})$, and $V = P_m$,

$$e_{P_{m-1}}(f^6)\deg P_{m-1} + e_{P_{m+k-1}}(f^6)\deg P_m \le q^6 \deg P_m.$$

Since $\gamma \le u_{m+k-1}$, $\deg P_{m-1} \ge 1$, and $\deg P_m \le 6$,

$$u_{m-1} \le \frac{\deg P_m}{\deg P_{m-1}}(1 - u_{m+k-1}) \le 6(1-\gamma).$$

Since $u_{m-1} > 0$, $\gamma < 1$, and $a_m = a_{m-1}u_{m-1} \le 6(1-\gamma)a_{m-1}$. By (5.8),

$$\sum_{n=m}^{+\infty} a_n \le 6k a_{m-1} \le 18.$$

In all three cases, $\sum_{n=0}^{+\infty} a_n < 8 + 18 = 26$. This proves (5.6) and hence (1).

(2) Let P be an irreducible component of C, so $\mathrm{mult}_P B_{288} > \frac{3}{5}$. Since $288 = 6 \cdot 48$ and

$$\frac{26}{48+1} = \frac{26}{49} < \frac{3}{5},$$

(1) for $n = 48$ implies $f^{6k}(P) = P$ for some $k \in \{1, 2, 3\}$. Since k divides 48,

$$f^{288}(P) = (f^{6k})^{48/k}(P) = P.$$

(3) Let $n \ge 8192$ be an integer and $x \in \mathrm{Nlc}(\mathbb{P}^2, B_{288n})$. By Construction-Proposition 5.3(3), $x \in C \cup \mathrm{Supp}\, Z$. If $x \notin C$, then $x \in \mathrm{Supp}\, Z \backslash C$. Suppose that $x \in C$, let P be an irreducible component of C containing x, and suppose that $h_P^2(z) \ne z$ for any $z \in \pi_P^{-1}(x)$. By Construction-Proposition 5.3, $\deg P \le \deg C \le 4$, and by (2), $f^{288}(P) = P$. Since $B_{288n} = \frac{1}{(q^{288})^n - 1} R((f^{288})^n)$, by Theorem 3.1 for the endomorphism f^{288} of degree $q^{288} \ge 4$ and the curve P, $(\mathbb{P}^2, B_{288n})$ is lc at x, a contradiction. Thus $x \in \pi_P(\{z \in \overline{P} \mid h_P^2(z) = z\})$. □

Proposition 5.5. *Notation and conditions as in Set-up 5.1. Then:*

(1) For any integer $n \ge 8192$ such that $(\mathbb{P}^2, B_{288n})$ is not lc, $\mathrm{Nlc}(\mathbb{P}^2, B_{288n}) = \{p\}$ for a closed point p, $f^{288(k+i)}(p) = f^{288k}(p)$ for some integers $0 \le k \le 10$ and $1 \le i \le 6$, and $f^{576}(p) = p$ if $p \in C$.

(2) For any integer $n \ge 137$ such that $(\mathbb{P}^2, B_{17280n})$ is not lc, $\mathrm{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ for a closed point p with $g^2(p) = g(p)$.

Proof. We first prove (1). Let $n \ge 8192$ be an integer such that $(\mathbb{P}^2, B_{288n})$ is not lc. Since $288n = 6 \cdot 48n$, Theorem 4.1(2) implies that $\mathrm{Nlc}(\mathbb{P}^2, B_{288n}) = \{p\}$ for a closed point p. Since $(\mathbb{P}^2, B_{288n})$ is not lc at p, there exists a divisorial valuation v over $\mathbb{P}^2$ whose center contains p with $v(B_{288n}) > A(v)$. By Theorem 4.1(1), $\mathrm{mult}_{P'} B_{288n} \le 1$ for any irreducible curve $P' \subset \mathbb{P}^2$, so the center of v is not a curve. Thus the center of v is a closed point at which $(\mathbb{P}^2, B_{288n})$ is not lc, that is, v is centered at p. By Lemma 2.4 for f^{288n},

$$A\big((f^{288n})_* v\big) = A(v) + (q^{288n} - 1)v(B_{288n}) > q^{288n} A(v). \tag{5.9}$$

For any integer $0 \le j \le n$, let

$$v_j := (f^{288j})_* v \quad \text{and} \quad p_j := f^{288j}(p).$$

By Lemma 2.4, v_j is a divisorial valuation over $\mathbb{P}^2$, the center of v_j is p_j, and $v_{j+1} = (f^{288})_* v_j$ for any $0 \le j < n$. Let Z be as in Construction-Proposition 5.3, and let

$$S := C \cup \mathrm{Supp}\, Z \quad \text{and} \quad V := \mathrm{Supp}\, Z \backslash C.$$

By Construction-Proposition 5.3(2),

$$\#V \le \mathrm{length}\, Z \le h^0\big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2 - \deg C)\big) \le 6. \tag{5.10}$$

For any irreducible component P of C, $f^{288}(P) = P$ by Proposition 5.4(2). Let $\pi_P : \overline{P} \to P$ be the normalization and $h_P : \overline{P} \to \overline{P}$ the morphism with $\pi_P \circ h_P = f^{288} \circ \pi_P$.

Step 1. In this step we prove

$$\#\{j \in \mathbb{Z} \mid 0 \le j < n,\ p_j \notin S\} \le 4. \tag{5.11}$$

For any integers $0 \le j \le j' \le n$,

$$A(v_{j'}) \le \frac{3}{2} q^{288(j'-j)} A(v_j). \tag{5.12}$$

If $j' = j$, then (5.12) is trivial. If $j' > j$, then $v_{j'} = (f^{6\cdot 48(j'-j)})_* v_j$, and Theorem 4.1(3) for $48(j'-j)$ in place of m implies

$$A(v_{j'}) \le \frac{3q^{288(j'-j)} - 1}{2} A(v_j),$$

hence (5.12).

Let $0 \le j < n$ be an integer with $p_j \notin S$. By Construction-Proposition 5.3(1), for any sufficiently small rational number $\epsilon > 0$, the multiplier ideal

$$\mathcal{J}\Big(\mathbb{P}^2, \frac{5}{3}(1-\epsilon)B_{288}\Big)$$

is trivial near p_j, so by Lemma 2.7(1), $\big(\mathbb{P}^2, \frac{5}{3}(1-\epsilon)B_{288}\big)$ is klt at p_j, and $\frac{5}{3}(1-\epsilon)v_j(B_{288}) < A(v_j)$ because v_j is centered at p_j. Letting ϵ tend to 0, we obtain $v_j(B_{288}) \le \frac{3}{5}A(v_j)$. Since $q^{288} \ge 36$, Lemma 2.4 for f^{288} implies

$$A(v_{j+1}) = A(v_j) + (q^{288}-1)v_j(B_{288}) \le \frac{3q^{288}+2}{5} A(v_j) \le \frac{11}{18} q^{288} A(v_j). \tag{5.13}$$

Suppose that there exist integers $0 \le j_1 < j_2 < \cdots < j_5 < n$ with $p_{j_t} \notin S$ for any $1 \le t \le 5$, and let $j_0 := -1$. We apply (5.12) to the pairs $(j_{t-1}+1, j_t)$ for $1 \le t \le 5$ and to the pair (j_5+1, n), and (5.13) to $j = j_t$ for $1 \le t \le 5$. Since $v_0 = v$ and

$$\sum_{t=1}^{5} (j_t - j_{t-1}) + (n - j_5 - 1) = n,$$

we obtain

$$A(v_n) \le \Big(\frac{3}{2}\Big)^6 \Big(\frac{11}{18}\Big)^5 q^{288n} A(v) = \frac{3}{2}\Big(\frac{11}{12}\Big)^5 q^{288n} A(v) = \frac{161051}{165888} q^{288n} A(v) < q^{288n} A(v).$$

Since $v_n = (f^{288n})_* v$, this contradicts (5.9). Thus (5.11) holds.

Step 2. In this step we prove the following. Assume that $p \notin C$, and let $1 \le m \le 10$ be an integer and P an irreducible component of C with $p_m \in P$. Then there exists $z \in \pi_P^{-1}(p_m)$ with $h_P^2(z) = z$, and in particular

$$f^{576}(p_m) = p_m. \tag{5.14}$$

Let $d := \deg P$ and $e := e_P(f^{288})$. By Construction-Proposition 5.3, $d \le \deg C \le 4$. Since $f^{288}(P) = P$, $P \not\subset \operatorname{Supp}(f^{288})^* D'$ for any divisor $D' \ge 0$ with $P \not\subset \operatorname{Supp} D'$. Thus, by induction on m,

$$(f^{288m})^* P = e^m P + D$$

for a divisor $D \ge 0$ with $P \not\subset \operatorname{Supp} D$. Since f^{288m} is an endomorphism of degree q^{288m}, $\deg (f^{288m})^* P = q^{288m} d$, and hence

$$\deg D = d(q^{288m} - e^m).$$

Since $(f^{288})^* P \ge eP$ and $\deg (f^{288})^* P = q^{288} d$, $e \le q^{288}$. Let $\theta := e/q^{288} \in (0, 1]$.

Since $p \notin C$, $v(P) = 0$. By Lemma 2.5, and since $\operatorname{mult}_p D \le \deg D$,

$$v_m(P) = v\big((f^{288m})^* P\big) = v(D) \le (\operatorname{mult}_p D)\big(A(v) - v(\mathfrak{m}_p)\big) \le d(q^{288m} - e^m) A(v). \tag{5.15}$$

Let $N := n-m$, so $N \geq 8182$, and write $R(f^{288N}) = (e^N-1)P + R'_N$. Suppose that $h_P^2(z) \neq z$ for any $z \in \pi_P^{-1}(p_m)$. By Lemma 3.5 for the endomorphism f^{288} of degree $q^{288} \geq 4$, the curve P, the point p_m, the integer $N \geq 2$ and the valuation v_m centered at p_m,

$$v_m(R'_N) < \Big(\frac{8+32d^2}{N} q^{288N} + 6\Big) A(v_m). \tag{5.16}$$

Since $v_n = (f^{288N})_* v_m$, Lemma 2.4 for f^{288N} implies

$$A(v_n) = A(v_m) + v_m\big(R(f^{288N})\big) = A(v_m) + (e^N-1)v_m(P) + v_m(R'_N).$$

We bound $A(v_m) \leq \frac{3}{2} q^{288m} A(v)$ by (5.12), $(e^N-1)v_m(P) \leq e^N v_m(P)$ by (5.15), and $v_m(R'_N)$ by (5.16). Since $n = m+N$, we obtain

$$\begin{aligned}\frac{A(v_n)}{q^{288n}A(v)} &< \frac{3}{2}q^{-288N} + d\theta^N(1-\theta^m) + \frac{3}{2}\Big(\frac{8+32d^2}{N} + 6q^{-288N}\Big) \\ &= d\theta^N(1-\theta^m) + \frac{12+48d^2}{N} + \frac{21}{2}q^{-288N}.\end{aligned}$$

Since $1-\theta^m \leq m(1-\theta)$, $\theta^N(1-\theta) \leq \frac{1}{N+1}$ for $\theta \in [0,1]$, $d \leq 4$, and $m \leq 10$, we have $d\theta^N(1-\theta^m) \leq \frac{40}{N+1} < \frac{40}{N}$. Since $d \leq 4$, $12+48d^2 \leq 780$. Since $q \geq 2$, $\frac{21}{2}q^{-288N} < \frac{1}{N}$. Thus

$$\frac{A(v_n)}{q^{288n}A(v)} < \frac{821}{N} \leq \frac{821}{8182} < 1,$$

which contradicts (5.9). Hence there exists $z \in \pi_P^{-1}(p_m)$ with $h_P^2(z) = z$, and

$$f^{576}(p_m) = f^{576}\big(\pi_P(z)\big) = \pi_P\big(h_P^2(z)\big) = \pi_P(z) = p_m.$$

Step 3. In this step we prove (1). We consider three cases.

Case 1: $p \in C$. Since $p \notin \operatorname{Supp} Z \backslash C$, Proposition 5.4(3) implies that there exist an irreducible component P of C and $z \in \overline{P}$ with $h_P^2(z) = z$ and $\pi_P(z) = p$. Thus $f^{576}(p) = \pi_P(h_P^2(z)) = p$, which is the last conclusion of (1), and (1) holds with $k = 0$ and $i = 2$.

Case 2: $p \notin C$, and $p_j \in C$ for some integer $1 \leq j < n$. Let m be the smallest such j. Then $p_0, \ldots, p_{m-1}$ are pairwise distinct: if $p_a = p_b$ for some integers $0 \leq a < b \leq m-1$, then $p_{j'+b-a} = p_{j'}$ for any integer $j' \geq a$, so $p_m \in \{p_a, \ldots, p_{b-1}\}$, which is disjoint from C, a contradiction. Since $p_j \notin C$ for any $0 \leq j < m$, either $p_j \in V$ or $p_j \notin S$. Since $p_0, \ldots, p_{m-1}$ are pairwise distinct, $\#\{0 \leq j < m \mid p_j \in V\} \leq \#V \leq 6$ by (5.10), and since $m < n$, $\#\{0 \leq j < m \mid p_j \notin S\} \leq 4$ by (5.11). Thus $m \leq 10$. Let P be an irreducible component of C with $p_m \in P$. By **Step 2**, (5.14) holds, that is, $f^{288(m+2)}(p) = f^{288m}(p)$. Thus (1) holds with $k = m \leq 10$ and $i = 2$.

Case 3: $p_j \notin C$ for any integer $0 \leq j < n$. Then $p_j \in V$ for any $0 \leq j < n$ with $p_j \in S$. Since $10 < n$, (5.11) implies

$$\#\{0 \leq j \leq 10 \mid p_j \in V\} \geq 11 - 4 = 7 > 6 \geq \#V.$$

Thus there exist integers $0 \leq a < b \leq 10$ with $p_a = p_b$. Let $i := \min\{i' \in \mathbb{N}^+ \mid p_{a+i'} = p_a\}$. Then $a \leq 9$, $i \leq b-a \leq 10$, $p_{j+i} = p_j$ for any $j \geq a$, and $p_a, \ldots, p_{a+i-1}$ are pairwise distinct. Let $a \leq j \leq a+i-1$ be an integer. If $p_j \notin S$, then $p_{j+ti} = p_j \notin S$ for any $0 \leq t \leq 4$, and $j + 4i \leq a + 5i - 1 \leq 58 < n$. This contradicts (5.11). Thus $p_j \in V$. Since $p_a, \ldots, p_{a+i-1}$ are pairwise distinct points of V, (5.10) implies $i \leq 6$. Hence $f^{288(a+i)}(p) = f^{288a}(p)$, and (1) holds with $k = a \leq 9$ and this i.

We prove (2). Let $n \geq 137$ be an integer such that $(\mathbb{P}^2, B_{17280n})$ is not lc. Since

$$17280n = 288 \cdot 60n \quad \text{and} \quad 60n \geq 8220 \geq 8192,$$

(1) for $60n$ in place of n implies that $\operatorname{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ for a closed point p, and $f^{288(k+i)}(p) = f^{288k}(p)$ for some integers $0 \leq k \leq 10$ and $1 \leq i \leq 6$. Thus $f^{288(j+i)}(p) = f^{288j}(p)$ for any $j \geq k$.

Since i divides 60, $f^{288(j+60)}(p) = f^{288j}(p)$ for any $j \geq k$. Since $60 \geq k$ and $g = f^{288\cdot 60}$, taking $j = 60$ we obtain

$$g^2(p) = f^{288\cdot 120}(p) = f^{288\cdot 60}(p) = g(p).$$

□

5.2. **The non-lc point is a fixed point.** In this subsection we prove Proposition 5.2.

Lemma 5.6. *Let $D \geq 0$ be an $\mathbb{R}$-divisor on $\mathbb{P}^2$ of degree 3. Let ν be a normalized divisorial valuation at a closed point $p \in \mathbb{P}^2$ with $A(\nu) > 10^4$ and $\nu(D) > \frac{13}{20}A(\nu)$. Then there exists an irreducible component C of D through p with*

$$\deg C \leq 4, \quad \operatorname{mult}_C D > \frac{3}{5}, \quad \nu(C) > \frac{13 \deg C}{60} A(\nu), \quad \text{and} \quad \nu(D) \leq \operatorname{mult}_C D \cdot \nu(C) + 14.$$

Proof. Write $D = \sum_{i=1}^m c_i C_i$, where $C_1, \dots, C_m$ are distinct irreducible curves and $c_i > 0$ for any i. Since $\deg D = 3$, $\sum_{i=1}^m c_i \deg C_i = 3$. If $\nu(C_i) \leq \frac{13 \deg C_i}{60} A(\nu)$ for any i, then

$$\nu(D) = \sum_{i=1}^m c_i \nu(C_i) \leq \frac{13}{60} A(\nu) \sum_{i=1}^m c_i \deg C_i = \frac{13}{20} A(\nu),$$

a contradiction. Thus there exists i with $\nu(C_i) > \frac{13 \deg C_i}{60} A(\nu)$. We let $C := C_i$, $d := \deg C$, and $c := \operatorname{mult}_C D = c_i$. Then

$$\nu(C) > \frac{13d}{60} A(\nu), \quad c > 0, \quad \text{and} \quad cd \leq 3. \tag{5.17}$$

Since $\nu(C) > 0$ and the center of ν is p, $p \in C$.

Since ν is divisorial, $A(\nu) < +\infty$, and Theorem 2.9 (3) implies

$$\alpha(\nu) \leq A(\nu) - 1 < A(\nu).$$

Let $D' := D - cC$. Then $D' \geq 0$, C is not a component of D', and $\deg D' = 3 - cd$. By Lemma 2.10 (3), $\nu(C)\nu(D') \leq \alpha(\nu) d(3 - cd)$. By (5.17),

$$\nu(D') \leq \frac{\alpha(\nu) d(3 - cd)}{\nu(C)} \leq \frac{60(3 - cd)}{13} \leq \frac{180}{13} < 14.$$

Since $\nu(D) = c\nu(C) + \nu(D')$, we obtain

$$\nu(D) < c\nu(C) + 14. \tag{5.18}$$

We show that $c > \frac{3}{5}$. We have

$$\nu(C) \leq A(\nu) - 1 + (d-1)\sqrt{A(\nu) - 1}\,:$$

if $\nu(C) > A(\nu) - 1$, this is Lemma 4.5. If $\nu(C) \leq A(\nu) - 1$, this follows from $d \geq 1$. Since $cd \leq 3$,

$$c\nu(C) \leq c\big(A(\nu) - 1\big) + c(d-1)\sqrt{A(\nu) - 1} < cA(\nu) + 3\sqrt{A(\nu)}.$$

Suppose that $c \leq \frac{3}{5}$. Then by (5.18) and $A(\nu) > 10^4$,

$$\frac{\nu(D)}{A(\nu)} < c + \frac{3}{\sqrt{A(\nu)}} + \frac{14}{A(\nu)} < \frac{3}{5} + \frac{3}{100} + \frac{14}{10^4} < \frac{13}{20},$$

which contradicts $\nu(D) > \frac{13}{20} A(\nu)$. Thus $c > \frac{3}{5}$. Since $cd \leq 3$,

$$d \leq \frac{3}{c} < 5,$$

so $d \leq 4$. Together with (5.17) and (5.18), the lemma follows. □

Proof of Proposition 5.2. Since $n \geq 137$, Proposition 5.5 (2) implies $\mathrm{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ for a closed point p with $g^2(p) = g(p)$. Let $y := g(p)$. Then $g(y) = y$, and $g^i(p) = y$ for any integer $i \geq 1$. Since $f^{17280i} = g^i$ and $q^{17280i} = Q^i$,

$$B_{17280i} = \frac{1}{Q^i - 1} R(g^i) \quad \text{for any integer } i \geq 1.$$

Suppose that $y \neq p$.

Step 1. In this step we bound the thinness of the pushforwards of a valuation centered at p by the normalized volume.

For a divisorial valuation u centered at a closed point $x \in \mathbb{P}^2$ and a real number $c > 0$, let

$$\mathfrak{a}_c(u) := \{\phi \in \widehat{\mathcal{O}}_{\mathbb{P}^2,x} \mid u(\phi) \geq c\},$$

and let

$$\mathrm{vol}(u) := \limsup_{c \to +\infty} 2c^{-2} \dim_{\mathbb{C}} \widehat{\mathcal{O}}_{\mathbb{P}^2,x}/\mathfrak{a}_c(u)$$

be the volume of u [FJ04, Remark 3.33]. If u is normalized, then $\mathrm{vol}(u) = 1/\alpha(u)$ [FJ04, Remark 3.33]. For any real number $a > 0$, $\mathfrak{a}_c(au) = \mathfrak{a}_{c/a}(u)$, so $\mathrm{vol}(au) = a^{-2}\,\mathrm{vol}(u)$. We let

$$\widehat{\mathrm{vol}}(u) := A(u)^2\,\mathrm{vol}(u)$$

be the normalized volume of u. Since $A(au) = aA(u)$, $\widehat{\mathrm{vol}}(au) = \widehat{\mathrm{vol}}(u)$ for any real number $a > 0$. If u is normalized, then Theorem 2.9 (3), (4) imply

$$\widehat{\mathrm{vol}}(u) = \frac{A(u)^2}{\alpha(u)} \geq \frac{\big(1 + \alpha(u)\big)^2}{\alpha(u)} \geq 4.$$

Since $u/u(\mathfrak{m}_x)$ is normalized, we obtain

$$\widehat{\mathrm{vol}}(u) \geq 4 \quad \text{for any divisorial valuation } u \text{ centered at a closed point of } \mathbb{P}^2. \tag{5.19}$$

Since g is finite and flat (Definition 2.2), $\widehat{\mathcal{O}}_{\mathbb{P}^2,x}$ is a free $\widehat{\mathcal{O}}_{\mathbb{P}^2,g(x)}$-module of rank $\deg_x g$ for any closed point $x \in \mathbb{P}^2$, and $\sum_{x \in g^{-1}(y)} \deg_x g = Q^2$. Since p and y are distinct points of $g^{-1}(y)$,

$$\deg_p g + \deg_y g \leq Q^2.$$

Let $\sigma := \deg_y g/Q^2$. Then $0 < \sigma < 1$ and $\deg_p g \leq (1 - \sigma)Q^2$. Since $g^i(p) = y$ for any $i \geq 1$ and ranks of free modules multiply under composition, for any integer $j \geq 1$,

$$\deg_p(g^j) = \deg_p g \cdot (\deg_y g)^{j-1} \leq (1 - \sigma)\sigma^{j-1}Q^{2j}. \tag{5.20}$$

For a divisorial valuation v centered at p and an integer $j \geq 1$, let $w_j := (g^j)_* v$. By Lemma 2.4, w_j is a divisorial valuation. Since $g^j(p) = y$, w_j is centered at y, and $v((g^j)^*\phi) = w_j(\phi)$ for any $\phi \in \widehat{\mathcal{O}}_{\mathbb{P}^2,y}$. Thus for any $c > 0$,

$$(g^j)^* \mathfrak{a}_c(w_j) \subset \mathfrak{a}_c(v).$$

Since $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$ is a free $\widehat{\mathcal{O}}_{\mathbb{P}^2,y}$-module of rank $\deg_p(g^j)$ via $(g^j)^*$, $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}/\mathfrak{a}_c(v)$ is a quotient of

$$\widehat{\mathcal{O}}_{\mathbb{P}^2,p} \otimes_{\widehat{\mathcal{O}}_{\mathbb{P}^2,y}} \big(\widehat{\mathcal{O}}_{\mathbb{P}^2,y}/\mathfrak{a}_c(w_j)\big),$$

so

$$\dim_{\mathbb{C}} \widehat{\mathcal{O}}_{\mathbb{P}^2,p}/\mathfrak{a}_c(v) \leq \deg_p(g^j) \dim_{\mathbb{C}} \widehat{\mathcal{O}}_{\mathbb{P}^2,y}/\mathfrak{a}_c(w_j).$$

Multiplying by $2c^{-2}$ and taking $\limsup_{c \to +\infty}$, we obtain $\mathrm{vol}(v) \leq \deg_p(g^j)\,\mathrm{vol}(w_j)$. By (5.19) and (5.20),

$$\left(\frac{A(w_j)}{Q^j A(v)}\right)^2 = \frac{\widehat{\mathrm{vol}}(w_j)}{\widehat{\mathrm{vol}}(v)} \cdot \frac{\mathrm{vol}(v)}{Q^{2j}\,\mathrm{vol}(w_j)} \leq \frac{\widehat{\mathrm{vol}}(w_j)}{4} \cdot \frac{\deg_p(g^j)}{Q^{2j}} \leq \frac{\widehat{\mathrm{vol}}(w_j)}{4}(1 - \sigma)\sigma^{j-1}.$$

Since $0 < \sigma < 1$,

$$j(1-\sigma)\sigma^{j-1} \le (1-\sigma)\sum_{i=0}^{j-1}\sigma^i = 1-\sigma^j \le 1.$$

Thus

$$\left(\frac{A((g^j)_*v)}{Q^jA(v)}\right)^2 \le \frac{\widehat{\mathrm{vol}}((g^j)_*v)}{4j} \tag{5.21}$$

for any divisorial valuation v centered at p and any integer $j \ge 1$.

Step 2. In this step we find a pushforward w of a valuation centered at p with $A(w) > 10^4 w(\mathfrak{m}_y)$, and a curve P by Lemma 5.6.

Since $(\mathbb{P}^2, B_{17280n})$ is not lc at p, there exists a divisorial valuation v over $\mathbb{P}^2$ whose center contains p with $v(B_{17280n}) > A(v)$. By Theorem 4.1 (1) with $m = 2880n$, $\mathrm{mult}_\Gamma B_{17280n} \le 1$ for any irreducible curve $\Gamma \subset \mathbb{P}^2$, so the center of v is not a curve. Thus the center of v is p. By Lemma 2.4,

$$A\big((g^n)_*v\big) = A(v) + v\big(R(g^n)\big) = A(v) + (Q^n-1)v(B_{17280n}) > Q^nA(v). \tag{5.22}$$

Let $j := 10^8$ and $k := n - j$. Since $n \ge 4\cdot 10^8$, $k \ge 3j$. Let $w := w_j = (g^j)_*v$, a divisorial valuation centered at y. Then $(g^k)_*w = (g^n)_*v$. By Theorem 4.1 (3) with $m = 2880j$ and $m = 2880k$,

$$A(w) \le \frac{3}{2}Q^jA(v) \quad \text{and} \quad A\big((g^n)_*v\big) \le \frac{3}{2}Q^kA(w).$$

By (5.22), $Q^nA(v) < \frac{3}{2}Q^kA(w)$, so

$$\frac{2}{3} < \frac{A(w)}{Q^jA(v)} \le \frac{3}{2}.$$

By (5.21), $\widehat{\mathrm{vol}}(w) > \frac{16j}{9} > 10^8$. Let $\nu := w/w(\mathfrak{m}_y)$, a normalized divisorial valuation at y. Since $\alpha(\nu) \ge 1$ (Theorem 2.9 (1)) and $\mathrm{vol}(\nu) = 1/\alpha(\nu)$,

$$A(\nu)^2 \ge \frac{A(\nu)^2}{\alpha(\nu)} = \widehat{\mathrm{vol}}(\nu) = \widehat{\mathrm{vol}}(w) > 10^8.$$

Thus

$$A(w) > 10^4 w(\mathfrak{m}_y). \tag{5.23}$$

By Lemma 2.4 and (5.22),

$$A(w) + (Q^k-1)w(B_{17280k}) = A\big((g^k)_*w\big) > Q^nA(v) = Q^k\cdot Q^jA(v) \ge \frac{2}{3}Q^kA(w).$$

Since $Q^k > 21$,

$$\frac{w(B_{17280k})}{A(w)} > \frac{\frac{2}{3}Q^k-1}{Q^k-1} = \frac{2}{3} - \frac{1}{3(Q^k-1)} > \frac{13}{20}.$$

Since $\deg R(g^k) = 3(Q^k-1)$, $B_{17280k} \ge 0$ is a $\mathbb{Q}$-divisor of degree 3. By (5.23), $A(\nu) > 10^4$, and $\nu(B_{17280k}) > \frac{13}{20}A(\nu)$. By Lemma 5.6 for $D = B_{17280k}$ and ν, there exists an irreducible component P of B_{17280k} through y with

$$d := \deg P \le 4, \quad b := \mathrm{mult}_P B_{17280k} > \frac{3}{5}, \quad \text{and} \quad w(P) > \frac{13d}{60}A(w). \tag{5.24}$$

Step 3. In this step we show that P is an irreducible component of C and that $p \notin C$.

Since $f^{6\cdot 2880k} = g^k$, Proposition 5.4 (1) with $2880k$ in place of n implies that if $f^{6i}(P) \ne P$ for $i = 1, 2, 3$, then $b < \frac{26}{2880k+1} < \frac{3}{5}$, which contradicts (5.24). Thus $f^{6i}(P) = P$ for some $i \in \{1,2,3\}$. Since $6i$ divides 288, $f^{288}(P) = P$, and since $g = (f^{288})^{60}$, $g(P) = P$.

Let $e' := e_P(f^{288})$. By Theorem 4.1 (1) with $m = 48$,

$$\frac{e'-1}{q^{288}-1} = \operatorname{mult}_P B_{288} \le 1,$$

so $e' \le q^{288}$. Since $f^{288}(P) = P$, $e_P(f^{288i}) = e'^i$ for any integer $i \ge 1$. In particular, $e_P(g^k) = e'^{60k}$ and

$$b = \frac{e'^{60k}-1}{q^{288\cdot 60k}-1}.$$

Since $b > 0$, $e' \ge 2$. Since $e' \le q^{288}$,

$$\frac{e'^{60k}-1}{e'-1} = \sum_{i=0}^{60k-1} e'^i \le \sum_{i=0}^{60k-1} q^{288i} = \frac{q^{288\cdot 60k}-1}{q^{288}-1}.$$

Thus $\operatorname{mult}_P B_{288} \ge b > \frac{3}{5}$, and P is an irreducible component of C.

Since $B_{288\cdot 60n} = B_{17280n}$ and $60n \ge 8192$, $p \in \operatorname{Nlc}(\mathbb{P}^2, B_{288\cdot 60n})$. Suppose that $p \in C$. By Proposition 5.5 (1) for $60n$ in place of n, $f^{576}(p) = p$. Since 576 divides 17280, $g(p) = p$, which contradicts $g(p) = y \ne p$. Thus $p \notin C$. In particular, $p \notin P$.

Step 4. In this step we bound $w(P)$ from above by the degree of $(g^j)^*P$ off P and reach a contradiction.

Let $e := e_P(g) = e'^{60}$ and $t := e/Q$. Since $e' \le q^{288}$, $e \le Q$, so $0 < t \le 1$. Since $g(P) = P$, $e_P(g^i) = e^i$ for any integer $i \ge 1$. Thus $\operatorname{mult}_P(g^j)^*P = e^j$, the divisor $(g^j)^*P - e^jP \ge 0$ does not contain P, and

$$\deg\big((g^j)^*P - e^jP\big) = d(Q^j - e^j).$$

By the definition of $w = (g^j)_*v$,

$$w(P) = v\big((g^j)^*P\big) = e^j v(P) + v\big((g^j)^*P - e^jP\big).$$

Since $p \notin P$, $v(P) = 0$.

By Lemma 2.5, and since $\operatorname{mult}_p D \le \deg D$, $v(D) \le \deg D \cdot A(v)$ for any divisor $D \ge 0$ on $\mathbb{P}^2$. For $D = (g^j)^*P - e^jP$, we obtain

$$w(P) \le d(Q^j - e^j)A(v). \tag{5.25}$$

By (5.23), $A(w) \ge 100w(\mathfrak{m}_y)$, so Theorem 4.1 (4) with $m = 2880k$ implies $w(B_{17280k}) \le A(w)$. By Lemma 2.4 and (5.22),

$$Q^n A(v) < A\big((g^k)_*w\big) = A(w) + (Q^k - 1)w(B_{17280k}) \le Q^k A(w),$$

so $A(w) > Q^j A(v)$. By (5.24) and (5.25),

$$\frac{13d}{60}Q^j A(v) < \frac{13d}{60}A(w) < w(P) \le d(Q^j - e^j)A(v).$$

Dividing by $dQ^jA(v) > 0$, we obtain

$$t^j < \frac{47}{60}.$$

On the other hand, since $e \le Q$, $k \ge 3j$, and $0 < t \le 1$,

$$\frac{3}{5} < b = \frac{e^k-1}{Q^k-1} \le \frac{e^k}{Q^k} = t^k \le (t^j)^3 < \left(\frac{47}{60}\right)^3 = \frac{103823}{216000} < \frac{3}{5},$$

a contradiction. Thus $y = p$, that is, $g(p) = p$. The proposition follows. □

6. Components of the curve C

In this section, we prove Proposition 6.1, which lists the possible components of C and locates the non-lc points of $(\mathbb{P}^2, B_{17280n})$ for $n \gg 0$.

Proposition 6.1. *Notation and conditions as in Set-up 5.1, and let Z be as in Construction-Proposition 5.3. Then:*

(1) Any irreducible component of C is a line, a smooth cubic, or a nodal cubic.
(2) For any integer $n \geq 8192$, the pair $(\mathbb{P}^2, B_{288n})$ is lc at any closed point of any component of C of degree 3, and at any closed point lying on two components of C.
(3) For any integer $n \geq 4 \cdot 10^8$ and any closed point $p \in \mathrm{Nlc}(\mathbb{P}^2, B_{17280n})$, we have $g(p) = p$, and either $p \in \mathrm{Supp}\, Z \backslash C$, or exactly one irreducible component of C contains p and it is a line.

Lemma 6.2. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$ and $C \subset \mathbb{P}^2$ a smooth conic with $g(C) = C$. Then $e_C(g) \leq \lceil Q/2 \rceil$. In particular, $\mathrm{mult}_C \frac{1}{Q^n-1} R(g^n) \leq \frac{1}{2}$ for any integer $n \geq 1$.*

The proof follows the method of symmetric determinantal presentations and duality in [Bea00, Section 2]. The result [Bea00, Theorem B] assumes that the hypersurface is integral, which fails for the multiple conics C_r below with $r \geq 2$.

Proof of Lemma 6.2. We choose homogeneous coordinates $[x_0 : x_1 : x_2]$ on $\mathbb{P}^2$ such that

$$C = \{c = 0\}, \qquad c := x_0 x_2 - x_1^2.$$

For any integer $r \geq 1$, we let $C_r \subset \mathbb{P}^2$ be the closed subscheme defined by c^r, so that $C_1 = C$. Let $\iota_r : C_r \to \mathbb{P}^2$ be the inclusion. For any line bundle $\mathcal{M}$ on C_r and any integer k, we let $\mathcal{O}_{C_r}(k) := \mathcal{O}_{\mathbb{P}^2}(k)|_{C_r}$ and $\mathcal{M}(k) := \mathcal{M} \otimes \mathcal{O}_{C_r}(k)$. We let $\psi : \mathbb{P}^1 \to C$ be the isomorphism given by

$$\psi([y_0 : y_1]) = [y_0^2 : y_0 y_1 : y_1^2],$$

so that $\psi^* \mathcal{O}_C(1) \cong \mathcal{O}_{\mathbb{P}^1}(2)$, with $\psi^* x_0 = y_0^2$, $\psi^* x_1 = y_0 y_1$, and $\psi^* x_2 = y_1^2$.

Step 1. We show that for any integer $r \geq 1$,

$$H^0(C_r, \mathcal{O}_{C_r}) = \mathbb{C}, \tag{6.1}$$

and $\mathrm{Pic}(C_r)$ has no non-zero element of order 2. In particular, $\mathcal{M} \cong \mathcal{M}'$ for any line bundles $\mathcal{M}, \mathcal{M}'$ on C_r with $\mathcal{M}^2 \cong \mathcal{M}'^2$, since $(\mathcal{M} \otimes \mathcal{M}'^{-1})^2 \cong \mathcal{O}_{C_r}$.

Since $C_1 = C \cong \mathbb{P}^1$ and, for $r \geq 2$, the kernel $\mathcal{N}_r$ of $\mathcal{O}_{C_r} \to \mathcal{O}_{C_{r-1}}$ satisfies $\mathcal{N}_r^2 = 0$ and $\mathcal{N}_r \cong \mathcal{O}_C(-2(r-1))$, induction on r with the exact sequences

$$0 \to H^0(C, \mathcal{N}_r) \to H^0(C_r, \mathcal{O}_{C_r}) \to H^0(C_{r-1}, \mathcal{O}_{C_{r-1}})$$

and

$$0 \to H^1(C, \mathcal{N}_r) \to \mathrm{Pic}(C_r) \to \mathrm{Pic}(C_{r-1}) \to 0$$

gives (6.1), and $\mathrm{Pic}(C_r)$ has no non-zero element of order 2 since $H^1(C, \mathcal{N}_r)$ is a $\mathbb{C}$-vector space.

Step 2. Let $r \geq 1$ be an integer and $m := 2r - 1$. We construct a symmetric $2r \times 2r$ matrix Φ_r whose entries are linear forms in x_0, x_1, x_2, such that

$$\det \Phi_r = \gamma_r c^r \text{ for some } \gamma_r \in \mathbb{C}^*, \tag{6.2}$$

and

$$\mathrm{rank}\, \Phi_r(\mathbf{z}) = 2r - 1 \text{ for any } \mathbf{z} \in \mathbb{C}^3 \backslash \{0\} \text{ with } c(\mathbf{z}) = 0. \tag{6.3}$$

Let u_1, u_2 be the standard basis of $\mathbb{C}^2$, and let $w_j := u_1^{m-j} u_2^j \in \mathrm{Sym}^m(\mathbb{C}^2)$ for $0 \leq j \leq m$. Then $w_0, \ldots, w_m$ is a basis of $\mathrm{Sym}^m(\mathbb{C}^2)$, and $m + 1 = 2r$. Let

$$\rho_m : \mathfrak{gl}_2(\mathbb{C}) \to \mathfrak{gl}(\mathrm{Sym}^m(\mathbb{C}^2))$$

be the infinitesimal action

$$\rho_m(\xi)(z_1\cdots z_m)=\sum_{i=1}^{m} z_1\cdots z_{i-1}(\xi z_i)z_{i+1}\cdots z_m,\quad z_1,\dots,z_m\in\mathbb{C}^2.$$

We identify $\rho_m(\xi)$ with the matrix of $\rho_m(\xi)$ in the basis $w_0,\dots,w_m$, and extend ρ_m linearly to 2×2 matrices with entries in $\mathbb{C}[x_0,x_1,x_2]$. For any $\tau\in GL_2(\mathbb{C})$ and $\xi\in\mathfrak{gl}_2(\mathbb{C})$, we have

$$\rho_m(\tau\xi\tau^{-1})=\operatorname{Sym}^m(\tau)\rho_m(\xi)\operatorname{Sym}^m(\tau)^{-1}.$$

Let

$$\eta_0:=\begin{pmatrix}1&0\\0&-1\end{pmatrix},\quad \eta_+:=\begin{pmatrix}0&1\\0&0\end{pmatrix},\quad \eta_-:=\begin{pmatrix}0&0\\1&0\end{pmatrix}.$$

Then for any $0\le j\le m$,

$$\rho_m(\eta_0)w_j=(m-2j)w_j,\quad \rho_m(\eta_+)w_j=jw_{j-1},\quad\text{and}\quad \rho_m(\eta_-)w_j=(m-j)w_{j+1},$$

where $w_{-1}:=0$ and $w_{m+1}:=0$. Let

$$N_0:=\begin{pmatrix}x_1&x_0\\-x_2&-x_1\end{pmatrix}=x_1\eta_0+x_0\eta_+-x_2\eta_-.$$

Then $\operatorname{tr}N_0=0$ and $\det N_0=c$.

Let $\Omega=(\Omega_{ij})_{0\le i,j\le m}$ be the matrix with

$$\Omega_{i,m-i}=\frac{(-1)^i}{\binom{m}{i}}\text{ for }0\le i\le m,\quad \Omega_{ij}=0\text{ if }i+j\ne m,$$

and let $\langle w_i,w_j\rangle:=\Omega_{ij}$ be the bilinear form on $\operatorname{Sym}^m(\mathbb{C}^2)$ with matrix Ω. Since m is odd, $\Omega_{m-i,i}=-\Omega_{i,m-i}$, so $\Omega^t=-\Omega$, and $\det\Omega\ne 0$. We show that

$$\rho_m(\xi)^t\Omega+\Omega\rho_m(\xi)=0\text{ for any }\xi\in\{\eta_0,\eta_+,\eta_-\},\tag{6.4}$$

that is, $\langle\rho_m(\xi)w_i,w_j\rangle+\langle w_i,\rho_m(\xi)w_j\rangle=0$ for any $0\le i,j\le m$. For $\xi=\eta_0$, the left side is $(2m-2i-2j)\Omega_{ij}$, which is 0 because $\Omega_{ij}=0$ unless $i+j=m$. For $\xi=\eta_+$, the left side is 0 unless $i+j=m+1$, in which case $1\le i\le m$, $j=m+1-i$, and the left side is

$$i\Omega_{i-1,j}+j\Omega_{i,j-1}=\frac{(-1)^{i-1}i}{\binom{m}{i-1}}+\frac{(-1)^i(m+1-i)}{\binom{m}{i}}=0,$$

because $(m+1-i)\binom{m}{i-1}=i\binom{m}{i}$. For $\xi=\eta_-$, the left side is 0 unless $i+j=m-1$, in which case $m-j=i+1$ and the left side is

$$(m-i)\Omega_{i+1,j}+(m-j)\Omega_{i,j+1}=\frac{(-1)^{i+1}(m-i)}{\binom{m}{i+1}}+\frac{(-1)^i(i+1)}{\binom{m}{i}}=0,$$

because $(m-i)\binom{m}{i}=(i+1)\binom{m}{i+1}$. This proves (6.4).

We let

$$\Phi_r:=\Omega\rho_m(N_0)=x_1\Omega\rho_m(\eta_0)+x_0\Omega\rho_m(\eta_+)-x_2\Omega\rho_m(\eta_-).$$

The entries of Φ_r are linear forms in x_0,x_1,x_2, and by (6.4) and $\Omega^t=-\Omega$,

$$\Phi_r^t=\rho_m(N_0)^t\Omega^t=-\rho_m(N_0)^t\Omega=\Omega\rho_m(N_0)=\Phi_r.$$

Let $\mathbf{z}\in\mathbb{C}^3$ with $c(\mathbf{z})\ne 0$. The eigenvalues of $N_0(\mathbf{z})$ are $\pm\mu$, where $\mu^2=-c(\mathbf{z})\ne 0$, so

$$N_0(\mathbf{z})=\tau\operatorname{diag}(\mu,-\mu)\tau^{-1}=\tau(\mu\eta_0)\tau^{-1}$$

for some $\tau\in GL_2(\mathbb{C})$. Thus $\rho_m(N_0(\mathbf{z}))$ is conjugate to $\mu\rho_m(\eta_0)$, and

$$\det\rho_m(N_0(\mathbf{z}))=\prod_{j=0}^{m}(m-2j)\mu=(-1)^r\prod_{j=0}^{r-1}(m-2j)^2\cdot\mu^{2r}=\prod_{j=0}^{r-1}(m-2j)^2\cdot c(\mathbf{z})^r,$$

where we pair j with $m-j$. Since both sides are polynomial functions of $\mathbf{z}$ and the set $\{c\ne 0\}$ is dense in $\mathbb{C}^3$, (6.2) holds with $\gamma_r:=\det\Omega\cdot\prod_{j=0}^{r-1}(m-2j)^2\ne 0$.

Let $\mathbf{z} \in \mathbb{C}^3 \backslash \{0\}$ with $c(\mathbf{z}) = 0$. Then $\operatorname{tr} N_0(\mathbf{z}) = \det N_0(\mathbf{z}) = 0$, and $N_0(\mathbf{z}) \neq 0$ because the entries of N_0 include x_0, x_1, x_2. Thus $N_0(\mathbf{z})$ is a non-zero nilpotent matrix, so $N_0(\mathbf{z}) = \tau \eta_+ \tau^{-1}$ for some $\tau \in GL_2(\mathbb{C})$, and $\rho_m(N_0(\mathbf{z}))$ is conjugate to $\rho_m(\eta_+)$. Since $\rho_m(\eta_+)w_j = jw_{j-1}$ and $j \neq 0$ for $1 \leq j \leq m$, $\operatorname{rank} \rho_m(\eta_+) = m = 2r-1$. Since $\det \Omega \neq 0$, (6.3) holds.

Step 3. Let $r \geq 1$ be an integer. We show that there exists a line bundle $\mathcal{L}_r$ on C_r with $\mathcal{L}_r^2 \cong \mathcal{O}_{C_r}(1)$ and

$$H^0(C_r, \mathcal{L}_r(k)) = 0 \text{ for any integer } k < r-1. \tag{6.5}$$

By (6.2), $\det \Phi_r \neq 0$. Since the local rings of $\mathbb{P}^2$ are integral domains, the map

$$\mathcal{O}_{\mathbb{P}^2}(-1)^{2r} \xrightarrow{\Phi_r} \mathcal{O}_{\mathbb{P}^2}^{2r}$$

is injective. Let $z \in \mathbb{P}^2$ be a closed point. We trivialize $\mathcal{O}_{\mathbb{P}^2}(1)$ near z by a coordinate x_j with $x_j(z) \neq 0$, so that the entries of Φ_r and c^r become elements of $\mathcal{O}_{\mathbb{P}^2,z}$. If $z \notin C$, then $\det \Phi_r = \gamma_r c^r$ is a unit of $\mathcal{O}_{\mathbb{P}^2,z}$, and the cokernel of Φ_r vanishes at z. If $z \in C$, then by (6.3), some $(2r-1) \times (2r-1)$ minor of Φ_r is a unit of $\mathcal{O}_{\mathbb{P}^2,z}$. By row and column operations over $\mathcal{O}_{\mathbb{P}^2,z}$, there exist invertible matrices κ_1, κ_2 over $\mathcal{O}_{\mathbb{P}^2,z}$ with

$$\kappa_1 \Phi_r \kappa_2 = \operatorname{diag}(1, \dots, 1, \det(\kappa_1)\det(\kappa_2)\gamma_r c^r).$$

Thus

$$(\operatorname{coker} \Phi_r)_z \cong \mathcal{O}_{\mathbb{P}^2,z}/(c^r) = \mathcal{O}_{C_r,z}.$$

Hence there exist a line bundle $\mathcal{K}_r$ on C_r and an exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^2}(-1)^{2r} \xrightarrow{\Phi_r} \mathcal{O}_{\mathbb{P}^2}^{2r} \to (\iota_r)_* \mathcal{K}_r \to 0. \tag{6.6}$$

We apply $\mathcal{H}om_{\mathbb{P}^2}(-, \mathcal{O}_{\mathbb{P}^2}(-1))$ to (6.6). Since $(\iota_r)_*\mathcal{K}_r$ is a torsion sheaf,

$$\mathcal{H}om_{\mathbb{P}^2}((\iota_r)_*\mathcal{K}_r, \mathcal{O}_{\mathbb{P}^2}(-1)) = 0,$$

so

$$\mathcal{E}xt^1_{\mathbb{P}^2}((\iota_r)_*\mathcal{K}_r, \mathcal{O}_{\mathbb{P}^2}(-1)) \cong \operatorname{coker}\left(\mathcal{O}_{\mathbb{P}^2}(-1)^{2r} \xrightarrow{\Phi_r^t} \mathcal{O}_{\mathbb{P}^2}^{2r}\right).$$

Since $\Phi_r^t = \Phi_r$, we have

$$\mathcal{E}xt^1_{\mathbb{P}^2}((\iota_r)_*\mathcal{K}_r, \mathcal{O}_{\mathbb{P}^2}(-1)) \cong (\iota_r)_*\mathcal{K}_r. \tag{6.7}$$

On the other hand, applying $\mathcal{H}om_{\mathbb{P}^2}(-, \mathcal{O}_{\mathbb{P}^2}(-1))$ to

$$0 \to \mathcal{O}_{\mathbb{P}^2}(-2r) \xrightarrow{c^r} \mathcal{O}_{\mathbb{P}^2} \to (\iota_r)_*\mathcal{O}_{C_r} \to 0,$$

we obtain

$$\mathcal{E}xt^1_{\mathbb{P}^2}((\iota_r)_*\mathcal{O}_{C_r}, \mathcal{O}_{\mathbb{P}^2}(-1)) \cong \operatorname{coker}\left(\mathcal{O}_{\mathbb{P}^2}(-1) \xrightarrow{c^r} \mathcal{O}_{\mathbb{P}^2}(2r-1)\right) = (\iota_r)_*\mathcal{O}_{C_r}(2r-1).$$

Since $\mathcal{M}$ is locally isomorphic to $\mathcal{O}_{C_r}$, the last display gives

$$\mathcal{E}xt^1_{\mathbb{P}^2}((\iota_r)_*\mathcal{M}, \mathcal{O}_{\mathbb{P}^2}(-1)) \cong (\iota_r)_*(\mathcal{M}^{-1}(2r-1))$$

for any line bundle $\mathcal{M}$ on C_r. With $\mathcal{M} = \mathcal{K}_r$ and (6.7), we obtain $\mathcal{K}_r \cong \mathcal{K}_r^{-1}(2r-1)$, that is,

$$\mathcal{K}_r^2 \cong \mathcal{O}_{C_r}(2r-1).$$

We let $\mathcal{L}_r := \mathcal{K}_r(1-r)$. Then $\mathcal{L}_r^2 \cong \mathcal{O}_{C_r}(1)$, and twisting (6.6) by $\mathcal{O}_{\mathbb{P}^2}(1-r)$, we obtain the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^2}(-r)^{2r} \to \mathcal{O}_{\mathbb{P}^2}(1-r)^{2r} \to (\iota_r)_*\mathcal{L}_r \to 0.$$

Let $k < r-1$ be an integer. Twisting by $\mathcal{O}_{\mathbb{P}^2}(k)$, since $k+1-r < 0$ and $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(k-r)) = 0$, we have

$$H^0(C_r, \mathcal{L}_r(k)) = H^0(\mathbb{P}^2, (\iota_r)_*\mathcal{L}_r(k)) = 0.$$

This proves (6.5).

Step 4. We prove $e_C(g) \le \lceil Q/2 \rceil$. Let $e := e_C(g)$. Since

$$g^*\mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^2}(Q),$$

there exist homogeneous polynomials g_0, g_1, g_2 of degree Q without common zero on $\mathbb{P}^2$ with $g = [g_0 : g_1 : g_2]$. Since $g(C) = C$, c is irreducible, and $g^*C = g^*g(C)$ is the divisor of $g_0g_2 - g_1^2$, we have

$$e = \max\{i \in \mathbb{N} \mid c^i \text{ divides } g_0g_2 - g_1^2\}.$$

Thus $g|_{C_e} : C_e \to \mathbb{P}^2$ factors through C. Let $\widetilde{g} : C_e \to C$ be the induced morphism. We let

$$\mathcal{M} := \widetilde{g}^*(\psi^{-1})^*\mathcal{O}_{\mathbb{P}^1}(1) \quad \text{and} \quad \sigma_j := \widetilde{g}^*(\psi^{-1})^*y_j \in H^0(C_e, \mathcal{M}) \text{ for } j \in \{0,1\}.$$

Since y_0, y_1 generate $\mathcal{O}_{\mathbb{P}^1}(1)$, σ_0, σ_1 generate $\mathcal{M}$, so $H^0(C_e, \mathcal{M}) \neq 0$. Let

$$\alpha_C : ((\psi^{-1})^*\mathcal{O}_{\mathbb{P}^1}(1))^2 \to \mathcal{O}_C(1)$$

be the isomorphism sending y_0^2, y_0y_1, y_1^2 to $x_0|_C, x_1|_C, x_2|_C$. Under

$$\widetilde{g}^*\mathcal{O}_C(1) = g^*\mathcal{O}_{\mathbb{P}^2}(1)|_{C_e} \cong \mathcal{O}_{C_e}(Q),$$

we have $\widetilde{g}^*(x_j|_C) = g_j|_{C_e}$ for $j \in \{0,1,2\}$. Thus

$$\alpha := \widetilde{g}^*\alpha_C : \mathcal{M}^2 \to \mathcal{O}_{C_e}(Q)$$

is an isomorphism with

$$\alpha(\sigma_0^2) = g_0|_{C_e}, \quad \alpha(\sigma_0\sigma_1) = g_1|_{C_e}, \quad \text{and} \quad \alpha(\sigma_1^2) = g_2|_{C_e}. \tag{6.8}$$

Assume that $Q = 2k+1$ is odd. Since

$$\mathcal{L}_e(k)^2 \cong \mathcal{O}_{C_e}(2k+1) \cong \mathcal{M}^2,$$

by **Step 1**, $\mathcal{M} \cong \mathcal{L}_e(k)$. If $e > k+1$, then $k < e-1$, and by (6.5), $H^0(C_e, \mathcal{M}) = 0$, a contradiction. Thus $e \le k+1 = \lceil Q/2 \rceil$.

Assume that $Q = 2k$ is even, and suppose that $e > k$. Since $\mathcal{O}_{C_e}(k)^2 \cong \mathcal{M}^2$, by **Step 1**, there exists an isomorphism $\beta : \mathcal{M} \to \mathcal{O}_{C_e}(k)$. The restriction

$$H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(k)) \to H^0(C_e, \mathcal{O}_{C_e}(k))$$

is surjective, because the cokernel of the restriction injects into $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(k-2e)) = 0$. Let G_1, G_2 be homogeneous polynomials of degree k with

$$G_1|_{C_e} = \beta(\sigma_0) \quad \text{and} \quad G_2|_{C_e} = \beta(\sigma_1).$$

The isomorphisms α and $\beta^{\otimes 2}$ from $\mathcal{M}^2$ to $\mathcal{O}_{C_e}(2k)$ differ by multiplication by a unit of $H^0(C_e, \mathcal{O}_{C_e})$, which is a constant $a \in \mathbb{C}^*$ by (6.1): $\alpha = a\beta^{\otimes 2}$. Possibly replacing β with $\sqrt{a}\beta$ and G_1, G_2 with $\sqrt{a}G_1, \sqrt{a}G_2$, we may assume that $a = 1$. By (6.8), the homogeneous polynomials

$$g_0 - G_1^2, \quad g_1 - G_1G_2, \quad \text{and} \quad g_2 - G_2^2$$

of degree $2k$ vanish on C_e, so they are divisible by c^e. Since $\deg c^e = 2e > 2k$, they are 0. Thus $g = [G_1^2 : G_1G_2 : G_2^2]$, and $g(\mathbb{P}^2) \subset C$, which contradicts the surjectivity of g. Hence $e \le k = \lceil Q/2 \rceil$.

Step 5. We prove the last claim. Let $n \ge 1$ be an integer. Since g^n is an endomorphism of $\mathbb{P}^2$ of degree $Q^n \ge 2$ and $g^n(C) = C$, **Step 4** for g^n implies $e_C(g^n) \le \lceil Q^n/2 \rceil \le \frac{Q^n+1}{2}$. By Definition 2.2,

$$\operatorname{mult}_C \frac{1}{Q^n-1}R(g^n) = \frac{e_C(g^n)-1}{Q^n-1} \le \frac{\frac{Q^n+1}{2}-1}{Q^n-1} = \frac{1}{2}.$$

The lemma follows. □

Lemma 6.3. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 2$ and $p \in \mathbb{P}^2$ a closed point with $g(p) = p$. Assume that one of the following holds.*

(1) There exist irreducible curves $C_1, C_2 \subset \mathbb{P}^2$ with $g(C_1) = C_1$ and $g(C_2) = C_2$, which are smooth at p and intersect transversally at p, such that $e_{C_1}(g) + e_{C_2}(g) > Q$.

(2) There exists an irreducible nodal cubic $C \subset \mathbb{P}^2$ with node p and $g(C) = C$ such that $2e_C(g) > Q$, and, writing $\pi : \overline{C} \to C$ for the normalization, $h : \overline{C} \to \overline{C}$ for the morphism with $\pi \circ h = g \circ \pi$, and $\pi^{-1}(p) = \{z_1, z_2\}$, we have $h(\{z_1, z_2\}) = \{z_1, z_2\}$.

Then $\left(\mathbb{P}^2, \frac{1}{Q^n-1}R(g^n)\right)$ *is lc at p for any integer $n \geq 1$.*

Proof. We first prove that

$$v(R(g)) \leq (Q-1)A(v) \tag{6.9}$$

for any divisorial valuation v over $\mathbb{P}^2$ centered at p.

Step 1. In this step we write g and $R(g)$ in holomorphic coordinates at p.

For $i \in \{1,2\}$ we define Γ_i, $\overline{C}_i$, π_i, h_i, and z_i as follows. In case (1), Γ_i is the germ of C_i at p, $\pi_i : \overline{C}_i \to C_i$ is the normalization, $h_i : \overline{C}_i \to \overline{C}_i$ is the morphism with $\pi_i \circ h_i = g \circ \pi_i$, and z_i is the unique point of $\pi_i^{-1}(p)$. In case (2), $\overline{C}_i := \overline{C}$, $\pi_i := \pi$, $h_i := h$, and Γ_i is the image under π of a small neighborhood of z_i in $\overline{C}$. In both cases Γ_1 and Γ_2 are smooth analytic curve germs at p intersecting transversally, and π_i maps a neighborhood of z_i isomorphically onto Γ_i. Since $\pi_i(h_i(z_i)) = g(p) = p$, there exists a permutation σ of $\{1,2\}$ with

$$h_1(z_1) = z_{\sigma(1)} \quad \text{and} \quad h_2(z_2) = z_{\sigma(2)} :$$

in case (1), $\sigma = \mathrm{id}_{\{1,2\}}$, and in case (2), σ exists because $h(\{z_1, z_2\}) = \{z_1, z_2\}$. Thus $g(\Gamma_i) = \Gamma_{\sigma(i)}$ as germs for $i \in \{1,2\}$. Let $e_i := e_{C_i}(g)$ for $i \in \{1,2\}$ in case (1), and $e_1 = e_2 := e_C(g)$ in case (2), so that

$$e_1 + e_2 \geq Q + 1$$

in cases (1) and (2). For $i \in \{1,2\}$, let λ_i be the local degree of h_i at z_i. By Lemma 3.2, $\deg h_i = Q$, so $\lambda_i \leq Q$.

Let x, y be holomorphic coordinates at p with $\Gamma_1 = \{x = 0\}$ and $\Gamma_2 = \{y = 0\}$. Let $(x', y') := (x, y)$ if $\sigma = \mathrm{id}_{\{1,2\}}$ and $(x', y') := (y, x)$ otherwise, so that $\Gamma_{\sigma(1)} = \{x' = 0\}$ and $\Gamma_{\sigma(2)} = \{y' = 0\}$. We show that

$$g^*x' = x^{e_1}U \quad \text{and} \quad g^*y' = y^{e_2}V \tag{6.10}$$

for holomorphic germs U, V at p divisible by neither x nor y. In case (1), $\mathrm{div}(x') = C_{\sigma(1)} = C_1$ near p, so

$$\mathrm{mult}_{\Gamma_1} \mathrm{div}(g^*x') = \mathrm{mult}_{C_1} g^*C_1 = e_1.$$

In case (2), $\mathrm{div}(x'y') = C$ near p, so

$$\mathrm{mult}_{\Gamma_1} \mathrm{div}(g^*x' \cdot g^*y') = \mathrm{mult}_C g^*C = e_1.$$

As $g(\Gamma_1) = \{x' = 0\} \not\subset \{y' = 0\}$, g^*y' does not vanish identically on Γ_1, and thus $\mathrm{mult}_{\Gamma_1} \mathrm{div}(g^*x') = e_1$ in case (2) as well. In cases (1) and (2), since $g(\Gamma_2) = \{y' = 0\} \not\subset \{x' = 0\}$, g^*x' does not vanish identically on Γ_2. This proves the first equality of (6.10), and the second one follows by exchanging the roles of (x, x', Γ_1) and (y, y', Γ_2).

Since x' restricts to a coordinate on $\{y' = 0\} = \Gamma_{\sigma(2)}$, $x' \circ \pi_{\sigma(2)}$ is a uniformizer of $\overline{C}_{\sigma(2)}$ at $z_{\sigma(2)} = h_2(z_2)$, and $x \circ \pi_2$ is a uniformizer of $\overline{C}_2$ at z_2. By (6.10), $h_2^*(x' \circ \pi_{\sigma(2)}) = (g^*x') \circ \pi_2$ is $x^{e_1}U(x, 0)$ written in the uniformizer $x \circ \pi_2$. Thus $\lambda_2 = e_1 + m_1$, where $m_1 := (\mathrm{div}(U) \cdot \Gamma_2)_p$ is the order of vanishing of $U(x, 0)$ at $x = 0$. Exchanging the roles of $(x, x', \Gamma_1, z_1, U)$ and $(y, y', \Gamma_2, z_2, V)$ in the computation of λ_2 above, we obtain $\lambda_1 = e_2 + m_2$, where $m_2 := (\mathrm{div}(V) \cdot \Gamma_1)_p$ is the order of vanishing of $V(0, y)$ at $y = 0$. Let

$$\kappa_1 := Q - e_1, \qquad \kappa_2 := Q - e_2, \quad \text{and} \quad \kappa := \kappa_1 + \kappa_2.$$

Since $\lambda_1, \lambda_2 \leq Q$ and $e_1 + e_2 \geq Q + 1$, we have

$$m_1 \leq \kappa_1, \qquad m_2 \leq \kappa_2, \quad \text{and} \quad 0 \leq \kappa \leq Q - 1. \tag{6.11}$$

Let $D := R(g) - (e_1-1)C_1 - (e_2-1)C_2$ in case (1) and $D := R(g) - (e_1-1)C$ in case (2). By Definition 2.2, $D \geq 0$ and D does not contain C_1, C_2, resp. C. Let

$$J := \partial_x(g^*x')\partial_y(g^*y') - \partial_y(g^*x')\partial_x(g^*y').$$

Since $g^*(dx' \wedge dy') = Jdx \wedge dy$ and $K_{\mathbb{P}^2} = g^*K_{\mathbb{P}^2} + R(g)$, we have $R(g) = \mathrm{div}(J)$ near p. By (6.10),

$$J = x^{e_1-1}y^{e_2-1}W, \qquad W := (e_1U + x\partial_xU)(e_2V + y\partial_yV) - (y\partial_yU)(x\partial_xV). \tag{6.12}$$

Near p, both $(e_1-1)C_1+(e_2-1)C_2$ in case (1) and $(e_1-1)C$ in case (2) are equal to $\mathrm{div}(x^{e_1-1}y^{e_2-1})$, so $D = \mathrm{div}(W)$ near p.

Step 2. In this step we fix coprime positive integers a, b and compute the part of lowest weight of W.

Let a, b be coprime positive integers. Let $\nu_0 := \mathrm{mon}_{x,y}(a, b)$. We call $ai + bj = \nu_0(x^iy^j)$ the *weight* of the monomial x^iy^j, so $\nu_0(\phi)$ is the smallest weight of a monomial of a holomorphic germ $\phi \neq 0$. Let $u := \nu_0(U)$ and $w := \nu_0(V)$, and write

$$U_0 = \sum_{(i,j)\in I_U} c_{ij}x^iy^j \quad \text{and} \quad V_0 = \sum_{(i',j')\in I_V} c'_{i'j'}x^{i'}y^{j'}$$

for the sums of the terms of U of weight u and of the terms of V of weight w, where $I_U, I_V \subset \mathbb{N}^2$ are the sets of exponents with $c_{ij} \neq 0$, resp. $c'_{i'j'} \neq 0$. Since U contains the monomial x^{m_1} and V contains the monomial y^{m_2}, (6.11) implies $u \leq \kappa_1 a$ and $w \leq \kappa_2 b$. Thus, for any $(i, j) \in I_U$ and $(i', j') \in I_V$,

$$0 \leq i \leq \kappa_1, \qquad bj \leq \kappa_1 a, \qquad ai' \leq \kappa_2 b, \quad \text{and} \quad 0 \leq j' \leq \kappa_2. \tag{6.13}$$

In particular $abji' \leq ab\kappa_1\kappa_2$, and

$$(e_1 + i)(e_2 + j') - ji' \geq e_1e_2 - \kappa_1\kappa_2 = Q(e_1 + e_2 - Q) > 0. \tag{6.14}$$

The operators $x\partial_x$ and $y\partial_y$ multiply each monomial by a non-negative integer, so they do not decrease weights and preserve the set of monomials of a given weight. By (6.12), any monomial of $W - W_0$ has weight at least $u + w + 1$, where

$$W_0 := \sum_{(i,j)\in I_U} \sum_{(i',j')\in I_V} \big((e_1 + i)(e_2 + j') - ji'\big)c_{ij}c'_{i'j'}x^{i+i'}y^{j+j'}.$$

We show that $W_0 \neq 0$. Let $(i_*, j_*) \in I_U$ and $(i'_*, j'_*) \in I_V$ be the exponents with the largest x-exponents. For $(i, j) \in I_U$ and $(i', j') \in I_V$ with $i + i' = i_* + i'_*$, we have $i = i_*$ and $i' = i'_*$, and then $j = j_*$ and $j' = j'_*$ since $bj = u - ai$ and $bj' = w - ai'$. Thus the pair $\big((i_*, j_*), (i'_*, j'_*)\big)$ is the only pair in $I_U \times I_V$ contributing to the monomial $x^{i_*+i'_*}y^{j_*+j'_*}$ of W_0, and the coefficient of $x^{i_*+i'_*}y^{j_*+j'_*}$ in W_0 is non-zero by (6.14). Thus, by (6.12) and $u \leq \kappa_1 a$, $w \leq \kappa_2 b$,

$$\nu_0(W) = u + w, \qquad N := \nu_0(J) = (e_1 - 1)a + (e_2 - 1)b + u + w \leq (Q - 1)(a + b). \tag{6.15}$$

Moreover, by (6.13), the x-exponents of the monomials of W_0 belong to the interval $[0, \kappa_1 + \kappa_2 b/a]$.

Step 3. In this step we construct a chart in which ν_0 becomes the order along a divisor.

Claim 6.4. *Let X be a smooth surface, $q \in X$ a closed point, ξ, η holomorphic coordinates at q, and a, b coprime positive integers. Then there exist a composition of point blow-ups $\mu : Y \to X$ over q, a μ-exceptional prime divisor E, a finite subset $\Sigma \subset E$, an open neighborhood Ω of $E\backslash\Sigma$ in Y in the Euclidean topology, holomorphic functions s, τ on Ω, and integers $a', b' \geq 0$ with $ab' - a'b = 1$, satisfying the following.*

- *(i) τ has no zero on Ω, $E \cap \Omega = \{s = 0\}$, and $(s, \tau - \tau(q'))$ are holomorphic coordinates at q' for any $q' \in E\backslash\Sigma$.*
- *(ii) $\xi \circ \mu = s^a\tau^{a'}$ and $\eta \circ \mu = s^b\tau^{b'}$ on Ω.*
- *(iii) For any divisorial valuation v over X centered at q with $bv(\xi) = av(\eta)$, the center of v on Y is E or a closed point of $E\backslash\Sigma$.*

Proof. We use induction on $a+b$. If $a+b=2$, then $a=b=1$. Let μ be the blow-up of q and E the exceptional curve, let $\Sigma := E \cap \mu_*^{-1}\{\xi\eta = 0\}$, and let $(a', b') := (0, 1)$. Let $Y_0 \subset Y$ be the chart with holomorphic coordinates s, τ satisfying $\xi \circ \mu = s$ and $\eta \circ \mu = s\tau$. The chart Y_0 contains $E \backslash \mu_*^{-1}\{\xi = 0\}$. Let $\Omega := Y_0 \backslash \{\tau = 0\}$. Then (i) and (ii) hold. For (iii), $v(\eta/\xi) = v(\xi/\eta) = 0$. The function $\eta/\xi = \tau$ is regular and vanishes at the point $E \cap \mu_*^{-1}\{\eta = 0\}$, and ξ/η is regular and vanishes at the point $E \cap \mu_*^{-1}\{\xi = 0\}$. Thus neither point of Σ is the center of v, and the center of v, contained in $E = \mu^{-1}(q)$, is E or a point of $E \backslash \Sigma$.

Assume that $a+b \geq 3$, so that $a \neq b$ as a, b are coprime. By symmetry, after replacing (s, τ) with $(s\tau^j, \tau^{-1})$ for an integer $j \gg 0$ in the case $a < b$, we may assume that $a > b$. Let $\mu_1 : X_1 \to X$ be the blow-up of q, and let $q_1 \in \mu_1^{-1}(q)$ be the point at which $\xi_1 := \xi/\eta$ and $\eta_1 := \eta$ are holomorphic coordinates. By induction for $(X_1, q_1, \xi_1, \eta_1, a-b, b)$, there exist $\mu_2 : Y \to X_1$, E, Σ, Ω, s, τ and integers $a_1', b_1' \geq 0$ with $(a-b)b_1' - a_1'b = 1$ satisfying (i)–(iii) for $(X_1, q_1, \xi_1, \eta_1, a-b, b)$. Let $\mu := \mu_1 \circ \mu_2$ and $(a', b') := (a_1' + b_1', b_1')$. Then $ab' - a'b = (a-b)b_1' - a_1'b = 1$, (i) holds, and on Ω,

$$\xi \circ \mu = (\xi_1\eta_1) \circ \mu_2 = s^a \tau^{a_1' + b_1'}, \qquad \eta \circ \mu = \eta_1 \circ \mu_2 = s^b \tau^{b_1'}.$$

For (iii), let v be a divisorial valuation over X centered at q with $bv(\xi) = av(\eta)$. Then

$$v(\eta_1) = v(\eta) > 0, \qquad v(\xi_1) = v(\xi) - v(\eta) = \frac{a-b}{b} v(\eta) > 0.$$

The functions ξ_1, η_1 are regular on X_1 near any point of $\mu_1^{-1}(q)$ except one point q_∞, at which η/ξ is regular and vanishes. Since $v(\eta/\xi) < 0$, the center of v on X_1 does not contain q_∞, and since $v(\xi_1), v(\eta_1) > 0$, the center of v on X_1 is q_1. As $bv(\xi_1) = (a-b)v(\eta_1)$, (iii) for $(X_1, q_1, \xi_1, \eta_1, a-b, b)$ implies (iii) for v. □

Let X be a smooth surface, $q \in X$ a closed point, ξ, η holomorphic coordinates at q, and a, b coprime positive integers, and let $\mu : Y \to X$, E, Σ, Ω, s, τ, a', b' satisfy (i)–(iii) of Claim 6.4. By (ii), on Ω,

$$d(\xi \circ \mu) \wedge d(\eta \circ \mu) = (ab' - a'b) s^{a+b-1} \tau^{a'+b'-1} ds \wedge d\tau = s^{a+b-1} \tau^{a'+b'-1} ds \wedge d\tau.$$

Since τ has no zero on Ω, $K_{Y/X} = (a+b-1)E$ on Ω. Thus $\mathrm{mult}_E \operatorname{div}(\xi) = a$, $\mathrm{mult}_E \operatorname{div}(\eta) = b$, $a(E, X, 0) = a+b$, and

$$A_X(v) = A_Y(v) + (a+b-1)v(s) \tag{6.16}$$

for any divisorial valuation v over Y centered at a closed point of $E \backslash \Sigma$.

Step 4. In this step we prove (6.9).

Let v be a divisorial valuation over $\mathbb{P}^2$ centered at p. Since $v(x), v(y) > 0$ and $v(x)/v(y) \in \mathbb{Q}$ (Definition 2.1), there exist coprime positive integers a, b and a real number $t > 0$ with

$$v(x) = ta \quad \text{and} \quad v(y) = tb.$$

Let ν_0, u, w, W_0, and N be as in **Step 2** for (a, b), and, by Claim 6.4 for $(\mathbb{P}^2, p, x, y, a, b)$, let $\mu : Y \to \mathbb{P}^2$, E, Σ, Ω, s, τ, a', b' satisfy (i)–(iii) of Claim 6.4. Possibly replacing Ω with a smaller open neighborhood of $E \backslash \Sigma$, we may assume that $U \circ \mu$, $V \circ \mu$, and $W \circ \mu$ are holomorphic on Ω. Let

$$P_W(\tau) := W_0(\tau^{a'}, \tau^{b'}),$$

a Laurent polynomial in τ. All monomials of W_0 have weight $u+w$, so the exponents of any two of them differ by $k(b, -a)$ for an integer k, and their images in P_W have exponents differing by $k(a'b - ab') = -k$. Thus $P_W \neq 0$, and by the end of **Step 2**, the largest exponent of P_W minus the smallest one is at most

$$\frac{1}{b}\left(\kappa_1 + \frac{\kappa_2 b}{a}\right) = \frac{\kappa_1}{b} + \frac{\kappa_2}{a} \leq \kappa. \tag{6.17}$$

By Claim 6.4(ii) and the weighted homogeneity of W_0, $W_0(x\circ\mu, y\circ\mu) = s^{u+w}P_W(\tau)$. Since $W - W_0$ belongs to the ideal generated by the monomials of weight at least $u+w+1$, $(W-W_0)\circ\mu$ is divisible by s^{u+w+1} on Ω. Hence, by (6.12),

$$J\circ\mu = s^N\tau^{(e_1-1)a'+(e_2-1)b'}\big(P_W(\tau)+sW_+\big) \quad\text{and}\quad W\circ\mu = s^{u+w}\big(P_W(\tau)+sW_+\big) \tag{6.18}$$

for a holomorphic function W_+ on Ω. As $P_W(\tau)$ does not vanish identically on E, $\operatorname{mult}_E\operatorname{div}(J) = N$. By Claim 6.4(iii), the center of v on Y is E or a closed point $q'\in E\backslash\Sigma$.

If the center of v on Y is E, then $v(B) = t\operatorname{mult}_E B$ for any $\mathbb{R}$-divisor B near p and $A(v) = t(a+b)$, since v is a positive multiple of the valuation of E (Definition 2.1), $v(x) = ta$, $\operatorname{mult}_E\operatorname{div}(x) = a$, and $a(E,\mathbb{P}^2,0) = a+b$. By (6.15),

$$v(R(g)) = t\operatorname{mult}_E\operatorname{div}(J) = tN \le t(Q-1)(a+b) = (Q-1)A(v).$$

Assume that the center of v on Y is a closed point $q'\in E\backslash\Sigma$. Since $\tau(q')\neq 0$,

$$ta = v(x) = v(s^a\tau^{a'}) = av(s),$$

so $v(s) = t$. Let $D_Y := \mu_*^{-1}D$. By (6.18), near q',

$$\mu^*R(g) = NE + D_Y,$$

and D_Y is defined by $P_W(\tau)+sW_+$. Thus $D_Y|_E$ is defined near q' by $P_W(\tau)$. Since $\tau-\tau(q')$ is a coordinate of E at q' and $\tau(q')\neq 0$, and since P_W times a power of τ is a non-zero polynomial of degree at most κ by (6.17), we obtain

$$(D_Y\cdot E)_{q'} \le \kappa.$$

Since (Y,E) is lc near q', $A_Y(v)\ge v(E) = v(s) = t$. By Lemma 2.5 for D_Y and the smooth curve E at q',

$$v(D_Y) \le (D_Y\cdot E)_{q'}\big(A_Y(v)-v(E)\big) \le \kappa\big(A_Y(v)-t\big).$$

By (6.16), $A(v) = (a+b-1)t + A_Y(v)$. Hence, by (6.15) and (6.11),

$$\begin{aligned} v(R(g)) - (Q-1)A(v) &= Nt + v(D_Y) - (Q-1)(a+b)t - (Q-1)\big(A_Y(v)-t\big)\\ &\le \big(\kappa-(Q-1)\big)\big(A_Y(v)-t\big) \le 0.\end{aligned}$$

This proves (6.9).

Step 5. In this step we iterate (6.9) and prove the lemma.

Let $n\ge 1$ be an integer and v a divisorial valuation over $\mathbb{P}^2$ centered at p. For any integer $j\ge 0$, $(g^j)_*v$ is a divisorial valuation over $\mathbb{P}^2$ centered at $g^j(p) = p$ by Lemma 2.4, and $(g^{j+1})_*v = g_*\big((g^j)_*v\big)$. By Lemma 2.4 and (6.9) for $(g^j)_*v$,

$$A\big((g^{j+1})_*v\big) = A\big((g^j)_*v\big) + \big((g^j)_*v\big)(R(g)) \le QA\big((g^j)_*v\big).$$

Thus $A((g^n)_*v)\le Q^nA(v)$. By Lemma 2.4 for g^n, $A((g^n)_*v) = A(v) + v(R(g^n))$, so

$$v(R(g^n)) \le (Q^n-1)A(v) \tag{6.19}$$

for any divisorial valuation v over $\mathbb{P}^2$ centered at p.

Let P be a prime divisor over $\mathbb{P}^2$ whose center on $\mathbb{P}^2$ contains p. If the center of P is p, then $a\big(P,\mathbb{P}^2,\frac{1}{Q^n-1}R(g^n)\big)\ge 0$ by (6.19) and Definition 2.1. Otherwise P is a prime divisor on $\mathbb{P}^2$ with $p\in P$ and $a\big(P,\mathbb{P}^2,\frac{1}{Q^n-1}R(g^n)\big) = 1-b_P$, where $b_P := \frac{1}{Q^n-1}\operatorname{mult}_P R(g^n)$. We show that $b_P\le 1$. By the embedded resolution of curves on smooth surfaces, there exists a composition of point blow-ups $\mu' : Y'\to\mathbb{P}^2$ over p, not an isomorphism, such that $\mu'^{-1}(p)\cup\mu'^{-1}_*(\operatorname{Supp}R(g^n)\cup P)$ is a simple normal crossing divisor near $\mu'^{-1}(p)$. Let $P' := \mu'^{-1}_*P$ and $q\in P'$ a closed point with $\mu'(q) = p$. Then exactly one μ'-exceptional prime divisor F contains q, and near q,

$$\frac{1}{Q^n-1}\mu'^*R(g^n) = b_PP' + \frac{\operatorname{mult}_F R(g^n)}{Q^n-1}F \quad\text{and}\quad K_{Y'/\mathbb{P}^2} = \big(a(F,\mathbb{P}^2,0)-1\big)F.$$

Let ξ, η be holomorphic coordinates at q with $F = \{\xi = 0\}$ and $P' = \{\eta = 0\}$. For any integer $k \geq 1$, let E_k be the prime divisor E of Claim 6.4 for $(Y', q, \xi, \eta, 1, k)$. By **Step 3**, $\operatorname{mult}_{E_k} \operatorname{div}(\xi) = 1$, $\operatorname{mult}_{E_k} \operatorname{div}(\eta) = k$, and $a(E_k, Y', 0) = k + 1$. Thus E_k has center p on $\mathbb{P}^2$,

$$\frac{\operatorname{mult}_{E_k} R(g^n)}{Q^n - 1} = b_P k + \frac{\operatorname{mult}_F R(g^n)}{Q^n - 1},$$
$$a(E_k, \mathbb{P}^2, 0) = a(E_k, Y', 0) + \operatorname{mult}_{E_k} K_{Y'/\mathbb{P}^2} = k + a(F, \mathbb{P}^2, 0),$$

and (6.19) and Definition 2.1 imply $b_P k + \frac{\operatorname{mult}_F R(g^n)}{Q^n - 1} \leq k + a(F, \mathbb{P}^2, 0)$ for any integer $k \geq 1$. Thus $b_P \leq 1$, and $\left(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n)\right)$ is lc at p. The lemma follows. □

Lemma 6.5. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$ and $C \subset \mathbb{P}^2$ an irreducible nodal cubic with node p such that $g(C) = C$ and $g(p) = p$. Let $\pi : \overline{C} \to C$ be the normalization, $h : \overline{C} \to \overline{C}$ the morphism with $\pi \circ h = g \circ \pi$, and $\pi^{-1}(p) = \{z_1, z_2\}$. Assume that $h(z_1) = h(z_2)$ and $e_C(g) > \frac{12}{13} Q$. Then $\left(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n)\right)$ is klt at p for any integer $n \geq 72$.*

Proof. Let $e := e_C(g)$ and write $g^* C = eC + D$. Since $\pi(h(z_1)) = g(p) = p$, $h(z_1) \in \{z_1, z_2\}$. Possibly reordering indices, we may assume that $h(z_1) = h(z_2) = z_1$. For $i \in \{1, 2\}$, let Γ_i be the image under π of a small neighborhood of z_i in $\overline{C}$, and let x, y be holomorphic coordinates at p with $\Gamma_1 = \{x = 0\}$ and $\Gamma_2 = \{y = 0\}$. Let $\lambda := \deg_{z_1} h$, $\alpha_{z_1} := \operatorname{mult}_{z_1} \pi^*(D|_C)$, and $t := e/Q$.

Since $h(z_1) = h(z_2) = z_1$, $g(\Gamma_1) = g(\Gamma_2) = \Gamma_1$ as germs. Thus $g^* x$ vanishes on $\Gamma_1 \cup \Gamma_2$ and $g^* y$ vanishes identically on neither Γ_1 nor Γ_2. Since $\operatorname{div}(xy) = C$ near p and $\operatorname{mult}_C g^* C = e$, we obtain

$$g^* x = x^e y^e U \quad \text{and} \quad g^* y = V$$

for holomorphic germs U, V at p divisible by neither x nor y, and $D = \operatorname{div}(UV)$ near p. The function $y \circ \pi$ is a uniformizer of $\overline{C}$ at z_1, and $h^*(y \circ \pi) = (g^* y) \circ \pi$ is $V(0, y)$ written in the uniformizer $y \circ \pi$, so $\lambda = (\operatorname{div}(V) \cdot \Gamma_1)_p$. Since π maps a neighborhood of z_1 isomorphically onto Γ_1 and $D = \operatorname{div}(UV)$ near p, $\alpha_{z_1} = (D \cdot \Gamma_1)_p \geq \lambda$, and since $\deg \pi^*(D|_C) = 3 \deg D = 9(Q - e)$, we have

$$1 \leq \lambda \leq \alpha_{z_1} \leq 9(Q - e). \tag{6.20}$$

In particular $e < Q$, so $\frac{12}{13} < t < 1$.

Let $n \geq 1$ be an integer, $D_n := (g^n)^* C - e^n C$, and $R'_n := R(g^n) - (e^n - 1)C$. By Lemma 3.2(1), $e_C(g^n) = e^n$, $R'_n \geq 0$, and $C \not\subset \operatorname{Supp} R'_n$. Let $\rho_n := \operatorname{mult}_{z_1} \pi^*(R'_n|_C)$. Since p is a node of C and C is smooth away from p, the different of $(\mathbb{P}^2, C)$ on $\overline{C}$ is $z_1 + z_2$. Since $h^n(z_1) = z_1$, the local degree of h^n at z_1 is λ^n, so

$$\operatorname{mult}_{z_1} R(h^n) = \lambda^n - 1 \quad \text{and} \quad \operatorname{mult}_{z_1} (h^n)^*(z_1 + z_2) = \lambda^n.$$

By Lemma 3.2(1), and since $h^i(z_1) = z_1$ for any $i \geq 0$,

$$\rho_n = \lambda^n - 1 + 1 - \lambda^n + \operatorname{mult}_{z_1} \pi^*(D_n|_C) = \sum_{i=0}^{n-1} e^{n-1-i} \lambda^i \alpha_{z_1}.$$

By (6.20), $\frac{\alpha_{z_1}}{Q} \leq 9(1 - t)$ and $\frac{\lambda}{Q} \leq 9(1 - t) < t$, where the last inequality holds as $t > \frac{9}{10}$. Thus

$$\frac{\rho_n}{Q^n} = \frac{\alpha_{z_1}}{Q} \sum_{i=0}^{n-1} t^{n-1-i} \left(\frac{\lambda}{Q}\right)^i \leq 9(1-t) t^{n-1} \sum_{i=0}^{+\infty} \left(\frac{9(1-t)}{t}\right)^i = \frac{9t}{10t - 9} (1-t) t^{n-1} < 36(1-t) t^{n-1},$$

where the last inequality is equivalent to $t > \frac{12}{13}$. Since $1 - t^n = (1 - t) \sum_{i=0}^{n-1} t^i \geq n(1 - t) t^{n-1}$, we obtain, for any integer $n \geq 72$,

$$\rho_n < \frac{36}{n} (Q^n - e^n) \leq \frac{1}{2} (Q^n - e^n). \tag{6.21}$$

Let $n \geq 72$ be an integer. Since π maps a neighborhood of z_1 isomorphically onto Γ_1, $(R'_n \cdot \Gamma_1)_p = \rho_n$. Let v be a divisorial valuation over $\mathbb{P}^2$ centered at p. Since p is a node of C, $(\mathbb{P}^2, C)$ is lc near p, so $v(C) \leq A(v)$. By Lemma 2.5 for R'_n and the smooth curve germ $\Gamma_1 \not\subset \operatorname{Supp} R'_n$,

$$v(R'_n) \leq (R'_n \cdot \Gamma_1)_p\big(A(v) - v(\Gamma_1)\big) \leq \rho_n A(v).$$

Thus, by $R(g^n) = (e^n - 1)C + R'_n$ and (6.21),

$$v(R(g^n)) \leq (e^n - 1 + \rho_n)A(v) \leq \left(Q^n - 1 - \frac{1}{2}(Q^n - e^n)\right) A(v),$$

so

$$A(v) - \frac{v(R(g^n))}{Q^n - 1} \geq \frac{Q^n - e^n}{2(Q^n - 1)} A(v) > 0.$$

Thus $a\big(E, \mathbb{P}^2, \frac{1}{Q^n-1}R(g^n)\big) > 0$ for any prime divisor E over $\mathbb{P}^2$ with center p (Definition 2.1). Let P be a prime divisor on $\mathbb{P}^2$ with $p \in P$. If $P = C$, then $\operatorname{mult}_P R(g^n) = e^n - 1 < Q^n - 1$ as $C \not\subset \operatorname{Supp} R'_n$ and $e < Q$. If $P \neq C$, then P does not contain Γ_1 and $(P \cdot \Gamma_1)_p \geq 1$, so, by (6.21),

$$\operatorname{mult}_P R'_n \leq \operatorname{mult}_P R'_n \cdot (P \cdot \Gamma_1)_p \leq (R'_n \cdot \Gamma_1)_p = \rho_n < \frac{1}{2}(Q^n - e^n) < Q^n - 1,$$

and $\operatorname{mult}_P R(g^n) = \operatorname{mult}_P R'_n < Q^n - 1$. Thus $\left(\mathbb{P}^2, \frac{1}{Q^n-1}R(g^n)\right)$ is klt at p. The lemma follows. □

Proof of Proposition 6.1. For any integer $m \geq 1$ and any irreducible curve $P \subset \mathbb{P}^2$ with $f^m(P) = P$, we have $e_P(f^m) \leq q^m$ since $\deg(f^m)^* P = q^m \deg P$, and $e_P(f^{mk}) = e_P(f^m)^k$ for any integer $k \geq 1$ since $e_P(h_2 \circ h_1) = e_P(h_1)e_{h_1(P)}(h_2)$ for finite surjective endomorphisms h_1, h_2 of $\mathbb{P}^2$.

Step 1. In this step we show that any irreducible component of C is preserved by f^{6r} for some $r \in \{1, 2, 3\}$, and we bound the index of f^{6r} along it from below.

Let P be an irreducible component of C. Since $\operatorname{mult}_P B_{288} > \frac{3}{5} > \frac{26}{49}$, Proposition 5.4 (1) for $n = 48$ implies that there exists an integer $r \in \{1, 2, 3\}$ with $f^{6r}(P) = P$. We fix such an integer r for any irreducible component P of C, and we let

$$s := \frac{e_P(f^{6r})}{q^{6r}}.$$

Then $q^{6r} \geq 64$ and $s \leq 1$. Since r divides 48, $e_P(f^{288}) = e_P(f^{6r})^{48/r} \leq q^{288}$, so

$$\frac{3}{5} < \operatorname{mult}_P B_{288} = \frac{e_P(f^{6r})^{48/r} - 1}{q^{288} - 1} \leq s^{48/r}. \tag{6.22}$$

Step 2. In this step we prove (1).

Let P be an irreducible component of C, and let r and s be as in **Step 1**. By Construction-Proposition 5.3, $\deg P \leq \deg C \leq 4$.

If $\deg P = 2$, then P is a smooth conic and $f^{288}(P) = P$. By Lemma 6.2 for the endomorphism f^{288} of degree q^{288} and the curve P, $\operatorname{mult}_P B_{288} \leq \frac{1}{2}$, which contradicts (6.22).

Assume that $\deg P \in \{3, 4\}$ and that P is neither a smooth cubic nor a nodal cubic. Since an irreducible cubic is smooth, nodal, or cuspidal, either P is a cuspidal cubic or $\deg P = 4$. We show

$$s \leq \frac{17}{18}, \tag{6.23}$$

which contradicts (6.22), since $\frac{48}{r} \geq 16$, $\left(\frac{18}{17}\right)^{16} \geq \frac{33}{17}$, and so

$$s^{48/r} \leq \left(\frac{17}{18}\right)^{16} \leq \frac{17}{33} < \frac{3}{5}.$$

Let $\pi : \overline{P} \to P$ be the normalization and $\Delta := \mathrm{Diff}_{\overline{P}}(0)$. We use Lemma 3.2 for the endomorphism f^{6r} of degree $q^{6r} \geq 64$ and the curve P. One of the following holds.

(1) P is a cuspidal cubic. Since $(\mathbb{P}^2, P)$ is not lc at the cusp of P, Lemma 3.2 (3) with $d = 3$ implies $e_P(f^{6r}) \leq (1 - \frac{1}{18})q^{6r}$, that is, $s \leq 1 - \frac{1}{18}$.

(2) $\deg P = 4$ and $(\mathbb{P}^2, P)$ is lc. Then Lemma 3.2 (4) with $d = 4$ implies $e_P(f^{6r}) \leq (1 - \frac{1}{16})q^{6r}$, so $s \leq \frac{15}{16} < \frac{17}{18}$.

(3) $\deg P = 4$ and $(\mathbb{P}^2, P)$ is not lc. Since $(\mathbb{P}^2, 0)$ is lc, $(\mathbb{P}^2, P)$ is not lc near P, so $(\overline{P}, \Delta)$ is not lc by [Kaw07, Theorem]. Since $K_{\mathbb{P}^2} + P$ is Cartier, Δ has integer coefficients, so there exists $z \in \overline{P}$ with $\mathrm{mult}_z \Delta \geq 2$. Since $\deg \Delta = 6 - 2h^1(\overline{P}, \mathcal{O}_{\overline{P}}) \leq 6$, Lemma 3.2 (2) with $d = 4$ implies
$$16q^{6r}(1 - s) \geq q^{6r} + 1 - 6.$$
Since $q^{6r} \geq 64$, we obtain
$$1 - s \geq \frac{1}{16} - \frac{5}{16q^{6r}} \geq \frac{59}{1024} > \frac{1}{18}.$$

In any case (6.23) holds. This proves (1).

Step 3. In this step we prove that $(\mathbb{P}^2, B_{288n})$ is lc at any closed point of any smooth cubic component of C for any integer $n \geq 8192$.

Let P be a smooth cubic component of C and $x \in P$. By **Step 1**, $f^{288}(P) = P$. Let
$$w := \frac{e_P(f^{288})}{q^{288}} \leq 1.$$
We use the notation c_i, α_i, D of Lemma 3.3 for the endomorphism f^{288} of degree q^{288} and the curve P, so that $(f^{288})^*P = e_P(f^{288})P + D$. Since P has genus 1, the Riemann–Hurwitz formula implies that $f^{288}|_P$ is étale, so $c_i = q^{-288i}$ for any $i \geq 0$. Since $\deg(D|_P) = 3 \deg D = 9(q^{288} - e_P(f^{288}))$, we have
$$0 \leq \alpha_i \leq 9(1 - w) \leq 9 \quad \text{for any } i \geq 0.$$
Since $n(1 - w)w^{n-1} \leq (1 - w)\sum_{j=0}^{n-1} w^j = 1 - w^n \leq 1$ and $\sum_{i \geq 1} q^{-288i} = \frac{1}{q^{288} - 1} \leq 2^{-287}$, the right side of the formula in Lemma 3.3 (2) satisfies
$$q^{-288n} + \sum_{i=0}^{n-1} \alpha_i w^{n-1-i} q^{-288i} \leq 2^{-288} + 9(1 - w)w^{n-1} + 9\sum_{i \geq 1} q^{-288i} \leq 2^{-288} + \frac{9}{n} + 9 \cdot 2^{-287} < 1.$$
Since $R((f^{288})^n)/(q^{288n} - 1) = B_{288n}$, Lemma 3.3 (3) implies that $(\mathbb{P}^2, B_{288n})$ is lc at x.

Step 4. In this step we bound the local degrees $\deg_z h$ for $z \in \overline{P} \backslash h^{-1}(\pi^{-1}(o))$, where h is the lift of f^{12r} to the normalization of a nodal cubic component of C with node o.

Let P be a nodal cubic component of C with node o, let r and s be as in **Step 1**, and let
$$e := e_P(f^{12r}), \quad t := \frac{e}{q^{12r}}, \quad \text{and} \quad (f^{12r})^*P = eP + D.$$
Since $e = e_P(f^{6r})^2$, $t = s^2$, so (6.22) implies $t^{24/r} > \frac{3}{5}$. Since $\frac{24}{r} \geq 8$ and $\left(\frac{13}{12}\right)^8 > 1 + \frac{8}{12} = \frac{5}{3}$, we have $\left(\frac{12}{13}\right)^{24/r} < \frac{3}{5}$. Thus
$$t > \frac{12}{13}. \tag{6.24}$$

Let $\pi : \overline{P} \to P$ be the normalization, let $h : \overline{P} \to \overline{P}$ be the morphism with $\pi \circ h = f^{12r} \circ \pi$, let $\pi^{-1}(o) = \{z_1, z_2\}$, and let $\Delta := \mathrm{Diff}_{\overline{P}}(0)$. Since π is an isomorphism over $P \backslash \{o\}$ and P has 2 smooth branches intersecting transversally at o, $\Delta = z_1 + z_2$. By Lemma 3.2 for the endomorphism f^{12r} of degree q^{12r} and the curve P, we have $\deg h = q^{12r}$ and $2 = \deg \Delta = 2 - 2h^1(\overline{P}, \mathcal{O}_{\overline{P}})$, so $\overline{P} \cong \mathbb{P}^1$. Let
$$\overline{D} := \pi^*(D|_P), \quad V := h^{-1}(\{z_1, z_2\}), \quad \text{and} \quad \lambda_z := \deg_z h \text{ for any } z \in \overline{P}.$$

Then $\deg \overline{D} = 3 \deg D = 9(q^{12r} - e)$.

Let $z \in V \backslash \{z_1, z_2\}$. Then $\operatorname{mult}_z \Delta = 0$, $\operatorname{mult}_z h^* \Delta = \lambda_z$, and $\operatorname{mult}_z R(h) = \lambda_z - 1$. Since $R(f^{12r}) - (e-1)P \geq 0$ does not contain P, Lemma 3.2 (1) implies

$$0 \leq \operatorname{mult}_z \pi^* \left((R(f^{12r}) - (e-1)P)|_P \right) = \operatorname{mult}_z \overline{D} - 1.$$

Thus $\#V - 2 \leq \deg \overline{D} = 9(q^{12r} - e)$. By the Riemann–Hurwitz formula, $\sum_{z \in \overline{P}} (\lambda_z - 1) = 2q^{12r} - 2$, and $\sum_{z \in V} \lambda_z = 2q^{12r}$ since $z_1 \neq z_2$. Hence for any $z \in \overline{P} \backslash V$,

$$\lambda_z - 1 \leq (2q^{12r} - 2) - (2q^{12r} - \#V) = \#V - 2 \leq 9(q^{12r} - e),$$

that is,

$$\frac{\lambda_z}{q^{12r}} \leq 9(1-t) + q^{-12r} \quad \text{for any } z \in \overline{P} \backslash V. \tag{6.25}$$

Step 5. In this step we prove that, in the notation of **Step 4**, $(\mathbb{P}^2, B_{12rk})$ is lc at any closed point $x \in P \backslash \{o\}$ with $f^{12r}(x) = x$, for any integer $k \geq 100$.

Let $x \in P \backslash \{o\}$ with $f^{12r}(x) = x$, and let $z \in \overline{P}$ be the point with $\pi(z) = x$. Since $\pi(h(z)) = f^{12r}(x) = x$, $h(z) = z$, so $z \notin V$. Let

$$\lambda := \lambda_z, \quad u := \frac{\lambda}{q^{12r}}, \quad \text{and} \quad a := \operatorname{mult}_x (D|_P).$$

Since $q^{12r} \geq 4096$, (6.24) and (6.25) imply $u \leq \frac{9}{13} + \frac{1}{4096} < \frac{3}{4}$. Since π is an isomorphism near z, $a = \operatorname{mult}_z \overline{D} \leq 9(1-t)q^{12r}$.

For any integer $k \geq 1$, let $R'_k := R(f^{12rk}) - (e^k - 1)P$ and $\rho_k := \operatorname{mult}_x (R'_k|_P)$. By Lemma 3.2 (1) for the endomorphism f^{12r} and the curve P, we have $R'_k \geq 0$ and $P \not\subset \operatorname{Supp} R'_k$. Since $h(z) = z$, $\deg_z h = \lambda$, $\operatorname{mult}_z \Delta = 0$, and π is an isomorphism near z, taking coefficients at z in the formula of Lemma 3.2 (1) gives $1 + \rho_k = \lambda^k + a \sum_{i=0}^{k-1} e^{k-1-i} \lambda^i$, that is,

$$\frac{1 + \rho_k}{q^{12rk}} = u^k + \frac{a}{q^{12r}} \sum_{i=0}^{k-1} t^{k-1-i} u^i. \tag{6.26}$$

Since $u < \frac{3}{4} < t$ and $\frac{t}{t - 3/4}$ is decreasing in $t > \frac{3}{4}$ with value $\frac{16}{3}$ at $t = \frac{12}{13}$, we have

$$\sum_{i=0}^{k-1} t^{k-1-i} u^i \leq \frac{t^k}{t-u} \leq \frac{16}{3} t^{k-1}.$$

Since $k(1-t)t^{k-1} \leq 1 - t^k \leq 1$, (6.26) implies

$$\frac{1 + \rho_k}{q^{12rk}} \leq \left(\frac{3}{4}\right)^k + 48(1-t)t^{k-1} \leq \left(\frac{3}{4}\right)^k + \frac{48}{k} < 1 \quad \text{for any } k \geq 100.$$

Thus $\frac{\rho_k}{q^{12rk} - 1} < 1$. On a neighborhood of x on which P is smooth and $\operatorname{Supp}(R'_k|_P) \subset \{x\}$, the pair $\left(P, \frac{1}{q^{12rk}-1} R'_k|_P\right)$ is klt, so $\left(\mathbb{P}^2, P + \frac{1}{q^{12rk}-1} R'_k\right)$ is plt near x by inversion of adjunction. Since $\frac{e^k - 1}{q^{12rk} - 1} \leq 1$, $(\mathbb{P}^2, B_{12rk})$ is lc at x.

Step 6. In this step we prove that, in the notation of **Step 4**, if $f^{12r}(o) = o$, then $(\mathbb{P}^2, B_{12rk})$ is lc at o for any integer $k \geq 72$.

Since $\pi(h(z_j)) = f^{12r}(o) = o$ for $j = 1, 2$, $h(\{z_1, z_2\}) \subset \{z_1, z_2\}$. If $h(\{z_1, z_2\}) = \{z_1, z_2\}$, then, since $2e > q^{12r}$ by (6.24), Lemma 6.3 (2) for the endomorphism f^{12r} of degree q^{12r} and the curve P implies that $(\mathbb{P}^2, B_{12rk})$ is lc at o for any $k \geq 1$, since $\frac{1}{q^{12rk}-1} R((f^{12r})^k) = B_{12rk}$. Otherwise $h(z_1) = h(z_2)$, and since $e > \frac{12}{13} q^{12r}$ by (6.24), Lemma 6.5 for the endomorphism f^{12r} of degree q^{12r} and the curve P implies that $(\mathbb{P}^2, B_{12rk})$ is klt at o for any $k \geq 72$.

Step 7. In this step we prove that $(\mathbb{P}^2, B_{288n})$ is lc at any closed point of any nodal cubic component of C for any integer $n \geq 8192$.

Let P be a nodal cubic component of C, use the notation of **Step 4**, and let $x \in P$. Let $h' : \overline{P} \to \overline{P}$ be the morphism with $\pi \circ h' = f^{6r} \circ \pi$. Since π is birational, $h'^2 = h$. Since $f^{288n} = (f^{6r})^{48n/r}$ and $\frac{48n}{r} \geq 16 \cdot 8192 = 131072$, Theorem 3.1 for the endomorphism f^{6r} of degree $q^{6r} \geq 64$ and the curve P implies that $(\mathbb{P}^2, B_{288n})$ is lc at x unless $h(z) = z$ for some $z \in \pi^{-1}(x)$. Assume that $h(z) = z$ for some $z \in \pi^{-1}(x)$. Then $f^{12r}(x) = \pi(h(z)) = x$. Since $f^{288n} = (f^{12r})^{24n/r}$ and $\frac{24n}{r} \geq 65536$, **Step 5** (if $x \neq o$) or **Step 6** (if $x = o$) for $k = \frac{24n}{r}$ implies that $(\mathbb{P}^2, B_{288n})$ is lc at x.

Step 8. In this step we prove (2).

Let $n \geq 8192$ be an integer. By (1), **Step 3**, and **Step 7**, $(\mathbb{P}^2, B_{288n})$ is lc at any closed point of any component of C of degree 3. Let p be a closed point contained in 2 distinct irreducible components P_1, P_2 of C. If $\deg P_1 = 3$ or $\deg P_2 = 3$, we are done. Otherwise, by (1), P_1 and P_2 are lines, so $P_1 \cap P_2 = \{p\}$ and P_1, P_2 intersect transversally at p. By **Step 1**, $f^{288}(P_j) = P_j$ for $j = 1, 2$, so $f^{288}(p) = p$. Moreover,

$$\frac{e_{P_j}(f^{288})}{q^{288}} \geq \frac{e_{P_j}(f^{288}) - 1}{q^{288} - 1} = \operatorname{mult}_{P_j} B_{288} > \frac{3}{5} \quad \text{for } j = 1, 2,$$

so

$$e_{P_1}(f^{288}) + e_{P_2}(f^{288}) > \frac{6}{5} q^{288} > q^{288}.$$

By Lemma 6.3 (1) for the endomorphism f^{288} of degree q^{288} and the curves P_1, P_2, $(\mathbb{P}^2, B_{288n})$ is lc at p. This proves (2).

We prove (3). Let $n \geq 4 \cdot 10^8$ be an integer and $p \in \operatorname{Nlc}(\mathbb{P}^2, B_{17280n})$. Since $(\mathbb{P}^2, B_{17280n})$ is not lc, Proposition 5.2 implies $\operatorname{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ and $g(p) = p$. Since $B_{17280n} = B_{288 \cdot 60n}$, Construction-Proposition 5.3 (3) implies $p \in C \cup \operatorname{Supp} Z$. If $p \notin C$, then $p \in \operatorname{Supp} Z \backslash C$. Assume that $p \in C$. Since $60n \geq 8192$, (2) with $60n$ in place of n implies that no component of C of degree 3 contains p and no 2 distinct components of C contain p. Thus exactly one irreducible component of C contains p, and by (1) it is a line. The proposition follows. □

7. The asymptotic attraction rate at a non-lc point

Theorem 7.1. *Notation and conditions as in Set-up 5.1. Let $n \geq 4 \cdot 10^8$ be an integer and v a divisorial valuation over $\mathbb{P}^2$ centered at a closed point p with $v(B_{17280n}) > A(v)$. Then the following hold.*

(1) $g(p) = p$, and $\lambda := c_\infty(g, p)$ satisfies

$$\frac{9}{10} Q < \lambda < Q \quad \text{and} \quad \left(\frac{\lambda}{Q}\right)^{\lfloor n/1000 \rfloor} > \frac{1}{15000}.$$

(2) If $p \notin C$, then there does not exist an irreducible analytic curve germ ϕ at p satisfying the following.

(i) ν_ϕ is an eigenvaluation of g at p.

(ii) There exists a normalized valuation $\mu < \nu_\phi$ with $c(g, \nu) = \lambda$ and $g_\bullet \nu > \nu$ for any $\nu \in [\mu, \nu_\phi)$.

7.1. Totally invariant curves and pencils.

Lemma 7.2 (cf. [Men23, Theorem 4.2])**.** *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$ and $C \subset \mathbb{P}^2$ an irreducible curve with $g^{-1}(C) = C$. Then $\left(\mathbb{P}^2, \frac{1}{Q-1} R(g)\right)$ is lc.*

Proof. Since $g^{-1}(C) = C$, we have $g(C) = C$ and g^*C is supported on C, so $g^*C = e_C(g)C$. Since g has degree Q,

$$e_C(g) \deg C = \deg g^*C = Q \deg C,$$

so $e_C(g) = Q$. By Definition 2.2, $\operatorname{mult}_C R(g) = Q - 1$. Let

$$R' := R(g) - (Q-1)C.$$

Then $R' \geq 0$ and $C \not\subset \operatorname{Supp} R'$. Since $\deg R(g) = 3(Q-1)$,

$$\deg R' = (Q-1)(3 - \deg C) \geq 0.$$

Thus $\deg C \leq 3$ (cf. [FS94, Proposition 4.2]). We let

$$D := \frac{1}{Q-1}R' \quad \text{and} \quad \Delta := \frac{1}{Q-1}R(g) = C + D.$$

It remains to show that $(\mathbb{P}^2, \Delta)$ is lc.

Assume that $\deg C \leq 2$. Then C is a line or a smooth conic. Since $C \cong \mathbb{P}^1$ and $\deg C \leq 2$, the different in Lemma 3.2 has degree $(\deg C - 1)(\deg C - 2) = 0$, so it is 0. Since $g^*C = QC$, $\deg(g|_C) = Q$ and $R'|_C = R(g|_C)$ by Lemma 3.2. Let $x \in C$ be a closed point. Then $(R' \cdot C)_x = \deg_x(g|_C) - 1 \leq Q - 1$. Thus $(D \cdot C)_x \leq 1$. Since C is smooth and $(D \cdot C)_x \leq 1$ for any closed point $x \in C$, the pair $(C, D|_C)$ is lc, so $(\mathbb{P}^2, \Delta) = (\mathbb{P}^2, C + D)$ is lc at any closed point of C by inversion of adjunction.

We show that $(\mathbb{P}^2, \Delta)$ is lc when $\deg C \leq 2$. Suppose not. Since $\operatorname{Nlc}(\mathbb{P}^2, \Delta)$ is closed and disjoint from C, and any curve on $\mathbb{P}^2$ intersects C, $\operatorname{Nlc}(\mathbb{P}^2, \Delta)$ is a non-empty finite set. Let $U := \mathbb{P}^2 \setminus \operatorname{Nlc}(\mathbb{P}^2, \Delta)$, let v be a divisorial valuation over $\mathbb{P}^2$ with $v(\Delta) > A(v)$, let $\epsilon \in (0,1)$ be a rational number with $v(C + (1-\epsilon)D) > A(v)$, and let

$$\Delta_\epsilon := C + (1-\epsilon)D.$$

Let $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon)$ be the set of points at which $(\mathbb{P}^2, \Delta_\epsilon)$ is not klt. Let w be a divisorial valuation over $\mathbb{P}^2$ whose center intersects U and is not C. Since C is smooth, $A(w) > w(C)$, and since $(\mathbb{P}^2, \Delta)$ is lc on U, $A(w) \geq w(\Delta)$. Thus

$$A(w) - w(\Delta_\epsilon) = \epsilon\,(A(w) - w(C)) + (1-\epsilon)\,(A(w) - w(\Delta)) > 0.$$

Hence $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon) \cap U = C$. Since $v(\Delta_\epsilon) > A(v)$ and $v(\Delta) > A(v)$, the center of v is contained in $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon)\backslash U$. Thus $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon)$ is the disjoint union of C and the non-empty finite set $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon)\backslash U$, which is contained in $\operatorname{Nlc}(\mathbb{P}^2, \Delta)$, so $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon)$ is not connected. On the other hand, $\Delta_\epsilon \geq 0$ is a $\mathbb{Q}$-divisor, and since $\deg D = 3 - \deg C$,

$$\deg\left(-(K_{\mathbb{P}^2} + \Delta_\epsilon)\right) = 3 - \deg C - (1-\epsilon)(3 - \deg C) = \epsilon(3 - \deg C) > 0,$$

so $-(K_{\mathbb{P}^2} + \Delta_\epsilon)$ is ample. By [Hac04, Lemma 3.7], $\operatorname{Nklt}(\mathbb{P}^2, \Delta_\epsilon)$ is connected, a contradiction. Thus $(\mathbb{P}^2, \Delta)$ is lc.

Assume that $\deg C = 3$. Then $\deg R' = 0$, so $R' = 0$ and $\Delta = C$. Since C is reduced and $g^{-1}(C) = C$, $(\mathbb{P}^2, C)$ is lc by [BH14, Corollary 3.3]. The lemma follows. □

Lemma 7.3. *Let f be an endomorphism of $\mathbb{P}^2$ of degree $q \geq 2$. Assume that there exist a dominant rational map $\pi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ and a morphism $h : \mathbb{P}^1 \to \mathbb{P}^1$ with $\pi \circ f = h \circ \pi$. Then $\left(\mathbb{P}^2, \frac{1}{q-1}R(f)\right)$ is lc.*

Proof. Let $\Delta := \frac{1}{q-1}R(f)$. Since f is surjective and π is dominant, $h \circ \pi = \pi \circ f$ is dominant, so h is not constant. Let τ be an affine coordinate on $\mathbb{P}^1$ and let $\mathcal{F}$ be the foliation on $\mathbb{P}^2$ defined by the rational 1-form $\pi^* d\tau$ [FP11, Sections 2.1 and 2.3]. Since

$$f^*(\pi^* d\tau) = d(\tau \circ h \circ \pi) = \left(\frac{d(\tau \circ h)}{d\tau} \circ \pi\right) \pi^* d\tau,$$

and $\frac{d(\tau \circ h)}{d\tau} \circ \pi$ is a non-zero rational function, $f^*\mathcal{F} = \mathcal{F}$. By [FP11, Theorem C], there exist homogeneous coordinates $[x : y : z]$ on $\mathbb{P}^2$ in which one of the following holds:

(1) $f = [F_0(x,y) : F_1(x,y) : F_2(x,y,z)]$, where F_0, F_1, and F_2 are homogeneous polynomials of degree q (cases (1), (2), (4), and (5) of [FP11, Theorem C]).
(2) $f = [z^q : x^q : y^q]$ (case (3) of [FP11, Theorem C]).

If $f = [F_0 : F_1 : F_2]$ for homogeneous polynomials F_0, F_1, F_2 of degree q, then by the Euler formula,

$$R(f) = \operatorname{div} \det \frac{\partial(F_0, F_1, F_2)}{\partial(x, y, z)}.$$

Assume that $f = [z^q : x^q : y^q]$. Then

$$\det \frac{\partial(z^q, x^q, y^q)}{\partial(x, y, z)} = q^3 (xyz)^{q-1},$$

so $\Delta = \{x = 0\} + \{y = 0\} + \{z = 0\}$. Since Δ is a reduced simple normal crossing divisor, $(\mathbb{P}^2, \Delta)$ is lc.

Assume that $f = [F_0(x, y) : F_1(x, y) : F_2(x, y, z)]$. Let $x_0 := [0 : 0 : 1]$. Since f is a morphism and $F_0(x_0) = F_1(x_0) = 0$, $F_2(x_0) \neq 0$. If $F_0(a, b) = F_1(a, b) = 0$ for some $(a, b) \neq (0, 0)$, then F_0 and F_1 vanish on the line $\{bx - ay = 0\}$, and F_2 vanishes at a point of $\{bx - ay = 0\}$, which is a base point of f. Thus F_0 and F_1 have no common zero on $\mathbb{P}^1$, so

$$\chi := [F_0 : F_1] : \mathbb{P}^1 \to \mathbb{P}^1$$

is a morphism of degree q. For any $t = [a : b] \in \mathbb{P}^1$, let $L_t := \{bx - ay = 0\}$ be the line through x_0 corresponding to t. Let

$$W := \frac{\partial F_0}{\partial x}\frac{\partial F_1}{\partial y} - \frac{\partial F_0}{\partial y}\frac{\partial F_1}{\partial x}.$$

Since the third column of $\partial(F_0, F_1, F_2)/\partial(x, y, z)$ is $(0, 0, \partial F_2/\partial z)^T$,

$$\det \frac{\partial(F_0, F_1, F_2)}{\partial(x, y, z)} = W \cdot \frac{\partial F_2}{\partial z}, \quad \text{so} \quad R(f) = \operatorname{div}(W) + C,$$

where $C := \operatorname{div}(\partial F_2/\partial z)$. We show that

$$\operatorname{div}(W) = \sum_{t \in \mathbb{P}^1} (\deg_t \chi - 1) L_t.$$

Since W is a binary form, $\operatorname{div}(W)$ is a sum of lines L_t, and $\operatorname{mult}_{L_t} \operatorname{div}(W)$ is the order of vanishing of W at t. By the Euler formula,

$$W(1, u) = q\big(F_0(1, u)\partial_u F_1(1, u) - \partial_u F_0(1, u) F_1(1, u)\big),$$

and similarly on the other chart, so the order of vanishing of W at t is $\deg_t \chi - 1$. For any $t \in \mathbb{P}^1$, let

$$b_t := \frac{\deg_t \chi - 1}{q - 1}.$$

Since $1 \leq \deg_t \chi \leq q$ and $\deg W = 2(q - 1)$,

$$0 \leq b_t \leq 1 \quad \text{and} \quad \sum_{t \in \mathbb{P}^1} b_t = 2.$$

By the Euler formula at $(0, 0, 1)$, $\frac{\partial F_2}{\partial z}(x_0) = q F_2(x_0) \neq 0$, so $C \geq 0$ is a divisor of degree $q - 1$ with $x_0 \notin C$. In particular, no line through x_0 is a component of C. Thus

$$\Delta = \sum_{t \in \mathbb{P}^1} b_t L_t + \frac{1}{q - 1} C,$$

where $b_t \neq 0$ for only finitely many t.

We show that $(\mathbb{P}^2, \Delta)$ is lc at x_0. Near x_0, $\Delta = \sum_t b_t L_t$. Let $\mu : Y \to \mathbb{P}^2$ be the blow-up of x_0 and E_1 the exceptional prime divisor. Then near E_1,

$$K_Y + \sum_{t \in \mathbb{P}^1} b_t \mu_*^{-1} L_t + \left(\sum_{t \in \mathbb{P}^1} b_t - 1 \right) E_1 = \mu^*(K_{\mathbb{P}^2} + \Delta), \quad \text{where} \quad \sum_{t \in \mathbb{P}^1} b_t - 1 = 1.$$

Since the lines L_t have distinct tangent directions at x_0, the curves $\mu_*^{-1}L_t$ intersect E_1 transversally at distinct points. Thus $E_1 + \sum_{b_t \neq 0} \mu_*^{-1}L_t$ is a simple normal crossing divisor near E_1, and since $b_t \leq 1$ for any t and the coefficient of E_1 is 1, $(\mathbb{P}^2, \Delta)$ is lc at x_0.

Let $o \neq x_0$ be a closed point, L the line through x_0 and o, and $D := \frac{1}{q-1}C$. Since $L \not\subset \operatorname{Supp} C$, $D|_L \geq 0$ is a $\mathbb{Q}$-divisor of degree $D \cdot L = \frac{\deg C}{q-1} = 1$ on L, so $(L, D|_L)$ is lc, and $(\mathbb{P}^2, L + D)$ is lc at o by inversion of adjunction. Since two distinct lines through x_0 intersect only at x_0, near o we have $\Delta = b_t L + D$, where $t \in \mathbb{P}^1$ is the point with $L = L_t$. Since $b_t \leq 1$, $\Delta \leq L + D$ near o, so $(\mathbb{P}^2, \Delta)$ is lc at o. The lemma follows. □

7.2. **Points of maximal asymptotic attraction rate.**

Proposition 7.4. *Let f be an endomorphism of $\mathbb{P}^2$ of degree $q \geq 2$. Assume that there exist a closed point $p \in \mathbb{P}^2$ with $f^{-1}(p) = \{p\}$ and a divisorial valuation v centered at p with $f_*v = qv$. Then for any integer $n \geq 1$,*

$$\left(\mathbb{P}^2, \frac{1}{q^n - 1} R(f^n)\right) \text{ is lc.}$$

Proof. **Step 1.** Possibly replacing v with a positive real multiple, we may assume that there exists a prime divisor F over $\mathbb{P}^2$ with $v(\zeta) = \operatorname{mult}_F \operatorname{div}(\zeta)$ for any non-zero rational function ζ on $\mathbb{P}^2$. Since $f^*\mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^2}(q)$, by [LM26, Corollary 4.5] there exist a normal $\mathbb{Q}$-factorial projective surface Y, a birational morphism $h : Y \to \mathbb{P}^2$, a finite morphism $f_Y : Y \to Y$, and a prime divisor E on Y satisfying the following.

(1) $h \circ f_Y = f \circ h$.
(2) $\operatorname{Exc}(h) = h^{-1}(p) = E$ as sets, and $v(\zeta) = \operatorname{mult}_E \operatorname{div}(h^*\zeta)$ for any non-zero rational function ζ on $\mathbb{P}^2$.
(3) $f_Y^* E = qE$ and $f_Y^{-1}(E) = E$.

In particular, Y is rational and h is an isomorphism over $\mathbb{P}^2 \backslash \{p\}$. Let H be the pullback under h of a line in $\mathbb{P}^2$, and let $a := A_{\mathbb{P}^2}(v) - 1$. Then

$$K_Y = h^* K_{\mathbb{P}^2} + aE \sim_{\mathbb{Q}} -3H + aE.$$

Any prime divisor on Y other than E is $h_*^{-1}C'$ for an irreducible curve $C' \subset \mathbb{P}^2$, and if $d := \deg C'$, then

$$h_*^{-1}C' = h^*C' - (\operatorname{mult}_E h^*C')E \sim_{\mathbb{Q}} dH - (\operatorname{mult}_E h^*C')E.$$

Thus any $\mathbb{Q}$-divisor on Y is $\mathbb{Q}$-linearly equivalent to $c_1 H + c_2 E$ for some $c_1, c_2 \in \mathbb{Q}$. Since $H^2 = 1$ and $H \cdot E = 0$, by the Hodge index theorem $E^2 < 0$. Therefore H and E are numerically independent, $\dim N^1(Y)_{\mathbb{R}} = 2$, and $\mathbb{Q}$-linear equivalence and numerical equivalence coincide for $\mathbb{Q}$-divisors on Y. Since $f_Y^* H \sim qH$ by (1) and $f_Y^* E = qE$ by (3),

$$f_Y^* D \sim_{\mathbb{Q}} qD \quad \text{for any } \mathbb{Q}\text{-divisor } D \text{ on } Y. \tag{7.1}$$

Step 2. We show that Y is a Mori dream space. Since Y is normal, for any integer $m \geq 1$ let $R(f_Y^m)$ be the Weil divisor with $K_Y = (f_Y^m)^* K_Y + R(f_Y^m)$. Since Y is smooth at the generic point of any prime divisor P, the coefficient of $R(f_Y^m)$ along P is $e_P(f_Y^m) - 1 \geq 0$ as in Definition 2.2. For any integer $m \geq 1$, $(f_Y^m)^* E = q^m E$, so the ramification index of f_Y^m along E is q^m and

$$R(f_Y^m) = (q^m - 1)E + S_m$$

for some divisor $S_m \geq 0$. By (7.1),

$$q^m(-K_Y) \sim_{\mathbb{Q}} (f_Y^m)^*(-K_Y) = -K_Y + R(f_Y^m) \sim_{\mathbb{Q}} 3H + (q^m - 1 - a)E + S_m.$$

We fix an integer $m \geq 1$ with $q^m - 1 \geq a$. Since H is nef and big and $(q^m - 1 - a)E + S_m \geq 0$, $-K_Y$ is big. Since Y is a normal $\mathbb{Q}$-factorial rational projective surface with $-K_Y$ big, Y is a Mori dream space [MY21, Lemma 5.1] in the sense of [HK00, Definition 1.10]. In particular, $\operatorname{Nef}(Y)$ is the convex cone spanned by the classes of finitely many semi-ample Cartier divisors [HK00, Definition 1.10(2)].

Step 3. We construct a contraction $\psi : Y \to Z$ and a morphism $f_Z : Z \to Z$ with $\psi \circ f_Y = f_Z \circ \psi$. Since $\dim N^1(Y)_{\mathbb{R}} = 2$ and $\mathrm{Nef}(Y)$ is a closed convex cone with non-empty interior which contains no line, $\mathrm{Nef}(Y)$ is spanned by two rays, and the ample cone of Y is the interior of $\mathrm{Nef}(Y)$. Since $H \cdot E = 0$, H is nef and not ample, so $\mathbb{R}_{\geq 0}H$ is one of the rays spanning $\mathrm{Nef}(Y)$. By **Step 2**, the other ray contains the class of a semi-ample Cartier divisor L. Since $E^2 < 0$ and $H \cdot E = 0$, any $D \in N^1(Y)_{\mathbb{R}}$ with $D \cdot E = 0$ is a multiple of H. Since L is nef and not a multiple of H,

$$L \cdot E > 0.$$

Possibly replacing L with a multiple, $|L|$ is base point free. Let $\psi : Y \to Z$ be the Stein factorization of the morphism defined by $|L|$. Then Z is a normal projective variety, ψ is a contraction, and an irreducible curve $\Gamma \subset Y$ is contracted by ψ if and only if $L \cdot \Gamma = 0$. Since $L \not\equiv 0$, $\dim Z \in \{1, 2\}$. Since L is not ample, ψ is not finite, so ψ contracts at least one irreducible curve.

By (7.1), $f_Y^* L \sim_{\mathbb{Q}} qL$, so $L \cdot f_Y(\Gamma) = 0$ for any irreducible curve $\Gamma \subset Y$ with $L \cdot \Gamma = 0$. Thus $\psi \circ f_Y$ contracts any curve contracted by ψ, and by the rigidity lemma there exists a morphism $f_Z : Z \to Z$ with

$$\psi \circ f_Y = f_Z \circ \psi.$$

Step 4. We prove the proposition when $\dim Z = 1$. Since Z is a normal curve dominated by the rational surface Y, $Z \cong \mathbb{P}^1$ by Lüroth's theorem. By (1), for any integer $n \geq 1$, $h^{-1} \circ f^n = f_Y^n \circ h^{-1}$ as rational maps, so

$$(\psi \circ h^{-1}) \circ f^n = f_Z^n \circ (\psi \circ h^{-1})$$

as rational maps $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$. Since f^n is an endomorphism of $\mathbb{P}^2$ of degree $q^n \geq 2$, $\psi \circ h^{-1} : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ is dominant, and $f_Z^n : \mathbb{P}^1 \to \mathbb{P}^1$ is a morphism, by Lemma 7.3 with f^n, q^n in place of f, q, the pair $\left(\mathbb{P}^2, \frac{1}{q^n - 1} R(f^n)\right)$ is lc.

Step 5. We prove the proposition when $\dim Z = 2$, following [Men23, Corollary 4.3]. Since ψ is generically finite with connected fibers, ψ is birational. Since $\dim Z = 2$, L is nef and big, so $L^2 > 0$, and by the Hodge index theorem the intersection form is negative definite on

$$L^{\perp} := \{D \in N^1(Y)_{\mathbb{R}} \mid D \cdot L = 0\},$$

where $\dim L^{\perp} = 1$. Suppose that ψ contracts two distinct irreducible curves Γ and Γ'. Then the classes of Γ and Γ' are non-zero elements of $L^{\perp}$ with positive degree with respect to an ample divisor, so $\Gamma' \equiv c\Gamma$ for some real number $c > 0$, and

$$\Gamma \cdot \Gamma' = c\Gamma^2 < 0,$$

which is not possible for distinct irreducible curves. Hence ψ contracts exactly one irreducible curve Γ_0. Since $L \cdot E > 0$, $\Gamma_0 \neq E$.

We show that $f_Y^{-1}(\Gamma_0) = \Gamma_0$. Since Γ_0 is $\mathbb{Q}$-Cartier and f_Y is finite and surjective, $f_Y^{-1}(\Gamma_0)$ is a non-empty closed subset of pure dimension 1. Let Γ_1 be an irreducible component of $f_Y^{-1}(\Gamma_0)$. Then $f_Y(\Gamma_1) = \Gamma_0$, so $(f_Y)_*\Gamma_1$ is a positive multiple of Γ_0 and

$$qL \cdot \Gamma_1 = f_Y^* L \cdot \Gamma_1 = L \cdot (f_Y)_*\Gamma_1 = 0.$$

Thus ψ contracts Γ_1, so $\Gamma_1 = \Gamma_0$ and $f_Y^{-1}(\Gamma_0) = \Gamma_0$.

Let $C := h(\Gamma_0)$, which is an irreducible curve in $\mathbb{P}^2$ because $\Gamma_0 \neq E$. Let $z \in f^{-1}(C) \backslash \{p\}$ and $y := h^{-1}(z) \in Y \backslash E$. Since $f^{-1}(p) = \{p\}$, $f(z) \in C \backslash \{p\}$. By (1), $h(f_Y(y)) = f(z)$, so $f_Y(y) \in h^{-1}(C \backslash \{p\}) \subset \Gamma_0$. Thus $y \in f_Y^{-1}(\Gamma_0) = \Gamma_0$ and $z \in C$. Since $f^{-1}(C)$ is a non-empty closed subset of pure dimension 1 and $f^{-1}(C) \backslash \{p\} \subset C$, we have $f^{-1}(C) \subset C$, so

$$f^{-1}(C) = C.$$

Thus $(f^n)^{-1}(C) = C$ for any integer $n \geq 1$. By Lemma 7.2 with f^n, q^n in place of g, Q, the pair $\left(\mathbb{P}^2, \frac{1}{q^n - 1} R(f^n)\right)$ is lc. The proposition follows. □

Proposition 7.5. *Let f be an endomorphism of $\mathbb{P}^2$ of degree $q \geq 2$, k a positive integer, and $x \in \mathbb{P}^2$ a closed point. Assume that $f^k(x) = x$ and $c_\infty(f^k, x) = q^k$. Then for any positive integer m,*

$$\left(\mathbb{P}^2, \frac{1}{q^{12m}-1} R(f^{12m})\right) \text{ is lc.}$$

Proof. Let $N := \min\{r \in \mathbb{N}^+ \mid f^r(x) = x\}$. Then N divides k, and by Definition 2.11,

$$c_\infty(f^k, x) = \lim_{j \to +\infty} c\big((f^N)^{kj/N}, \mathrm{mult}_x\big)^{1/j} = c_\infty(f^N, x)^{k/N}.$$

Since $c_\infty(f^k, x) = q^k$,

$$c_\infty(f^N, x) = q^N.$$

Since $f^{kj}(x) = x$ for any j,

$$\lim_{j \to +\infty} c(f^{kj}, \mathrm{mult}_x)^{1/(kj)} = c_\infty(f^k, x)^{1/k} = q,$$

where the limit exists by [FJ03, Proposition 3.11]. Thus $x \in E_2(f)$ [LM26, Definitions 2.12–2.13]. By [LM26, Proposition 2.14(ii)], $E_2(f)$ is finite, $f^{-1}(E_2(f)) = E_2(f)$, and the germ of f^N at x is superattracting, that is, $c(f^{Nj}, \mathrm{mult}_x) \to +\infty$ as $j \to +\infty$ [LM26, Section 2.3].

Since f is surjective and $f^{-1}(E_2(f)) = E_2(f)$, f restricts to a surjection, hence a bijection, of the finite set $E_2(f)$, and $f^{-1}(x')$ is a single point for any $x' \in E_2(f)$. Thus for any $i \in \{0, 1, \ldots, N-1\}$,

$$(f^N)^{-1}(f^i(x)) = \{f^i(x)\},$$

and $x, f(x), \ldots, f^{N-1}(x)$ are N distinct closed points. By [CL26, Theorem 1.3] (the case of points on $\mathbb{P}^2$), $N \leq 3$. In particular, $2N$ divides 12.

Since f^N is an endomorphism of $\mathbb{P}^2$ of degree q^N and $(f^N)^{-1}(x) = \{x\}$, Definition 2.2 gives

$$\deg_x(f^N) = q^{2N} = c_\infty(f^N, x)^2,$$

and $\deg_x(f^N)$ is $d(f^N)$ in [LM26, Theorem 4.2]. Since the germ of f^N at x is a finite superattracting germ, by [LM26, Theorem 4.2] there exists a divisorial valuation v centered at x with

$$(f^{2N})_* v = q^{2N} v.$$

Since v is realized by a finite sequence of blow-ups of points over x, v is a divisorial valuation over $\mathbb{P}^2$. Since $(f^{2N})^{-1}(x) = \{x\}$ and f^{2N} is an endomorphism of $\mathbb{P}^2$ of degree $q^{2N} \geq 2$, by Proposition 7.4 with f^{2N}, q^{2N}, x in place of f, q, p, the pair $\left(\mathbb{P}^2, \frac{1}{q^{2Nn}-1} R(f^{2Nn})\right)$ is lc for any integer $n \geq 1$. Since $2N$ divides $12m$, taking $n = 6m/N$, the proposition follows. □

7.3. **Curve valuations as eigenvaluations.**

Lemma 7.6. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$, $p \in \mathbb{P}^2$ a closed point, and ϕ an irreducible analytic curve germ at p. Assume that the following hold.*

(1) $g(p) = p$ and $\lambda := c_\infty(g, p) > 1$.
(2) ν_ϕ is an eigenvaluation of g at p.
(3) There exists a normalized valuation $\mu < \nu_\phi$ such that $c(g, \nu) = \lambda$ and $g_\bullet \nu > \nu$ for any $\nu \in [\mu, \nu_\phi)$.

Then ϕ is a branch at p of an irreducible curve $C \subset \mathbb{P}^2$ with $g(C) = C$ and

$$e_C(g) \geq \lambda.$$

Proof. Let x, y be regular parameters at p, and let

$$\psi := \frac{\partial(g^*x)}{\partial x}\frac{\partial(g^*y)}{\partial y} - \frac{\partial(g^*x)}{\partial y}\frac{\partial(g^*y)}{\partial x} \in \widehat{\mathcal{O}}_{\mathbb{P}^2, p}$$

be the Jacobian determinant of g at p. Since $g(p) = p$ and $K_{\mathbb{P}^2} = g^* K_{\mathbb{P}^2} + R(g)$, ψ is a local equation of $R(g)$ at p. We denote an equation of ϕ in $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$ again by ϕ, and write

$$\psi = u\phi^b \prod_{j=1}^{n} \psi_j,$$

where u is a unit, b and n are non-negative integers, and $\psi_1, \dots, \psi_n$ are irreducible formal curve germs at p different from ϕ.

Step 1. We show that $b \geq \lambda - 1$. For any j, since $(\phi \cdot \psi_j)_p < +\infty$, by Theorem 2.9(2)(5),

$$\alpha(\nu_\phi \wedge \nu_{\psi_j}) < +\infty = \alpha(\nu_\phi),$$

so $\nu_\phi \wedge \nu_{\psi_j} < \nu_\phi$. Since $[\mathrm{mult}_p, \nu_\phi]$ is totally ordered, by Theorem 2.9(5), possibly replacing μ with a larger element of $[\mu, \nu_\phi)$, we may assume that

$$m(\nu) = \mathrm{mult}_p\, \phi \text{ for any } \nu \in [\mu, \nu_\phi), \quad \text{and} \quad \nu_\phi \wedge \nu_{\psi_j} \leq \mu \text{ for any } j.$$

Since α restricts to a strictly increasing bijection of $[\mu, \nu_\phi]$ onto $[\alpha(\mu), +\infty]$ (Theorem 2.9(1)(5)), for any real number $t > \alpha(\mu)$ we let ν_t be the element of $[\mu, \nu_\phi)$ with $\alpha(\nu_t) = t$. By Theorem 2.9(3),

$$A(\nu_t) = A(\mu) + \int_\mu^{\nu_t} m(\nu) d\alpha(\nu) = \mathrm{mult}_p\, \phi \cdot t + c_1, \quad \text{where } c_1 := A(\mu) - \mathrm{mult}_p\, \phi \cdot \alpha(\mu).$$

For any j, since $\nu_\phi \wedge \nu_{\psi_j} \leq \mu \leq \nu_t \leq \nu_\phi$, we have $\nu_t \wedge \nu_{\psi_j} = \nu_\phi \wedge \nu_{\psi_j}$. Also $\nu_t \wedge \nu_\phi = \nu_t$. By Theorem 2.9(2),

$$\nu_t(\psi) = b\nu_t(\phi) + \sum_{j=1}^{n} \nu_t(\psi_j) = b\, \mathrm{mult}_p\, \phi \cdot t + c_2, \quad \text{where } c_2 := \sum_{j=1}^{n} \mathrm{mult}_p\, \psi_j \cdot \alpha(\nu_\phi \wedge \nu_{\psi_j}).$$

Since $g_\bullet \nu_t > \nu_t$ and $m \geq 1$, by Theorem 2.9(1)(3), $A(g_\bullet \nu_t) \geq A(\nu_t)$. Since $c(g, \nu_t) = \lambda$, by Theorem 2.12(2),

$$\nu_t(\psi) = \lambda A(g_\bullet \nu_t) - A(\nu_t) \geq (\lambda - 1) A(\nu_t).$$

Thus for any real number $t > \alpha(\mu)$,

$$b\, \mathrm{mult}_p\, \phi \cdot t + c_2 \geq (\lambda - 1)(\mathrm{mult}_p\, \phi \cdot t + c_1).$$

Dividing by $\mathrm{mult}_p\, \phi \cdot t$ and letting $t \to +\infty$, we obtain $b \geq \lambda - 1$. In particular, $b > 0$ since $\lambda > 1$.

Step 2. We construct C. Write $R(g) = \sum_{i=1}^{N} r_i C_i$, where $C_1, \dots, C_N$ are distinct irreducible curves and $r_1, \dots, r_N$ are positive integers. For any i with $p \in C_i$, a local equation of C_i at p is, up to a unit in $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$, the product of the equations of the branches of C_i at p, which are pairwise distinct since C_i is reduced. Since ψ is a local equation of $R(g)$ at p and ϕ divides ψ by **Step 1**, ϕ is a branch at p of C_i for some i. Since two distinct irreducible curves in $\mathbb{P}^2$ have finitely many common points, they have no common branch, so such i is unique and $b = r_i$. Let $C := C_i$. Then the coefficient of C in $R(g)$ is $e_C(g) - 1$, so

$$e_C(g) = b + 1 \geq \lambda.$$

Since g is finite, by [FJ07, Proposition 2.6], $g(\phi)$ is an irreducible analytic curve germ at p and $g_\bullet \nu_\phi = \nu_{g(\phi)}$. Since ν_ϕ is an eigenvaluation of g at p, $g_\bullet \nu_\phi = \nu_\phi$, so $\nu_{g(\phi)} = \nu_\phi$. By Theorem 2.9(2), $\nu_\phi(\phi') < +\infty$ for any irreducible formal curve germ $\phi' \neq \phi$ at p, while $\nu_{g(\phi)}(g(\phi)) = +\infty$. Thus $g(\phi) = \phi$. Since $g(C)$ is an irreducible curve with branch $g(\phi) = \phi$ at p, $g(C)$ and C have the common branch ϕ, so $g(C) = C$. The lemma follows. □

7.4. **The asymptotic attraction rate.**

Lemma 7.7. *Notation and conditions as in Set-up 5.1. Then:*

(1) For any integer $i \geq 1$ and any divisorial valuation w over $\mathbb{P}^2$,

$$A\left((g^i)_* w\right) = A(w) + (Q^i - 1)w(B_{17280i}) \leq \frac{3}{2} Q^i A(w).$$

(2) Let $n \geq 1$ be an integer and v a divisorial valuation over $\mathbb{P}^2$ with $v(B_{17280n}) > A(v)$. Then $A((g^n)_* v) > Q^n A(v)$, and

$$A\left((g^j)_* v\right) > \frac{2}{3} Q^j A(v)$$

for any integer $0 \leq j \leq n$.

Proof. Let $i \geq 1$ be an integer and w a divisorial valuation over $\mathbb{P}^2$. Since $g^i = f^{17280i}$ has degree $Q^i = q^{17280i}$, we have $R(g^i) = (Q^i - 1)B_{17280i}$, and the equality in (1) follows from Lemma 2.4. Since $B_{17280i} = B_{6m}$ and $Q^i = q^{6m}$ for $m := 2880i$, Theorem 4.1(3) implies

$$A\left((g^i)_* w\right) \leq \frac{3Q^i - 1}{2} A(w) \leq \frac{3}{2} Q^i A(w).$$

This proves (1).

We prove (2). By (1) with $i = n$ and $w = v$, and since $v(B_{17280n}) > A(v)$,

$$A\left((g^n)_* v\right) > A(v) + (Q^n - 1)A(v) = Q^n A(v).$$

Let $0 \leq j < n$ be an integer. By Lemma 2.4, $(g^j)_* v$ is a divisorial valuation over $\mathbb{P}^2$, and $(g^n)_* v = (g^{n-j})_* \left((g^j)_* v\right)$. By (1) with $i = n - j$ and $w = (g^j)_* v$,

$$Q^n A(v) < A\left((g^n)_* v\right) \leq \frac{3}{2} Q^{n-j} A\left((g^j)_* v\right),$$

so $A((g^j)_* v) > \frac{2}{3} Q^j A(v)$. For $j = n$ the inequality follows from $A((g^n)_* v) > Q^n A(v)$. The lemma follows. □

Proposition 7.8. *Notation and conditions as in Set-up 5.1. Let $n \geq 4 \cdot 10^8$ be an integer and v a divisorial valuation over $\mathbb{P}^2$ centered at a closed point p. Assume that $v(B_{17280n}) > A(v)$. Then $g(p) = p$, and*

$$A\left((g^j)_* v\right) \leq 10^4 \left((g^j)_* v\right)(\mathfrak{m}_p)$$

for any integer $0 \leq j \leq n/1000$.

Proof. Since v is centered at p and $v(B_{17280n}) > A(v)$, we have $p \in \mathrm{Nlc}(\mathbb{P}^2, B_{17280n})$. By Proposition 5.2, $\mathrm{Nlc}(\mathbb{P}^2, B_{17280n}) = \{p\}$ and $g(p) = p$. Since $B_{17280n} = B_{6m}$ for $m := 2880n$, Theorem 4.1(4) implies

$$A(v) < 100 v(\mathfrak{m}_p). \tag{7.2}$$

This proves the inequality for $j = 0$.

Fix an integer $1 \leq j \leq n/1000$. Let $k := n - j$ and $w := (g^j)_* v$. Since $n \geq 1000j$,

$$k \geq 999j \geq 999.$$

By Lemma 2.4, w is a divisorial valuation over $\mathbb{P}^2$ centered at $g^j(p) = p$. Let

$$M := \frac{A(w)}{w(\mathfrak{m}_p)}.$$

Suppose that $M > 10^4$. We derive a contradiction.

Step 1. In this step we find an irreducible curve $P \ni p$ with $g(P) = P$ and $e_P(g) > \frac{99}{100} Q$.

Since $(g^n)_* v = (g^k)_* w$, Lemma 7.7 implies

$$Q^n A(v) < A\left((g^k)_* w\right) = A(w) + (Q^k - 1)w(B_{17280k}) \quad \text{and} \quad A(w) \leq \frac{3}{2} Q^j A(v).$$

Thus

$$(Q^k-1)w(B_{17280k}) > Q^n A(v) - A(w) \geq \left(\frac{2}{3}Q^k - 1\right) A(w),$$

and since $Q^k > 21$,

$$\frac{w(B_{17280k})}{A(w)} > \frac{\frac{2}{3}Q^k-1}{Q^k-1} = \frac{2}{3} - \frac{1}{3(Q^k-1)} > \frac{13}{20}. \tag{7.3}$$

Since $\deg R(f^i) = 3(q^i-1)$ for any integer $i \geq 1$, $B_{17280k} \geq 0$ is a $\mathbb{Q}$-divisor of degree 3. Let $\nu := w/w(\mathfrak{m}_p)$. Then ν is a normalized divisorial valuation at p with $A(\nu) = M > 10^4$, and $\nu(B_{17280k}) > \frac{13}{20}A(\nu)$ by (7.3). By Lemma 5.6, there exists an irreducible component P of B_{17280k} with $p \in P$, $d := \deg P \leq 4$, $b := \operatorname{mult}_P B_{17280k} > \frac{3}{5}$, and

$$w(B_{17280k}) \leq bw(P) + 14w(\mathfrak{m}_p). \tag{7.4}$$

Since $B_{17280k} = B_{6\cdot 2880k}$ and $\frac{26}{2880k+1} < \frac{3}{5} < b$, Proposition 5.4(1) implies $f^{6r}(P) = P$ for some $r \in \{1,2,3\}$. Since r divides 2880, $g(P) = (f^{6r})^{2880/r}(P) = P$. Let

$$e := e_P(g) \quad \text{and} \quad t := \frac{e}{Q}.$$

Since $g(P) = P$, Lemma 3.2(1) gives $e_P(g^j) = e^j$ and $e_P(g^k) = e^k$. Thus

$$(g^j)^*P = e^jP + D, \quad D \geq 0, \quad P \notin \operatorname{Supp} D, \quad \text{and} \quad \deg D = d(Q^j - e^j), \tag{7.5}$$

and $\operatorname{mult}_P R(g^k) = e^k - 1$, that is, $(Q^k-1)b = e^k - 1$. Since $\deg D \geq 0$ in (7.5), $e \leq Q$, and

$$\frac{3}{5} < b = \frac{e^k-1}{Q^k-1} \leq t^k. \tag{7.6}$$

If $t \leq \frac{99}{100}$, then

$$t^{-k} \geq \left(1 + \frac{1}{99}\right)^k \geq 1 + \frac{k}{99} > 11,$$

contradicting (7.6). Thus

$$t > \frac{99}{100}.$$

Step 2. In this step we bound $A((g^n)_*v)$ from above. Let

$$s := \frac{v(P)}{A(v)}.$$

By Lemma 2.5, $\operatorname{mult}_p D \leq \deg D$, and (7.5),

$$v(D) \leq \deg D\big(A(v) - v(\mathfrak{m}_p)\big) \leq d(Q^j - e^j)A(v).$$

By (7.5),

$$w(P) = v\left((g^j)^*P\right) = e^j v(P) + v(D) \leq e^j v(P) + d(Q^j - e^j)A(v). \tag{7.7}$$

By Lemma 7.7(1), (7.4) and $(Q^k-1)b = e^k - 1$,

$$A\left((g^n)_*v\right) = A(w) + (Q^k-1)w(B_{17280k}) \leq A(w) + e^k w(P) + 14Q^k w(\mathfrak{m}_p).$$

Since $A(w) \leq \frac{3}{2}Q^j A(v)$ and $w(\mathfrak{m}_p) = A(w)/M$, (7.7) implies

$$\frac{A\left((g^n)_*v\right)}{Q^n A(v)} \leq \frac{3}{2}Q^{-k} + \frac{21}{M} + t^n s + dt^k(1-t^j). \tag{7.8}$$

Step 3. In this step we show that $d = 3$ and P has a node at p.

If $(\mathbb{P}^2, P)$ is not lc, then Lemma 3.2(3) implies $t \leq 1 - \frac{1}{2d^2} \leq \frac{31}{32}$. If $d = 4$ and $(\mathbb{P}^2, P)$ is lc, then Lemma 3.2(4) implies $t \leq \frac{15}{16}$. Both contradict $t > \frac{99}{100}$. Thus $d \leq 3$ and $(\mathbb{P}^2, P)$ is lc. An irreducible plane curve of degree at most 2 is smooth, an irreducible plane cubic singular at p

has a node or a cusp at p, and $(\mathbb{P}^2, P)$ is not lc at a cusp of P. Thus either P is smooth at p, or $d = 3$ and P has a node at p.

Suppose that P is smooth at p. Let $H \neq P$ be a line through p that is not tangent to P at p. Then $(H \cdot P)_p = 1$. Since P is smooth at p, Lemma 2.5 implies $v(H) \leq A(v) - v(P)$. Since $v(H) \geq v(\mathfrak{m}_p)$, (7.2) implies

$$s \leq 1 - \frac{v(\mathfrak{m}_p)}{A(v)} < \frac{99}{100}.$$

Since $0 < t \leq 1$,

$$(1-t^j)t^k \leq j(1-t)t^k \leq \frac{j}{k+1} < \frac{1}{999},$$

where the second inequality follows from $(k+1)(1-t)t^k \leq (1-t)\sum_{i=0}^{k} t^i = 1 - t^{k+1} \leq 1$. Since $t \leq 1$, $d \leq 3$, $M > 10^4$, and $Q^k \geq 2^{999}$, (7.8) implies

$$\frac{A\left((g^n)_* v\right)}{Q^n A(v)} < \frac{1}{10^4} + \frac{21}{10^4} + \frac{99}{100} + \frac{3}{999} < \frac{1 + 21 + 9900 + 31}{10^4} < 1,$$

contradicting Lemma 7.7(2). Thus $d = 3$ and P has a node at p.

Step 4. Let $\pi : \overline{P} \to P$ be the normalization, $h : \overline{P} \to \overline{P}$ the morphism with $\pi \circ h = g \circ \pi$, and $\pi^{-1}(p) = \{z_1, z_2\}$. Since $g(p) = p$, $h(\{z_1, z_2\}) \subset \{z_1, z_2\}$. In this step we show that $h(z_1) = h(z_2)$.

Suppose that $h(z_1) \neq h(z_2)$. Then h^j restricts to a permutation of $\{z_1, z_2\}$, and we let σ be the permutation of $\{1, 2\}$ with $h^j(z_i) = z_{\sigma(i)}$ for $i \in \{1, 2\}$. Let x_1, x_2 be analytic local coordinates of $\mathbb{P}^2$ at p such that, near p, $P = \Gamma_1 \cup \Gamma_2$, where $\Gamma_i := \{x_i = 0\}$ is the branch of P at p corresponding to z_i. Then g^j maps Γ_i onto $\Gamma_{\sigma(i)}$. Since $(g^j)^{-1}(p)$ is finite, g^j maps $\Gamma_i \backslash \{p\}$ near p into $\Gamma_{\sigma(i)} \backslash \{p\}$, near which $P = \{x_{\sigma(i)} = 0\}$. Thus (7.5) implies that $x_i^{e^j}$ divides $(g^j)^* x_{\sigma(i)}$. Thus

$$w(x_{\sigma(i)}) = v\left((g^j)^* x_{\sigma(i)}\right) \geq e^j v(x_i) \geq e^j v(\mathfrak{m}_p)$$

for $i \in \{1, 2\}$. Since x_1, x_2 generate the maximal ideal of $\widehat{\mathcal{O}}_{\mathbb{P}^2, p}$,

$$w(\mathfrak{m}_p) = \min\{w(x_1), w(x_2)\} \geq e^j v(\mathfrak{m}_p).$$

Since $t \leq 1$, $j \leq k$ and $t^k > \frac{3}{5}$, Lemma 7.7(1) and (7.2) imply

$$M = \frac{A(w)}{w(\mathfrak{m}_p)} \leq \frac{\frac{3}{2} Q^j A(v)}{e^j v(\mathfrak{m}_p)} < 150 t^{-j} \leq 150 t^{-k} < 250,$$

contradicting $M > 10^4$. Thus $h(z_1) = h(z_2)$.

Step 5. In this step we derive the contradiction. By **Step 3**, P is an irreducible nodal cubic with node p. We have $g(P) = P$, $g(p) = p$, $h(z_1) = h(z_2)$ by **Step 4**, and

$$e > \frac{99}{100} Q > \frac{12}{13} Q.$$

Since $n \geq 72$, Lemma 6.5 implies that

$$\left(\mathbb{P}^2, \frac{1}{Q^n - 1} R(g^n)\right) = \left(\mathbb{P}^2, B_{17280n}\right)$$

is klt at p, contradicting $v(B_{17280n}) > A(v)$.

Thus $M \leq 10^4$ for any integer $1 \leq j \leq n/1000$. The proposition follows. □

Proof of Theorem 7.1. By Proposition 7.8, $g(p) = p$. For any integer $j \geq 1$, let

$$c_j := c(g^j, \mathrm{mult}_p) = \mathrm{mult}_p\left((g^j)^* \mathfrak{m}_p\right).$$

By Definition 2.11, $\lambda = \lim_{j \to +\infty} c_j^{1/j}$.

Since $(g^k)^* \mathfrak{m}_p \subset \mathfrak{m}_p^{c_k}$, we have $c_{j+k} \geq c_j c_k$ for any integers $j, k \geq 1$, so $\lambda = \sup_{j \geq 1} c_j^{1/j}$, that is,

$$c_j \leq \lambda^j \quad \text{for any integer } j \geq 1. \tag{7.9}$$

Let H be a line through p. Since a local equation of H at p is an element of $\mathfrak{m}_p$,

$$1 \le c_j \le \operatorname{mult}_p\left((g^j)^*H\right) \le \deg(g^j)^*H = Q^j \quad \text{for any integer } j \ge 1.$$

Thus $1 \le \lambda \le Q$.

Let $N := \lfloor n/1000 \rfloor$ and $\nu_0 := v/v(\mathfrak{m}_p)$. Since ν_0 is a normalized divisorial valuation at p, Theorem 2.9(3)(4) implies $\alpha(\nu_0) \le A(\nu_0) - 1 < +\infty$. Let $w := (g^N)_* v$ and let $\psi \in (g^N)^*\mathfrak{m}_p$ with $\operatorname{mult}_p \psi = c_N$. By Lemma 2.10(1) and (7.9),

$$w(\mathfrak{m}_p) = v\left((g^N)^*\mathfrak{m}_p\right) \le v(\psi) = v(\mathfrak{m}_p)\nu_0(\psi) \le v(\mathfrak{m}_p)\alpha(\nu_0)c_N \le \left(A(v) - v(\mathfrak{m}_p)\right)\lambda^N < A(v)\lambda^N.$$

By Lemma 7.7(2) and Proposition 7.8,

$$\frac{2}{3}Q^N A(v) < A(w) \le 10^4 w(\mathfrak{m}_p) < 10^4 \lambda^N A(v).$$

Thus

$$\left(\frac{\lambda}{Q}\right)^N > \frac{1}{15000}.$$

Since $N \ge 4 \cdot 10^5 \ge 98$, if $\lambda \le \frac{9}{10}Q$, then $(\lambda/Q)^N \le ((9/10)^7)^{14} < 2^{-14} < \frac{1}{15000}$, a contradiction. Thus $\lambda > \frac{9}{10}Q$.

Suppose that $\lambda = Q$. Then $f^{17280}(p) = g(p) = p$ and $c_\infty(f^{17280}, p) = \lambda = q^{17280}$. By Proposition 7.5 with $k = 17280$ and $x = p$, $(\mathbb{P}^2, B_{12m})$ is lc for any integer $m \ge 1$. With $m = 1440n$, $(\mathbb{P}^2, B_{17280n})$ is lc, contradicting $v(B_{17280n}) > A(v)$. Thus

$$\frac{9}{10}Q < \lambda < Q.$$

Assume that $p \notin C$, and suppose that there exists an irreducible analytic curve germ ϕ at p such that ν_ϕ is an eigenvaluation of g at p and there exists a normalized valuation $\mu < \nu_\phi$ with $c(g, \nu) = \lambda$ and $g_\bullet \nu > \nu$ for any $\nu \in [\mu, \nu_\phi)$. Since $\lambda > \frac{9}{10}Q > 1$, Lemma 7.6 implies that ϕ is a branch at p of an irreducible curve $P \subset \mathbb{P}^2$ with $g(P) = P$ and $e_P(g) \ge \lambda$. Since $\operatorname{mult}_P R(g) = e_P(g) - 1$ and $Q > \frac{4}{3}$,

$$\operatorname{mult}_P B_{17280} = \frac{e_P(g) - 1}{Q - 1} \ge \frac{\lambda - 1}{Q - 1} > \frac{\frac{9}{10}Q - 1}{Q - 1} > \frac{3}{5}.$$

Since $B_{17280} = B_{6 \cdot 2880}$ and $\frac{26}{2881} < \frac{3}{5}$, Proposition 5.4(1) implies $f^{6r}(P) = P$ for some $r \in \{1, 2, 3\}$. Since r divides 48, $f^{288}(P) = P$. Let $Q' := q^{288}$ and $e' := e_P(f^{288})$. Since $g = (f^{288})^{60}$ and $f^{288}(P) = P$, we have $Q = Q'^{60}$ and $e_P(g) = e'^{60}$. Thus

$$\left(\frac{e'}{Q'}\right)^{60} = \frac{e_P(g)}{Q} \ge \frac{\lambda}{Q} > \frac{9}{10},$$

so $\frac{e'}{Q'} > \left(\frac{9}{10}\right)^{1/60} > \frac{9}{10}$. Since $Q' > \frac{4}{3}$,

$$\operatorname{mult}_P B_{288} = \frac{e' - 1}{Q' - 1} > \frac{\frac{9}{10}Q' - 1}{Q' - 1} > \frac{3}{5}.$$

By the definition of C in Set-up 5.1, $P \subset C$. Since $p \in P$, $p \in C$, a contradiction. The theorem follows. □

8. Skewness and inner product bounds in the valuative tree

8.1. **Tree measures of plane divisors.**

Definition 8.1. Let $p \in \mathbb{P}^2$ be a closed point. Let $\mathcal{M}$ be the real vector space with basis $\{\delta_\mu\}$, where μ ranges over the normalized valuations at p with $\alpha(\mu) < +\infty$. Following [FJ04, Section 7.12], let $\langle \cdot, \cdot \rangle$ be the symmetric bilinear form on $\mathcal{M}$ with $\langle \delta_\mu, \delta_\nu \rangle = \alpha(\mu \wedge \nu)$, which is positive definite [FJ04, Remark 7.78 and Theorem 7.83], and let $\|\xi\| := \langle \xi, \xi \rangle^{1/2}$. For a finite set

S of normalized valuations at p with finite skewness, let $\Gamma_S := \{\mathrm{mult}_p\} \cup \bigcup_{\nu\in S}[\mathrm{mult}_p, \nu]$, and for a normalized valuation μ at p let

$$\pi_S(\mu) := \max\big(\{\mathrm{mult}_p\} \cup \{\nu \wedge \mu \mid \nu \in S\}\big),$$

the maximum in the totally ordered set $[\mathrm{mult}_p, \mu]$. By Theorem 2.9(1), every element of Γ_S, in particular $\pi_S(\mu)$, has finite skewness. For an $\mathbb{R}$-divisor $D = \sum_i c_i P_i$ on $\mathbb{P}^2$ with distinct irreducible curves P_i and $c_i \in \mathbb{R}$, let

$$\rho_S(D) := \sum_i c_i \sum_\phi \mathrm{mult}_p \phi \cdot \delta_{\pi_S(\nu_\phi)},$$

where the inner sum is over the irreducible formal branches ϕ of P_i at p.

Theorem 8.2. *Let $p \in \mathbb{P}^2$ be a closed point, S a finite set of normalized valuations at p with finite thinness, and $D = \sum_i c_i P_i \geq 0$ an $\mathbb{R}$-divisor on $\mathbb{P}^2$, where the P_i are distinct irreducible curves and $c_i \in \mathbb{R}_{>0}$. Let $\epsilon \geq 0$ be a real number with $c_i \leq \epsilon$ for any P_i through p. Then*

$$\langle \delta_\mu, \rho_S(D)\rangle = \mu(D) \quad \text{for any } \mu \in \Gamma_S,$$

and the following hold.

(1) If $\kappa \geq 2$ is a real number and $A(\nu) \leq \kappa$ for any $\nu \in S$, then

$$\|\rho_S(D)\|^2 \leq (\deg D)^2 + (\kappa - 2)\epsilon \deg D.$$

(2) If L is a line through p with $L \not\subset \operatorname{Supp} D$, $\kappa \geq 1$ is a real number, and $A(\nu) - \nu(L) \leq \kappa$ for any $\nu \in S$, then

$$\|\rho_S(D)\|^2 \leq (\deg D)^2 + (\kappa - 1)\epsilon \deg D.$$

(3) For any $\mathbb{R}$-divisor $D' \geq 0$ on $\mathbb{P}^2$ without common component with D,

$$\langle \rho_S(D), \rho_S(D')\rangle \leq (D \cdot D')_p.$$

Proof. By Theorem 2.9(3), $\alpha(\nu) \leq A(\nu) - 1 < +\infty$ for any $\nu \in S$, so Γ_S, π_S, and $\rho_S(D)$ are defined. Let $\Lambda_p := \{i \mid p \in P_i\}$. For $i \in \Lambda_p$, let $d_i := \deg P_i$. Since P_i is reduced, a local equation of P_i at p is the product of a unit and the distinct irreducible formal branches β of P_i at p. For such a branch β, let $t_\beta := \pi_S(\nu_\beta)$. Then

$$\rho_S(D) = \sum_{i\in\Lambda_p} c_i \rho_S(P_i) \quad \text{and} \quad \rho_S(P_i) = \sum_\beta \mathrm{mult}_p \beta \cdot \delta_{t_\beta},$$

where β runs over the irreducible formal branches of P_i at p.

Step 1. In this step we prove $\langle \delta_\mu, \rho_S(D)\rangle = \mu(D)$ for any $\mu \in \Gamma_S$.

Let β be an irreducible formal branch of some P_i at p. We show $\mu \wedge t_\beta = \mu \wedge \nu_\beta$. Since $t_\beta \leq \nu_\beta$, we have $\mu \wedge t_\beta \leq \mu \wedge \nu_\beta$. Conversely, let $\omega := \mu \wedge \nu_\beta$. Since $\mu \in \Gamma_S$, either $\omega \leq \mu = \mathrm{mult}_p$, or $\omega \leq \mu \leq \nu$ for some $\nu \in S$ and then $\omega \leq \nu \wedge \nu_\beta$. In both cases, $\omega \leq t_\beta$ by the definition of π_S, so $\omega \leq \mu \wedge t_\beta$. By Theorem 2.9(2) and $\mu(D) = \sum_{i\in\Lambda_p} c_i \mu(P_i)$ (Definition 2.8),

$$\langle \delta_\mu, \rho_S(D)\rangle = \sum_{i\in\Lambda_p} c_i \sum_\beta \mathrm{mult}_p \beta \cdot \alpha(\mu \wedge \nu_\beta) = \sum_{i\in\Lambda_p} c_i \sum_\beta \mu(\beta) = \sum_{i\in\Lambda_p} c_i \mu(P_i) = \mu(D).$$

Step 2. In this step we bound the contribution of one branch. For a formal curve germ γ at p and a point q infinitely near to p (here and below, a point counts as infinitely near to itself, so $q = p$ is allowed), let $\mathrm{mult}_q \gamma$ be the multiplicity at q of the strict transform of γ ($\mathrm{mult}_q \gamma = 0$ if the strict transform of γ does not pass through q). For an irreducible formal branch β at p, let

$$s(\beta) := \sum_q \mathrm{mult}_q \beta(\mathrm{mult}_q \beta - 1),$$

where q runs over the points infinitely near to p. We show that for any irreducible formal branch β at p and any normalized valuation $\mu \le \nu_\beta$ with $\alpha(\mu) < +\infty$,

$$(\mathrm{mult}_p \beta)^2 \alpha(\mu) - \mathrm{mult}_p \beta \cdot A(\mu) \le s(\beta) - 1. \tag{8.1}$$

Let $Y_0 := \mathbb{P}^2$ and $q_1 := p$. For any integer $j \ge 1$, let $Y_j \to Y_{j-1}$ be the blow-up of q_j with exceptional curve F_j, let $h_j : Y_j \to \mathbb{P}^2$ be the composition, let $\beta_j := (h_j)_*^{-1}\beta$, let q_{j+1} be the point of $\beta_j \cap F_j$, and let $r_j := \mathrm{mult}_{q_j} \beta$. Then $r_1 = \mathrm{mult}_p \beta$, and $\mathrm{mult}_q \beta = 0$ for any $q \notin \{q_j\}_{j\ge1}$. By embedded resolution of β, there exists an integer $k_0 \ge 1$ such that for any $k \ge k_0$, β_k is smooth, q_{k+1} is contained in no exceptional curve of h_k other than F_k, and β_k is transverse to F_k at q_{k+1}. In particular, $r_j = 1$ for any $j > k_0$, and

$$s(\beta) = \sum_{j=1}^{k} r_j(r_j - 1) \quad \text{for any } k \ge k_0.$$

Let $k \ge k_0$ be an integer. Let F_j^* be the pullback of F_j to Y_k for $1 \le j \le k$. A *curvette* of F_k is the image in $\mathbb{P}^2$ of a smooth formal curve germ $\widetilde{\phi}$ on Y_k at a point $y \in F_k$ such that y is not contained in any other exceptional curve of h_k and $\widetilde{\phi}$ is transverse to F_k at y. By the choice of k_0, β is a curvette of F_k. Let ϕ be a curvette of F_k, let $\widetilde{\phi}$ and y be the germ and the point in the definition of a curvette, and let $\phi_j := (h_j)_*^{-1}\phi$. Since $\widetilde{\phi}$ intersects the exceptional locus of h_k only at y, transversally to F_k, the projection formula implies

$$\mathrm{mult}_{q_j} \phi = (\phi_j \cdot F_j)_{Y_j} = (\widetilde{\phi} \cdot F_j^*)_y = \mathrm{mult}_{F_k} F_j^* \quad \text{for any } 1 \le j \le k.$$

Taking $\phi = \beta$, we obtain $\mathrm{mult}_{q_j} \phi = r_j$ for any curvette ϕ of F_k and any $1 \le j \le k$. In particular, $\mathrm{mult}_p \phi = r_1$ for any curvette ϕ of F_k. Since $\mathfrak{m}_p \mathcal{O}_{Y_1} = \mathcal{O}_{Y_1}(-F_1)$ and $K_{Y_k} = h_k^* K_{\mathbb{P}^2} + \sum_{j=1}^k F_j^*$, we have

$$\mathrm{mult}_{F_k} \mathfrak{m}_p = \mathrm{mult}_{F_k} F_1^* = r_1 \quad \text{and} \quad a(F_k, \mathbb{P}^2, 0) = 1 + \sum_{j=1}^{k} r_j.$$

Let ν_k be the divisorial normalized valuation at p with $r_1 \nu_k(\zeta) = \mathrm{mult}_{F_k} \mathrm{div}(h_k^* \zeta)$ for any non-zero $\zeta \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p}$. By Theorem 2.9(4),

$$r_1 A(\nu_k) = 1 + \sum_{j=1}^{k} r_j.$$

For any curvette ϕ of F_k and any non-zero $\zeta \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p}$ not divisible by ϕ, we have

$$(\phi \cdot \zeta)_p = (\widetilde{\phi} \cdot h_k^* \zeta)_y \ge r_1 \nu_k(\zeta),$$

so $\nu_\phi(\zeta) \ge \nu_k(\zeta)$. Thus $\nu_k \le \nu_\phi$. In particular, $\nu_k \le \nu_\beta$. Let ϕ, ψ be curvettes of F_k with $\widetilde{\phi}, \widetilde{\psi}$ at distinct points y, y' of F_k. Since $\psi_k := (h_k)_*^{-1}\psi$ does not pass through y and

$$h_k^* \psi = \psi_k + \sum_{j=1}^{k} \mathrm{mult}_{q_j} \psi \cdot F_j^*,$$

we have

$$(\phi \cdot \psi)_p = (\widetilde{\phi} \cdot h_k^* \psi)_y = r_1 \nu_k(\psi) = \sum_{j=1}^{k} r_j \, \mathrm{mult}_{F_k} F_j^* = \sum_{j=1}^{k} r_j^2.$$

Since $\nu_k \le \nu_\phi \wedge \nu_\psi$ and $\mathrm{mult}_p \psi = r_1$, the definition of skewness and Theorem 2.9(1)(2) imply

$$\frac{\nu_k(\psi)}{r_1} \le \alpha(\nu_k) \le \alpha(\nu_\phi \wedge \nu_\psi) = \frac{(\phi \cdot \psi)_p}{r_1^2} = \frac{\nu_k(\psi)}{r_1}.$$

Therefore

$$r_1^2\alpha(\nu_k)=\sum_{j=1}^{k}r_j^2,\quad\text{so}\quad r_1^2\alpha(\nu_k)-r_1A(\nu_k)=\sum_{j=1}^{k}r_j(r_j-1)-1=s(\beta)-1\quad\text{and}\quad\alpha(\nu_k)\geq\frac{k}{r_1^2}.$$

For a normalized valuation $\omega\leq\nu_\beta$ with $\alpha(\omega)<+\infty$, let $\Phi_\beta(\omega):=r_1^2\alpha(\omega)-r_1A(\omega)$. By Theorem 2.9(3)(5), $m(\tau)\leq\operatorname{mult}_p\beta=r_1$ for any $\tau<\nu_\beta$. Thus for any $\omega\leq\omega'\leq\nu_\beta$ with $\alpha(\omega')<+\infty$, Theorem 2.9(3) implies

$$\Phi_\beta(\omega')-\Phi_\beta(\omega)=\int_\omega^{\omega'}r_1\big(r_1-m(\tau)\big)d\alpha(\tau)\geq 0.$$

Now let $\mu\leq\nu_\beta$ with $\alpha(\mu)<+\infty$. Since $\alpha(\nu_k)\geq k/r_1^2$, there exists an integer $k'\geq k_0$ with $\alpha(\nu_{k'})>\alpha(\mu)$. Since μ and $\nu_{k'}$ are elements of the totally ordered set $[\operatorname{mult}_p,\nu_\beta]$ and α is increasing on it, $\mu\leq\nu_{k'}$. Thus $\Phi_\beta(\mu)\leq\Phi_\beta(\nu_{k'})=s(\beta)-1$, which is (8.1).

Step 3. In this step we bound $\|\rho_S(P_i)\|^2$ for $i\in\Lambda_p$ in terms of $A(t_\beta)$. We show

$$\|\rho_S(P_i)\|^2\leq(d_i-1)(d_i-2)-1+\sum_\beta\operatorname{mult}_p\beta\cdot A(t_\beta).\tag{8.2}$$

Since $t_\beta\leq\nu_\beta$ and $\alpha(t_\beta)<+\infty$, (8.1) with $\mu=t_\beta$ implies

$$(\operatorname{mult}_p\beta)^2\alpha(t_\beta)\leq s(\beta)-1+\operatorname{mult}_p\beta\cdot A(t_\beta).$$

For distinct branches $\beta\neq\gamma$ of P_i at p, we have

$$t_\beta\wedge t_\gamma\leq\nu_\beta\wedge\nu_\gamma\leq\nu_\beta,$$

so Theorem 2.9(1)(2) implies

$$\operatorname{mult}_p\beta\cdot\operatorname{mult}_p\gamma\cdot\alpha(t_\beta\wedge t_\gamma)\leq\operatorname{mult}_p\beta\cdot\operatorname{mult}_p\gamma\cdot\alpha(\nu_\beta\wedge\nu_\gamma)=(\beta\cdot\gamma)_p.$$

Thus, since P_i has at least one branch at p,

$$\|\rho_S(P_i)\|^2\leq\sum_\beta s(\beta)+\sum_{\beta\neq\gamma}(\beta\cdot\gamma)_p-1+\sum_\beta\operatorname{mult}_p\beta\cdot A(t_\beta),$$

where (β,γ) runs over the ordered pairs of distinct branches of P_i at p. For any q infinitely near to p, $\operatorname{mult}_q P_i=\sum_\beta\operatorname{mult}_q\beta$, and by Noether's formula [CA00, Theorem 3.3.1], $(\beta\cdot\gamma)_p=\sum_q\operatorname{mult}_q\beta\cdot\operatorname{mult}_q\gamma$ for $\beta\neq\gamma$. Hence

$$\sum_q\operatorname{mult}_q P_i\big(\operatorname{mult}_q P_i-1\big)=\sum_\beta s(\beta)+\sum_{\beta\neq\gamma}(\beta\cdot\gamma)_p,$$

where q runs over the points infinitely near to p. By the genus formula for plane curves,

$$\sum_{q'}\operatorname{mult}_{q'}P_i\big(\operatorname{mult}_{q'}P_i-1\big)\leq(d_i-1)(d_i-2),$$

where q' runs over the points infinitely near to the points of P_i. Since $\operatorname{mult}_{q'}P_i(\operatorname{mult}_{q'}P_i-1)\geq 0$ for any q', (8.2) follows.

Step 4. In this step we prove (3). Let $D'=\sum_j c'_jP'_j$, where the P'_j are distinct irreducible curves and $c'_j\in\mathbb{R}_{>0}$. Let β be an irreducible formal branch of some P_i at p, let γ be an irreducible formal branch of some P'_j at p, and let $t_\gamma:=\pi_S(\nu_\gamma)$. Since $P_i\neq P'_j$, $(P_i\cdot P'_j)_p<+\infty$, so $\beta\neq\gamma$. Since $t_\beta\wedge t_\gamma\leq\nu_\beta\wedge\nu_\gamma$, Theorem 2.9(1)(2) implies

$$\operatorname{mult}_p\beta\cdot\operatorname{mult}_p\gamma\cdot\alpha(t_\beta\wedge t_\gamma)\leq\operatorname{mult}_p\beta\cdot\operatorname{mult}_p\gamma\cdot\alpha(\nu_\beta\wedge\nu_\gamma)=(\beta\cdot\gamma)_p.$$

Since $c_ic'_j>0$ for any i,j, we obtain

$$\langle\rho_S(D),\rho_S(D')\rangle\leq\sum_{i,j}c_ic'_j\sum_{\beta,\gamma}(\beta\cdot\gamma)_p=\sum_{i,j}c_ic'_j(P_i\cdot P'_j)_p=(D\cdot D')_p.$$

Step 5. In this step we prove (1) and (2). In case (1) let $\kappa' := \kappa$, and in case (2) let $\kappa' := \kappa + 1$. Then $\kappa' \geq 2$. We show

$$\sum_{\beta} \operatorname{mult}_p \beta \cdot A(t_\beta) \leq \kappa' d_i \quad \text{for any } i \in \Lambda_p. \tag{8.3}$$

By Bezout's theorem with a general line through p, $\sum_\beta \operatorname{mult}_p \beta = \operatorname{mult}_p P_i \leq d_i$. By Theorem 2.9(3) and $m \geq 1$, A is increasing on rooted segments. Either $t_\beta = \operatorname{mult}_p$, or $t_\beta \leq \nu$ for some $\nu \in S$.

In case (1), $A(\operatorname{mult}_p) = 2 \leq \kappa$ and $A(t_\beta) \leq A(\nu) \leq \kappa$ in the second case. Thus

$$\sum_{\beta} \operatorname{mult}_p \beta \cdot A(t_\beta) \leq \kappa \operatorname{mult}_p P_i \leq \kappa d_i.$$

In case (2), we first show that $\tau \to A(\tau) - \tau(L)$ is increasing on rooted segments. Let $\tau \leq \tau'$ be normalized valuations at p with $\alpha(\tau') < +\infty$. Since L is smooth at p, Theorem 2.9(2) implies $\tau(L) = \alpha(\tau \wedge \nu_L)$ and $\tau'(L) = \alpha(\tau' \wedge \nu_L)$. The elements τ and $\tau' \wedge \nu_L$ of the totally ordered set $[\operatorname{mult}_p, \tau']$ are comparable. If $\tau \leq \tau' \wedge \nu_L$, then $\tau \wedge \nu_L = \tau$ and

$$\tau'(L) - \tau(L) = \alpha(\tau' \wedge \nu_L) - \alpha(\tau) \leq \alpha(\tau') - \alpha(\tau).$$

If $\tau' \wedge \nu_L < \tau$, then $\tau \wedge \nu_L = \tau' \wedge \nu_L$ and

$$\tau'(L) - \tau(L) = 0 \leq \alpha(\tau') - \alpha(\tau).$$

In both cases, Theorem 2.9(3) and $m \geq 1$ imply

$$\tau'(L) - \tau(L) \leq \alpha(\tau') - \alpha(\tau) \leq \int_\tau^{\tau'} m d\alpha = A(\tau') - A(\tau).$$

If $t_\beta = \operatorname{mult}_p$, then $A(t_\beta) - t_\beta(L) = 2 - 1 \leq \kappa$. Otherwise $t_\beta \leq \nu$ for some $\nu \in S$, and hence $A(t_\beta) - t_\beta(L) \leq A(\nu) - \nu(L) \leq \kappa$. Since $L \not\subset \operatorname{Supp} D$, $P_i \neq L$, so $(P_i \cdot L)_p < +\infty$ and no branch of P_i at p is the branch of L at p. Since $t_\beta \wedge \nu_L \leq \nu_\beta \wedge \nu_L$, Theorem 2.9(1)(2) implies

$$\operatorname{mult}_p \beta \cdot t_\beta(L) = \operatorname{mult}_p \beta \cdot \alpha(t_\beta \wedge \nu_L) \leq \operatorname{mult}_p \beta \cdot \alpha(\nu_\beta \wedge \nu_L) = (\beta \cdot L)_p.$$

By Bezout's theorem, $\sum_\beta (\beta \cdot L)_p = (P_i \cdot L)_p \leq d_i$. Thus

$$\sum_{\beta} \operatorname{mult}_p \beta \cdot A(t_\beta) \leq \kappa \operatorname{mult}_p P_i + (P_i \cdot L)_p \leq (\kappa + 1) d_i.$$

This proves (8.3).

By (8.2), (8.3), and $d_i \geq 1$, for any $i \in \Lambda_p$,

$$\|\rho_S(P_i)\|^2 \leq (d_i - 1)(d_i - 2) - 1 + \kappa' d_i = d_i^2 + (\kappa' - 3) d_i + 1 \leq d_i^2 + (\kappa' - 2) d_i.$$

For distinct $i, j \in \Lambda_p$, by (3) and Bezout's theorem,

$$\langle \rho_S(P_i), \rho_S(P_j) \rangle \leq (P_i \cdot P_j)_p \leq d_i d_j.$$

Let $d_p := \sum_{i \in \Lambda_p} c_i d_i \leq \deg D$. Since $c_i > 0$ for any i, $c_i \leq \epsilon$ for any $i \in \Lambda_p$, and $\kappa' \geq 2$, we obtain

$$\|\rho_S(D)\|^2 \leq \sum_{i \in \Lambda_p} c_i^2 \big(d_i^2 + (\kappa' - 2) d_i\big) + \sum_{i \neq j \in \Lambda_p} c_i c_j d_i d_j = d_p^2 + (\kappa' - 2) \sum_{i \in \Lambda_p} c_i^2 d_i \leq d_p^2 + (\kappa' - 2) \epsilon d_p,$$

and therefore

$$\|\rho_S(D)\|^2 \leq (\deg D)^2 + (\kappa' - 2) \epsilon \deg D.$$

Since $\kappa' - 2 = \kappa - 2$ in case (1) and $\kappa' - 2 = \kappa - 1$ in case (2), the theorem follows. □

8.2. **Pushforward of valuations under finite germs.** Compare the global statement [BFJ08, Proposition 2.3].

Lemma 8.3. *Let X, X' be smooth surfaces, $p \in X$ and $p' \in X'$ closed points, and $h : (X, p) \to (X', p')$ a finite morphism of germs, that is, a finite local $\mathbb{C}$-algebra homomorphism*

$$h^* : \widehat{\mathcal{O}}_{X',p'} \to \widehat{\mathcal{O}}_{X,p}.$$

Let

$$d := \deg_p h := \dim_{\mathbb{C}} \widehat{\mathcal{O}}_{X,p}/h^*(\mathfrak{m}_{p'})\widehat{\mathcal{O}}_{X,p}.$$

Let $\nu_1, \dots, \nu_N$ be normalized valuations at p with finite skewness, and write $h_*\nu_i = m_i\nu_i'$ with $m_i := (h_*\nu_i)(\mathfrak{m}_{p'})$. Then any ν_i' has finite skewness, and the real symmetric matrix

$$\big(d\alpha(\nu_i \wedge \nu_j) - m_i m_j \alpha(\nu_i' \wedge \nu_j')\big)_{1 \le i,j \le N}$$

is positive semidefinite. In particular, $m_1^2\alpha(\nu_1') \le d\alpha(\nu_1)$.

Proof. Let $\mathcal{O} := \widehat{\mathcal{O}}_{X,p}$ and $\mathcal{O}' := \widehat{\mathcal{O}}_{X',p'}$, with maximal ideals $\mathfrak{m}$ and $\mathfrak{m}'$. Then $\mathcal{O} \cong \mathcal{O}' \cong \mathbb{C}[[x, y]]$, so the results of [FJ04] hold on $\mathcal{O}$ and on $\mathcal{O}'$. For an ideal $I' \subset \mathcal{O}'$, write $I'\mathcal{O}$ for the ideal of $\mathcal{O}$ generated by $h^*(I')$. Let $\mathcal{M}$ and $\mathcal{M}'$ be the real vector spaces with bases $\{\delta_\mu\}$, where μ ranges over the normalized valuations at p and at p' respectively with $\alpha(\mu) < +\infty$, with the bilinear forms $\langle \delta_\mu, \delta_\nu \rangle = \alpha(\mu \wedge \nu)$ as in Definition 8.1. The forms on $\mathcal{M}$ and on $\mathcal{M}'$ are positive definite [FJ04, Remark 7.78 and Theorem 7.83]. For a divisorial normalized valuation μ at p or at p', let $b(\mu)$ be the positive integer such that, for some prime divisor F over X or over X' respectively, $b(\mu)\mu(\zeta) = \mathrm{mult}_F \operatorname{div}(\zeta)$ for any non-zero ζ in $\mathcal{O}$ or in $\mathcal{O}'$.

We use the following facts (i)–(iv) on $\mathcal{O}$, and facts (i)–(iv) with $\mathcal{O}'$, $\mathfrak{m}'$, $\mathcal{M}'$, and p' in place of $\mathcal{O}$, $\mathfrak{m}$, $\mathcal{M}$, and p. For an $\mathfrak{m}$-primary ideal $I \subset \mathcal{O}$, let ρ_I be the tree measure of I [FJ04, Definition 8.3]. For a positive tree measure ρ, following [FJ04, (8.3)], let

$$I_\rho := \{\phi \in \mathcal{O} \mid \nu(\phi) \ge \int \alpha(\nu \wedge \sigma) d\rho(\sigma) \text{ for any quasimonomial normalized valuation } \nu \text{ at } p\}.$$

For $\rho = \sum_k c_k \delta_{\omega_k}$, the integral is $\sum_k c_k \alpha(\nu \wedge \omega_k)$.

(i) Since $I \subset I_{\rho_I}$ by definition, the ideal I_{ρ_I} is $\mathfrak{m}$-primary, so $\rho_I = \sum_k n_k b(\omega_k)\delta_{\omega_k}$ for finitely many divisorial normalized valuations ω_k at p and $n_k \in \mathbb{N}^+$ [FJ04, Theorem 8.2 and Proposition 8.8]. In particular, $\rho_I \in \mathcal{M}$.

(ii) $\langle \delta_\nu, \rho_I \rangle = \nu(I)$ for any normalized valuation ν at p with $\alpha(\nu) < +\infty$, by [FJ04, Theorem 7.50 and Theorem 8.2] for quasimonomial ν. If ν is not quasimonomial, then ν is an end of the tree [FJ04, Proposition 3.20], so we may choose a quasimonomial $\nu_0 < \nu$ with $\nu_0 \ge \nu \wedge \omega_k$ for any k and $\nu_0 \ge \nu \wedge \nu_\psi$ for the branches ψ of finitely many generators of I, and Theorem 2.9(1)(2) gives

$$\langle \delta_\nu, \rho_I \rangle = \langle \delta_{\nu_0}, \rho_I \rangle \quad \text{and} \quad \nu(I) = \nu_0(I).$$

(iii) $\langle \rho_I, \rho_J \rangle = e(I, J)$ for any $\mathfrak{m}$-primary ideals $I, J \subset \mathcal{O}$ [FJ04, Theorem 8.13], where $e(I, J)$ is the mixed multiplicity of I and J, that is, the coefficient of ab in the polynomial $\chi_{I,J}(a, b)$ with $\dim_{\mathbb{C}} \mathcal{O}/I^a J^b = \chi_{I,J}(a, b)$ for any integers $a, b \gg 0$ [FJ04, Section 8.1.4] (such a polynomial exists by [HS06, Theorem 17.4.2]).

(iv) For a divisorial normalized valuation μ at p, the ideal $I_\mu := I_{b(\mu)\delta_\mu}$ is $\mathfrak{m}$-primary [FJ04, Proposition 8.8], and $\rho_{I_\mu} = b(\mu)\delta_\mu$ [FJ04, (8.3) and Theorem 8.2].

Since $\mathcal{O}'$ and $\mathcal{O}$ are regular local rings of dimension 2 and $\mathcal{O}$ is a finite $\mathcal{O}'$-module via h^*, $\mathcal{O}$ is a free $\mathcal{O}'$-module, of rank $\dim_{\mathbb{C}} \mathcal{O}/\mathfrak{m}'\mathcal{O} = d$. Let I', J' be $\mathfrak{m}'$-primary ideals of $\mathcal{O}'$. For any integers $a, b \ge 0$, we have $(I'\mathcal{O})^a(J'\mathcal{O})^b = (I'^a J'^b)\mathcal{O}$ and

$$\mathcal{O}/(I'^a J'^b)\mathcal{O} \cong (\mathcal{O}'/I'^a J'^b)^{\oplus d} \quad \text{as } \mathcal{O}'\text{-modules},$$

so

$$\dim_{\mathbb{C}} \mathcal{O}/(I'\mathcal{O})^a(J'\mathcal{O})^b = d \dim_{\mathbb{C}} \mathcal{O}'/I'^a J'^b.$$

In particular $I'\mathcal{O}$ and $J'\mathcal{O}$ are $\mathfrak{m}$-primary, and

$$e(I'\mathcal{O}, J'\mathcal{O}) = de(I', J'). \tag{8.4}$$

Let $\mathcal{M}'_{\mathrm{div}} \subset \mathcal{M}'$ be the subspace spanned by the δ_μ with μ divisorial. For $y = \sum_\mu y_\mu \delta_\mu \in \mathcal{M}'_{\mathrm{div}}$, let

$$\widetilde{y} := \sum_\mu \frac{y_\mu}{b(\mu)} \rho_{I_\mu \mathcal{O}} \in \mathcal{M}.$$

By (iii), (8.4) and (iv), for divisorial μ, η at p',

$$\langle \rho_{I_\mu\mathcal{O}}, \rho_{I_\eta\mathcal{O}} \rangle = e(I_\mu\mathcal{O}, I_\eta\mathcal{O}) = de(I_\mu, I_\eta) = d\langle \rho_{I_\mu}, \rho_{I_\eta} \rangle = db(\mu)b(\eta)\alpha(\mu \wedge \eta),$$

so

$$\|\widetilde{y}\|^2 = d\|y\|^2 \quad \text{for any } y \in \mathcal{M}'_{\mathrm{div}}. \tag{8.5}$$

Let ν be a normalized valuation at p with $\alpha(\nu) < +\infty$, let $m := (h_*\nu)(\mathfrak{m}')$, and let $\nu' := h_*\nu/m$. Since $\mathfrak{m}'\mathcal{O} \subset \mathfrak{m}$, $m \geq \nu(\mathfrak{m}) = 1$. Since $\mathfrak{m}'\mathcal{O}$ is $\mathfrak{m}$-primary, $\mathfrak{m}^c \subset \mathfrak{m}'\mathcal{O}$ for some positive integer c, so $m \leq \nu(\mathfrak{m}^c) = c$. Thus ν' is a normalized valuation at p'. For any $\mathfrak{m}'$-primary ideal $I' \subset \mathcal{O}'$, the ideal $I'\mathcal{O}$ is generated by $h^*(I')$, so $\nu(I'\mathcal{O}) = (h_*\nu)(I') = m\nu'(I')$. By (ii), for any divisorial normalized valuation μ at p',

$$\langle \delta_\nu, \widetilde{\delta_\mu} \rangle = \frac{\nu(I_\mu\mathcal{O})}{b(\mu)} = \frac{m\nu'(I_\mu)}{b(\mu)}. \tag{8.6}$$

We show $m^2\alpha(\nu') \leq d\alpha(\nu)$. Let $\mu \leq \nu'$ be a divisorial normalized valuation at p'. Then $\nu'(I_\mu) \geq \mu(I_\mu) = b(\mu)\alpha(\mu)$ by (ii) and (iv). By (8.6), the Cauchy–Schwarz inequality in $\mathcal{M}$ and (8.5),

$$m\alpha(\mu) \leq \langle \delta_\nu, \widetilde{\delta_\mu} \rangle \leq \|\delta_\nu\| \cdot \|\widetilde{\delta_\mu}\| = \alpha(\nu)^{1/2}\big(d\alpha(\mu)\big)^{1/2},$$

so $m^2\alpha(\mu) \leq d\alpha(\nu)$. Since α maps $[\mathrm{mult}_{p'}, \nu']$ bijectively onto an interval (Theorem 2.9(1)), the points of $[\mathrm{mult}_{p'}, \nu')$ are quasimonomial, and quasimonomial valuations of rational skewness are divisorial [FJ04, Proposition 3.20 and Theorem 3.26], there exist divisorial normalized valuations $\mu_k \leq \nu'$ at p' with

$$\lim_{k\to+\infty} \alpha(\mu_k) = \alpha(\nu') \in [1, +\infty].$$

Thus

$$m^2\alpha(\nu') \leq d\alpha(\nu) < +\infty.$$

Applied to $\nu = \nu_i$, this shows that ν'_i has finite skewness and $m_i^2\alpha(\nu'_i) \leq d\alpha(\nu_i)$ for any i.

Since $\alpha(\nu'_i) < +\infty$, (ii) and (iv) imply

$$\nu'_i(I_\mu) = \langle \delta_{\nu'_i}, b(\mu)\delta_\mu \rangle = b(\mu)\alpha(\nu'_i \wedge \mu)$$

for any divisorial normalized valuation μ at p', and (8.6) becomes

$$\langle \delta_{\nu_i}, \widetilde{\delta_\mu} \rangle = m_i\alpha(\nu'_i \wedge \mu).$$

Let $(a_1, \ldots, a_N) \in \mathbb{R}^N$, and let

$$V := \sum_{i=1}^N a_i\delta_{\nu_i} \in \mathcal{M} \quad \text{and} \quad W := \sum_{i=1}^N a_i m_i \delta_{\nu'_i} \in \mathcal{M}'.$$

By linearity, $\langle W, y \rangle = \langle V, \widetilde{y} \rangle$ for any $y \in \mathcal{M}'_{\mathrm{div}}$, so the Cauchy–Schwarz inequality and (8.5) imply

$$\langle W, y \rangle \leq \|V\| \cdot \|\widetilde{y}\| = d^{1/2}\|V\| \cdot \|y\| \quad \text{for any } y \in \mathcal{M}'_{\mathrm{div}}. \tag{8.7}$$

For each i, by Theorem 2.9(1) and [FJ04, Proposition 3.20 and Theorem 3.26], choose divisorial normalized valuations $\mu_{i,k} \leq \nu'_i$ at p' with $\lim_{k\to+\infty} \alpha(\mu_{i,k}) = \alpha(\nu'_i)$, and let

$$y_k := \sum_{i=1}^N a_i m_i \delta_{\mu_{i,k}} \in \mathcal{M}'_{\mathrm{div}}.$$

For normalized valuations $\mu \le \nu$ with $\alpha(\nu) < +\infty$,

$$\|\delta_\nu - \delta_\mu\|^2 = \alpha(\nu) - 2\alpha(\mu) + \alpha(\mu) = \alpha(\nu) - \alpha(\mu).$$

Thus

$$\|y_k - W\| \le \sum_{i=1}^{N} |a_i| m_i \big(\alpha(\nu_i') - \alpha(\mu_{i,k})\big)^{1/2} \to 0 \quad \text{as } k \to +\infty,$$

so $\langle W, y_k\rangle \to \|W\|^2$ and $\|y_k\| \to \|W\|$. By (8.7) with $y = y_k$, $\|W\|^2 \le d^{1/2}\|V\| \cdot \|W\|$, so $\|W\|^2 \le d\|V\|^2$. Therefore

$$\sum_{i,j=1}^{N} a_i a_j \big(d\alpha(\nu_i \wedge \nu_j) - m_i m_j \alpha(\nu_i' \wedge \nu_j')\big) = d\|V\|^2 - \|W\|^2 \ge 0.$$

Since $(a_1, \dots, a_N) \in \mathbb{R}^N$ is arbitrary, the matrix is positive semidefinite. The lemma follows. □

8.3. **Values on lines through a point.**

Lemma 8.4. *Let $p \in \mathbb{P}^2$ be a closed point, let L be a germ at p of a smooth curve, and let τ be a normalized valuation at p. Assume that $A(\tau) < +\infty$ and $\beta := \tau(L) > 1$. Then:*

(1) If $\tau \ne \tau \wedge \nu_L$, then $\beta = a/b$ for coprime integers $a > b \ge 1$, and

$$A(\tau) \ge 1 + \beta + b\,(\alpha(\tau) - \beta)\,.$$

(2) $\beta A(\tau) - 2\alpha(\tau) \ge \beta(\beta - 1)$.

(3) $\alpha(\tau) - \beta \le (\beta - 1)\,(A(\tau) - 1 - \beta)$ and $\alpha(\tau) - 1 \le (\beta - 1)\,(A(\tau) - \beta)$.

Proof. Let $\mu := \tau \wedge \nu_L$. Since $\operatorname{mult}_p L = 1$, Theorem 2.9(2) implies $\alpha(\mu) = \tau(L) = \beta$. By Theorem 2.9(1)(3),

$$\beta = \alpha(\mu) \le \alpha(\tau) \le A(\tau) - 1 < +\infty.$$

By Theorem 2.9(5), there exists a normalized valuation $\mu' < \nu_L$ with $m \equiv 1$ on $[\mu', \nu_L)$. Since $m \ge 1$ and m is increasing on the totally ordered segment $[\operatorname{mult}_p, \nu_L]$ (Theorem 2.9(3)), $m \equiv 1$ on $[\operatorname{mult}_p, \nu_L)$. Since $\alpha(\mu) < +\infty = \alpha(\nu_L)$, $\mu \in [\operatorname{mult}_p, \nu_L)$, so Theorem 2.9(3) implies

$$A(\tau) = 2 + (\beta - 1) + \int_\mu^\tau m(\nu) d\alpha(\nu) = 1 + \beta + \int_\mu^\tau m(\nu) d\alpha(\nu). \tag{8.8}$$

If $\tau = \mu$, then $\alpha(\tau) = \beta$ and $A(\tau) = 1 + \beta$, and (2) and (3) hold with equality. From now on we assume that $\tau \ne \mu$, so $\mu < \tau$.

(1) Let $\nu \in (\mu, \tau) := [\mu, \tau)\backslash\{\mu\}$. Let ϕ be an irreducible formal curve germ at p with $\nu_\phi \ge \nu$. Since $\mu \le \nu \le \tau$, $\nu \wedge \nu_L = \mu$. Since $\nu \wedge \nu_L = \mu < \nu$, $\nu \not\le \nu_L$. Since $\nu_\phi \wedge \nu_L, \nu \in [\operatorname{mult}_p, \nu_\phi]$ and $[\operatorname{mult}_p, \nu_\phi]$ is totally ordered, $\nu_\phi \wedge \nu_L < \nu$, so

$$\nu_\phi \wedge \nu_L = \nu_\phi \wedge \nu_L \wedge \nu = \mu \wedge \nu_\phi = \mu \ne \nu_L.$$

In particular ϕ is not the germ L, and Theorem 2.9(2) implies

$$(\phi \cdot L)_p = \operatorname{mult}_p \phi \cdot \alpha(\mu) = \beta \operatorname{mult}_p \phi.$$

Thus $\beta \in \mathbb{Q}$. Since $\beta > 1$, $\beta = a/b$ for coprime integers $a > b \ge 1$. Since $b(\phi \cdot L)_p = a \operatorname{mult}_p \phi$ and $\gcd(a, b) = 1$, b divides $\operatorname{mult}_p \phi$, so $\operatorname{mult}_p \phi \ge b$. Thus $m(\nu) \ge b$ for any $\nu \in (\mu, \tau)$ by Definition 2.8, and (8.8) implies

$$A(\tau) \ge 1 + \beta + b\,(\alpha(\tau) - \beta)\,.$$

(2) If $\beta \ge 2$, then $\beta m(\nu) - 2 \ge 0$ for any $\nu \in (\mu, \tau)$, since $m \ge 1$. If $\beta < 2$, then $b \ge 2$ by (1), so $\beta m(\nu) \ge \beta b = a > b \ge 2$ for any $\nu \in (\mu, \tau)$. Since $\alpha(\tau) = \beta + \int_\mu^\tau d\alpha$, (8.8) implies

$$\beta A(\tau) - 2\alpha(\tau) = \beta(1 + \beta) - 2\beta + \int_\mu^\tau (\beta m(\nu) - 2)\, d\alpha(\nu) \ge \beta(\beta - 1).$$

(3) By (1), $A(\tau)-1-\beta \geq b(\alpha(\tau)-\beta) \geq 0$. Since $b(\beta-1)=a-b\geq 1$, we have $\frac{1}{b}\leq \beta-1$, so

$$\alpha(\tau)-\beta \leq \frac{1}{b}\left(A(\tau)-1-\beta\right) \leq (\beta-1)\left(A(\tau)-1-\beta\right).$$

Adding $\beta-1$ to both sides, we obtain $\alpha(\tau)-1\leq(\beta-1)(A(\tau)-\beta)$. □

Lemma 8.5. *Let $p\in\mathbb{P}^2$ be a closed point, let τ be a normalized valuation at p with $A(\tau)<+\infty$, and let*

$$\beta := \max\{\tau(M) \mid M\subset\mathbb{P}^2 \text{ is a line through } p\}.$$

Then:

(1) If $\beta=1$, then $\tau=\mathrm{mult}_p$.
(2) $\alpha(\tau)-\beta\leq(\beta-1)\left(A(\tau)-1-\beta\right)$ and $\alpha(\tau)-1\leq(\beta-1)\left(A(\tau)-1\right)$.
(3) If $A(\tau)<8$, then

$$\beta A(\tau)\geq \alpha(\tau)\left(2+\frac{A(\tau)-2}{21}\right).$$

(4) If M is a line through p with $\tau(M)=\beta$ and x,y are affine linear coordinates centered at p with $M=\{y=0\}$, then $\tau\wedge\nu_M=\mathrm{mon}_{x,y}(1,\beta)$, $\alpha(\tau\wedge\nu_M)=\beta$, and $A(\tau\wedge\nu_M)=1+\beta$.

Proof. Let x,y be affine linear coordinates centered at p, and let $V:=\mathbb{C}x+\mathbb{C}y$. The lines through p are the curves $\{\ell=0\}$ with $\ell\in V\backslash\{0\}$. For $\ell\in V\backslash\{0\}$, $\mathrm{mult}_p\,\ell=1$, so

$$1\leq\tau(\ell)\leq\alpha(\tau)\leq A(\tau)-1<+\infty$$

by the definition of skewness and Theorem 2.9(3). For $c\in\mathbb{R}$, $V_c:=\{\ell\in V\mid\tau(\ell)\geq c\}\cup\{0\}$ is a linear subspace of V. If τ took 3 distinct values $c_1<c_2<c_3$ on $V\backslash\{0\}$, then $V_{c_1}\supsetneq V_{c_2}\supsetneq V_{c_3}\supsetneq\{0\}$, contradicting $\dim V=2$. Thus τ takes at most 2 values on $V\backslash\{0\}$, the maximum β exists, and $1\leq\beta<+\infty$.

(1) Assume that $\beta=1$, so $\tau(\ell)=1$ for any $\ell\in V\backslash\{0\}$. Let $\phi\in\widehat{\mathcal{O}}_{\mathbb{P}^2,p}\backslash\{0\}$, let $k:=\mathrm{mult}_p\,\phi$, and let ϕ_k be the homogeneous part of ϕ of degree k in x,y. Since ϕ_k is a product of k elements of $V\backslash\{0\}$, $\tau(\phi_k)=k$. Since $\phi-\phi_k\in\mathfrak{m}_p^{k+1}$, $\tau(\phi-\phi_k)\geq k+1$. Thus $\tau(\phi)=k=\mathrm{mult}_p\,\phi$, so $\tau=\mathrm{mult}_p$.

(4) Since x,y are regular parameters at p and y is a local equation of M, Theorem 2.9(2) implies $\alpha(\tau\wedge\nu_M)=\tau(y)=\beta$. Let $w:=\mathrm{mon}_{x,y}(1,\beta)$, and let $\phi\in\widehat{\mathcal{O}}_{\mathbb{P}^2,p}\backslash\{0\}$. A monomial x^iy^j with $i+j=\mathrm{mult}_p\,\phi$ occurs in ϕ, and $\beta\geq1$, so

$$w(\phi)\leq i+\beta j\leq\beta\,\mathrm{mult}_p\,\phi.$$

Since moreover $w(y)=\beta$, $\alpha(w)=\beta$ by the definition of skewness. If $y\nmid\phi$, then $\nu_M(\phi)=(y\cdot\phi)_p$ is the order in x of $\phi(x,0)\neq0$, so $x^{\nu_M(\phi)}$ occurs in ϕ and $w(\phi)\leq\nu_M(\phi)$. If $y\mid\phi$, then $\nu_M(\phi)=+\infty$. Thus $w\leq\nu_M$, so $w,\tau\wedge\nu_M\in[\mathrm{mult}_p,\nu_M]$. Since α is injective on $[\mathrm{mult}_p,\nu_M]$ (Theorem 2.9(1)) and $\alpha(w)=\beta=\alpha(\tau\wedge\nu_M)$, $\tau\wedge\nu_M=w=\mathrm{mon}_{x,y}(1,\beta)$. Since $m\geq1$ is increasing on $[\mathrm{mult}_p,\nu_M]$, Theorem 2.9(5) implies $m\equiv1$ on $[\mathrm{mult}_p,\nu_M)$, so Theorem 2.9(3) implies $A(\tau\wedge\nu_M)=1+\beta$.

(2) If $\beta=1$, then $\tau=\mathrm{mult}_p$ by (1), so $\alpha(\tau)=\beta=1$, and $\alpha(\tau)-\beta=\alpha(\tau)-1=\beta-1=0$. If $\beta>1$, let M be a line through p with $\tau(M)=\beta$. By Lemma 8.4(3) for the germ of M at p,

$$\alpha(\tau)-\beta\leq(\beta-1)\left(A(\tau)-1-\beta\right)\quad\text{and}\quad\alpha(\tau)-1\leq(\beta-1)\left(A(\tau)-\beta\right)\leq(\beta-1)\left(A(\tau)-1\right).$$

(3) Assume that $A(\tau)<8$. By Theorem 2.9(1)(3), $1\leq\alpha(\tau)\leq A(\tau)-1<7$, so $0\leq A(\tau)-2<6$ and

$$\frac{\alpha(\tau)\left(A(\tau)-2\right)}{21}\leq\frac{A(\tau)-2}{2}\quad\text{and}\quad\frac{\alpha(\tau)\left(A(\tau)-2\right)}{21}<\frac{7\cdot6}{21}=2.$$

Thus it suffices to prove that $\beta A(\tau)-2\alpha(\tau)\geq2$ if $\beta\geq2$, and that $\beta A(\tau)-2\alpha(\tau)\geq\frac{A(\tau)-2}{2}$ if $\beta<2$.

If $\beta\geq2$, then Theorem 2.9(3) implies

$$\beta A(\tau)-2\alpha(\tau)\geq2(A(\tau)-\alpha(\tau))\geq2.$$

If $\beta = 1$, then $\tau = \mathrm{mult}_p$ by (1), so $A(\tau) = 2$ and $\alpha(\tau) = 1$, and $\beta A(\tau) - 2\alpha(\tau) = 0 = \frac{A(\tau)-2}{2}$. Assume that $1 < \beta < 2$. Let M be a line through p with $\tau(M) = \beta$. If $\tau = \tau \wedge \nu_M$, then $\alpha(\tau) = \beta$ and $A(\tau) = 1 + \beta$ by (4), so

$$\beta A(\tau) - 2\alpha(\tau) = \beta(\beta - 1) \geq \frac{\beta - 1}{2} = \frac{A(\tau) - 2}{2}.$$

If $\tau \neq \tau \wedge \nu_M$, then by Lemma 8.4(1) for the germ of M at p, $\beta = a/b$ for coprime integers $a > b \geq 1$, and $A(\tau) \geq A_0 := 1 + \beta + b(\alpha(\tau) - \beta)$. Since $1 < \beta < 2$, $b \geq 2$ and $a \geq b + 1$, so $a - 2 \geq b - 1 \geq \frac{b}{2}$. Since $\beta b = a$ and $\alpha(\tau) \geq \beta$,

$$\beta A_0 - 2\alpha(\tau) = \beta(\beta - 1) + (a - 2)\,(\alpha(\tau) - \beta) \geq \frac{\beta - 1}{2} + \frac{b}{2}\,(\alpha(\tau) - \beta) = \frac{A_0 - 2}{2}.$$

Since $\beta \geq \frac{1}{2}$ and $A(\tau) \geq A_0$,

$$\beta A(\tau) - 2\alpha(\tau) = \beta A_0 - 2\alpha(\tau) + \beta\,(A(\tau) - A_0) \geq \frac{A_0 - 2}{2} + \frac{A(\tau) - A_0}{2} = \frac{A(\tau) - 2}{2}. \quad \square$$

Lemma 8.6. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 1$, $p \in \mathbb{P}^2$ a closed point, x, y affine linear coordinates centered at p, $\beta \geq 1$ a real number, $w := \mathrm{mon}_{x,y}(1, \beta)$, and $C := \{y = 0\}$. Then for any line $M \subset \mathbb{P}^2$,*

$$w(g^*M) \leq Q + (\beta - 1)\,\mathrm{mult}_C\, g^*M.$$

Proof. Let x_0, x_1, x_2 be homogeneous coordinates on $\mathbb{P}^2$ with $x = x_1/x_0$ and $y = x_2/x_0$, and let ℓ be a linear form defining M. Since g is finite, $g^*\ell$ is a non-zero homogeneous polynomial of degree Q, so $F(x, y) := (g^*\ell)(1, x, y)$ is a non-zero polynomial of degree at most Q, and F is a local equation of g^*M on $\{x_0 \neq 0\} \ni p$. Write $F = y^k F_0$, where $F_0 \in \mathbb{C}[x, y]$ is not divisible by y. Then $k = \mathrm{mult}_C\, g^*M$, and $F_0(x, 0)$ is a non-zero polynomial of degree at most $\deg F_0 \leq Q - k$. Let cx^j with $c \in \mathbb{C}\backslash\{0\}$ be a term of $F_0(x, 0)$. Then $j \leq Q - k$, and cx^j is a term of F_0. Since w is monomial in x, y, $w(F_0) \leq w(x^j) = j \leq Q - k$. Thus

$$w(g^*M) = w(F) = k\beta + w(F_0) \leq k\beta + Q - k = Q + (\beta - 1)\,\mathrm{mult}_C\, g^*M. \quad \square$$

8.4. **Polar estimates.**

Lemma 8.7 (cf. [Tei73, Chapter II, Proposition 1.2] and [GP18, Corollary 3.4])**.** *Let X be a smooth surface, $p \in X$ a closed point, ϕ a reduced holomorphic germ at p with $\phi(p) = 0$, and ψ a holomorphic germ at p with $\psi(p) = 0$ and $(\phi \cdot \psi)_p < +\infty$. Let $\mu_p(\phi)$ be the Milnor number of ϕ, and let $\mathrm{Jac}(\phi, \psi)$ be the Jacobian determinant of (ϕ, ψ) with respect to local coordinates at p. Then*

$$(\phi \cdot \mathrm{Jac}(\phi, \psi))_p = \mu_p(\phi) + (\phi \cdot \psi)_p - 1.$$

Proof. By the chain rule, a change of local coordinates multiplies $\mathrm{Jac}(\phi, \psi)$ by a unit. Since ϕ has finitely many branches at p, we may fix local coordinates x, y at p with $(\phi \cdot x)_p < +\infty$. Then

$$\mathrm{Jac}(\phi, \psi) = \phi_x\psi_y - \phi_y\psi_x \quad \text{and} \quad \mathrm{Jac}(\phi, x) = -\phi_y.$$

By [GP18, Corollary 3.4] for the regular parameter x,

$$(\phi \cdot \phi_y)_p = \mu_p(\phi) + (\phi \cdot x)_p - 1 < +\infty. \tag{8.9}$$

Let Γ be a branch of $\{\phi = 0\}$ at p, and let $\gamma(t) = (x(t), y(t))$ be the normalization of Γ, that is, an injective parametrization of Γ by a germ of $\mathbb{C}$ at 0 with $\gamma(0) = p$, so that

$$(\Gamma \cdot \chi)_p = \mathrm{mult}_0\, \mathrm{div}(\chi \circ \gamma)$$

for any holomorphic germ χ at p. Differentiating $\phi(\gamma(t)) = 0$, we obtain

$$\phi_x(\gamma(t))x'(t) + \phi_y(\gamma(t))y'(t) = 0,$$

so

$$x'(t)\,\mathrm{Jac}(\phi, \psi)(\gamma(t)) = -\phi_y(\gamma(t))\frac{d}{dt}\psi(\gamma(t)).$$

Since $x(p) = \psi(p) = 0$, the integers

$$\operatorname{mult}_0 \operatorname{div}(x(t)) = (\Gamma \cdot x)_p \quad \text{and} \quad \operatorname{mult}_0 \operatorname{div}(\psi \circ \gamma) = (\Gamma \cdot \psi)_p$$

are positive, so

$$\operatorname{mult}_0 \operatorname{div}(x'(t)) = (\Gamma \cdot x)_p - 1 \quad \text{and} \quad \operatorname{mult}_0 \operatorname{div}((\psi \circ \gamma)') = (\Gamma \cdot \psi)_p - 1.$$

Thus

$$(\Gamma \cdot \operatorname{Jac}(\phi, \psi))_p = (\Gamma \cdot \psi)_p - (\Gamma \cdot x)_p + (\Gamma \cdot \phi_y)_p.$$

Since $(\phi \cdot \chi)_p$ is the sum of $(\Gamma \cdot \chi)_p$ over the branches Γ of $\{\phi = 0\}$ at p for any holomorphic germ χ at p, (8.9) implies

$$(\phi \cdot \operatorname{Jac}(\phi, \psi))_p = (\phi \cdot \psi)_p - (\phi \cdot x)_p + (\phi \cdot \phi_y)_p = \mu_p(\phi) + (\phi \cdot \psi)_p - 1. \quad \square$$

Proposition 8.8. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \geq 2$, $p \in \mathbb{P}^2$ a closed point, $\epsilon \geq 0$ and $\kappa \geq 2$ real numbers, and S a finite set of normalized valuations at p with $A(\nu) \leq \kappa$ for any $\nu \in S$. Let $\xi := \rho_S\big(\frac{1}{Q-1}R(g)\big)$, and for a line M through $g(p)$, let $\eta_M := \rho_S\big(\frac{1}{Q}g^*M\big)$. Assume that $\operatorname{mult}_P R(g) \leq \epsilon(Q-1)$ for any irreducible curve P through p. Then for any line M through $g(p)$,*

$$\|\eta_M\|^2 \leq 1 + (\kappa - 2)\left(\epsilon + \frac{1}{Q}\right) \quad \textit{and} \quad \langle \xi, \eta_M \rangle \leq \frac{Q^2 - 2Q + \deg_p g}{Q(Q-1)} + (\kappa - 2)\epsilon.$$

The upper bound for $\langle \xi, \eta_M \rangle$ holds under the weaker assumption that $\operatorname{mult}_P R(g) \leq \epsilon(Q-1)$ for any irreducible component P of g^*M through p. If moreover $g(p) = p$ and $\omega \in S$ satisfies $g_*\omega = \Lambda\omega$ for a real number $\Lambda > 0$, then

$$\langle \xi, \delta_\omega \rangle = \frac{\Lambda - 1}{Q - 1} A(\omega) \quad \textit{and} \quad \langle \eta_M, \delta_\omega \rangle = \frac{\Lambda}{Q}\omega(M).$$

Proof. **Step 1.** *Set-up.* Fix a line M through $g(p)$. Write

$$g^*M = \sum_{i \in I_0} e_i C_i,$$

where I_0 is a finite set, the C_i are distinct irreducible curves, and e_i are positive integers. Let $d_i := \deg C_i$, $I := \{i \in I_0 \mid p \in C_i\}$,

$$N := \sum_{i \in I} e_i d_i, \quad \text{and} \quad N_0 := \sum_{i \in I} d_i.$$

Since $p \in \operatorname{Supp} g^*M$ and $\deg g^*M = Q$, $1 \leq N_0 \leq N \leq Q$. Since g is finite, $g(C_i) = M$ for any $i \in I_0$, so Definition 2.2 implies

$$e_i = e_{C_i}(g) \quad \text{and} \quad \operatorname{mult}_{C_i} R(g) = e_i - 1 \quad \text{for any } i \in I_0. \tag{8.10}$$

Step 2. *The norm of η_M.* For any $i \in I$, (8.10) implies

$$\frac{e_i}{Q} = \frac{\operatorname{mult}_{C_i} R(g) + 1}{Q} \leq \frac{\epsilon(Q-1) + 1}{Q} \leq \epsilon + \frac{1}{Q}.$$

Since $\deg \frac{1}{Q}g^*M = 1$, Theorem 8.2(1) implies $\|\eta_M\|^2 \leq 1 + (\kappa - 2)(\epsilon + \frac{1}{Q})$.

Step 3. *The ramification divisor along the pullback of M.* Let x, y be local coordinates at p, and let u, z be affine linear coordinates centered at $g(p)$ with $M = \{u = 0\}$. Let $u' := g^*u$ and $z' := g^*z$. Since g is finite, the ideal (u', z') is $\mathfrak{m}_p$-primary, so Definition 2.2 implies

$$\deg_p g = \dim_{\mathbb{C}} \mathcal{O}_{\mathbb{P}^2,p}/(u', z') = (u' \cdot z')_p.$$

For $i \in I$, let ϕ_i be a reduced local equation of C_i at p. Then $u' = U_1 \prod_{i \in I} \phi_i^{e_i}$ for a unit U_1, so

$$\sum_{i \in I} e_i (\phi_i \cdot z')_p = (u' \cdot z')_p = \deg_p g. \tag{8.11}$$

Let

$$R' := R(g) - \sum_{i\in I_0}(e_i - 1)C_i.$$

By (8.10), $R' \geq 0$ is a divisor without common component with g^*M. The Jacobian determinant $\mathrm{Jac}(u', z')$ of g with respect to x, y and u, z is a local equation of $R(g)$ at p, so

$$\mathrm{Jac}(u', z') = U_2\chi \prod_{i\in I} \phi_i^{e_i-1}$$

for a local equation χ of R' at p and a unit U_2. By the product rule,

$$\mathrm{Jac}(u', z') = \Big(\prod_{k\in I} \phi_k^{e_k}\Big)\,\mathrm{Jac}(U_1, z') + U_1 \sum_{j\in I} e_j \phi_j^{e_j-1}\Big(\prod_{k\in I\setminus\{j\}} \phi_k^{e_k}\Big)\,\mathrm{Jac}(\phi_j, z'),$$

so

$$U_2\chi = \Big(\prod_{k\in I} \phi_k\Big)\,\mathrm{Jac}(U_1, z') + U_1 \sum_{j\in I} e_j \Big(\prod_{k\in I\setminus\{j\}} \phi_k\Big)\,\mathrm{Jac}(\phi_j, z').$$

Let $i \in I$. Modulo ϕ_i, all terms on the right-hand side except the term with $j = i$ vanish, so

$$U_2\chi \equiv U_1 e_i \Big(\prod_{k\in I\setminus\{i\}} \phi_k\Big)\,\mathrm{Jac}(\phi_i, z') \mod \phi_i.$$

Since U_1, U_2 are units and $e_i \neq 0$,

$$(C_i \cdot R')_p = (\phi_i \cdot \chi)_p = \sum_{k\in I\setminus\{i\}} (C_i \cdot C_k)_p + \big(\phi_i \cdot \mathrm{Jac}(\phi_i, z')\big)_p.$$

Since ϕ_i divides u', $(\phi_i \cdot z')_p \leq (u' \cdot z')_p < +\infty$, and $z'(p) = 0$. Thus Lemma 8.7 implies

$$(C_i \cdot R')_p = \sum_{k\in I\setminus\{i\}} (C_i \cdot C_k)_p + \mu_p(\phi_i) + (\phi_i \cdot z')_p - 1. \tag{8.12}$$

Step 4. *Bounding the local terms by degrees.* For $i \in I$, let $r_p(C_i) \geq 1$ be the number of branches of C_i at p, and let $\overline{\mathcal{O}}_{C_i,p}$ be the integral closure of $\mathcal{O}_{C_i,p}$. By Milnor's formula [Mil68, Theorem 10.5], see [GP18, formula (M) and Proposition 4.2],

$$\mu_p(\phi_i) + r_p(C_i) - 1 = 2\dim_{\mathbb{C}} \overline{\mathcal{O}}_{C_i,p}/\mathcal{O}_{C_i,p}.$$

Since C_i is an irreducible plane curve of degree d_i, the arithmetic genus of C_i is $\frac{(d_i-1)(d_i-2)}{2}$, and the genus formula implies

$$\dim_{\mathbb{C}} \overline{\mathcal{O}}_{C_i,p}/\mathcal{O}_{C_i,p} \leq \frac{(d_i - 1)(d_i - 2)}{2}.$$

Thus

$$\mu_p(\phi_i) + r_p(C_i) - 1 \leq (d_i - 1)(d_i - 2). \tag{8.13}$$

By Bezout's theorem, $(C_i \cdot C_k)_p \leq d_i d_k$ for any distinct $i, k \in I$. Thus (8.12), (8.13), and $r_p(C_i) \geq 1$ imply

$$(C_i \cdot R')_p \leq d_i(N_0 - d_i) + d_i^2 - 3d_i + 2 - r_p(C_i) + (\phi_i \cdot z')_p \leq d_i(N_0 - 3) + 1 + (\phi_i \cdot z')_p.$$

Multiplying by e_i and adding over $i \in I$, by (8.11) and $\sum_{i\in I} e_i \leq \sum_{i\in I} e_i d_i = N$, we obtain

$$(R' \cdot g^*M)_p = \sum_{i\in I} e_i (C_i \cdot R')_p \leq N(N_0 - 3) + N + \deg_p g = N(N_0 - 2) + \deg_p g. \tag{8.14}$$

Step 5. *The pairing in the tree.* For $i \in I$, let $Z_i := \rho_S(C_i)$. Curves not through p have no branch at p, so

$$\rho_S(g^*M) = \sum_{i\in I} e_i Z_i \quad \text{and} \quad \rho_S(R(g)) = \rho_S(R') + \sum_{i\in I}(e_i - 1)Z_i.$$

By Theorem 8.2(3) and Bezout's theorem, $\langle Z_i, Z_k\rangle \le d_i d_k$ for any distinct $i,k \in I$. By Theorem 8.2(1) for $D = C_i$, whose only coefficient is 1, $\|Z_i\|^2 \le d_i^2 + (\kappa-2)d_i$. Since $e_i - 1 \ge 0$ and $e_k > 0$,

$$\Big\langle \sum_{i\in I}(e_i-1)Z_i, \sum_{k\in I} e_k Z_k\Big\rangle \le \sum_{i,k\in I}(e_i-1)e_k d_i d_k + (\kappa-2)\sum_{i\in I} e_i(e_i-1)d_i$$
$$= (N-N_0)N + (\kappa-2)\sum_{i\in I} e_i(e_i-1)d_i.$$

By Theorem 8.2(3) and (8.14),

$$\langle \rho_S(R'), \rho_S(g^*M)\rangle \le (R'\cdot g^*M)_p \le N(N_0-2) + \deg_p g.$$

Thus

$$\langle \rho_S(R(g)), \rho_S(g^*M)\rangle \le N^2 - 2N + \deg_p g + (\kappa-2)\sum_{i\in I} e_i(e_i-1)d_i. \tag{8.15}$$

By (8.10) and the assumption $\mathrm{mult}_{C_i} R(g) \le \epsilon(Q-1)$ for $i \in I$, $e_i - 1 \le \epsilon(Q-1)$ for any $i \in I$, so

$$\sum_{i\in I} e_i(e_i-1)d_i \le \epsilon(Q-1)N \le \epsilon(Q-1)Q.$$

Since $1 \le N \le Q$, $(Q^2-2Q) - (N^2-2N) = (Q-N)(Q+N-2) \ge 0$. Since $\kappa \ge 2$, dividing (8.15) by $Q(Q-1)$, we obtain

$$\langle \xi, \eta_M\rangle \le \frac{Q^2 - 2Q + \deg_p g}{Q(Q-1)} + (\kappa-2)\epsilon.$$

The proof of the upper bound for $\langle \xi, \eta_M\rangle$ uses the assumption on $R(g)$ only in **Step 5**, for the curves C_i with $i \in I$, which are the irreducible components of g^*M through p. Thus the upper bound for $\langle \xi, \eta_M\rangle$ holds under the weaker assumption in the proposition.

Step 6. *Values at an eigenvaluation.* Assume that $g(p) = p$ and that $\omega \in S$ satisfies $g_*\omega = \Lambda\omega$ for a real number $\Lambda > 0$. Since $g_*\omega = \Lambda\omega$, we have $c(g,\omega) = (g_*\omega)(\mathfrak{m}_p) = \Lambda$ and $g_\bullet\omega = \omega$ by Definition 2.11. Let x, y be regular parameters at p. The Jacobian determinant of g with respect to x, y is a local equation of $R(g)$ at p, so Theorem 2.12(2), with $\nu = \omega$, implies

$$\Lambda A(\omega) = A(\omega) + \omega(R(g)).$$

Since $A(\omega) \le \kappa < +\infty$, $\omega(R(g)) = (\Lambda-1)A(\omega)$. Since $\omega \in S \subset \Gamma_S$, Theorem 8.2 implies

$$\langle \delta_\omega, \rho_S(D)\rangle = \omega(D)$$

for $D = \frac{1}{Q-1}R(g)$ and $D = \frac{1}{Q}g^*M$, so

$$\langle \xi, \delta_\omega\rangle = \frac{\omega(R(g))}{Q-1} = \frac{\Lambda-1}{Q-1}A(\omega) \quad \text{and}$$
$$\langle \eta_M, \delta_\omega\rangle = \frac{\omega(g^*M)}{Q} = \frac{(g_*\omega)(M)}{Q} = \frac{\Lambda}{Q}\omega(M). \qquad \square$$

9. Points off the curve C

Theorem 9.1. *Notation and conditions as in Set-up 5.1. Then for any integer $n \ge 4\cdot 10^8$, the pair $(\mathbb{P}^2, B_{17280n})$ is lc at any closed point $p \in \mathbb{P}^2\backslash C$.*

Lemma 9.2. *Let g be an endomorphism of $\mathbb{P}^2$ of degree at least 2 and $p \in \mathbb{P}^2$ a closed point with $g(p) = p$ and $\lambda := c_\infty(g,p) > 1$. Let ω be an eigenvaluation of g at p for which (a) or (b) of Theorem 2.12(1) holds, and assume that ω is quasimonomial or infinitely singular. Then $A(\omega) < +\infty$, $\alpha(\omega) < +\infty$, and*

$$(g^j)_*\omega = \lambda^j\omega \quad \text{and} \quad \omega(R(g^j)) = (\lambda^j - 1)A(\omega) \quad \text{for any integer } j \ge 1.$$

Proof. By Definition 2.11, $g_\bullet\omega = \omega$ and $c(g,\omega) = \lambda$, so $g_*\omega = \lambda\omega$. Thus for any integer $j \geq 1$, $(g^j)_*\omega = \lambda^j\omega$, so $(g^j)_\bullet\omega = \omega$ and $c(g^j,\omega) = \lambda^j$.

Let x, y be regular parameters at p. For any integer $j \geq 1$, $g^j(p) = p$. Let $\psi_j \in \mathcal{O}_{\mathbb{P}^2,p}$ be the Jacobian determinant of g^j at p with respect to x, y, that is, $(g^j)^*(dx \wedge dy) = \psi_j dx \wedge dy$. Since $K_{\mathbb{P}^2} = (g^j)^*K_{\mathbb{P}^2} + R(g^j)$, ψ_j is a local equation of $R(g^j)$ at p. Thus Theorem 2.12(2) implies that for any integer $j \geq 1$ and any normalized valuation ν at p,

$$c(g^j,\nu)A((g^j)_\bullet\nu) = A(\nu) + \nu(R(g^j)). \tag{9.1}$$

The classes of quasimonomial valuations, curve valuations and infinitely singular valuations are pairwise disjoint [FJ04, Definition 2.23]. If ν is a quasimonomial normalized valuation at p, then $\alpha(\nu) < +\infty$ [FJ04, Theorem 3.26 (i), (ii)] and $m(\nu) < +\infty$ [FJ04, Proposition 3.37(i)]. Since $m \leq m(\nu)$ on $[\mathrm{mult}_p, \nu]$, Theorem 2.9(3) implies

$$A(\nu) \leq 2 + m(\nu)(\alpha(\nu) - 1) < +\infty. \tag{9.2}$$

If ω is quasimonomial, then $A(\omega) < +\infty$ by (9.2).

Assume that ω is infinitely singular. Then ω is not quasimonomial, so (a) does not hold and (b) holds: there exists a normalized valuation $\mu < \omega$ with $c(g,\nu) = \lambda$ and $g_\bullet\nu > \nu$ for any $\nu \in [\mu,\omega)$. The curve valuations and the infinitely singular valuations are the maximal normalized valuations at p, and any other normalized valuation at p is quasimonomial [FJ04, Proposition 3.20 (ii)]. We have the following.

- $\omega(R(g)) < +\infty$, since ψ_1 is a unit times a product of finitely many irreducible $\phi \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p}$ and $\omega(\phi) < +\infty$ for any such ϕ. For any such ϕ, $\omega(\phi) = \mathrm{mult}_p\,\phi \cdot \alpha(\omega \wedge \nu_\phi)$ by Theorem 2.9(2). If $\omega \wedge \nu_\phi = \omega$, then $\omega \leq \nu_\phi$, so $\omega = \nu_\phi$ by the maximality of ω, which is impossible as ω is not a curve valuation. Thus $\omega \wedge \nu_\phi < \omega$, so $\omega \wedge \nu_\phi$ is quasimonomial and $\alpha(\omega \wedge \nu_\phi) < +\infty$.
- For any $\nu \in [\mu,\omega)$, $\nu < \omega$, so ν is quasimonomial and $A(\nu) < +\infty$ by (9.2). Since $\nu < g_\bullet\nu$, Theorem 2.9(3) implies $A(g_\bullet\nu) \geq A(\nu)$. By (9.1) with $j = 1$ and $c(g,\nu) = \lambda$,
$$A(\nu) + \nu(R(g)) = \lambda A(g_\bullet\nu) \geq \lambda A(\nu).$$
Since $A(\nu) < +\infty$ and $\nu \leq \omega$ (Definition 2.8),
$$(\lambda - 1)A(\nu) \leq \nu(R(g)) \leq \omega(R(g)).$$

By Theorem 2.9(3), $A(\omega)$ is the limit of $A(\nu)$ as ν increases to ω along $[\mu,\omega)$. Since $\lambda > 1$,

$$A(\omega) \leq \frac{\omega(R(g))}{\lambda - 1} < +\infty.$$

Thus $A(\omega) < +\infty$ whether ω is quasimonomial or infinitely singular, and by Theorem 2.9(3),

$$\alpha(\omega) \leq A(\omega) - 1 < +\infty.$$

Let $j \geq 1$ be an integer. Since $(g^j)_\bullet\omega = \omega$ and $c(g^j,\omega) = \lambda^j$, (9.1) with $\nu = \omega$ implies

$$\lambda^j A(\omega) = A(\omega) + \omega(R(g^j)).$$

Since $A(\omega) < +\infty$, $\omega(R(g^j)) = (\lambda^j - 1)A(\omega)$. □

Lemma 9.3. *Notation and conditions as in Set-up 5.1, and let $p \in \mathbb{P}^2 \backslash C$ be a closed point. Then*

$$\mathrm{mult}_P B_{17280n} < \frac{26}{2880n + 1}$$

for any integer $n \geq 1$ and any irreducible curve P through p.

Proof. Let $n \geq 1$ be an integer and P an irreducible curve through p. Since $17280n = 6 \cdot 2880n$, if $f^{6k}(P) \neq P$ for $k = 1, 2, 3$, then Proposition 5.4(1) with $2880n$ in place of n implies $\mathrm{mult}_P B_{17280n} < \frac{26}{2880n+1}$.

Thus we may assume that $f^{6k}(P) = P$ for some $k \in \{1, 2, 3\}$. Since k divides 48 and $288 = 6 \cdot 48$, we have $f^{288}(P) = P$, so Definition 2.2 implies

$$e_P(f^{288j}) = e_P(f^{288})^j \quad \text{for any integer } j \geq 1.$$

Since $p \in P$ and $p \notin C$, P is not an irreducible component of C, so $\operatorname{mult}_P B_{288} \leq \frac{3}{5}$ by Set-up 5.1(3), that is, $e_P(f^{288}) - 1 \leq \frac{3}{5}(q^{288} - 1)$ by Definition 2.2. Since $q^{288} \geq 4$,

$$\frac{e_P(f^{288})}{q^{288}} \leq \frac{3}{5} + \frac{2}{5q^{288}} \leq \frac{7}{10}.$$

In particular, $1 \leq e_P(f^{288})^{60n} \leq q^{17280n}$. Since $17280n = 288 \cdot 60n$ and $\frac{a-1}{b-1} \leq \frac{a}{b}$ for any real numbers a, b with $1 \leq a \leq b$ and $b > 1$,

$$\operatorname{mult}_P B_{17280n} = \frac{e_P(f^{288})^{60n} - 1}{q^{17280n} - 1} \leq \left(\frac{e_P(f^{288})}{q^{288}}\right)^{60n} \leq \left(\frac{7}{10}\right)^{60n}.$$

Since $7^7 = 823543$, we have $(\frac{7}{10})^7 < \frac{1}{10}$, so

$$\left(\frac{7}{10}\right)^{60} < \left(\frac{7}{10}\right)^{56} < 10^{-8} < \frac{1}{128}.$$

Since $128^{n-1} \geq 2^{n-1} \geq n$ and $26 \cdot 128n = 3328n > 2880n + 1$,

$$\left(\frac{7}{10}\right)^{60n} < \frac{1}{128^n} \leq \frac{1}{128n} < \frac{26}{2880n + 1}.$$

The lemma follows. □

Proposition 9.4. *Notation and conditions as in Set-up 5.1. Let $n \geq 4 \cdot 10^8$ be an integer and v a divisorial valuation centered at a closed point $p \notin C$ with $v(B_{17280n}) > A(v)$. Then $g(p) = p$, and $\lambda := c_\infty(g, p)$ satisfies $\frac{9999}{10000}Q < \lambda < Q$. Moreover:*

(1) For any irreducible curve P through p, $e_P(g) < \frac{1}{100}Q$.
(2) Any eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds is quasimonomial or infinitely singular, and $A(\omega) < 8$, $\alpha(\omega) < 7$, and $\omega(R(g)) = (\lambda-1)A(\omega)$.
(3) For any integer $j \geq 0$, $A((g^j)_* v) < 5600(g^j)_* v(\mathfrak{m}_p)$. Moreover, $(\lambda/Q)^n > \frac{1}{8}$.

Proof. **Step 1.** We show that $g(p) = p$ and $\frac{9999}{10000}Q < \lambda < Q$.

Let $N := \lfloor n/1000 \rfloor$. By Theorem 7.1(1), $g(p) = p$, and

$$\frac{9}{10}Q < \lambda < Q \quad \text{and} \quad \left(\frac{\lambda}{Q}\right)^N > \frac{1}{15000}.$$

Since $n \geq 4 \cdot 10^8$, $N \geq 4 \cdot 10^5 > 14 \cdot 10^4$. By the binomial theorem,

$$\left(\frac{10000}{9999}\right)^{10000} = \left(1 + \frac{1}{9999}\right)^{10000} > 1 + \frac{10000}{9999} > 2.$$

Suppose that $\lambda \leq \frac{9999}{10000}Q$. Since $0 < \lambda/Q < 1$,

$$\left(\frac{\lambda}{Q}\right)^N \leq \left(\frac{9999}{10000}\right)^{14 \cdot 10^4} < 2^{-14} = \frac{1}{16384} < \frac{1}{15000},$$

a contradiction. Thus

$$\frac{9999}{10000}Q < \lambda < Q.$$

In particular, since $Q \geq 2^{17280}$, we have $\lambda > 1$.

Step 2. We prove (1).

Let P be an irreducible curve through p. By Definition 2.2 and Lemma 9.3 with $n = 1$,

$$e_P(g) - 1 = (Q - 1) \operatorname{mult}_P B_{17280} < \frac{26}{2881}(Q - 1),$$

so

$$\frac{e_P(g)}{Q} < \frac{26}{2881} + \frac{1}{Q} < \frac{26}{2881} + \frac{1}{10^4} < \frac{1}{100},$$

since $Q \geq 2^{17280} > 10^4$. This proves (1).

Step 3. Let ω be an eigenvaluation of g at p for which (a) or (b) of Theorem 2.12(1) holds. We show that ω is quasimonomial or infinitely singular, $A(\omega) < +\infty$, $\alpha(\omega) < +\infty$, and $\omega(R(g)) = (\lambda - 1)A(\omega)$.

Suppose that (b) holds and $\omega = \nu_\phi$ for an irreducible analytic curve germ ϕ at p. Then ν_ϕ is an eigenvaluation of g at p, and there exists a normalized valuation $\mu < \nu_\phi$ with $c(g, \nu) = \lambda$ and $g_\bullet \nu > \nu$ for any $\nu \in [\mu, \nu_\phi)$. Since $p \notin C$, Theorem 7.1(2) excludes such a ϕ. Thus ω is quasimonomial or infinitely singular.

By **Step 1** and Lemma 9.2, $A(\omega) < +\infty$, $\alpha(\omega) < +\infty$,

$$(g^j)_*\omega = \lambda^j \omega \quad \text{for any integer } j \geq 1, \tag{9.3}$$

and

$$\omega(R(g)) = (\lambda - 1)A(\omega). \tag{9.4}$$

Step 4. Let ω be as in **Step 3**. We show that for any irreducible curve $P \subset \mathbb{P}^2$,

$$\omega(P)^2 \leq \frac{99Q}{100\lambda - Q}\alpha(\omega)(\deg P)^2. \tag{9.5}$$

If $p \notin P$, then $\omega(P) = 0$. Assume that $p \in P$. Since $g(p) = p$, (9.3) implies

$$\omega(g^*P) = (g_*\omega)(P) = \lambda\omega(P).$$

Moreover, $\deg g^*P = Q \deg P$ by Definition 2.2. By **Step 3**, $\alpha(\omega) < +\infty$.

If $g(P) \neq P$, then P is not an irreducible component of g^*P. By Lemma 2.10(3),

$$\lambda\omega(P)^2 = \omega(P)\omega(g^*P) \leq \alpha(\omega)Q(\deg P)^2.$$

If $g(P) = P$, write $g^*P = e_P(g)P + D$, where $D \geq 0$ and $P \not\subset \operatorname{Supp} D$. Then

$$\deg D = (Q - e_P(g)) \deg P \quad \text{and} \quad \omega(D) = (\lambda - e_P(g))\omega(P).$$

By (1) and **Step 1**, $e_P(g) < \frac{1}{100}Q < \lambda$. By Lemma 2.10(3),

$$(\lambda - e_P(g))\omega(P)^2 \leq \alpha(\omega)(Q - e_P(g))(\deg P)^2.$$

Let $e := e_P(g)$ if $g(P) = P$, and $e := 0$ if $g(P) \neq P$. In both cases $0 \leq e < \frac{1}{100}Q < \lambda$ and

$$\frac{\omega(P)^2}{\alpha(\omega)(\deg P)^2} \leq \frac{Q - e}{\lambda - e}.$$

Since $\lambda < Q$, $t \to \frac{Q-t}{\lambda-t}$ is strictly increasing on $[0, \lambda)$, so

$$\frac{Q - e}{\lambda - e} < \frac{Q - \frac{1}{100}Q}{\lambda - \frac{1}{100}Q} = \frac{99Q}{100\lambda - Q},$$

which proves (9.5).

Step 5. We finish the proof of (2).

Let ω be as in **Step 3**. By Definition 2.2, $R(g) = \sum_P (e_P(g) - 1)P$, where P runs over the irreducible curves in $\mathbb{P}^2$. Thus

$$\sum_P (e_P(g) - 1) \deg P = \deg R(g) = 3(Q - 1).$$

By (9.4) and (9.5),

$$(\lambda-1)A(\omega)=\sum_P(e_P(g)-1)\omega(P)$$
$$\le\left(\frac{99Q\alpha(\omega)}{100\lambda-Q}\right)^{1/2}\sum_P(e_P(g)-1)\deg P=3(Q-1)\left(\frac{99Q\alpha(\omega)}{100\lambda-Q}\right)^{1/2}.$$

Since $\lambda>1$,

$$\frac{A(\omega)^2}{\alpha(\omega)}\le 9\left(\frac{Q-1}{\lambda-1}\right)^2\frac{99Q}{100\lambda-Q}.$$

By **Step 1**, $\lambda>\frac{9999}{10000}Q$, and $Q\ge 10^4$, so

$$\lambda-1>\frac{9999}{10^4}Q-\frac{1}{10^4}Q=\frac{9998}{10^4}Q.$$

Thus

$$\frac{Q-1}{\lambda-1}<\frac{10^4}{9998}=\frac{5000}{4999}<\frac{1001}{1000}\quad\text{and}\quad\frac{99Q}{100\lambda-Q}<\frac{99}{\frac{9999}{100}-1}=\frac{9900}{9899}<\frac{1001}{1000}.$$

Since $9\cdot 1001^3=9027027009$,

$$\frac{A(\omega)^2}{\alpha(\omega)}<\frac{9\cdot 1001^3}{1000^3}<\frac{64}{7}.$$

By Theorem 2.9(1)(3), $\alpha(\omega)\ge 1$ and $A(\omega)\ge 1+\alpha(\omega)\ge 2$. The function $t\to\frac{t^2}{t-1}$ has derivative $\frac{t(t-2)}{(t-1)^2}\ge 0$ on $[2,+\infty)$, so it is increasing on $[2,+\infty)$. Since $\alpha(\omega)\le A(\omega)-1$,

$$\frac{A(\omega)^2}{A(\omega)-1}\le\frac{A(\omega)^2}{\alpha(\omega)}<\frac{64}{7}=\frac{8^2}{8-1},$$

so $A(\omega)<8$ and $\alpha(\omega)\le A(\omega)-1<7$. Together with **Step 3** and (9.4), this proves (2).

Step 6. We prove (3).

The inequality $v(B_{17280n})>A(v)$ and the inequalities in (3) are positively homogeneous of degree 1 in v. Possibly replacing v with $v/v(\mathfrak{m}_p)$, we may assume that $v(\mathfrak{m}_p)=1$. Then v is a divisorial normalized valuation at p, and $A(v)$ is the thinness of v by Theorem 2.9(4). Since $B_{17280n}=B_{6\cdot 2880n}$, Theorem 4.1(4) implies $A(v)<100$, so $\alpha(v)\le A(v)-1<99$ by Theorem 2.9(3).

Since $\lambda>1$, by Theorem 2.12(1) there exists an eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds. By (2), $A(\omega)<8$ and $\alpha(\omega)<7$. Let $j\ge 1$ be an integer. By **Step 1**, (2), and Lemma 9.2,

$$\omega(R(g^j))=(\lambda^j-1)A(\omega). \tag{9.6}$$

Since $\alpha(v)<+\infty$ and $\alpha(\omega)<+\infty$, Lemma 2.10(1) implies that for any non-zero $\phi\in\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$,

$$\frac{\omega(\phi)}{\alpha(\omega)}\le\operatorname{mult}_p\phi\le v(\phi)\le\alpha(v)\operatorname{mult}_p\phi\le\alpha(v)\omega(\phi). \tag{9.7}$$

For any non-zero $\phi\in\mathfrak{m}_p$, $(g^j)^*\phi\ne 0$ since g is finite, and $\omega(\phi)\ge\omega(\mathfrak{m}_p)=1$. By (9.7) and (9.3),

$$(g^j)_*v(\phi)=v((g^j)^*\phi)\ge\frac{\omega((g^j)^*\phi)}{\alpha(\omega)}=\frac{\lambda^j\omega(\phi)}{\alpha(\omega)}\ge\frac{\lambda^j}{\alpha(\omega)},$$

so

$$(g^j)_*v(\mathfrak{m}_p)\ge\frac{\lambda^j}{\alpha(\omega)}. \tag{9.8}$$

By (9.7) applied to a local equation of $R(g^j)$ at p, and by (9.6),

$$v(R(g^j))\le\alpha(v)\omega(R(g^j))=\alpha(v)(\lambda^j-1)A(\omega). \tag{9.9}$$

By Lemma 2.4 for g^j and by (9.9),

$$A((g^j)_*v) = A(v) + v(R(g^j)) \le A(v) + \alpha(v)(\lambda^j - 1)A(\omega).$$

For $j = 0$, $(g^0)_*v = v$, so

$$A((g^0)_*v) = A(v) + \alpha(v)(\lambda^0 - 1)A(\omega) \quad \text{and} \quad (g^0)_*v(\mathfrak{m}_p) = 1 \ge 1/\alpha(\omega),$$

so (9.8) holds for $j = 0$. By (9.8), for any integer $j \ge 0$,

$$\frac{A((g^j)_*v)}{(g^j)_*v(\mathfrak{m}_p)} \le \alpha(\omega)\left(\lambda^{-j}A(v) + (1-\lambda^{-j})\alpha(v)A(\omega)\right) \le \alpha(\omega)\max\{A(v), \alpha(v)A(\omega)\}.$$

Since $A(v) < 100 < 792$, $\alpha(v)A(\omega) < 99 \cdot 8 = 792$, and $\alpha(\omega) < 7$,

$$A((g^j)_*v) < 5544(g^j)_*v(\mathfrak{m}_p) < 5600(g^j)_*v(\mathfrak{m}_p).$$

Finally, $B_{17280n} = \frac{1}{Q^n-1}R(g^n)$ by Set-up 5.1. Since $v(B_{17280n}) > A(v)$, (9.9) with $j = n$ implies

$$(Q^n - 1)A(v) < v(R(g^n)) \le \alpha(v)(\lambda^n - 1)A(\omega).$$

Since $1 < \lambda^n < Q^n$, $\frac{\lambda^n-1}{Q^n-1} < \frac{\lambda^n}{Q^n}$. Since $A(v) \ge 1 + \alpha(v) > \alpha(v)$ (Theorem 2.9(3)) and $A(\omega) < 8$,

$$\left(\frac{\lambda}{Q}\right)^n > \frac{\lambda^n - 1}{Q^n - 1} > \frac{A(v)}{\alpha(v)A(\omega)} > \frac{1}{A(\omega)} > \frac{1}{8}.$$

This proves (3). □

Proposition 9.5. *Notation and conditions as in Set-up 5.1. Let $n \ge 4 \cdot 10^8$ be an integer and v a divisorial valuation centered at a closed point $p \notin C$. Assume that $v(B_{17280n}) > A(v)$. Then $g(p) = p$ by Proposition 9.4. Let $\lambda := c_\infty(g, p)$ and $\sigma := 1 - \lambda/Q$. Any eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds satisfies*

$$0 \le \alpha(\omega) - 1 \le A(\omega) - 2 < 168\sigma.$$

Proof. By Proposition 9.4, $g(p) = p$ and $\frac{9999}{10000}Q < \lambda < Q$. Since $Q \ge 2$,

$$0 < \sigma < \frac{1}{10000} \quad \text{and} \quad \lambda > 1.$$

Let ω be an eigenvaluation of g at p for which (a) or (b) of Theorem 2.12(1) holds. By Proposition 9.4(2), $A(\omega) < 8$ and $\alpha(\omega) < 7$. By Theorem 2.9(1) and (3), $1 \le \alpha(\omega)$ and $\alpha(\omega) + 1 \le A(\omega)$. Thus $0 \le \alpha(\omega) - 1 \le A(\omega) - 2$, in particular $A(\omega) \ge 2$, and it remains to prove $A(\omega) - 2 < 168\sigma$. By Proposition 9.4(2) and Lemma 9.2, $g_*\omega = \lambda\omega$.

Step 1. Let $S := \{\omega\}$, so that $\Gamma_S = [\mathrm{mult}_p, \omega]$. We show that for any $\mathbb{R}$-divisors $D \ge 0$ and $D' \ge 0$ on $\mathbb{P}^2$,

$$\langle \rho_S(D), \rho_S(D')\rangle \ge \frac{1}{\alpha(\omega)}\langle \rho_S(D), \delta_\omega\rangle\langle \rho_S(D'), \delta_\omega\rangle. \tag{9.10}$$

By Definition 8.1, $\rho_S(D)$ and $\rho_S(D')$ are linear combinations of δ_μ, $\mu \in \Gamma_S$, with non-negative coefficients. Let $\mu, \mu' \in \Gamma_S$. Then $\mu \wedge \mu' \in \{\mu, \mu'\}$ and $\mu \wedge \omega = \mu$. Since α is increasing on $[\mathrm{mult}_p, \omega]$ by Theorem 2.9(1), $\max\{\alpha(\mu), \alpha(\mu')\} \le \alpha(\omega)$, so

$$\langle \delta_\mu, \delta_{\mu'}\rangle = \min\{\alpha(\mu), \alpha(\mu')\} \ge \frac{\alpha(\mu)\alpha(\mu')}{\alpha(\omega)} = \frac{1}{\alpha(\omega)}\langle \delta_\mu, \delta_\omega\rangle\langle \delta_{\mu'}, \delta_\omega\rangle.$$

By bilinearity, (9.10) follows.

Step 2. Let M be a line through p with

$$\omega(M) = \max\{\omega(M') \mid M' \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

Since $A(\omega) < 8$, Lemma 8.5(3) with $\tau = \omega$ implies

$$\omega(M)A(\omega) \ge \alpha(\omega)\left(2 + \frac{A(\omega) - 2}{21}\right). \tag{9.11}$$

Step 3. Let $\epsilon := \frac{26}{2881}$. Since $R(g) = (Q-1)B_{17280}$, Lemma 9.3 with $n = 1$ implies that for any irreducible curve P through p,

$$\operatorname{mult}_P R(g) < \epsilon(Q-1).$$

Let

$$\xi := \rho_S\left(\frac{1}{Q-1}R(g)\right) \quad \text{and} \quad \eta_M := \rho_S\left(\frac{1}{Q}g^*M\right).$$

Since $\deg_p g \le Q^2$ (Definition 2.2), we have

$$\frac{Q^2 - 2Q + \deg_p g}{Q(Q-1)} \le 2,$$

so Proposition 8.8 with $\kappa = A(\omega) \ge 2$ and the line M through $g(p) = p$ implies

$$\langle \xi, \eta_M \rangle \le 2 + (A(\omega) - 2)\epsilon.$$

Since $g(p) = p$ and $g_*\omega = \lambda\omega$, the last sentence of Proposition 8.8 with $\Lambda = \lambda$ implies

$$\langle \xi, \delta_\omega \rangle = \frac{\lambda - 1}{Q-1}A(\omega) \quad \text{and} \quad \langle \eta_M, \delta_\omega \rangle = \frac{\lambda}{Q}\omega(M).$$

Since $R(g) \ge 0$, $g^*M \ge 0$, and $\lambda > 1$, (9.10) together with (9.11) implies

$$\frac{(\lambda-1)\lambda}{(Q-1)Q}\left(2 + \frac{A(\omega)-2}{21}\right) \le \frac{(\lambda-1)\lambda}{(Q-1)Q} \cdot \frac{\omega(M)A(\omega)}{\alpha(\omega)} \le \langle \xi, \eta_M \rangle \le 2 + (A(\omega)-2)\epsilon.$$

Equivalently,

$$\left(\frac{(\lambda-1)\lambda}{21(Q-1)Q} - \epsilon\right)(A(\omega)-2) \le 2\left(1 - \frac{(\lambda-1)\lambda}{(Q-1)Q}\right). \tag{9.12}$$

Step 4. Since $\lambda = (1-\sigma)Q$ and $Q \ge 2$,

$$1 - \frac{\lambda - 1}{Q-1} = \frac{Q\sigma}{Q-1} \le 2\sigma,$$

so

$$1 - \frac{(\lambda-1)\lambda}{(Q-1)Q} = \sigma + (1-\sigma)\left(1 - \frac{\lambda-1}{Q-1}\right) \le 3\sigma.$$

Since $\sigma < \frac{1}{10000}$, we have

$$\frac{(\lambda-1)\lambda}{(Q-1)Q} \ge 1 - 3\sigma > \frac{9997}{10000}.$$

Since moreover $\epsilon < \frac{1}{100}$,

$$\frac{(\lambda-1)\lambda}{21(Q-1)Q} - \epsilon > \frac{9997}{210000} - \frac{2100}{210000} = \frac{7897}{210000} > \frac{1}{28}.$$

Since $\sigma > 0$, (9.12) implies

$$A(\omega) - 2 \le 6\sigma\left(\frac{(\lambda-1)\lambda}{21(Q-1)Q} - \epsilon\right)^{-1} < 168\sigma.$$

The proposition follows. □

Proof of Theorem 9.1. Let $n \ge 4 \cdot 10^8$ be an integer and $p \in \mathbb{P}^2 \backslash C$ a closed point, and assume that $(\mathbb{P}^2, B_{17280n})$ is not lc at p. If a divisorial valuation u has center an irreducible curve $P \subset \mathbb{P}^2$, then $u(D) = A(u)\operatorname{mult}_P D$ for any $\mathbb{R}$-divisor D on $\mathbb{P}^2$. Since $17280n = 6 \cdot 2880n$, Theorem 4.1(1) with $m = 2880n$ implies that $\operatorname{mult}_P B_{17280n} \le 1$ for any irreducible curve $P \subset \mathbb{P}^2$, so $u(B_{17280n}) \le A(u)$ for any divisorial valuation u whose center is a curve. Thus there exists a divisorial valuation u centered at p with $u(B_{17280n}) > A(u)$. By Proposition 4.9 with $m = 2880n$, there exist affine linear coordinates x, y centered at p, a polynomial $\phi \in \mathbb{C}[x]$ with $\phi(0) = 0$, and positive integers $a \le b < 11a$ such that $a \deg \phi \le b$ and $w := \operatorname{mon}_{x,y-\phi(x)}(a,b)$ satisfies $w(B_{17280n}) > A(w)$.

Let $t := b/a$ and $v := \mathrm{mon}_{x,y-\phi(x)}(1,t) = w/a$. Then $1 \le t < 11$ and

$$v(B_{17280n}) - A(v) = \frac{w(B_{17280n}) - A(w)}{a} > 0.$$

Moreover:

(1) Since a and b are positive integers, $t \in \mathbb{Q}$, so v is a divisorial valuation centered at p with $v(\mathfrak{m}_p) = 1$.
(2) $v(y - \phi(x)) = t$ and $\mathrm{mult}_p(y - \phi(x)) = 1$, and any $\psi \in \widehat{\mathcal{O}}_{\mathbb{P}^2,p}$ with $\mathrm{mult}_p \psi = k$ has, as a power series in x and $y - \phi(x)$, a monomial $x^i(y - \phi(x))^j$ with $i + j = k$, so $v(\psi) \le i + tj \le tk$. Thus $\alpha(v) = t$.
(3) $A(v) = 1 + t$ by [FJ04, Propositions 3.37 and 3.48], since v is monomial.

Since $f^{17280n} = g^n$ and $q^{17280n} = Q^n$, we have $B_{17280n} = \frac{1}{Q^n - 1} R(g^n)$. By Lemma 2.4,

$$A((g^n)_* v) = 1 + t + v(R(g^n)) > 1 + t + (Q^n - 1)(1 + t) = Q^n(1 + t). \tag{9.13}$$

We show that $A((g^n)_* v) < Q^n(1 + t)$, which contradicts (9.13).

Step 1. In this step we fix an eigenvaluation ω of g at p and bound σ, $A(\omega)$, and $\alpha(\omega)$.

By Proposition 9.4, $g(p) = p$, and $\lambda := c_\infty(g, p)$ satisfies $\frac{9999}{10000} Q < \lambda < Q$ and $(\lambda/Q)^n > \frac{1}{8}$. Let $\sigma := 1 - \lambda/Q$. Then $\sigma > 0$, and since $(1 - \sigma)^n \le e^{-n\sigma}$,

$$n\sigma < \log 8 < 3, \quad \text{so} \quad \sigma < \frac{3}{n} < 10^{-8}. \tag{9.14}$$

Since $\lambda > 1$, Theorem 2.12(1) implies that there exists an eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds. By Proposition 9.4(2), $A(\omega) < 8$, and by Proposition 9.5, $0 \le \alpha(\omega) - 1 \le A(\omega) - 2 < 168\sigma$. Thus

$$1 \le \alpha(\omega) < 1 + 168\sigma \quad \text{and} \quad 2 \le A(\omega) < 2 + 168\sigma. \tag{9.15}$$

By Proposition 9.4(2) and Lemma 9.2, $(g^j)_* \omega = \lambda^j \omega$ for any integer $j \ge 1$, and

$$\omega(R(g^j)) = (\lambda^j - 1) A(\omega) \quad \text{for any integer } j \ge 1. \tag{9.16}$$

Since $\alpha(\omega) < +\infty$ and $\alpha(v) = t$, Lemma 2.10(1) implies that

$$\alpha(\omega)^{-1} \omega \le \mathrm{mult}_p \le v \le t\,\mathrm{mult}_p \le t\omega, \tag{9.17}$$

as functions on $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$, hence also on ideals and on divisors $D \ge 0$.

Since $\omega(\mathfrak{m}_p) = 1$ and $\mathfrak{m}_p = (x, y)$, there exists a line $M_1 \in \{\{x = 0\}, \{y = 0\}\}$ with $\omega(M_1) = 1$. By (9.17),

$$\mathrm{mult}_p(g^* M_1) \le \omega(g^* M_1) = (g_* \omega)(M_1) = \lambda < Q,$$

so $\mathrm{mult}_p(g^* M_1) \le Q - 1$ as $\mathrm{mult}_p(g^* M_1) \in \mathbb{Z}$. By (9.17) and (9.15),

$$Q(1 - \sigma) = \lambda = \omega(g^* M_1) \le \alpha(\omega)\,\mathrm{mult}_p(g^* M_1) \le (1 + 168\sigma)(Q - 1).$$

Thus $1 \le Q\sigma + 168\sigma(Q - 1) < 169Q\sigma$, that is,

$$\sigma > \frac{1}{169Q}. \tag{9.18}$$

Step 2. In this step we bound the value on $\mathfrak{m}_p$ and the log discrepancy of $(g^5)_* v$.

For any integer $j \ge 0$, let

$$w_j := (g^j)_* v, \quad m_j := w_j(\mathfrak{m}_p), \quad \text{and} \quad \nu_j := w_j / m_j.$$

By Lemma 2.4, w_j is a divisorial valuation, centered at $g^j(p) = p$, so ν_j is a divisorial normalized valuation at p, and $A(w_j) = 1 + t + v(R(g^j))$. By (9.17) and (9.16),

$$m_5 = v((g^5)^* \mathfrak{m}_p) \ge \alpha(\omega)^{-1} \omega((g^5)^* \mathfrak{m}_p) = \frac{\lambda^5}{\alpha(\omega)}$$

and

$$A(w_5) \le 1 + t + t\omega(R(g^5)) = 1 + t + t(\lambda^5 - 1)A(\omega).$$

Since $\lambda > 1$ and A is positively homogeneous,

$$A(\nu_5) = \frac{A(w_5)}{m_5} \le \alpha(\omega)\left((1+t)\lambda^{-5} + tA(\omega)(1-\lambda^{-5})\right) \le \alpha(\omega)\max\{1+t, tA(\omega)\}.$$

By (9.14) and (9.15), we have $\alpha(\omega) < 1.01$ and $A(\omega) < 2.01$. Since $1.01 \cdot \max\{12, 11 \cdot 2.01\} < 23$ and $1.01 \cdot \max\{3, 2 \cdot 2.01\} < 5$,

$$A(\nu_5) < 23, \quad \text{and} \quad A(\nu_5) < 5 \text{ if } t < 2. \tag{9.19}$$

Since $(1-\sigma)^5 \ge 1 - 5\sigma$ and $\alpha(\omega)^{-1} > (1 + 168\sigma)^{-1} \ge 1 - 168\sigma$,

$$\frac{m_5}{Q^5} \ge \frac{(1-\sigma)^5}{\alpha(\omega)} \ge (1 - 5\sigma)(1 - 168\sigma) \ge 1 - 173\sigma. \tag{9.20}$$

By Theorem 2.9(3) and (9.19), $\alpha(\nu_5) \le A(\nu_5) - 1 < 22$. Since g^5 is an endomorphism of $\mathbb{P}^2$ of degree Q^5, $\deg_p g^5 \le Q^{10}$ (Definition 2.2). Thus Lemma 8.3 for the germ of g^5 at p and the single valuation v, whose pushforward is $m_5\nu_5$, implies that

$$m_5^2 \alpha(\nu_5) \le Q^{10} t. \tag{9.21}$$

Since $\alpha(\nu_5) \ge 1$ and $t < 11$, $m_5 < 4Q^5$. Thus by (9.19),

$$A(w_5) = m_5 A(\nu_5) < 92Q^5. \tag{9.22}$$

Step 3. In this step we bound the coefficients of $R(g^5)$ along curves through p.

Let P be an irreducible curve through p. For any $0 \le i \le 4$, $g^i(P)$ is an irreducible curve through $g^i(p) = p$, so $e_{g^i(P)}(g) < \frac{Q}{100}$ by Proposition 9.4(1). Since ramification indices are multiplicative under composition, $e_P(g^5) = \prod_{i=0}^4 e_{g^i(P)}(g) < 10^{-10}Q^5$. In particular $e_P(g^5) \le Q^5$, so

$$\frac{\operatorname{mult}_P R(g^5)}{Q^5 - 1} = \frac{e_P(g^5) - 1}{Q^5 - 1} \le \frac{e_P(g^5)}{Q^5} < 10^{-10}. \tag{9.23}$$

Step 4. In this step we bound $w_5(M)$ for any line M through p.

Let $S := \{v\}$, so $\Gamma_S = [\operatorname{mult}_p, v]$, and for any line M through p let

$$\eta_M := \rho_S\left(\frac{1}{Q^5}(g^5)^*M\right) \quad \text{and} \quad c_M := \frac{\operatorname{mult}_p((g^5)^*M)}{Q^5}.$$

Since $A(v) = 1 + t < 12$, Proposition 8.8 with g^5 in place of g, Q^5 in place of Q, $\epsilon = 10^{-10}$ (see (9.23)) and $\kappa = 12$ implies that

$$\|\eta_M\|^2 \le 1 + 10\left(10^{-10} + Q^{-5}\right) < 1 + 2 \cdot 10^{-9}. \tag{9.24}$$

(a) By the first sentence of Theorem 8.2 with $D = Q^{-5}(g^5)^*M$ and $\mu \in \{\operatorname{mult}_p, v\} \subset \Gamma_S$,

$$\langle \delta_{\operatorname{mult}_p}, \eta_M\rangle = c_M \quad \text{and} \quad \langle \delta_v, \eta_M\rangle = \frac{v((g^5)^*M)}{Q^5} = \frac{w_5(M)}{Q^5}.$$

(b) Since $v \wedge \operatorname{mult}_p = \operatorname{mult}_p$ and $\alpha(\operatorname{mult}_p) = 1$ (see Theorem 2.9(1)), Definition 8.1 implies that

$$\|\delta_v - \delta_{\operatorname{mult}_p}\|^2 = \alpha(v) - 2\alpha(\operatorname{mult}_p) + \alpha(\operatorname{mult}_p) = t - 1 \quad \text{and} \quad \langle \delta_v - \delta_{\operatorname{mult}_p}, \delta_{\operatorname{mult}_p}\rangle = 0.$$

By (9.17), $(g^5)_*\omega = \lambda^5\omega$, and $\omega(M) \ge \omega(\mathfrak{m}_p) = 1$,

$$\operatorname{mult}_p((g^5)^*M) \ge \alpha(\omega)^{-1}\omega((g^5)^*M) = \alpha(\omega)^{-1}\lambda^5\omega(M) \ge \frac{\lambda^5}{\alpha(\omega)},$$

so $c_M \geq 1 - 173\sigma$ as in (9.20). Since $(g^5)^*M$ is a curve of degree Q^5, $c_M \leq 1$. Thus

$$1 - c_M^2 \leq 2(1 - c_M) \leq 346\sigma.$$

Let $\eta'_M := \eta_M - c_M \delta_{\mathrm{mult}_p}$. By (a) and $\|\delta_{\mathrm{mult}_p}\|^2 = 1$, $\langle \delta_{\mathrm{mult}_p}, \eta'_M \rangle = 0$, so $\|\eta'_M\|^2 = \|\eta_M\|^2 - c_M^2$, and by (b),

$$\langle \delta_v - \delta_{\mathrm{mult}_p}, \eta_M \rangle = \langle \delta_v - \delta_{\mathrm{mult}_p}, \eta'_M \rangle.$$

Thus by (a), (b), the Cauchy–Schwarz inequality (the form is positive definite by Definition 8.1), and (9.24),

$$\begin{aligned} \frac{w_5(M)}{Q^5} = c_M + \langle \delta_v - \delta_{\mathrm{mult}_p}, \eta'_M \rangle &\leq c_M + \sqrt{t-1}\sqrt{\|\eta_M\|^2 - c_M^2} \\ &\leq 1 + \sqrt{t-1}\sqrt{346\sigma + 2 \cdot 10^{-9}}. \end{aligned} \tag{9.25}$$

Now assume that $t < 2$. Since $a \deg \phi \leq b$, $\deg \phi \leq t < 2$, so ϕ is a scalar multiple of x. Thus x and $y - \phi(x)$ are affine linear coordinates centered at p, and $P_0 := \{y - \phi(x) = 0\}$ is a line through p. By Lemma 8.6 with g^5 in place of g, Q^5 in place of Q, the coordinates x and $y - \phi(x)$, $\beta = t$, and $\{y - \phi(x) = 0\} = P_0$ in place of C,

$$w_5(M) = v((g^5)^*M) \leq Q^5 + (t-1)\,\mathrm{mult}_{P_0}((g^5)^*M).$$

If $P_0 \subset \mathrm{Supp}(g^5)^*M$, then $g^5(P_0) = M$, so $\mathrm{mult}_{P_0}((g^5)^*M) = e_{P_0}(g^5) < 10^{-10}Q^5$ by **Step 3**. Thus

$$w_5(M) \leq Q^5\left(1 + 10^{-10}(t-1)\right) \quad \text{if } t < 2. \tag{9.26}$$

Step 5. In this step we bound $m_5\alpha(\nu_5)$.

Let

$$\beta := \max\{\nu_5(M) \mid M \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

Then $\beta \geq \nu_5(\mathfrak{m}_p) = 1$ and, since $w_5 = m_5\nu_5$,

$$m_5\beta = \max\{w_5(M) \mid M \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

Assume that $t < 2$. By (9.20), (9.26), and (9.14),

$$\beta \leq \frac{1 + 10^{-10}(t-1)}{1 - 173\sigma}.$$

By Lemma 8.5(2) with $\tau = \nu_5$ and by (9.19), $\alpha(\nu_5) - 1 \leq (\beta - 1)(A(\nu_5) - 1) \leq 4(\beta - 1)$. Thus by (9.26) and (9.20),

$$\begin{aligned} m_5\alpha(\nu_5) &\leq 4m_5\beta - 3m_5 \leq Q^5\left(4\left(1 + 10^{-10}(t-1)\right) - 3(1 - 173\sigma)\right) \\ &= Q^5\left(1 + 4 \cdot 10^{-10}(t-1) + 519\sigma\right). \end{aligned} \tag{9.27}$$

Assume that $t \geq 2$, so $1 \leq t - 1 < 10$. By (9.14), $346\sigma + 2 \cdot 10^{-9} < 4 \cdot 10^{-6}$ and $173\sigma < 2 \cdot 10^{-6}$. Thus by (9.20), (9.25), and $\sqrt{t-1} \geq 1$,

$$\beta - 1 \leq \frac{173\sigma + \sqrt{t-1}\sqrt{346\sigma + 2 \cdot 10^{-9}}}{1 - 173\sigma} < \sqrt{t-1} \cdot \frac{2 \cdot 10^{-6} + 2 \cdot 10^{-3}}{1 - 2 \cdot 10^{-6}} < \frac{\sqrt{t-1}}{400}.$$

By Lemma 8.5(2) with $\tau = \nu_5$, by (9.19), and since $\frac{22}{400} < \frac{1}{16}$ and $\sqrt{t-1} \leq t - 1$,

$$\alpha(\nu_5) - 1 \leq (\beta - 1)(A(\nu_5) - 1) \leq 22(\beta - 1) < \frac{22}{400}\sqrt{t-1} \leq \frac{t-1}{16}.$$

Thus by (9.21),

$$m_5\alpha(\nu_5) = \left(m_5^2\alpha(\nu_5)\right)^{1/2}\alpha(\nu_5)^{1/2} \leq Q^5\sqrt{t\alpha(\nu_5)} \leq Q^5\sqrt{t\left(1 + \frac{t-1}{16}\right)}. \tag{9.28}$$

Step 6. In this step we show that $A(w_n) < Q^n(1+t)$, which contradicts (9.13) since $w_n = (g^n)_*v$.

Since $\alpha(\nu_5) < 22$, Lemma 2.10(1) and (9.17) imply that

$$w_5 = m_5\nu_5 \le m_5\alpha(\nu_5)\,\mathrm{mult}_p \le m_5\alpha(\nu_5)\omega.$$

Since $w_n = (g^{n-5})_* w_5$ and $n-5 \ge 1$, Lemma 2.4 and (9.16) imply that

$$\begin{aligned} A(w_n) &= A(w_5) + w_5(R(g^{n-5})) \\ &\le A(w_5) + m_5\alpha(\nu_5)(\lambda^{n-5}-1)A(\omega) \\ &< A(w_5) + m_5\alpha(\nu_5)\lambda^{n-5}A(\omega). \end{aligned}$$

Since $\lambda/Q = 1-\sigma$,

$$\frac{A(w_n)}{Q^n} < \frac{A(w_5)}{Q^n} + (1-\sigma)^n \cdot \frac{A(\omega)}{(1-\sigma)^5} \cdot \frac{m_5\alpha(\nu_5)}{Q^5}. \tag{9.29}$$

We bound the terms of (9.29). Since $n \ge 8$ and $Q^2 > 15548 = 92 \cdot 169$, (9.22) and (9.18) imply that

$$\frac{A(w_5)}{Q^n} < 92Q^{5-n} \le \frac{92}{Q^3} < \frac{1}{169Q} < \sigma. \tag{9.30}$$

Since $(1-\sigma)^5 \ge 1-5\sigma$ and $(1-5\sigma)^{-1} \le 1+6\sigma$ for $\sigma \le \frac{1}{30}$, (9.15) and (9.14) imply that

$$\frac{A(\omega)}{(1-\sigma)^5} < (2+168\sigma)(1+6\sigma) = 2+180\sigma+1008\sigma^2 < 2+200\sigma. \tag{9.31}$$

Assume that $t < 2$. By (9.27), (9.31), $t-1<1$, and $\sigma < 10^{-8}$,

$$\begin{aligned} \frac{A(\omega)}{(1-\sigma)^5} \cdot \frac{m_5\alpha(\nu_5)}{Q^5} &< (2+200\sigma)\left(1+4\cdot 10^{-10}(t-1)+519\sigma\right) \\ &= 2+8\cdot 10^{-10}(t-1)+1238\sigma+\left(8\cdot 10^{-8}(t-1)+103800\sigma\right)\sigma \\ &< 2+(t-1)+1300\sigma \le (1+t)(1+650\sigma), \end{aligned}$$

where the last inequality uses $1+t \ge 2$. By Bernoulli's inequality, $(1-\sigma)^{-n} \ge (1+\sigma)^n \ge 1+n\sigma$. Since $n\sigma < 3$ by (9.14) and $n-650 > 4$,

$$(1-\sigma)^n(1+650\sigma) \le \frac{1+650\sigma}{1+n\sigma} = 1 - \frac{(n-650)\sigma}{1+n\sigma} < 1-\sigma.$$

Thus by (9.29) and (9.30),

$$\frac{A(w_n)}{Q^n} < \sigma + (1-\sigma)(1+t) = 1+t-t\sigma < 1+t.$$

Assume that $t \ge 2$, so $t-1 \ge 1$. Since

$$(1+t)^2 - 4t\left(1+\frac{t-1}{16}\right) = \frac{(t-1)(3t-4)}{4} \ge \frac{(t-1)^2}{2} \ge \frac{(1+t)^2}{18},$$

where the last two inequalities use $t-1 \ge 1$, we have $4t\left(1+\frac{t-1}{16}\right) \le \frac{17}{18}(1+t)^2$. Since

$$1+100\sigma < \frac{101}{100} \quad \text{and} \quad \left(\frac{101}{100}\right)^2 \cdot \frac{17}{18} < \left(\frac{99}{100}\right)^2 \quad \text{(that is, } 173417 < 176418\text{)},$$

(9.28) and (9.31) imply that

$$\frac{A(\omega)}{(1-\sigma)^5} \cdot \frac{m_5\alpha(\nu_5)}{Q^5} < (2+200\sigma)\cdot\frac{1}{2}\sqrt{\frac{17}{18}}(1+t) = (1+100\sigma)\sqrt{\frac{17}{18}}(1+t) < \frac{99}{100}(1+t).$$

Since $(1-\sigma)^n < 1$ and $\sigma < 10^{-8} < \frac{1+t}{100}$, (9.29) and (9.30) imply that

$$\frac{A(w_n)}{Q^n} < \sigma + \frac{99}{100}(1+t) < 1+t.$$

Thus $A((g^n)_*v) = A(w_n) < Q^n(1+t)$ both for $t<2$ and for $t \geq 2$, which contradicts (9.13). The theorem follows. □

10. Reduction to a monomial valuation transverse to an invariant line

In this section we study $(\mathbb{P}^2, B_{17280n})$ near a closed point $p \in \mathrm{Nlc}(\mathbb{P}^2, B_{17280n})$.

10.1. The invariant line.

Construction-Theorem 10.1. Notation and conditions as in Set-up 5.1. Let $n \geq 4 \cdot 10^8$ be an integer and let v be a divisorial valuation centered at a closed point p with $v(B_{17280n}) > A(v)$. Then $g(p) = p$, exactly one irreducible component L of C contains p, and L is a line with $g(L) = L$. Let $e := e_L(g)$, let J be the divisor with $g^*L = eL + J$, let r be the local degree of $g|_L : L \to L$ at p, and let $\lambda := c_\infty(g,p)$. Then:

(1) $1 \leq (J \cdot L)_p \leq Q - e$, $e < r \leq Q$, $r + (J \cdot L)_p > Q$, and $\frac{9}{10}Q < \lambda < Q$.
(2) For any irreducible curve $P \neq L$ through p, $e_P(g) \leq \frac{1}{100}Q$.
(3) Any eigenvaluation of g at p for which (a) or (b) of Theorem 2.12(1) holds is quasimonomial or infinitely singular.

Proof. **Step 1.** Since p is the center of v and

$$a(v, \mathbb{P}^2, B_{17280n}) = A(v) - v(B_{17280n}) < 0,$$

we have $p \in \mathrm{Nlc}(\mathbb{P}^2, B_{17280n})$. By Theorem 9.1, $p \in C$. Thus by Proposition 6.1(3), $g(p) = p$, exactly one irreducible component L of C contains p, and L is a line. By Proposition 5.4(2), $f^{288}(L) = L$. Since $g = f^{17280} = (f^{288})^{60}$, we have $g(L) = L$. By Theorem 7.1(1), $\frac{9}{10}Q < \lambda < Q$.

Step 2. We prove (2). Let $P \neq L$ be an irreducible curve through p. Since L is the only irreducible component of C containing p, P is not a component of C.

Assume first that $f^{6k}(P) \neq P$ for $k = 1, 2, 3$. By Proposition 5.4(1) for the integer 2880, $\mathrm{mult}_P B_{17280} < \frac{26}{2881}$. Thus, by Definition 2.2,

$$e_P(g) = 1 + (Q-1)\,\mathrm{mult}_P B_{17280} < 1 + \frac{26}{2881}Q \leq \frac{1}{100}Q,$$

where the last inequality is equivalent to $288100 \leq 281Q$, which holds as $Q = q^{17280} \geq 2^{17280}$.

Assume next that $f^{6k}(P) = P$ for some $k \in \{1,2,3\}$. Since $6k \mid 288$, $f^{288}(P) = P$. Since P is not a component of C, $\mathrm{mult}_P B_{288} \leq \frac{3}{5}$, so

$$e_P(f^{288}) = 1 + (q^{288} - 1)\,\mathrm{mult}_P B_{288} \leq 1 + \frac{3}{5}(q^{288} - 1) \leq \frac{7}{10}q^{288},$$

where the last inequality holds as $q^{288} \geq 4$. Since $f^{288}(P) = P$ and ramification indices are multiplicative under composition,

$$e_P(g) = e_P(f^{288})^{60} \leq \left(\frac{7}{10}\right)^{60} Q < \frac{1}{100}Q,$$

as $(\frac{7}{10})^7 < \frac{1}{10}$. Thus (2) holds.

Step 3. We prove (1). Since $\deg g^*L = Q$, we have $e \leq Q$ and $\deg J = Q - e$. Since $L \not\subset \mathrm{Supp}\, J$, by Bezout's theorem,

$$(J \cdot L)_p \leq \deg J = Q - e.$$

Since $g^*\mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_L(Q)$, the morphism $g|_L : L \to L$ has degree Q, so $1 \leq r \leq Q$.

We show that $r + (J \cdot L)_p > Q$. For any integer $k \geq 1$, write $R(g^k) = (e^k - 1)L + R'_k$, where $L \not\subset \mathrm{Supp}\, R'_k$. Since $g(p) = p$, Lemma 3.3 for g, with L and p in place of C and x, has $c_i = \left(\frac{r}{Q}\right)^i$ and $\alpha_i = \frac{(J \cdot L)_p}{Q}$ for any $i \geq 0$. Multiplying the formula in Lemma 3.3(2) by Q^k, we obtain

$$1 + \mathrm{mult}_p(R'_k|_L) = a_k := r^k + (J \cdot L)_p \sum_{i=0}^{k-1} e^{k-1-i} r^i,$$

and if $a_k \le Q^k$, then $(\mathbb{P}^2, \frac{1}{Q^k-1}R(g^k))$ is lc at p by Lemma 3.3(3). Since $B_{17280n} = \frac{1}{Q^n-1}R(g^n)$ and, by **Step 1**, $p \in \mathrm{Nlc}(\mathbb{P}^2, B_{17280n})$, we have

$$a_n > Q^n.$$

Let $a_0 := 1$. Then $a_{k+1} = ra_k + (J \cdot L)_p e^k$ for any integer $k \ge 0$. Suppose that $r + (J \cdot L)_p \le Q$. If $a_k \le Q^k$ for some $k \ge 0$, then since $e \le Q$,

$$a_{k+1} \le rQ^k + (J \cdot L)_p Q^k \le Q^{k+1}.$$

By induction on k, $a_n \le Q^n$, a contradiction. Thus $r + (J \cdot L)_p > Q$. Since $r \le Q$ and $(J \cdot L)_p \le Q - e$, we obtain

$$(J \cdot L)_p > Q - r \ge 0 \quad \text{and} \quad r > Q - (J \cdot L)_p \ge e.$$

Together with **Step 1**, (1) holds.

Step 4. We prove (3). Let ω be an eigenvaluation of g at p for which (a) or (b) of Theorem 2.12(1) holds. If (a) holds, then ω is quasimonomial. Suppose that (b) holds and ω is not infinitely singular. Then $\omega = \nu_\psi$ for an irreducible analytic curve germ ψ at p, and there exists a normalized valuation $\mu < \nu_\psi$ with $c(g, \nu) = \lambda$ and $g_\bullet \nu > \nu$ for any $\nu \in [\mu, \nu_\psi)$. Since $\lambda > \frac{9}{10}Q > 1$, by Lemma 7.6, ψ is a branch at p of an irreducible curve $P \subset \mathbb{P}^2$ with $g(P) = P$ and

$$e_P(g) \ge \lambda > \frac{1}{100}Q.$$

By (2), $P = L$. Since L is smooth at p, ψ is the germ of L at p, so $\nu_\psi = \nu_L$.

Since $g(L) = L$ and L is smooth at p, $c(g, \nu_L) = \nu_L(g^*\mathfrak{m}_p) = \deg_p(g|_L) = r$, where $\nu_L(\xi)$ is the order of $\xi|_L$ at p (Definition 2.8). Since ν_L is an eigenvaluation of g at p, $c(g, \nu_L) = c_\infty(g, p) = \lambda$ by Definition 2.11, so $\lambda = r$. Since $P = L$, $e = e_L(g) \ge \lambda = r$, which contradicts $e < r$ in (1). Thus (3) holds. □

10.2. **A monomial normal form.** Lemmas 10.2 and 10.3 bound the values of the ramification divisors of the iterates of an endomorphism of $\mathbb{P}^2$ preserving a line L by their contact with L.

Lemma 10.2. *Let $p \in \mathbb{P}^2$ be a closed point and L a line through p. Then:*

(1) For any normalized valuation ν at p with $\alpha(\nu) < +\infty$ and any $\mathbb{R}$-divisor $D \ge 0$ on $\mathbb{P}^2$ with $L \not\subset \operatorname{Supp} D$,

$$\nu(D) \le \frac{\alpha(\nu)}{\nu(L)}(D \cdot L)_p.$$

(2) If g is an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 2$ with $g(L) = L$, and we write $e := e_L(g)$, $g^*L = eL + J$, and $R(g) = (e-1)L + R'$, then

$$(R' \cdot L)_p = r - 1 + (J \cdot L)_p \le 2Q - e - 1,$$

where r is the local degree of $g|_L$ at p.

Proof. (1) Both sides are additive in D, and a prime divisor not through p contributes 0 to both sides. For an irreducible curve $P \ne L$ through p with local equation ψ, Lemma 2.10(2) for a local equation of L and ψ gives $\nu(\psi)\nu(L) \le \alpha(\nu)(\psi \cdot L)_p$. Since $\nu(L) \ge \nu(\mathfrak{m}_p) = 1$, (1) follows.

(2) The equality is Lemma 3.3(2) for g, with L, p, and 1 in place of C, x, and n, where $R'_1 = R'$, $c_1 = \frac{r}{Q}$, $\alpha_0 = \frac{(J \cdot L)_p}{Q}$, and $\mathrm{mult}_p(R'|_L) = (R' \cdot L)_p$. Lemma 3.3 does not require $g(p) = p$. Since $g|_L$ has degree Q, $r \le Q$, and by Bezout's theorem $(J \cdot L)_p \le \deg J = Q - e$. The inequality follows. □

Lemma 10.3. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 2$, L a line with $g(L) = L$, $e := e_L(g)$, and w a divisorial valuation centered at a closed point $p \in L$. Let $m := w(\mathfrak{m}_p)$, $\nu := w/m$, $\beta := \nu(L)$, and $\zeta := m^2\alpha(\nu)/w(L)$. Then $A(w) - w(L) - \zeta \ge 0$, and for any integer $j \ge 1$:*

(1) $\nu(R(g^j)) \le (e^j - 1)\beta + (2Q^j - e^j - 1)\alpha(\nu)/\beta$.

(2) If $\nu(R(g^j)) > (Q^j-1)A(\nu)$, then $\beta A(\nu) < 2\alpha(\nu)$.
(3) $A((g^j)_*w) \le e^j w(L) + (2Q^j - e^j)\zeta + A(w) - w(L) - \zeta$.

Proof. By Definition 2.8, ν is a divisorial normalized valuation at p. By Theorem 2.9(4) and the homogeneity in Definition 2.1, $A(w) = mA(\nu)$. By Theorem 2.9(3),

$$A(\nu) \ge 1 + \alpha(\nu);$$

in particular, $\alpha(\nu) < +\infty$. By Lemma 2.10(1), $1 \le \beta \le \alpha(\nu)$. Since $w(L) = m\beta$ and $\zeta = m\alpha(\nu)/\beta$,

$$\begin{aligned} A(w) - w(L) - \zeta &= m\Big(A(\nu) - \beta - \frac{\alpha(\nu)}{\beta}\Big) \\ &\ge m\Big(1 + \alpha(\nu) - \beta - \frac{\alpha(\nu)}{\beta}\Big) = \frac{m(\beta-1)(\alpha(\nu)-\beta)}{\beta} \ge 0. \end{aligned}$$

Fix an integer $j \ge 1$. Since $g(L) = L$, $e_L(g^j) = e^j$. Write

$$(g^j)^*L = e^j L + J_j \quad \text{and} \quad R(g^j) = (e^j - 1)L + R'_j,$$

where $L \not\subset \operatorname{Supp} R'_j$. By Lemma 10.2(2) for g^j, which has degree Q^j,

$$(R'_j \cdot L)_p \le 2Q^j - e^j - 1.$$

By Lemma 10.2(1) for ν and R'_j,

$$\nu(R(g^j)) = (e^j-1)\beta + \nu(R'_j) \le (e^j-1)\beta + (2Q^j - e^j - 1)\frac{\alpha(\nu)}{\beta}.$$

Thus (1) holds.

(2) Let $\gamma := \frac{e^j-1}{Q^j-1}$. Since $1 \le e \le \deg g^*L = Q$, we have $0 \le \gamma \le 1$. Since $\frac{2Q^j-e^j-1}{Q^j-1} = 2 - \gamma$, (1) implies

$$\frac{\nu(R(g^j))}{Q^j-1} \le \gamma\Big(\beta + \frac{\alpha(\nu)}{\beta}\Big) + (1-\gamma)\frac{2\alpha(\nu)}{\beta}.$$

By the first paragraph,

$$\beta + \frac{\alpha(\nu)}{\beta} \le 1 + \alpha(\nu) \le A(\nu).$$

If $\beta A(\nu) \ge 2\alpha(\nu)$, then $\frac{2\alpha(\nu)}{\beta} \le A(\nu)$, so $\nu(R(g^j)) \le (Q^j-1)A(\nu)$, a contradiction. Thus $\beta A(\nu) < 2\alpha(\nu)$.

(3) Since $w(R(g^j)) = m\nu(R(g^j))$, $m\beta = w(L)$ and $m\alpha(\nu)/\beta = \zeta$, by Lemma 2.4 and (1),

$$\begin{aligned} A((g^j)_*w) &= A(w) + w(R(g^j)) \le A(w) + (e^j-1)w(L) + (2Q^j - e^j - 1)\zeta \\ &= e^j w(L) + (2Q^j - e^j)\zeta + A(w) - w(L) - \zeta. \end{aligned}$$

□

Construction-Theorem 10.4. Notation and conditions as in Set-up 5.1. Let $n \ge 10^{16}$ be an integer such that $(\mathbb{P}^2, B_{17280n})$ is not lc. Then there exist a closed point p, affine linear coordinates x, y centered at p, a rational number $1 < t < 11$, and a polynomial $\phi \in \mathbb{C}[x]$ with $\phi(0) = 0$ and $\deg\phi \le t$ such that $v := \mathrm{mon}_{x,y-\phi(x)}(1,t)$ and $L := \{x = 0\}$ satisfy:

(1) $v(B_{17280n}) > A(v) = 1 + t$, $A((g^n)_*v) > Q^n(1+t)$, $\alpha(v) = t$, and $v(\mathfrak{m}_p) = v(L) = 1$.
(2) The conclusions of Construction-Theorem 10.1 hold for n and v, with this L.

Let $\sigma := 1 - \lambda/Q$, and for any integer $b \ge 0$ let $w_b := (g^b)_*v$, $m_b := w_b(\mathfrak{m}_p)$, and $\nu_b := w_b/m_b$.

Proof. **Step 1.** We find a monomial valuation v with $v(B_{17280n}) > A(v)$. Since $(\mathbb{P}^2, B_{17280n})$ is not lc, there exists an exceptional prime divisor E over $\mathbb{P}^2$ with negative log discrepancy [KM98, Corollary 2.31]. The center of E is a closed point p, and the divisorial valuation v' defined by E satisfies $v'(B_{17280n}) > A(v')$.

Since $17280n = 6 \cdot 2880n$, by Proposition 4.9 for f, the integer $2880n$ and v', there exist affine linear coordinates x', y' centered at p, a polynomial $\phi' \in \mathbb{C}[x']$ with $\phi'(0) = 0$, and

positive integers $a' \le b' < 11a'$ such that $a' \deg \phi' \le b'$ and $w' := \mathrm{mon}_{x',y'-\phi'(x')}(a', b')$ satisfies $w'(B_{17280n}) > A(w')$. Let

$$\eta := y' - \phi'(x'), \quad t := \frac{b'}{a'}, \quad \text{and} \quad v := \frac{w'}{a'} = \mathrm{mon}_{x',\eta}(1, t).$$

Since a', b' are positive integers, t is a rational number with $1 \le t < 11$. Since $\phi'(0) = 0$, x', η are regular parameters at p. By Definition 2.8, v is a divisorial normalized valuation at p: it is a positive multiple of the divisorial valuation of the weighted blow-up with weights (a', b') in the parameters x', η, and since $(\mathbb{P}^2, \{x'\eta = 0\})$ is log smooth near p and w' is monomial in x', η, $A(w') = w'(x'\eta) = a' + b'$, so by the homogeneity in Definition 2.1, $A(v) = A(w')/a' = 1 + t$. Hence

$$v(B_{17280n}) = \frac{w'(B_{17280n})}{a'} > \frac{A(w')}{a'} = A(v).$$

Since $\mathfrak{m}_p = (x', \eta)$, $v(\mathfrak{m}_p) = \min\{1, t\} = 1$. For any $\psi \in \mathfrak{m}_p \backslash \{0\}$ with expansion $\psi = \sum_{i,j} a_{ij} x'^i \eta^j$, since $i + tj \le t(i + j)$,

$$v(\psi) = \min\{i + tj \mid a_{ij} \ne 0\} \le t \min\{i + j \mid a_{ij} \ne 0\} = t \,\mathrm{mult}_p \psi,$$

and $v(\eta) = t = t \,\mathrm{mult}_p \eta$. Thus $\alpha(v) = t$.

Step 2. We show that $v(L) = 1$ and $t > 1$. Since $n \ge 10^{16} > 4 \cdot 10^8$, Construction-Theorem 10.1 for n and v implies that $g(p) = p$, that exactly one irreducible component L of C contains p, and that L is a line with $g(L) = L$. Let e, J, r, and λ be as in Construction-Theorem 10.1. Let ℓ be an affine linear equation of L. Then $\ell = \ell_1 x' + \ell_2 y'$ for some $(\ell_1, \ell_2) \in \mathbb{C}^2 \backslash \{(0, 0)\}$, so

$$\ell = \ell_0(x') + \ell_2 \eta, \quad \ell_0(x') := \ell_1 x' + \ell_2 \phi'(x') \in \mathbb{C}[x'].$$

Any monomial x'^k of ℓ_0 satisfies $1 \le k \le t$, since $k = 1$ or $k \le \deg \phi' \le b'/a' = t$. For regular parameters z_1, z_2 at p, we call $i + tj$ the *weight* of $z_1^i z_2^j$. Since $v = \mathrm{mon}_{x',\eta}(1, t)$, $v(\ell)$ is the smallest weight of the non-zero terms of $\ell_0(x') + \ell_2 \eta$, where x'^k has weight k and η has weight t. Thus

$$v(L) = v(\ell) \in \{1, \dots, \lfloor t \rfloor\} \cup \{t\}.$$

Since $v(R(g^n)) = (Q^n - 1) v(B_{17280n}) > (Q^n - 1) A(v)$, Lemma 10.3(2) for g, L, $w := v$, and $j := n$, together with **Step 1**, gives

$$v(L)(1 + t) < 2t.$$

If $v(L) \ge 2$, then $v(L)(1 + t) \ge 2 + 2t > 2t$. If $v(L) = t$, then $v(L)(1 + t) - 2t = t(t - 1) \ge 0$. Thus $v(L) = 1$, and $1 + t < 2t$, that is, $t > 1$.

Step 3. We change coordinates. We use the following observation.

(†) Let z_1, z_2 be regular parameters at p with $v = \mathrm{mon}_{z_1,z_2}(1, t)$, let $z_1', z_2' \in \mathcal{O}_{\mathbb{P}^2,p}$, and let $d_1, d_2 \in \mathbb{C}^*$. Assume that in the expansion in $\widehat{\mathcal{O}}_{\mathbb{P}^2,p} = \mathbb{C}[[z_1, z_2]]$, $z_1' - d_1 z_1$ is a sum of monomials of weight > 1 and $z_2' - d_2 z_2$ is a sum of monomials of weight $> t$. Then z_1', z_2' are regular parameters at p and $v = \mathrm{mon}_{z_1',z_2'}(1, t)$.

We prove (†). Since $t > 1$, $z_1' - d_1 z_1 \in \mathbb{C} z_2 + \mathfrak{m}_p^2$ and $z_2' - d_2 z_2 \in \mathfrak{m}_p^2$, so z_1', z_2' are regular parameters at p. Since $z_1' - d_1 z_1$ and $z_2' - d_2 z_2$ consist of monomials of weight > 1 and $> t$, for any $\psi = \sum_{i,j} a_{ij} z_1'^i z_2'^j \ne 0$ the part of ψ of weight $\le W$ in z_1, z_2, where $W := \min\{i + tj \mid a_{ij} \ne 0\}$, is $\sum_{i+tj=W} a_{ij} d_1^i d_2^j z_1^i z_2^j \ne 0$, so (†) holds.

Let $x := \ell$. By **Step 2**, $v(\ell) = 1$, and x' is the only monomial in x', η of weight 1, so the coefficient d of x' in ℓ_0 is non-zero. Then $x - dx' = (\ell_0(x') - dx') + \ell_2 \eta$ is a sum of monomials x'^k with $k \ge 2$ and of η, all of weight > 1. By (†) for $z_1 = x'$, $z_2 = z_2' = \eta$ and $z_1' = x$, $v = \mathrm{mon}_{x,\eta}(1, t)$.

Let y be an affine linear coordinate centered at p such that x, y are linearly independent. In $\widehat{\mathcal{O}}_{\mathbb{P}^2,p} = \mathbb{C}[[x, \eta]]$, write

$$y = \chi(x) + \eta u(x, \eta), \quad \chi \in \mathbb{C}[[x]], \ u \in \mathbb{C}[[x, \eta]].$$

Since $y(p)=0$, $\chi(0)=0$. Since x,y are regular parameters at p and $y\equiv\chi'(0)x+u(0,0)\eta \bmod \mathfrak{m}_p^2$, we have $u(0,0)\neq 0$. Let $\phi\in\mathbb{C}[x]$ be the sum of the terms of χ of degree at most $\lfloor t\rfloor$. Then $\phi(0)=0$, $\deg\phi\le t$, and

$$y-\phi(x)-u(0,0)\eta=(\chi(x)-\phi(x))+\eta\big(u(x,\eta)-u(0,0)\big)$$

is a sum of monomials x^k with $k>t$ and $x^i\eta^j$ with $j\ge 1$ and $i+j\ge 2$, all of weight $>t$. By (†) for $z_1=z_1'=x$, $z_2=\eta$ and $z_2'=y-\phi(x)$,

$$v=\mathrm{mon}_{x,y-\phi(x)}(1,t).$$

Step 4. We conclude. By **Steps 1 and 2**, $1<t<11$ and t is rational, and since $x=\ell$, $L=\{x=0\}$. By **Steps 1 and 2**, $v(B_{17280n})>A(v)=1+t$, $\alpha(v)=t$, and $v(\mathfrak{m}_p)=v(L)=1$. Since $B_{17280n}=\frac{1}{Q^n-1}R(g^n)$ and v is a divisorial valuation (**Step 1**), Lemma 2.4 for g^n gives

$$A((g^n)_*v)=A(v)+v(R(g^n))>Q^nA(v)=Q^n(1+t).$$

Thus (1) holds. By **Step 2**, the conclusions of Construction-Theorem 10.1 hold for n and v, with $L=\{x=0\}$, which is (2). □

11. Eigenvaluations at a fixed point of the invariant line

We prove the following theorem in Subsection 11.4.

Theorem 11.1. *Notation and conditions as in Construction-Theorem 10.4. Then:*

(1) $0<\sigma<\frac{\log 32}{n}$.

(2) Any eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds satisfies $1<\omega(L)<12$, $A(\omega)<32$, and

$$0\le\alpha(\omega)-\omega(L)\le A(\omega)-\omega(L)-1<30\sigma.$$

(3) If $n\ge 2^{100}$, then any ω as in (2) satisfies

$$\left(1-\frac{e}{Q}\right)(\omega(L)-1)\le\sigma\,(\omega(L)+1)\,.$$

(4) For any integer $j\ge 0$, $A(w_j)<11000m_j$.

11.1. Ramification off the invariant line.

Definition-Lemma 11.2. Notation and conditions as in Construction-Theorem 10.1. For any integer $j\ge 1$, write $(g^j)^*L=e^jL+J_j$. Let $E_1:=J_{\mathrm{red}}$ and let $D_1:=\sum_P(e_P(g)-1)\,P$, where the sum is over the irreducible curves $P\subset\mathbb{P}^2$ with $g(P)\neq L$. Then $J_j\ge 0$ and $L\not\subset\operatorname{Supp}J_j$ for any integer $j\ge 1$, and

$$R(g)-g^*L+L=D_1-E_1\quad\text{and}\quad\deg D_1=2(Q-1)+\deg E_1.$$

Moreover, $\mathrm{mult}_P D_1\le\frac{1}{100}(Q-1)$ and $\mathrm{mult}_P(g^j)^*M\le 100^{-j}Q^j$ for any integer $j\ge 1$, any irreducible curve P through p, and any line $M\neq L$ through p.

Proof. Fix an integer $j\ge 1$. For any irreducible curve $P\subset\mathbb{P}^2$, Definition 2.2 implies

$$\mathrm{mult}_P(g^j)^*L=\begin{cases}e_P(g^j) & \text{if } g^j(P)=L,\\ 0 & \text{if } g^j(P)\neq L,\end{cases}\qquad \mathrm{mult}_P R(g^j)=e_P(g^j)-1.$$

Since $e_L(g^j)=e^j$ by Definition 2.2, we have

$$J_j=\sum_{P\neq L,\ g^j(P)=L}e_P(g^j)P\ge 0,$$

so $L\not\subset\operatorname{Supp}J_j$.

We compare the coefficients of $R(g)-g^*L+L$ and D_1-E_1 along an irreducible curve P.

- If $P=L$, the first coefficient is $(e-1)-e+1=0$, and L is a component of neither D_1 nor E_1.

- If $P \neq L$ and $g(P) = L$, the first coefficient is -1, P is a component of J, $\operatorname{mult}_P E_1 = 1$, and $\operatorname{mult}_P D_1 = 0$.
- If $g(P) \neq L$, the first coefficient is $e_P(g) - 1 = \operatorname{mult}_P D_1$, and $\operatorname{mult}_P E_1 = 0$.

Thus $R(g) - g^*L + L = D_1 - E_1$. Since $\deg g^*L = Q$ and $\deg R(g) = 3(Q-1)$, we have $\deg D_1 - \deg E_1 = 2(Q-1)$.

Let P be an irreducible curve through p. If $g(P) \neq L$, then $P \neq L$, so by Construction-Theorem 10.1(2),

$$\operatorname{mult}_P D_1 = e_P(g) - 1 \leq \frac{Q}{100} - 1 \leq \frac{1}{100}(Q-1).$$

If $g(P) = L$, then $\operatorname{mult}_P D_1 = 0$.

Let P be an irreducible curve through p with $g^j(P) \neq L$. For any integer $0 \leq i \leq j-1$, $g^i(P)$ is an irreducible curve through p because $g(p) = p$, and $g^i(P) \neq L$ because otherwise $g^j(P) = g^{j-i}(L) = L$. By Construction-Theorem 10.1(2) and Definition 2.2,

$$e_P(g^j) \leq 100^{-j} Q^j.$$

Let $M \neq L$ be a line through p. If P is a component of $(g^j)^*M$, then $g^j(P) = M \neq L$, so $\operatorname{mult}_P (g^j)^*M = e_P(g^j) \leq 100^{-j}Q^j$. Otherwise, $\operatorname{mult}_P (g^j)^*M = 0$. The lemma follows. □

Lemma 11.3. *Notation and conditions as in Construction-Theorem 10.1 and Definition-Lemma 11.2. Let $\kappa \geq 2$ and S a finite set of normalized valuations at p with $A(\nu) \leq \kappa$ for any $\nu \in S$. Then for any integer $b \geq 1$,*

$$\left\| \rho_S \left(\frac{1}{Q^b} J_b \right) \right\|^2 \leq \left(1 - \frac{e^b}{Q^b} \right)^2 + \frac{\kappa - 2}{40b}.$$

Proof. **Step 1.** For any integer $i \geq 2$,

$$(g^i)^*L = (g^{i-1})^*(eL + J) = e(g^{i-1})^*L + (g^{i-1})^*J, \quad \text{so} \quad J_i = eJ_{i-1} + (g^{i-1})^*J.$$

Since $g^*L = eL + J$, we have $J_1 = J$, and induction on b implies

$$J_b = \sum_{i=1}^{b} e^{b-i} (g^{i-1})^* J. \tag{11.1}$$

For any $i \in \{1, \dots, b\}$, let $F_i := Q^{-b} e^{b-i} (g^{i-1})^* J$, so $\frac{1}{Q^b} J_b = \sum_{i=1}^b F_i$. Fix $i \in \{1, \dots, b\}$ and let P be an irreducible component of F_i. Then $g^{i-1}(P) \subset \operatorname{Supp} J$, so $g^{i-1}(P) \neq L$ by Definition-Lemma 11.2, and $g^i(P) = L$ because $\operatorname{Supp} g^*L = g^{-1}(L)$. If $g^k(P) = L$ for an integer $0 \leq k < i$, then $g^{i-1}(P) = g^{i-1-k}(L) = L$, which is impossible. Thus

$$i = \min\{k \in \mathbb{N} \mid g^k(P) = L\}. \tag{11.2}$$

In particular, for $i \neq j$, F_i and F_j have no common component.

Since $\deg J = Q - e$, we have

$$d_i := \deg F_i = Q^{-b} e^{b-i} Q^{i-1} (Q - e) = \left(1 - \frac{e}{Q}\right) \left(\frac{e}{Q}\right)^{b-i}.$$

Assume moreover that P passes through p. If $i \geq 2$, then by (11.1)

$$J_{i-1} = \sum_{k=1}^{i-1} e^{i-1-k} (g^{k-1})^* J,$$

so P is not a component of J_{i-1}, since any component P' of $(g^{k-1})^*J$ with $k \leq i-1$ is a component of F_k, so (11.2) for F_k implies $\min\{k' \in \mathbb{N} \mid g^{k'}(P') = L\} = k \neq i$. Since $J_1 = J$ and $J_i = eJ_{i-1} + (g^{i-1})^*J$ for $i \geq 2$, we have

$$\operatorname{mult}_P (g^{i-1})^* J = \operatorname{mult}_P J_i = \operatorname{mult}_P (g^i)^* L = e_P(g^i).$$

For any integer $0 \le k \le i-1$, $g^k(P)$ is an irreducible curve through p different from L, so Construction-Theorem 10.1(2) and Definition 2.2 imply $e_P(g^i) \le 100^{-i}Q^i$. Hence

$$\operatorname{mult}_P F_i \le Q^{-b}e^{b-i}100^{-i}Q^i = \epsilon_i := 100^{-i}\left(\frac{e}{Q}\right)^{b-i}.$$

Step 2. Since $F_i \ge 0$ and $\operatorname{mult}_P F_i \le \epsilon_i$ for any irreducible curve P through p, Theorem 8.2(1) implies

$$\|\rho_S(F_i)\|^2 \le d_i^2 + (\kappa-2)\epsilon_i d_i.$$

For $i \ne j$, F_i and F_j have no common component by **Step 1**, so Theorem 8.2(3) and Bezout's theorem imply

$$\langle \rho_S(F_i), \rho_S(F_j)\rangle \le (F_i \cdot F_j)_p \le F_i \cdot F_j = d_i d_j.$$

Since ρ_S is linear and $\frac{1}{Q^b}J_b = \sum_{i=1}^b F_i$, we have

$$\left\|\rho_S\left(\frac{1}{Q^b}J_b\right)\right\|^2 \le \sum_{i=1}^b d_i^2 + \sum_{i\ne j} d_i d_j + (\kappa-2)\sum_{i=1}^b \epsilon_i d_i = \left(\sum_{i=1}^b d_i\right)^2 + (\kappa-2)\sum_{i=1}^b \epsilon_i d_i.$$

Here $\sum_{i=1}^b d_i = \deg \frac{1}{Q^b}J_b = 1 - \frac{e^b}{Q^b}$ and

$$\sum_{i=1}^b \epsilon_i d_i = \left(1-\frac{e}{Q}\right)\sum_{i=1}^b 100^{-i}\left(\frac{e}{Q}\right)^{2(b-i)}.$$

Since $\kappa \ge 2$, it suffices to show $\sum_{i=1}^b \epsilon_i d_i < \frac{1}{40b}$.

Step 3. By Construction-Theorem 10.1(1), $0 < \frac{e}{Q} < 1$. For any integer $1 \le i \le \lfloor b/2\rfloor$, $2(b-i) \ge b$, so

$$\left(1-\frac{e}{Q}\right)\left(\frac{e}{Q}\right)^{2(b-i)} \le \left(1-\frac{e}{Q}\right)\left(\frac{e}{Q}\right)^{b} \le \max_{x\in[0,1]}(1-x)x^b = \frac{1}{b+1}\left(\frac{b}{b+1}\right)^b \le \frac{1}{b+1}.$$

Since $\sum_{i\ge 1} 100^{-i} = \frac{1}{99}$, the terms with $i \le \lfloor b/2\rfloor$ contribute at most $\frac{1}{99(b+1)} < \frac{1}{99b}$. For any integer $i > \lfloor b/2\rfloor$, $(1-e/Q)(e/Q)^{2(b-i)} \le 1$, and $100^{\lfloor b/2\rfloor} \ge 10^{b-1} \ge b$, so the terms with $i > \lfloor b/2\rfloor$ contribute at most

$$\sum_{i>\lfloor b/2\rfloor} 100^{-i} = \frac{100^{-\lfloor b/2\rfloor}}{99} \le \frac{1}{99b}.$$

Thus

$$\left(1-\frac{e}{Q}\right)\sum_{i=1}^b 100^{-i}\left(\frac{e}{Q}\right)^{2(b-i)} \le \frac{2}{99b} < \frac{1}{40b}.$$

The lemma follows. □

11.2. **Values of the images of the monomial valuation on lines.**

Proposition 11.4. *Notation and conditions as in Construction-Theorem 10.4. Then for any integer $b \ge 1$:*

(1) We have

$$w_b(L) \le Q^b\left(\frac{e^b}{Q^b} + \sqrt{t\left(\left(1-\frac{e^b}{Q^b}\right)^2 + \frac{t-1}{40b}\right)}\right).$$

(2) If $t < 2$, then $w_b(M) \le Q^b\left(1+\frac{t-1}{100}\right)$ for any line M through p.

Proof. For any $\mathbb{R}$-divisor $D \geq 0$ on $\mathbb{P}^2$, $w_b(D) = v((g^b)^*D)$. By Construction-Theorem 10.4(2), the conclusions of Construction-Theorem 10.1 hold for n and v.

Let $S := \{v\}$. By Construction-Theorem 10.4(1), v is a normalized valuation at p with

$$A(v) = 1 + t < +\infty \quad \text{and} \quad \alpha(v) = t,$$

and $v \in \Gamma_S = [\mathrm{mult}_p, v]$. By Definition 8.1, $\|\delta_v\|^2 = \alpha(v \wedge v) = \alpha(v) = t$. Thus for any $\mathbb{R}$-divisor $D \geq 0$ on $\mathbb{P}^2$, Theorem 8.2 and the Cauchy–Schwarz inequality [FJ04, Corollary 7.75] imply

$$v(D) = \langle \delta_v, \rho_S(D) \rangle \leq \sqrt{t} \|\rho_S(D)\|. \tag{11.3}$$

We let $\kappa := 1 + t$. Then $\kappa \geq 2$ and $A(v) \leq \kappa$.

(1) Write $(g^b)^*L = e^b L + J_b$ as in Definition-Lemma 11.2. Since $v(L) = 1$,

$$w_b(L) = e^b + v(J_b).$$

By Lemma 11.3,

$$\left\| \rho_S \left(\frac{1}{Q^b} J_b \right) \right\|^2 \leq \left(1 - \frac{e^b}{Q^b} \right)^2 + \frac{t-1}{40b}.$$

Since $J_b \geq 0$, (11.3) for $D = \frac{1}{Q^b} J_b$ implies (1).

(2) Since $\deg \phi \leq t < 2$ and $\phi(0) = 0$, we have $\phi = sx$ for some $s \in \mathbb{C}$, so $v = \mathrm{mon}_{x, y - sx}(1, t)$, and $x, y - sx$ are affine linear coordinates centered at p. Let $P := \{y = sx\}$. Then P is a line through p and $P \neq L = \{x = 0\}$, so $e_P(g) \leq \frac{1}{100} Q$ by Construction-Theorem 10.1(2). Let M be a line through p. Since $g(P)$ is a curve and $\deg (g^{b-1})^* M = Q^{b-1}$,

$$\mathrm{mult}_P (g^b)^* M = e_P(g) \cdot \mathrm{mult}_{g(P)} (g^{b-1})^* M \leq \frac{Q}{100} \cdot Q^{b-1} = \frac{Q^b}{100}.$$

Since $t \geq 1$, Lemma 8.6 with g^b, Q^b, $y - sx$, t, and P in place of g, Q, y, β, and C implies

$$w_b(M) = v((g^b)^* M) \leq Q^b + (t-1) \, \mathrm{mult}_P (g^b)^* M \leq Q^b \left(1 + \frac{t-1}{100} \right).$$

This proves (2). □

11.3. **Finite thinness and tangency.**

Lemma 11.5. *Notation and conditions as in Construction-Theorem 10.4. Then* $0 < n\sigma < 10010$ *and* $0 < \sigma < 1.001 \cdot 10^{-12}$.

Proof. Since t is rational, v is a divisorial valuation centered at p. Since $n \geq 10^{16} \geq 4 \cdot 10^8$ and $v(B_{17280n}) > A(v)$ by Construction-Theorem 10.4(1), Theorem 7.1(1) implies $\lambda < Q$ and

$$(1 - \sigma)^{\lfloor n/1000 \rfloor} = \left(\frac{\lambda}{Q} \right)^{\lfloor n/1000 \rfloor} > \frac{1}{15000}.$$

Since $\lambda < Q$, $\sigma > 0$. Since $\sigma \leq -\log(1 - \sigma)$ and $\lfloor n/1000 \rfloor \geq n/1001$,

$$\frac{n\sigma}{1001} \leq \left\lfloor \frac{n}{1000} \right\rfloor \sigma \leq - \left\lfloor \frac{n}{1000} \right\rfloor \log(1 - \sigma) < \log 15000 < 10.$$

Since $n \geq 10^{16}$, $\sigma < 10010/n \leq 1.001 \cdot 10^{-12}$. □

Lemma 11.6. *Notation and conditions as in Construction-Theorem 10.4, and let* ω *be an eigenvaluation of* g *at* p *for which (a) or (b) of Theorem 2.12(1) holds. Then* $\alpha(\omega) < +\infty$, *and the following hold.*

(1) The inequalities

$$\frac{1}{\alpha(\omega)} \omega \leq \mathrm{mult}_p \leq v \leq t \, \mathrm{mult}_p \leq t\omega$$

hold on $\widehat{\mathcal{O}}_{\mathbb{P}^2, p}$, *on ideals of* $\widehat{\mathcal{O}}_{\mathbb{P}^2, p}$, *and on* $\mathbb{R}$*-divisors* $D \geq 0$ *on* $\mathbb{P}^2$.

(2) $\lambda \leq \alpha(\omega)(Q - 1)$.

(3) For any integer $j \geq 0$ and any line M through p,

$$m_j \geq \frac{\lambda^j}{\alpha(\omega)}, \quad \operatorname{mult}_p\big((g^j)^*M\big) \geq \frac{\lambda^j}{\alpha(\omega)}, \quad \text{and} \quad A(w_j) \leq 1 + t + t(\lambda^j - 1)A(\omega),$$

$$\text{and } A(\nu_j) = A(w_j)/m_j \leq \alpha(\omega)\max\{1+t, tA(\omega)\}.$$

Proof. By Construction-Theorem 10.4(2), the conclusions of Construction-Theorem 10.1 hold for n and v. In particular, $g(p) = p$, ω is quasimonomial or infinitely singular by Construction-Theorem 10.1(3), and $\lambda > \frac{9}{10}Q > 1$ by Construction-Theorem 10.1(1). Thus Lemma 9.2 implies $A(\omega) < +\infty$, $\alpha(\omega) < +\infty$, and, since g^0 is the identity,

$$(g^j)_*\omega = \lambda^j\omega \quad \text{and} \quad \omega(R(g^j)) = (\lambda^j - 1)A(\omega) \quad \text{for any integer } j \geq 0.$$

(1) Since $v(\mathfrak{m}_p) = 1$ and $\alpha(v) = t$ by Construction-Theorem 10.4(1), Lemma 2.10(1) for ω and for v gives $\operatorname{mult}_p \leq \omega \leq \alpha(\omega)\operatorname{mult}_p$ and $\operatorname{mult}_p \leq v \leq t\operatorname{mult}_p$ on $\widehat{\mathcal{O}}_{\mathbb{P}^2,p}$. Taking minima over the elements of an ideal, and sums with non-negative coefficients over the components of D, gives (1) on ideals and on $\mathbb{R}$-divisors $D \geq 0$.

(2) Local equations of two distinct lines through p generate $\mathfrak{m}_p$, so some line M through p satisfies $\omega(M) = \omega(\mathfrak{m}_p) = 1$. By (1), $g_*\omega = \lambda\omega$, and $\lambda < Q$ (Construction-Theorem 10.1(1)), $\operatorname{mult}_p(g^*M) \leq \omega(g^*M) = \lambda < Q$, so $\operatorname{mult}_p(g^*M) \leq Q - 1$ and

$$\lambda = \omega(g^*M) \leq \alpha(\omega)\operatorname{mult}_p(g^*M) \leq \alpha(\omega)(Q-1).$$

(3) Let $j \geq 0$ be an integer and M a line through p. By (1), $(g^j)_*\omega = \lambda^j\omega$, and $\omega(\mathfrak{m}_p) = 1$,

$$m_j = v((g^j)^*\mathfrak{m}_p) \geq \frac{\omega((g^j)^*\mathfrak{m}_p)}{\alpha(\omega)} = \frac{\lambda^j}{\alpha(\omega)},$$

and, since $\omega(M) \geq \operatorname{mult}_p M = 1$,

$$\operatorname{mult}_p((g^j)^*M) \geq \frac{\lambda^j\omega(M)}{\alpha(\omega)} \geq \frac{\lambda^j}{\alpha(\omega)}.$$

Since t is rational, v is a divisorial valuation, so Lemma 2.4 for g^j, $A(v) = 1 + t$, and (1) imply

$$A(w_j) = 1 + t + v(R(g^j)) \leq 1 + t + t\omega(R(g^j)) = 1 + t + t(\lambda^j - 1)A(\omega).$$

By Lemma 2.4 and $g(p) = p$, ν_j is a divisorial normalized valuation at p, so $A(\nu_j) = A(w_j)/m_j$ by Theorem 2.9(4). Since $m_j \geq \lambda^j/\alpha(\omega)$ and $0 < \lambda^{-j} \leq 1$,

$$A(\nu_j) \leq \alpha(\omega)\big(\lambda^{-j}(1+t) + (1-\lambda^{-j})tA(\omega)\big) \leq \alpha(\omega)\max\{1+t, tA(\omega)\}.$$

□

Proposition 11.7. *Notation and conditions as in Construction-Theorem 10.4. Then any eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds satisfies*

$$A(\omega) < +\infty \quad \text{and} \quad \omega(L) > 1.$$

Proof. By Construction-Theorem 10.4(2), the conclusions of Construction-Theorem 10.1 hold for n and v. Since t is rational, v is divisorial, so Construction-Theorem 10.4(1) and Lemma 2.4 imply $A(v) + v(R(g^n)) = A((g^n)_*v) > Q^n(1+t)$, that is,

$$v(R(g^n)) > (Q^n - 1)(1+t). \tag{11.4}$$

Since $q \geq 2$ (Set-up 5.1), $Q = q^{17280} \geq 2^{17280}$. By Definition 2.11, $\omega(\mathfrak{m}_p) = 1$, $g_\bullet\omega = \omega$, and $c(g,\omega) = \lambda$, so $g_*\omega = \lambda\omega$.

Step 1. In this step we prove $A(\omega) < +\infty$ and

$$\omega(R(g^j)) = (\lambda^j - 1)A(\omega) \quad \text{for any integer } j \geq 1. \tag{11.5}$$

By Construction-Theorem 10.1(3), ω is quasimonomial or infinitely singular. Since $\lambda > \frac{9}{10}Q > 1$ by Construction-Theorem 10.1(1), Lemma 9.2 implies $A(\omega) < +\infty$, $\alpha(\omega) < +\infty$, and (11.5). Moreover, $\alpha(\omega) \le A(\omega) - 1$ by Theorem 2.9(3).

Since $\omega(\mathfrak{m}_p) = 1$ and $x \in \mathfrak{m}_p$, $\omega(L) \ge 1$. It remains to show $\omega(L) \ne 1$. From now on, suppose that $\omega(L) = 1$. We derive a contradiction.

Step 2. In this step we prove $A(\omega) < 8$. We first show that any irreducible curve $P \subset \mathbb{P}^2$ satisfies

$$\omega(P)^2 \le \frac{1001}{1000}\alpha(\omega)(\deg P)^2. \tag{11.6}$$

If $p \notin P$, then $\omega(P) = 0$. If $P = L$, then $\omega(L)^2 = 1 \le \alpha(\omega)$. Assume that $P \ne L$ and $p \in P$. Since $g_*\omega = \lambda\omega$, we have $\omega(g^*P) = \lambda\omega(P)$.

- If $g(P) \ne P$, then $P \not\subset \operatorname{Supp} g^*P$ and $\deg g^*P = Q \deg P$. By Lemma 2.10(3), $\lambda\omega(P)^2 \le \alpha(\omega)Q(\deg P)^2$, and $Q/\lambda = 1/(1-\sigma) < 1001/1000$ by Lemma 11.5.
- If $g(P) = P$, write $g^*P = e_P(g)P + D$ with $P \not\subset \operatorname{Supp} D$. Then $e_P(g) \le \frac{1}{100}Q$ by Construction-Theorem 10.1(2), $\deg D = (Q - e_P(g)) \deg P$, and $\omega(D) = (\lambda - e_P(g))\omega(P)$. By Lemma 2.10(3),

$$(\lambda - e_P(g))\omega(P)^2 \le \alpha(\omega)(Q - e_P(g))(\deg P)^2.$$

Since $\lambda - e_P(g) > \frac{89}{100}Q$,

$$\frac{Q - e_P(g)}{\lambda - e_P(g)} = 1 + \frac{\sigma Q}{\lambda - e_P(g)} < 1 + 2\sigma < \frac{1001}{1000}.$$

Thus (11.6) holds. Write $R(g) = \sum_i a_i P_i$ with distinct irreducible curves P_i and $a_i > 0$. Since $K_{\mathbb{P}^2} = g^*K_{\mathbb{P}^2} + R(g)$, $\sum_i a_i \deg P_i = \deg R(g) = 3(Q-1)$. By (11.6),

$$\omega(R(g)) = \sum_i a_i\omega(P_i) \le 3(Q-1)\sqrt{\frac{1001}{1000}\alpha(\omega)}.$$

Since $\lambda - 1 = (Q-1) - \sigma Q \ge (1-2\sigma)(Q-1)$, Lemma 11.5 implies $(Q-1)/(\lambda-1) < 1001/1000$. By (11.5) for $j = 1$ and $\alpha(\omega) \le A(\omega) - 1$,

$$A(\omega)^2 \le 9\left(\frac{Q-1}{\lambda-1}\right)^2 \frac{1001}{1000}\alpha(\omega) < 9\left(\frac{1001}{1000}\right)^3 (A(\omega)-1) < \frac{64}{7}(A(\omega)-1).$$

Thus $(7A(\omega) - 8)(A(\omega) - 8) < 0$, and

$$A(\omega) < 8. \tag{11.7}$$

Step 3. In this step we prove

$$0 \le \alpha(\omega) - 1 \le A(\omega) - 2 < 168\sigma \quad \text{and} \quad \sigma > \frac{1}{169Q}. \tag{11.8}$$

Let

$$\beta_\omega := \max\{\omega(M) \mid M \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

We choose a line $M \ne L$ through p with $\omega(M) = \beta_\omega$. If $\beta_\omega = 1$, any line $M \ne L$ through p satisfies $\omega(M) = 1$. If $\beta_\omega > 1$, a maximizing line is different from L because $\omega(L) = 1$.

Let $S := \{\omega\}$, so $\Gamma_S = [\operatorname{mult}_p, \omega]$. Any $\mu, \mu' \in \Gamma_S$ are comparable and satisfy $\alpha(\mu), \alpha(\mu') \le \alpha(\omega)$ by Theorem 2.9(1), so

$$\langle \delta_\mu, \delta_{\mu'} \rangle = \min\{\alpha(\mu), \alpha(\mu')\} \ge \frac{\alpha(\mu)\alpha(\mu')}{\alpha(\omega)} = \frac{\langle \delta_\mu, \delta_\omega \rangle \langle \delta_{\mu'}, \delta_\omega \rangle}{\alpha(\omega)}.$$

For any $\mathbb{R}$-divisor $D \geq 0$, $\rho_S(D)$ is a combination of the δ_μ, $\mu \in \Gamma_S$, with non-negative coefficients, and $\langle \delta_\omega, \rho_S(D)\rangle = \omega(D)$ by Theorem 8.2. Thus any $\mathbb{R}$-divisors $D, D' \geq 0$ satisfy

$$\langle \rho_S(D), \rho_S(D')\rangle \geq \frac{\omega(D)\omega(D')}{\alpha(\omega)}. \tag{11.9}$$

Let $D := R(g)/(Q-1)$ and $D' := g^*M/Q$. By (11.5), $\omega(D) = (\lambda-1)A(\omega)/(Q-1)$, and $\omega(D') = \lambda\beta_\omega/Q = (1-\sigma)\beta_\omega$. Any irreducible component P of g^*M through p satisfies $g(P) = M \neq L$, so $\mathrm{mult}_P R(g) = \mathrm{mult}_P D_1 \leq \frac{1}{100}(Q-1)$ by Definition-Lemma 11.2. Thus g, p, M, and $\epsilon = \frac{1}{100}$ satisfy the weaker assumption in Proposition 8.8, and since $A(\omega) \geq 1+\alpha(\omega) \geq 2$, Proposition 8.8 for $\kappa = A(\omega)$ and S, together with $\deg_p g \leq Q^2$, implies

$$\langle \rho_S(D), \rho_S(D')\rangle \leq \frac{Q^2 - 2Q + \deg_p g}{Q(Q-1)} + \frac{A(\omega)-2}{100} \leq 2 + \frac{A(\omega)-2}{100}.$$

Let $u := (1-\sigma)(\lambda-1)/(Q-1)$. By (11.9),

$$u\frac{A(\omega)\beta_\omega}{\alpha(\omega)} \leq 2 + \frac{A(\omega)-2}{100}.$$

By (11.7) and Lemma 8.5(3) for $\tau = \omega$, $A(\omega)\beta_\omega/\alpha(\omega) \geq 2 + (A(\omega)-2)/21$. Thus

$$\left(\frac{u}{21} - \frac{1}{100}\right)(A(\omega)-2) \leq 2(1-u).$$

Since $(\lambda-1)/(Q-1) \geq 1-2\sigma$, we have $1-u \leq 3\sigma$, and Lemma 11.5 implies $u/21 - 1/100 > 1/28$. If $A(\omega) > 2$, then $(A(\omega)-2)/28 < 2(1-u) \leq 6\sigma$. If $A(\omega) = 2$, then $A(\omega) - 2 < 168\sigma$ because $\sigma > 0$. Since $1 \leq \alpha(\omega) \leq A(\omega) - 1$, the first inequality of (11.8) holds.

For the second, Lemma 11.6(2) and the first inequality of (11.8) imply

$$(1-\sigma)Q = \lambda \leq \alpha(\omega)(Q-1) < (1+168\sigma)(Q-1),$$

so $1 < \sigma Q + 168\sigma(Q-1) < 169\sigma Q$.

Step 4. In this step we let $b := 10^6$ and bound m_b, $A(\nu_b)$, and the values of w_b on lines through p. By Lemma 11.5,

$$(b+168)\sigma < 1.0012 \cdot 10^{-6}. \tag{11.10}$$

By (11.8),

$$(1-\sigma)^b/\alpha(\omega) > (1-b\sigma)(1-168\sigma) > 1-(b+168)\sigma.$$

(i) By Lemma 11.6(3) for $j = b$, any line M through p satisfies

$$m_b \geq \frac{\lambda^b}{\alpha(\omega)} \quad \text{and} \quad \mathrm{mult}_p((g^b)^*M) \geq \frac{\lambda^b}{\alpha(\omega)}.$$

Since the multiplicity of a plane curve at a point is at most the degree of the curve,

$$\frac{m_b}{Q^b} > 1-(b+168)\sigma \quad \text{and} \quad 1-(b+168)\sigma < \frac{\mathrm{mult}_p((g^b)^*M)}{Q^b} \leq 1. \tag{11.11}$$

(ii) By Lemma 11.6(1) and (11.5), $v(R(g^b)) \leq t\omega(R(g^b))$, so

$$v(R(g^b)) \leq t(\lambda^b - 1)A(\omega) < 11Q^b(2+168\sigma) < 23Q^b. \tag{11.12}$$

By Lemma 11.6(3) and $1 + t < 2t \leq tA(\omega)$,

$$A(\nu_b) \leq tA(\omega)\alpha(\omega) < (2+168\sigma)(1+168\sigma)t.$$

Thus $A(\nu_b) < 23$, and $A(\nu_b) < 5$ if $t < 2$. By Lemma 8.3 for the germ of g^b at p and $\nu_1 = v$, $\alpha(\nu_b) < +\infty$.

(iii) We show that any line M through p satisfies

$$\frac{w_b(M)}{Q^b} < 1 + \frac{\sqrt{t-1}}{600}. \tag{11.13}$$

Let $S' := \{v\}$, so $\Gamma_{S'} = [\mathrm{mult}_p, v]$ and $A(v) = 1 + t < 12$. Let $\eta_M := \rho_{S'}((g^b)^* M/Q^b)$. We claim $\|\eta_M\|^2 \le 1 + \frac{1}{4b}$. If $M \ne L$, then $\deg((g^b)^* M/Q^b) = 1$, and any irreducible curve through p has coefficient at most 100^{-b} in $(g^b)^* M/Q^b$ by Definition-Lemma 11.2. Theorem 8.2(1) for $\kappa = 12$ and $\epsilon = 100^{-b}$ implies $\|\eta_M\|^2 \le 1 + 10 \cdot 100^{-b} < 1 + \frac{1}{4b}$. If $M = L$, then $(g^b)^* L = e^b L + J_b$. By Theorem 2.9(2), $\alpha(v \wedge \nu_L) = v(L) = 1$, so $v \wedge \nu_L = \mathrm{mult}_p$ by Lemma 2.10(1). Thus $\pi_{S'}(\nu_L) = \mathrm{mult}_p$ and $\rho_{S'}(L) = \delta_{\mathrm{mult}_p}$. By Theorem 8.2,

$$\langle \delta_{\mathrm{mult}_p}, \rho_{S'}(J_b) \rangle = \mathrm{mult}_p J_b \le \deg J_b = Q^b - e^b.$$

By Lemma 11.3 for $\kappa = 12$ and S', $\|\rho_{S'}(J_b/Q^b)\|^2 \le (1 - e^b/Q^b)^2 + \frac{10}{40b}$. Thus

$$\|\eta_L\|^2 \le \frac{e^{2b}}{Q^{2b}} + 2\frac{e^b}{Q^b}\left(1 - \frac{e^b}{Q^b}\right) + \left(1 - \frac{e^b}{Q^b}\right)^2 + \frac{1}{4b} = 1 + \frac{1}{4b}.$$

Let $c_M := \mathrm{mult}_p((g^b)^* M)/Q^b$. By Theorem 8.2, $\langle \eta_M, \delta_{\mathrm{mult}_p} \rangle = c_M$ and $\langle \eta_M, \delta_v \rangle = w_b(M)/Q^b$. Moreover,

$$\langle \delta_{\mathrm{mult}_p}, \delta_{\mathrm{mult}_p} \rangle = \langle \delta_{\mathrm{mult}_p}, \delta_v \rangle = 1 \quad \text{and} \quad \langle \delta_v, \delta_v \rangle = t.$$

By (11.11) and (11.10),

$$\|\eta_M - c_M \delta_{\mathrm{mult}_p}\|^2 = \|\eta_M\|^2 - c_M^2 < 2(b + 168)\sigma + \frac{1}{4b} < \frac{1}{600^2},$$
$$\|\delta_v - \delta_{\mathrm{mult}_p}\|^2 = t - 1, \quad \text{and}$$
$$\langle \eta_M - c_M \delta_{\mathrm{mult}_p}, \delta_v - \delta_{\mathrm{mult}_p} \rangle = \frac{w_b(M)}{Q^b} - c_M.$$

Since $\langle \cdot, \cdot \rangle$ is positive definite (Definition 8.1), the Cauchy–Schwarz inequality and $c_M \le 1$ imply (11.13).

(iv) If $t < 2$, then Proposition 11.4(2) implies that any line M through p satisfies

$$\frac{w_b(M)}{Q^b} \le 1 + \frac{t-1}{100}. \tag{11.14}$$

Step 5. In this step we bound $\alpha(\nu_b)$. Let

$$\beta_b := \max\{\nu_b(M) \mid M \subset \mathbb{P}^2 \text{ is a line through } p\},$$

so $\beta_b \ge 1$ and $m_b \beta_b = \max_M w_b(M)$. We show

$$m_b \alpha(\nu_b) \le Q^b \left(1 + \frac{t-1}{25} + 3(b + 168)\sigma\right) \qquad \text{if } t < 2, \tag{11.15}$$

$$\alpha(\nu_b) < 1 + \frac{t-1}{16} \qquad \text{if } t \ge 2. \tag{11.16}$$

Assume that $t < 2$. By (11.14) and (11.11), $\beta_b \le (1 + \frac{t-1}{100})/(1 - (b + 168)\sigma)$. Since $A(\nu_b) < 5$, Lemma 8.5(2) for $\tau = \nu_b$ implies $\alpha(\nu_b) - 1 \le (\beta_b - 1)(A(\nu_b) - 1) \le 4(\beta_b - 1)$. Thus

$$m_b \alpha(\nu_b) \le 4 \max_M w_b(M) - 3m_b \le 4Q^b \left(1 + \frac{t-1}{100}\right) - 3Q^b\big(1 - (b + 168)\sigma\big),$$

which is (11.15).

Assume that $t \ge 2$. By (11.13), (11.11), (11.10), and $\sqrt{t-1} \ge 1$,

$$\beta_b - 1 < \frac{\sqrt{t-1}/600 + (b + 168)\sigma}{1 - (b + 168)\sigma} < \frac{\sqrt{t-1}}{500}.$$

Since $A(\nu_b) < 23$, Lemma 8.5(2) for $\tau = \nu_b$ implies

$$\alpha(\nu_b) - 1 \le 22(\beta_b - 1) < \frac{22}{500}\sqrt{t-1} \le \frac{t-1}{16},$$

where the last inequality holds because $\sqrt{t-1} \ge 1 > \frac{352}{500}$.

Step 6. In this step we contradict (11.4). Since $K_{\mathbb{P}^2} = h^*K_{\mathbb{P}^2} + R(h)$ for any finite morphism $h : \mathbb{P}^2 \to \mathbb{P}^2$, we have $R(g^n) = R(g^b) + (g^b)^*R(g^{n-b})$, so

$$v(R(g^n)) = v(R(g^b)) + w_b(R(g^{n-b})).$$

By Lemma 2.10(1),

$$w_b = m_b\nu_b \le m_b\alpha(\nu_b)\,\mathrm{mult}_p \le m_b\alpha(\nu_b)\omega,$$

so (11.5) implies $w_b(R(g^{n-b})) < m_b\alpha(\nu_b)\lambda^{n-b}A(\omega)$. By (11.4) and (11.12), dividing by Q^n,

$$1 + t < \frac{1 + t + 23Q^b}{Q^n} + (1-\sigma)^{n-b}A(\omega)\frac{m_b\alpha(\nu_b)}{Q^b}.$$

Since $n - b \ge 3$, $Q \ge 2^{17280}$, and $\sigma > 1/(169Q)$ by (11.8),

$$\frac{1 + t + 23Q^b}{Q^n} < 35Q^{-3} < \frac{1}{169Q} < \sigma.$$

By (11.8), Lemma 11.5, and $(1-\sigma)^b \ge 1 - b\sigma$, we have

$$(1-\sigma)^{-b}A(\omega) < (1 + 2b\sigma)(2 + 168\sigma) < 2 + 5b\sigma.$$

Thus

$$1 + t < \sigma + (1-\sigma)^n(2 + 5b\sigma)\frac{m_b\alpha(\nu_b)}{Q^b}. \tag{11.17}$$

Assume that $t < 2$. By (11.15),

$$\begin{aligned}(2 + 5b\sigma)\frac{m_b\alpha(\nu_b)}{Q^b} &\le 2 + \frac{2(t-1)}{25} + \left(6(b + 168) + 5b\left(1 + \frac{t-1}{25}\right) + 15b(b + 168)\sigma\right)\sigma \\ &< 2 + \frac{2(t-1)}{25} + 12b\sigma,\end{aligned}$$

where the last inequality holds by (11.10). Since $1 + t - 2 - \frac{2(t-1)}{25} = \frac{23(t-1)}{25} > 0$ and $1 + t > 2$, we have $2 + \frac{2(t-1)}{25} + 12b\sigma < (1 + t)(1 + 6b\sigma)$. Since $(1-\sigma)^{-n} \ge (1 + \sigma)^n \ge 1 + n\sigma$ and $n - 6b > 10011 > 1 + n\sigma$ by Lemma 11.5,

$$(1-\sigma)^n(1 + 6b\sigma) \le \frac{1 + 6b\sigma}{1 + n\sigma} = 1 - \frac{(n - 6b)\sigma}{1 + n\sigma} < 1 - \sigma.$$

Thus

$$\sigma + (1-\sigma)^n(2 + 5b\sigma)\frac{m_b\alpha(\nu_b)}{Q^b} < \sigma + (1 + t)(1-\sigma) = 1 + t - t\sigma < 1 + t,$$

which contradicts (11.17).

Assume that $t \ge 2$. By Lemma 8.3 for the germ of g^b at p and $\nu_1 = v$, with $\deg_p(g^b) \le Q^{2b}$, we have $m_b^2\alpha(\nu_b) \le Q^{2b}t$, so by (11.16),

$$\frac{m_b\alpha(\nu_b)}{Q^b} \le \sqrt{t\alpha(\nu_b)} < \sqrt{t\left(1 + \frac{t-1}{16}\right)}.$$

Since

$$(1 + t)^2 - 4t\left(1 + \frac{t-1}{16}\right) - \frac{(1 + t)^2}{18} = \frac{(t-1)(3t-4)}{4} - \frac{(1 + t)^2}{18} = \frac{(t-2)(50t-34)}{72} \ge 0,$$

we have $\sqrt{t(1 + \frac{t-1}{16})} \le \sqrt{17/18}(1 + t)/2$. By Lemma 11.5, $(2 + 5b\sigma)/2 < \frac{101}{100}$, and $(\frac{101}{100})^2 \cdot \frac{17}{18} < (\frac{99}{100})^2$. Thus

$$\sigma + (1-\sigma)^n(2 + 5b\sigma)\frac{m_b\alpha(\nu_b)}{Q^b} < \sigma + \frac{99}{100}(1 + t) < 1 + t,$$

which contradicts (11.17).

Thus $\omega(L) \ne 1$, so $\omega(L) > 1$. The proposition follows. □

Proposition 11.8. *Notation and conditions as in Construction-Theorem 10.4. Then any eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds satisfies*

$$e < \lambda \le r \le Q, \quad 1 \le A(\omega) - \omega(L) < 20, \quad \text{and} \quad 0 \le \alpha(\omega) - \omega(L) < 19.$$

Proof. By Construction-Theorem 10.4(2), the conclusions of Construction-Theorem 10.1 hold. In particular, $r \le Q$ and $\frac{9}{10}Q < \lambda < Q$. By Definition 2.11, $g_*\omega = \lambda\omega$. Since $\lambda > 1$, Construction-Theorem 10.1(3) and Lemma 9.2 imply $A(\omega) < +\infty$, $\alpha(\omega) < +\infty$, and

$$\omega(R(g)) = (\lambda - 1)A(\omega). \tag{11.18}$$

We prove $e < \lambda \le r$. Since $(J \cdot L)_p \ge 1$, $p \in \operatorname{Supp} J$, so $\omega(J) \ge \operatorname{mult}_p J \ge 1$ by Lemma 2.10(1). Since $g^*L = eL + J$ and $\omega(g^*L) = \lambda\omega(L)$, we have $(\lambda - e)\omega(L) = \omega(J) > 0$. Since $\omega(L) > 0$ and $\omega(L) \le \alpha(\omega) < +\infty$, we have $\lambda > e$. Let $M \ne L$ be a line through p with affine equation $\ell_M \in \mathfrak{m}_p$, and let $j \ge 1$ be an integer. Since $g^j(L) = L \ne M$, $L \not\subset \operatorname{Supp}(g^j)^*M$. Since M and L intersect transversally at p and $(g|_L)^j$ has local degree r^j at p, $((g^j)^*M \cdot L)_p = r^j$. Thus

$$c(g^j, \operatorname{mult}_p) = \operatorname{mult}_p((g^j)^*\mathfrak{m}_p) \le \operatorname{mult}_p((g^j)^*\ell_M) \le ((g^j)^*M \cdot L)_p = r^j,$$

and

$$\lambda = \lim_{j \to +\infty} c(g^j, \operatorname{mult}_p)^{1/j} \le r.$$

We prove the lower bounds. Since L is smooth, $\omega(L) \le \alpha(\omega)$ by the definition of skewness, and $A(\omega) \ge 1 + \alpha(\omega)$ by Theorem 2.9(3). Thus

$$1 \le A(\omega) - \omega(L), \qquad 0 \le \alpha(\omega) - \omega(L) \le A(\omega) - \omega(L) - 1. \tag{11.19}$$

By (11.19), it suffices to show $A(\omega) - \omega(L) < 20$.

By Definition-Lemma 11.2, $R(g) - g^*L + L = D_1 - E_1$. Since ω is additive on $\mathbb{R}$-divisors and $\omega(g^*L) = \lambda\omega(L)$, (11.18) implies

$$(\lambda - 1)\big(A(\omega) - \omega(L)\big) = \omega(D_1) - \omega(E_1) \le \omega(D_1). \tag{11.20}$$

Let $F := D_1/(Q-1)$. Then $F \ge 0$, $L \not\subset \operatorname{Supp} F$ because $g(L) = L$, and any irreducible curve through p has coefficient at most $\frac{1}{100}$ in F by Definition-Lemma 11.2. Since $E_1 = J_{\mathrm{red}}$ and $e \ge 1$, $\deg E_1 \le \deg J = Q - e \le Q - 1$, so

$$\deg F = 2 + \frac{\deg E_1}{Q-1} \le 3.$$

Let $\mu := \omega \wedge \nu_L$. By Theorem 2.9(2), $\alpha(\mu) = \omega(L)$ and, since $\mu \le \nu_L$, $\mu(L) = \alpha(\mu)$. Thus Lemma 10.2(1) for $\nu = \mu$ and $D = F$ implies $\mu(F) \le (F \cdot L)_p \le \deg F$.

Let $S := \{\omega\}$, so $\mu \in \Gamma_S = [\operatorname{mult}_p, \omega]$. By Theorem 8.2,

$$\langle \delta_\omega, \rho_S(F) \rangle = \omega(F) \quad \text{and} \quad \langle \delta_\mu, \rho_S(F) \rangle = \mu(F).$$

By Theorem 8.2(2) for $\kappa = A(\omega) - \omega(L)$ and $\epsilon = \frac{1}{100}$,

$$\|\rho_S(F)\|^2 \le (\deg F)^2 + \frac{A(\omega) - \omega(L) - 1}{100} \deg F.$$

Since $\mu \le \omega$, (11.19) implies

$$\|\delta_\omega - \delta_\mu\|^2 = \alpha(\omega) - 2\alpha(\mu) + \alpha(\mu) = \alpha(\omega) - \omega(L) \le A(\omega) - \omega(L) - 1.$$

Since $\langle \cdot, \cdot \rangle$ is positive definite (Definition 8.1), the Cauchy–Schwarz inequality and $\deg F \le 3$ imply

$$\omega(F) = \mu(F) + \langle \delta_\omega - \delta_\mu, \rho_S(F) \rangle \le 3 + \sqrt{9\big(A(\omega) - \omega(L) - 1\big) + \frac{3}{100}\big(A(\omega) - \omega(L) - 1\big)^2}.$$

Since $q \ge 2$ (Set-up 5.1), $Q \ge 11$, so $\lambda > \frac{9}{10}Q$ implies $(\lambda - 1)/(Q - 1) > \frac{89}{100}$. By (11.20),

$$\frac{89}{100}\big(A(\omega) - \omega(L)\big) < 3 + \sqrt{9\big(A(\omega) - \omega(L) - 1\big) + \frac{3}{100}\big(A(\omega) - \omega(L) - 1\big)^2}.$$

Suppose that $A(\omega) - \omega(L) \geq 20$. Dividing by $A(\omega) - \omega(L)$, the right-hand side becomes at most

$$\frac{3}{20} + \sqrt{\frac{9}{20} + \frac{3}{100}} < \frac{3}{20} + \frac{7}{10} = \frac{17}{20} < \frac{89}{100},$$

a contradiction. Thus $A(\omega) - \omega(L) < 20$, and (11.19) implies $\alpha(\omega) - \omega(L) < 19$. □

11.4. **Bounds for the eigenvaluations.**

Proof of Theorem 11.1. By Construction-Theorem 10.1(1), $\lambda > \frac{9}{10}Q > 1$, so by Theorem 2.12(1) there exists an eigenvaluation of g at p for which (a) or (b) of Theorem 2.12(1) holds. Let ω be any such eigenvaluation, and let $\beta := \omega(L)$.

Step 1. Since $Q = q^{17280} \geq 2^{17280}$, $\frac{Q}{Q-1} < \frac{101}{100}$.

By Definition 2.11, $g_\bullet \omega = \omega$ and $c(g, \omega) = \lambda$, so $g_*\omega = \lambda\omega$. By Construction-Theorem 10.1(3) and Lemma 9.2,

$$\omega(R(g^j)) = (\lambda^j - 1)A(\omega) \quad \text{for any integer } j \geq 0, \tag{11.21}$$

where $R(g^0) = R(\mathrm{id}_{\mathbb{P}^2}) = 0$. Since t is rational, v is a divisorial valuation centered at p, so Lemma 2.4 for g^j and $A(v) = 1 + t$ imply

$$A(w_j) = 1 + t + v(R(g^j)) \quad \text{for any integer } j \geq 0. \tag{11.22}$$

By Construction-Theorem 10.4(1),

$$A(w_n) > Q^n(1 + t). \tag{11.23}$$

Let $\mu := \omega \wedge \nu_L$. Then $\alpha(\mu) = \beta$ and $A(\mu) = 1 + \beta$ by Theorem 2.9(2)(3)(5), and (11.19) gives

$$0 \leq \alpha(\omega) - \beta \leq A(\omega) - \beta - 1. \tag{11.24}$$

By Propositions 11.7 and 11.8,

$$\beta > 1, \qquad e < \lambda \leq r \leq Q, \qquad A(\omega) - \beta < 20, \quad \text{and} \quad \alpha(\omega) - \beta < 19. \tag{11.25}$$

Since $g^*L = eL + J$ and $g_*\omega = \lambda\omega$, $\omega(J) = \omega(g^*L) - e\beta = (\lambda - e)\beta$. Since $L \not\subset \operatorname{Supp} J$ and $(J \cdot L)_p \leq Q - e$ (Construction-Theorem 10.1(1)), Lemma 10.2(1) implies

$$(\lambda - e)\beta = \omega(J) \leq \frac{\alpha(\omega)}{\beta}(J \cdot L)_p \leq \frac{\alpha(\omega)}{\beta}(Q - e).$$

Multiplying by β/Q and using $\frac{\lambda - e}{Q} = 1 - \frac{e}{Q} - \sigma$, we obtain

$$\left(1 - \frac{e}{Q}\right)\left(\beta^2 - \alpha(\omega)\right) \leq \sigma\beta^2. \tag{11.26}$$

The coefficient of L in $R(g)$ is $e-1$. We write $R(g) = (e-1)L + R_1'$, so $R_1' \geq 0$ and $L \not\subset \operatorname{Supp} R_1'$. By (11.21) for $j = 1$,

$$\omega(R_1') = (\lambda - 1)A(\omega) - (e - 1)\beta = (\lambda - 1)(A(\omega) - \beta) + (\lambda - e)\beta.$$

By Lemma 10.2(2), $(R_1' \cdot L)_p \leq 2Q - e - 1$, so Lemma 10.2(1) implies

$$(\lambda - 1)(A(\omega) - \beta) + (\lambda - e)\beta = \omega(R_1') \leq \frac{\alpha(\omega)}{\beta}(2Q - e - 1). \tag{11.27}$$

We let

$$\gamma := \frac{e - 1}{Q - 1} \quad \text{and} \quad \eta := \frac{\lambda - 1}{Q - 1}.$$

Since $1 \leq e < \lambda < Q$, $0 \leq \gamma < \eta < 1$. Moreover,

$$1 - \eta = \frac{Q}{Q - 1}\sigma < \frac{101}{100}\sigma, \qquad 1 - \gamma = \frac{Q}{Q - 1}\left(1 - \frac{e}{Q}\right), \quad \text{and} \quad \frac{2Q - e - 1}{Q - 1} = 2 - \gamma. \tag{11.28}$$

Since the left side of (11.27) equals $(\lambda-1)A(\omega)-(e-1)\beta$, dividing (11.27) by $Q-1$ implies

$$\eta A(\omega) \le \gamma\beta + (2-\gamma)\frac{\alpha(\omega)}{\beta}. \tag{11.29}$$

Step 2. In this step we prove $\beta < 12$ and $A(\omega) < 32$. Assume that $\beta \ge 12$.

By (11.25), $\alpha(\omega) < \beta + 19$. Since $\frac{\beta^2}{2} - \beta - 19 \ge 41$ for $\beta \ge 12$,

$$\beta^2 - \alpha(\omega) > \beta^2 - \beta - 19 > \frac{\beta^2}{2} > 0.$$

Thus (11.26) implies

$$0 < 1 - \frac{e}{Q} < 2\sigma. \tag{11.30}$$

Since $\lambda > e$, (11.27) implies

$$(\lambda-1)(A(\omega)-\beta) \le \frac{\alpha(\omega)}{\beta}(2Q-e-1), \quad \text{that is,} \quad \eta(A(\omega)-\beta) \le (2-\gamma)\frac{\alpha(\omega)}{\beta}.$$

By (11.24),

$$\frac{\alpha(\omega)}{\beta} \le 1 + \frac{A(\omega)-\beta-1}{\beta}.$$

Hence

$$\eta\left(1+(A(\omega)-\beta-1)\right) \le (2-\gamma)\left(1+\frac{A(\omega)-\beta-1}{\beta}\right),$$

that is,

$$\left(\eta - \frac{2-\gamma}{\beta}\right)(A(\omega)-\beta-1) \le (1-\eta)+(1-\gamma). \tag{11.31}$$

By Lemma 11.5, (11.28), and (11.30),

$$1-\eta < \frac{101}{100}\sigma < \frac{101}{10000} \quad \text{and} \quad 1-\gamma < \frac{101}{100}\cdot 2\sigma < \frac{101}{5000}.$$

Since $\beta \ge 12$,

$$\eta - \frac{2-\gamma}{\beta} > 1 - \frac{101}{10000} - \frac{1}{12}\left(1+\frac{101}{5000}\right) = \frac{54293}{60000} > \frac{9}{10},$$

and $(1-\eta)+(1-\gamma) < \frac{303}{100}\sigma$. Thus (11.24), (11.31), and Lemma 11.5 imply

$$0 \le A(\omega)-\beta-1 < \frac{10}{9}\cdot\frac{303}{100}\sigma < 4\sigma \quad \text{and} \quad \alpha(\omega) < \beta + 4\sigma < \beta + \frac{1}{25}. \tag{11.32}$$

By Lemma 11.6(1), $v(D) \le t\omega(D)$ for any $\mathbb{R}$-divisor $D \ge 0$ on $\mathbb{P}^2$. Let $j \ge 1$ be an integer. Since $g(L) = L$, $e_L(g^j) = e^j$, and the coefficient of L in $R(g^j)$ is $e^j - 1$. We write $R(g^j) = (e^j-1)L + R'_j$, so $R'_j \ge 0$ and $L \notin \operatorname{Supp} R'_j$. By (11.21),

$$\omega(R'_j) = (\lambda^j-1)A(\omega) - (e^j-1)\beta = (\lambda^j - e^j)\beta + (\lambda^j-1)(A(\omega)-\beta).$$

Since $v(L) = 1$, (11.22) implies

$$\begin{aligned} A(w_j) &= 1 + t + (e^j-1) + v(R'_j) \le e^j + t + t\omega(R'_j) \\ &= e^j + t\left[(\lambda^j-e^j)\beta + \lambda^j(A(\omega)-\beta)\right] - t\left(A(\omega)-\beta-1\right) \\ &\le e^j + t\left[(\lambda^j-e^j)\beta + \lambda^j(A(\omega)-\beta)\right]. \end{aligned}$$

For $j = n$, dividing by Q^n, we obtain

$$\frac{A(w_n)}{Q^n} \le \left(\frac{e}{Q}\right)^n + t\left[\left((1-\sigma)^n - \left(\frac{e}{Q}\right)^n\right)\beta + (1-\sigma)^n\left(A(\omega)-\beta\right)\right]. \tag{11.33}$$

By (11.25), $0 < \frac{e}{Q} < 1 - \sigma < 1$, and (11.26) is equivalent to

$$(1-\sigma) - \frac{e}{Q} \leq \frac{\alpha(\omega)}{\beta^2}\left(1 - \frac{e}{Q}\right).$$

Thus

$$\begin{aligned}(1-\sigma)^n - \left(\frac{e}{Q}\right)^n &= \left((1-\sigma) - \frac{e}{Q}\right)\sum_{i=0}^{n-1}(1-\sigma)^{n-1-i}\left(\frac{e}{Q}\right)^i \\ &\leq \frac{\alpha(\omega)}{\beta^2}\left(1 - \frac{e}{Q}\right)\sum_{i=0}^{n-1}\left(\frac{e}{Q}\right)^i = \frac{\alpha(\omega)}{\beta^2}\left(1 - \left(\frac{e}{Q}\right)^n\right).\end{aligned}$$

By (11.24),

$$\frac{\alpha(\omega)}{\beta} \leq 1 + \frac{A(\omega) - \beta - 1}{\beta},$$

so

$$\left((1-\sigma)^n - \left(\frac{e}{Q}\right)^n\right)\beta \leq \left(1 + \frac{A(\omega) - \beta - 1}{\beta}\right)\left(1 - \left(\frac{e}{Q}\right)^n\right).$$

Since $\frac{e}{Q} < 1 - \sigma$,

$$1 - \left(\frac{e}{Q}\right)^n = \left(1 - \frac{e}{Q}\right)\sum_{i=0}^{n-1}\left(\frac{e}{Q}\right)^i \leq \frac{1 - e/Q}{\sigma}\cdot\sigma\sum_{i=0}^{n-1}(1-\sigma)^i = \frac{1-e/Q}{\sigma}\left(1 - (1-\sigma)^n\right).$$

Inserting the last two inequalities and $A(\omega) - \beta = 1 + (A(\omega) - \beta - 1)$ into (11.33), and since $t > 1$ and

$$\left(\frac{e}{Q}\right)^n + t\left(1 - \left(\frac{e}{Q}\right)^n\right) + t(1-\sigma)^n = 1 + t - t\left(1 - (1-\sigma)^n\right) + (t-1)\left(1 - \left(\frac{e}{Q}\right)^n\right),$$

we obtain

$$\begin{aligned}\frac{A(w_n)}{Q^n} \leq 1 + t &- \left(t - (t-1)\frac{1-e/Q}{\sigma}\right)\left(1 - (1-\sigma)^n\right) \\ &+ t\left(A(\omega) - \beta - 1\right)\left(\frac{1 - (e/Q)^n}{\beta} + (1-\sigma)^n\right).\end{aligned} \tag{11.34}$$

By (11.26),

$$\frac{1 - e/Q}{\sigma} \leq \frac{\beta^2}{\beta^2 - \alpha(\omega)},$$

so

$$t - (t-1)\frac{1-e/Q}{\sigma} \geq \frac{\beta^2 - t\alpha(\omega)}{\beta^2 - \alpha(\omega)}.$$

Since $t < 11$ and $\alpha(\omega) < \beta + \frac{1}{25}$ by (11.32), and since $19\beta^2 - 219\beta$ is strictly increasing for $\beta \geq 12$,

$$19\beta^2 - (20t-1)\alpha(\omega) > 19\beta^2 - 219\left(\beta + \frac{1}{25}\right) \geq 2736 - 2628 - \frac{219}{25} > 0.$$

Thus $20(\beta^2 - t\alpha(\omega)) > \beta^2 - \alpha(\omega) > 0$, and

$$t - (t-1)\frac{1-e/Q}{\sigma} > \frac{1}{20}. \tag{11.35}$$

By (11.32), $t < 11$, $\beta \geq 12$, $0 < \frac{e}{Q} < 1$, and $0 < 1 - \sigma < 1$,

$$t\left(A(\omega) - \beta - 1\right)\left(\frac{1 - (e/Q)^n}{\beta} + (1-\sigma)^n\right) < 44\sigma\left(\frac{1}{12} + 1\right) < 48\sigma. \tag{11.36}$$

By Bernoulli's inequality,

$$(1-\sigma)^{-n} = \left(1+\frac{\sigma}{1-\sigma}\right)^n \geq 1+n\sigma.$$

Since $n\sigma < 10010$ by Lemma 11.5 and $n \geq 10^{16} > 960 \cdot 10011$,

$$1-(1-\sigma)^n \geq \frac{n\sigma}{1+n\sigma} > \frac{n\sigma}{10011} > 960\sigma. \tag{11.37}$$

By (11.34), (11.35), (11.36), and (11.37),

$$\frac{A(w_n)}{Q^n} < 1+t-\frac{960\sigma}{20}+48\sigma = 1+t,$$

which contradicts (11.23). Hence $\beta < 12$. By (11.25), $\beta > 1$ and $A(\omega) < \beta + 20 < 32$.

Step 3. In this step we prove (4) and (1). Let $j \geq 0$ be an integer. By Lemma 11.6(3),

$$\frac{A(w_j)}{m_j} = A(\nu_j) \leq \alpha(\omega)\max\{1+t, tA(\omega)\}.$$

By (11.24) and **Step 2**, $\alpha(\omega) \leq A(\omega)-1 < 31$ and $tA(\omega) < 11 \cdot 32 = 352$, and $1+t < 12$. Thus

$$\frac{A(w_j)}{m_j} < 31 \cdot 352 = 10912 < 11000.$$

This proves (4).

By (11.24) and (11.25), $A(\omega) \geq \beta+1 > 2$, so $tA(\omega) > 2t > 1+t$. By (11.23) and Lemma 11.6(3) for $j = n$,

$$Q^n(1+t) < A(w_n) \leq t\lambda^n A(\omega) + (1+t-tA(\omega)) < t\lambda^n A(\omega).$$

Thus

$$(1-\sigma)^n > \frac{1+t}{tA(\omega)} > \frac{1}{A(\omega)} > \frac{1}{32},$$

so $-n\log(1-\sigma) < \log 32$. Since $\sigma > 0$ by Lemma 11.5 and $\sigma \leq -\log(1-\sigma)$,

$$0 < \sigma < \frac{\log 32}{n}.$$

This proves (1).

Step 4. In this step we prove $A(\omega) - \beta - 1 < 30\sigma$. By (11.28) and Lemma 11.5,

$$\eta > 1 - \frac{101}{100}\sigma > \frac{49}{50}. \tag{11.38}$$

We first show $\alpha(\omega) < \beta^2$. Assume that $\alpha(\omega) \geq \beta^2$. Then $\beta \leq \frac{\alpha(\omega)}{\beta}$, so the right side of (11.29) is at most $\frac{2\alpha(\omega)}{\beta}$, and

$$\eta \leq \frac{2\alpha(\omega)}{\beta A(\omega)}.$$

If $\beta \geq 2$, then Lemma 8.4(2), for $\tau = \omega$, implies $2\alpha(\omega) \leq \beta A(\omega) - \beta(\beta-1)$. Since $A(\omega) < 32$ by **Step 2**,

$$\eta \leq 1 - \frac{\beta-1}{A(\omega)} < 1 - \frac{1}{32} < \frac{49}{50},$$

which contradicts (11.38). If $1 < \beta < 2$, then $\alpha(\omega) \geq \beta^2 > \beta = \alpha(\mu)$, so $\omega \neq \mu$. By Lemma 8.4(1), $\beta = k_1/k_2$ for coprime integers $k_1 > k_2 \geq 1$, and

$$A(\omega) \geq 1 + \beta + k_2\left(\alpha(\omega) - \beta\right).$$

Since $1<\beta<2$, $k_2\geq 2$ and $k_2(\beta-1)=k_1-k_2\geq 1$. Since $k_2\beta\geq 2\beta>1+\beta$, the derivative of $\frac{2x}{\beta(1+\beta+k_2(x-\beta))}$ with respect to x, which equals $\frac{2(1+\beta-k_2\beta)}{\beta(1+\beta+k_2(x-\beta))^2}$, is negative for any $x\geq\beta^2$. Thus

$$\eta\leq\frac{2\alpha(\omega)}{\beta\left(1+\beta+k_2(\alpha(\omega)-\beta)\right)}\leq\frac{2\beta^2}{\beta\left(1+\beta+k_2\beta(\beta-1)\right)}\leq\frac{2\beta}{1+2\beta}<\frac{4}{5},$$

where the third inequality uses $k_2\beta(\beta-1)\geq\beta$. This contradicts (11.38). Hence $\alpha(\omega)<\beta^2$.

Since $\alpha(\omega)<\beta^2$, the right side of (11.29) is a strictly increasing function of γ. Since $\gamma\leq 1$, (11.29) implies

$$\eta A(\omega)\leq\beta+\frac{\alpha(\omega)}{\beta}. \tag{11.39}$$

If $\beta\geq 2$, then by (11.24),

$$\frac{\alpha(\omega)}{\beta}\leq 1+\frac{A(\omega)-\beta-1}{\beta},$$

so (11.39) implies

$$\left(\eta-\frac{1}{\beta}\right)(A(\omega)-\beta-1)\leq(1-\eta)(\beta+1).$$

Since $\beta+1<13$ and

$$\eta-\frac{1}{\beta}>\frac{49}{50}-\frac{1}{2}=\frac{12}{25},$$

(11.28) implies

$$A(\omega)-\beta-1<\frac{25}{12}\cdot 13\cdot\frac{101}{100}\sigma<30\sigma.$$

If $1<\beta<2$, then Lemma 8.4(3), for $\tau=\omega$, implies

$$\alpha(\omega)-\beta\leq(\beta-1)\left(A(\omega)-1-\beta\right),\quad\text{so}\quad\frac{\alpha(\omega)}{\beta}\leq 1+\frac{\beta-1}{\beta}\left(A(\omega)-\beta-1\right),$$

and (11.39) implies

$$\left(\eta-\frac{\beta-1}{\beta}\right)(A(\omega)-\beta-1)\leq(1-\eta)(\beta+1).$$

Since $\beta+1<3$ and

$$\eta-\frac{\beta-1}{\beta}>\frac{49}{50}-\frac{1}{2}=\frac{12}{25},$$

(11.28) implies

$$A(\omega)-\beta-1<\frac{25}{12}\cdot 3\cdot\frac{101}{100}\sigma<30\sigma.$$

Together with (11.24) and **Step 2**, this proves (2).

Step 5. In this step we prove (3). Assume that $n\geq 2^{100}$.

By Construction-Theorem 10.1, L is an irreducible component of C, so $\operatorname{mult}_L B_{288}>\frac{3}{5}$. By Proposition 5.4(2), $f^{288}(L)=L$, so the coefficient of L in $R(f^{288})$ is $e_L(f^{288})-1$. Since $B_{288}=\frac{1}{q^{288}-1}R(f^{288})$,

$$\frac{e_L(f^{288})-1}{q^{288}-1}>\frac{3}{5},\quad\text{so}\quad\frac{e_L(f^{288})}{q^{288}}>\frac{3}{5}>\frac{1}{2}.$$

Since $g=(f^{288})^{60}$, $Q=(q^{288})^{60}$ and $f^{288}(L)=L$, we have $e=e_L(f^{288})^{60}$ and $\frac{e}{Q}>2^{-60}$. Since $Q\geq 2^{17280}$, $\frac{1}{Q}<2^{-61}$. Thus

$$\gamma=\frac{e-1}{Q-1}>\frac{e-1}{Q}=\frac{e}{Q}-\frac{1}{Q}>2^{-61}. \tag{11.40}$$

By (1), $\sigma<\frac{\log 32}{n}<\frac{4}{n}\leq 2^{-98}$, so by (11.28),

$$1-\eta<2\sigma<2^{-97}. \tag{11.41}$$

Inserting

$$A(\omega) = \beta + 1 + (A(\omega) - \beta - 1) \quad \text{and} \quad \frac{\alpha(\omega)}{\beta} = 1 + \frac{\alpha(\omega) - \beta}{\beta}$$

into (11.29), we obtain

$$\eta\,(A(\omega) - \beta - 1) - (2-\gamma)\frac{\alpha(\omega) - \beta}{\beta} + (1-\gamma)(\beta - 1) \le (1-\eta)(\beta + 1). \tag{11.42}$$

If $\omega = \mu$, then $\alpha(\omega) = \alpha(\mu) = \beta$ and $A(\omega) = A(\mu) = 1 + \beta$ by **Step 1**, so (11.42) implies

$$(1-\gamma)(\beta - 1) \le (1-\eta)(\beta + 1). \tag{11.43}$$

If $\omega \neq \mu$, then by Lemma 8.4(1), $\beta = k_1/k_2$ for coprime integers $k_1 > k_2 \ge 1$, and

$$\alpha(\omega) - \beta \le \frac{1}{k_2}\,(A(\omega) - \beta - 1)\,.$$

Since $k_2\beta = k_1$, (11.42) implies

$$\left(\eta - \frac{2-\gamma}{k_1}\right)(A(\omega) - \beta - 1) + (1-\gamma)(\beta - 1) \le (1-\eta)(\beta + 1).$$

Since $k_1 \ge 2$ and $2 - \gamma > 0$, (11.40) and (11.41) imply

$$\eta - \frac{2-\gamma}{k_1} \ge \eta - \frac{2-\gamma}{2} = \frac{\gamma}{2} - (1-\eta) > 2^{-62} - 2^{-97} > 0.$$

Since $A(\omega) - \beta - 1 \ge 0$ by (11.24), (11.43) holds in this case too. By (11.28), multiplying (11.43) by $\frac{Q-1}{Q}$, we obtain

$$\left(1 - \frac{e}{Q}\right)(\beta - 1) \le \sigma(\beta + 1).$$

This proves (3). □

12. The orbit of the monomial valuation

In this section, we prove the properties of the images $w_b = (g^b)_* v$ that are used in Section 13. Proposition 12.3, together with Theorem 12.1, bounds the skewness $\alpha(\nu_b)$ from below by a constant greater than 1 and independent of n and b.

Theorem 12.1. *Notation and conditions as in Construction-Theorem 10.4, and assume $n \ge 2^{200}$. Then:*

(1) $0 < 1 - \frac{e}{Q} \le \frac{10^{12}}{n}$.

(2) $2 < t < 11$.

(3) For any integer b with $10^6 \le b \le n/10^{20}$, $w_b(L) = m_b$ and

$$1 - \frac{10^{12} b}{n} \le \frac{m_b}{Q^b} \le 1 + \frac{10^{12} b}{n}(t-1).$$

To prove the equality $w_b(L) = m_b$ in Theorem 12.1(3), we need the following lemma.

Lemma 12.2. *Let g be an endomorphism of $\mathbb{P}^2$ of degree $Q \ge 2$, L a line with $g(L) = L$, $e := e_L(g)$, and w a divisorial valuation centered at a closed point $p \in L$ with $w(L) > w(\mathfrak{m}_p)$. Then for any integer $j \ge 1$,*

$$A((g^j)_* w) \le Q^j A(w) - (Q^j - e^j)\,(w(L) - w(\mathfrak{m}_p))\,.$$

Proof. Let $m := w(\mathfrak{m}_p)$, $\nu := w/m$, $\beta := \nu(L)$, and $\zeta := m^2\alpha(\nu)/w(L)$. Then $\beta > 1$, $A(w) = mA(\nu)$, and $\zeta = m\alpha(\nu)/\beta$. Fix an integer $j \ge 1$. A direct expansion shows

$$\begin{aligned} Q^j A(w) - (Q^j - e^j)(w(L) - m) - \Big(e^j w(L) + (2Q^j - e^j)\zeta + A(w) - w(L) - \zeta\Big) \\ = (e^j - 1)\big(A(w) - w(L) - \zeta\big) + (Q^j - e^j)\big(A(w) - 2\zeta - w(L) + m\big). \end{aligned}$$

By Lemma 10.3, $A(w) - w(L) - \zeta \geq 0$, and $e^j - 1 \geq 0$. Since ν is a normalized valuation at p with $A(\nu) < +\infty$ and $\nu(L) = \beta > 1$, Lemma 8.4(2) implies

$$A(w) - 2\zeta - w(L) + m = \frac{m}{\beta}\big(\beta A(\nu) - 2\alpha(\nu) - \beta(\beta - 1)\big) \geq 0,$$

and $Q^j - e^j \geq 0$. Thus the left side of the expansion is non-negative, that is,

$$Q^j A(w) - (Q^j - e^j)(w(L) - m) \geq e^j w(L) + (2Q^j - e^j)\zeta + A(w) - w(L) - \zeta \geq A((g^j)_* w),$$

where the last inequality is Lemma 10.3(3). The lemma follows. □

The second inequality of the next proposition is the lower bound for $m_b^2\alpha(\nu_b)/w_b(L)$ used in Section 13.

Proposition 12.3. *Notation and conditions as in Construction-Theorem 10.4. Then for any integer b with $10^6 \leq b \leq n-1$,*

$$\frac{A(w_b)}{Q^b} \leq 1 + t + \left(1 - \frac{e^b}{Q^b}\right)(t-1) \quad \textit{and} \quad \frac{m_b^2\alpha(\nu_b)}{w_b(L)Q^b} > \frac{1+t}{2} - \frac{\left(1 - e^b/Q^b\right)(t-1)}{2\left(Q^{n-b} - 1\right)}.$$

Proof. By Construction-Theorem 10.4(2), the conclusions of Construction-Theorem 10.1 hold. In particular, $g(p) = p$, $g(L) = L$, and $1 \leq e < r \leq Q$. Since $t \in \mathbb{Q}$, v is a divisorial valuation centered at p. For any integer $k \geq 0$, w_k is a divisorial valuation centered at $g^k(p) = p$ by Lemma 2.4.

We prove the first inequality. By Construction-Theorem 10.4(1),

$$v(\mathfrak{m}_p) = v(L) = 1, \quad \alpha(v) = t, \quad \text{and} \quad A(v) - v(L) - \frac{v(\mathfrak{m}_p)^2\alpha(v)}{v(L)} = 1 + t - 1 - t = 0.$$

Thus Lemma 10.3(3) for $w = v$ and $j = b$ implies

$$A(w_b) \leq e^b + (2Q^b - e^b)t.$$

Since

$$\frac{e^b}{Q^b} + \left(2 - \frac{e^b}{Q^b}\right)t = 1 + t + \left(1 - \frac{e^b}{Q^b}\right)(t-1),$$

the first inequality follows.

We prove the second inequality. Let $j := n - b$, so $j \geq 1$, and let

$$\zeta_b := \frac{m_b^2\alpha(\nu_b)}{w_b(L)}, \quad \Lambda_b := \max\{w_b(L) + \zeta_b, 2\zeta_b\}, \quad \text{and} \quad \epsilon_b := \frac{\left(1 - e^b/Q^b\right)(t-1)}{Q^j - 1}.$$

By Construction-Theorem 10.4(1),

$$A((g^n)_* v) > Q^n(1+t). \tag{12.1}$$

Since $(g^n)_* v = (g^j)_* w_b$, Lemma 10.3(3) for $w = w_b$ implies

$$\begin{aligned} A((g^n)_* v) &\leq e^j w_b(L) + (2Q^j - e^j)\zeta_b + A(w_b) - w_b(L) - \zeta_b \\ &= A(w_b) + (e^j - 1)\big(w_b(L) + \zeta_b\big) + 2(Q^j - e^j)\zeta_b \\ &\leq A(w_b) + (Q^j - 1)\Lambda_b, \end{aligned}$$

where the last inequality holds because $e^j - 1 \geq 0$, $Q^j - e^j \geq 0$, and $(e^j - 1) + (Q^j - e^j) = Q^j - 1$. Combining with (12.1) and the first inequality, since $Q^n = Q^b Q^j$, we obtain

$$\frac{\Lambda_b}{Q^b} > 1 + t - \epsilon_b. \tag{12.2}$$

We show

$$\frac{w_b(L)}{Q^b} < \frac{1+t}{2} - \frac{\epsilon_b}{2}. \tag{12.3}$$

Since $0 \le 1 - e^b/Q^b \le 1$, $j \ge 1$, and $Q = q^{17280} \ge 2^{17280}$,

$$\frac{\epsilon_b}{2} \le \frac{t-1}{2(Q-1)}.$$

Assume that $t \ge 2$. Then $0 < \frac{e^b}{Q^b} \le 1 \le \sqrt{t}$. Since $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ for any $x, y \ge 0$, Proposition 11.4(1) implies

$$\frac{w_b(L)}{Q^b} \le \frac{e^b}{Q^b} + \left(1 - \frac{e^b}{Q^b}\right)\sqrt{t} + \sqrt{\frac{t(t-1)}{40b}} \le \sqrt{t} + \sqrt{\frac{t(t-1)}{40b}} < \sqrt{t} + \frac{1}{100},$$

where the last inequality holds because $t(t-1) < 110$ and $b \ge 10^6$. Since $\frac{1+t}{2} - \sqrt{t} = \frac{1}{2}(\sqrt{t}-1)^2$ is increasing in $t \ge 1$ and $\sqrt{2} < \frac{17}{12}$,

$$\frac{1+t}{2} - \sqrt{t} \ge \frac{3}{2} - \sqrt{2} > \frac{1}{12}.$$

Since $t < 11$, $\frac{\epsilon_b}{2} < \frac{5}{Q-1} < \frac{1}{100}$. Thus

$$\frac{w_b(L)}{Q^b} < \frac{1+t}{2} - \frac{1}{12} + \frac{1}{100} < \frac{1+t}{2} - \frac{1}{100} < \frac{1+t}{2} - \frac{\epsilon_b}{2}.$$

Assume that $1 < t < 2$. Proposition 11.4(2) for $M = L$ implies

$$\frac{w_b(L)}{Q^b} \le 1 + \frac{t-1}{100} = \frac{1+t}{2} - \frac{49}{100}(t-1),$$

and

$$\frac{\epsilon_b}{2} \le \frac{t-1}{2(Q-1)} < \frac{49}{100}(t-1).$$

This proves (12.3).

Suppose that $\zeta_b/Q^b \le \frac{1+t}{2} - \frac{\epsilon_b}{2}$. Then by (12.3),

$$\frac{w_b(L) + \zeta_b}{Q^b} < 1 + t - \epsilon_b \quad \text{and} \quad \frac{2\zeta_b}{Q^b} \le 1 + t - \epsilon_b,$$

so $\Lambda_b/Q^b \le 1 + t - \epsilon_b$, which contradicts (12.2). Thus

$$\frac{\zeta_b}{Q^b} > \frac{1+t}{2} - \frac{\epsilon_b}{2},$$

which is the second inequality. □

Proof of Theorem 12.1(1) and (3). By Construction-Theorem 10.4(1),

$$A\left((g^n)_* v\right) > Q^n(1+t). \tag{12.4}$$

Let $\gamma := 1 - \frac{e}{Q}$. Since $1 \le e < r \le Q$ by Construction-Theorem 10.1(1),

$$0 < \gamma < 1.$$

Since $c_\infty(g,p) = \lambda > \frac{9}{10}Q > 1$, by Theorem 2.12(1) there exists an eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds. We fix such an ω. By Construction-Theorem 10.1(3) and Lemma 9.2,

$$\omega(R(g^j)) = (\lambda^j - 1)A(\omega) \quad \text{for any integer } j \ge 1. \tag{12.5}$$

Since $n \ge 2^{100}$ and $\log 32 < 4$, Theorem 11.1(1)–(3) implies

$$\begin{gathered} 0 < \sigma < \frac{4}{n}, \qquad 1 < \omega(L) < 12, \qquad \gamma\left(\omega(L) - 1\right) \le \sigma\left(\omega(L) + 1\right), \\ 0 \le \alpha(\omega) - \omega(L) \le A(\omega) - \omega(L) - 1 < 30\sigma. \end{gathered} \tag{12.6}$$

Step 1. In this step we prove (1). Assume that $n\gamma > 10^{12}$. Let $\theta := \sigma/\gamma$. Since $\sigma < \frac{4}{n} < 4\cdot 10^{-12}\gamma$ and $\gamma < 1$,

$$0 < \theta < 4 \cdot 10^{-12} \quad \text{and} \quad \sigma \le \theta.$$

By (12.6), $\omega(L) - 1 \le \theta(\omega(L)+1) < 13\theta$ and $A(\omega) - \omega(L) - 1 < 30\theta$, so

$$0 \le \alpha(\omega) - 1 \le A(\omega) - 2 = (\omega(L) - 1) + (A(\omega) - \omega(L) - 1) < 43\theta. \tag{12.7}$$

Let $N := \lceil 100/\gamma \rceil$. Since $\gamma < 1$ and $n \ge 2^{200}$,

$$100 \le N \le \frac{101}{\gamma}, \qquad N\sigma \le 101\theta, \qquad \text{and} \qquad N < \frac{101n}{10^{12}} \le n - 3. \tag{12.8}$$

Let P be an irreducible curve through p. Since $g(p) = p$, for any integer $j \ge 0$ the curve $g^j(P)$ passes through p, so $e_{g^j(P)}(g) \le \max\{e, \frac{1}{100}Q\}$ by Construction-Theorem 10.1(2). Since $e_P(g^N) = \prod_{j=0}^{N-1} e_{g^j(P)}(g)$,

$$\begin{aligned} \frac{e_P(g^N)}{Q^N} &\le \max\left\{1-\gamma, \frac{1}{100}\right\}^N \le \max\left\{\exp(-\gamma N), 100^{-N}\right\} \\ &\le \max\left\{\exp(-100), 100^{-100}\right\} < 10^{-40}. \end{aligned} \tag{12.9}$$

Since $e_P(g^N) \le Q^N$,

$$\operatorname{mult}_P R(g^N) = e_P(g^N) - 1 \le \frac{e_P(g^N)}{Q^N}\left(Q^N - 1\right) < 10^{-40}\left(Q^N - 1\right). \tag{12.10}$$

We bound $w_N = (g^N)_*v$, $m_N = w_N(\mathfrak{m}_p)$, and $\nu_N = w_N/m_N$. By Lemma 11.6(3) for $j = N$,

$$m_N \ge \frac{\lambda^N}{\alpha(\omega)} \quad \text{and} \quad A(\nu_N) = \frac{A(w_N)}{m_N} \le \alpha(\omega)\max\{1+t, tA(\omega)\}.$$

Since $1 < t < 11$, (12.7) and $\theta < 4\cdot 10^{-12}$ imply

$$A(\nu_N) < 23, \quad \text{and} \quad A(\nu_N) < 5 \ \text{ if } t < 2. \tag{12.11}$$

For a line M through p, let $a_M := \operatorname{mult}_p\left((g^N)^*M\right)$. Since $(g^N)^*M$ is a curve of degree Q^N, $a_M \le Q^N$, and $a_M \ge \lambda^N/\alpha(\omega)$ by Lemma 11.6(3). Since $\lambda^N/Q^N = (1-\sigma)^N \ge 1 - N\sigma \ge 1 - 101\theta$ and $1/\alpha(\omega) > 1/(1+43\theta) > 1 - 43\theta$,

$$1 - 144\theta \le \frac{a_M}{Q^N} \le 1 \quad \text{and} \quad \frac{m_N}{Q^N} \ge 1 - 144\theta. \tag{12.12}$$

By Lemma 8.3 for the germ of g^N at p and $\nu_1 = v$, where $m_1 = m_N$ and $\nu_1' = \nu_N$, we have $\alpha(\nu_N) < +\infty$ and, since $\deg_p(g^N) \le Q^{2N}$ by Definition 2.2 for the endomorphism g^N of degree Q^N,

$$m_N^2\alpha(\nu_N) \le \deg_p(g^N)\alpha(v) \le Q^{2N}t. \tag{12.13}$$

Since $\alpha(\nu_N) \ge 1$ and $t < 11$, $m_N < 4Q^N$, so $A(w_N) = m_N A(\nu_N) < 92Q^N$ by (12.11).

We bound θ from below. By Lemma 11.6(2) and (12.7),

$$Q(1-\sigma) = \lambda \le \alpha(\omega)(Q-1) \le (1+43\theta)(Q-1).$$

Thus $1 \le Q\sigma + 43\theta(Q-1) < 44Q\theta$. Since $n \ge N + 3$ by (12.8) and $Q = q^{17280} \ge 2^{17280}$,

$$\frac{A(w_N)}{Q^n} < 92Q^{N-n} \le \frac{92}{Q^3} < \frac{1}{44Q} < \theta. \tag{12.14}$$

We bound $A((g^n)_*v)$ through w_N. Since t is rational, v is divisorial, so w_N is a divisorial valuation by Lemma 2.4 for g^N, and Lemma 2.4 for g^{n-N} implies

$$A\left((g^n)_*v\right) = A(w_N) + w_N\left(R(g^{n-N})\right).$$

By Lemma 2.10(1) for ν_N and for ω, $\nu_N \le \alpha(\nu_N)\operatorname{mult}_p \le \alpha(\nu_N)\omega$, so by (12.5),

$$w_N\left(R(g^{n-N})\right) \le m_N\alpha(\nu_N)\omega\left(R(g^{n-N})\right) < m_N\alpha(\nu_N)\lambda^{n-N}A(\omega).$$

Since $\lambda^{n-N}/Q^n = (1-\sigma)^n(1-\sigma)^{-N}/Q^N$, (12.14) implies

$$\frac{A\left((g^n)_*v\right)}{Q^n} < (1-\sigma)^n \cdot A(\omega)(1-\sigma)^{-N} \cdot \frac{m_N\alpha(\nu_N)}{Q^N} + \theta. \tag{12.15}$$

By (12.7), (12.8) and $\theta < 4\cdot 10^{-12}$,

$$A(\omega)(1-\sigma)^{-N} < \frac{2+43\theta}{1-101\theta} < 2+400\theta. \tag{12.16}$$

Let

$$\beta_N := \max\{\nu_N(M) \mid M \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

Since $\nu_N(M) \geq \nu_N(\mathfrak{m}_p) = 1$ for any line M through p, $\beta_N \geq 1$, and $m_N\beta_N = \max_M w_N(M)$.

Case 1: $1 < t < 2$**.** Since $\deg\phi \leq t < 2$ and $\phi(0) = 0$, the functions x and $y - \phi(x)$ are affine linear coordinates centered at p, and $v = \mathrm{mon}_{x,y-\phi(x)}(1,t)$. Let $P_0 := \{y = \phi(x)\}$. For any line M through p, Lemma 8.6 with g^N, Q^N, $y-\phi(x)$, t, and P_0 in place of g, Q, y, β, and C implies

$$w_N(M) = v\left((g^N)^*M\right) \leq Q^N + (t-1)\,\mathrm{mult}_{P_0}(g^N)^*M.$$

Since $\mathrm{mult}_{P_0}(g^N)^*M$ equals $e_{P_0}(g^N)$ if $g^N(P_0) = M$ and 0 otherwise, (12.9) for $P = P_0$ implies

$$w_N(M) \leq Q^N\left(1 + 10^{-40}(t-1)\right). \tag{12.17}$$

By Lemma 8.5(2) for $\tau = \nu_N$ and (12.11),

$$\alpha(\nu_N) - 1 \leq (\beta_N - 1)\,(A(\nu_N) - 1) \leq 4(\beta_N - 1).$$

Multiplying by m_N, by (12.17) and (12.12),

$$m_N\alpha(\nu_N) \leq 4\max_M w_N(M) - 3m_N \leq Q^N\left(1 + 4\cdot 10^{-40}(t-1) + 432\theta\right).$$

By (12.16), since $t - 1 < 1$ and $\theta < 4\cdot 10^{-12}$,

$$\begin{aligned} A(\omega)(1-\sigma)^{-N}\frac{m_N\alpha(\nu_N)}{Q^N} &< (2+400\theta)\left(1 + 4\cdot 10^{-40}(t-1) + 432\theta\right) \\ &< 2 + 8\cdot 10^{-40}(t-1) + 1300\theta \leq (1+t)(1+650\theta). \end{aligned}$$

Since $(1-\sigma)^n \leq 1/(1+n\sigma)$, $n\sigma = n\gamma\theta > 10^{12}\theta$ and $n\sigma < 4$,

$$1 - (1-\sigma)^n(1+650\theta) \geq \frac{n\sigma - 650\theta}{1+n\sigma} > \frac{(10^{12}-650)\theta}{5} > \theta.$$

Thus (12.15) implies

$$\frac{A\left((g^n)_*v\right)}{Q^n} < (1+t)(1-\theta) + \theta < 1 + t,$$

which contradicts (12.4).

Case 2: $2 \leq t < 11$**.** Let $S := \{\mathrm{mult}_p, v\}$. Since $A(\mathrm{mult}_p) = 2$ and $A(v) = 1+t$, by (12.10), Proposition 8.8 for g^N with $\epsilon = 10^{-40}$ and $\kappa = 1+t$ implies that for any line M through p, $\eta_M := \rho_S\left(\frac{1}{Q^N}(g^N)^*M\right)$ satisfies

$$\|\eta_M\|^2 \leq 1 + (t-1)\left(10^{-40} + Q^{-N}\right) < 1 + 2\cdot 10^{-39}.$$

Since $\mathrm{mult}_p, v \in \Gamma_S = [\mathrm{mult}_p, v]$, Theorem 8.2 implies

$$\langle \eta_M, \delta_{\mathrm{mult}_p}\rangle = \frac{a_M}{Q^N} \quad \text{and} \quad \langle \eta_M, \delta_v\rangle = \frac{w_N(M)}{Q^N}.$$

Since

$$\langle \delta_v, \delta_{\mathrm{mult}_p}\rangle = \langle \delta_{\mathrm{mult}_p}, \delta_{\mathrm{mult}_p}\rangle = \alpha(\mathrm{mult}_p) = 1 \quad \text{and} \quad \langle \delta_v, \delta_v\rangle = \alpha(v) = t,$$

we have

$$\frac{w_N(M)-a_M}{Q^N}=\left\langle \delta_v-\delta_{\mathrm{mult}_p},\eta_M-\frac{a_M}{Q^N}\delta_{\mathrm{mult}_p}\right\rangle,\qquad \|\delta_v-\delta_{\mathrm{mult}_p}\|^2=t-1,$$

$$\text{and}\quad \left\|\eta_M-\frac{a_M}{Q^N}\delta_{\mathrm{mult}_p}\right\|^2=\|\eta_M\|^2-\frac{a_M^2}{Q^{2N}}.$$

By the Cauchy–Schwarz inequality, (12.12) and $\theta<4\cdot10^{-12}$,

$$\frac{w_N(M)}{Q^N}\le 1+\sqrt{t-1}\sqrt{288\theta+2\cdot10^{-39}}<1+\frac{\sqrt{t-1}}{500}.$$

Since $144\theta<10^{-9}$ and $\sqrt{t-1}\ge1$, (12.12) implies

$$\beta_N-1\le\frac{1+\sqrt{t-1}/500}{1-144\theta}-1<\frac{\sqrt{t-1}}{400}.$$

By Lemma 8.5(2) for $\tau=\nu_N$, (12.11), and $t-1\ge1$,

$$\alpha(\nu_N)-1\le(\beta_N-1)\,(A(\nu_N)-1)<\frac{22\sqrt{t-1}}{400}\le\frac{t-1}{16}.$$

By (12.13),

$$\frac{m_N\alpha(\nu_N)}{Q^N}=\sqrt{\frac{m_N^2\alpha(\nu_N)}{Q^{2N}}\alpha(\nu_N)}\le\sqrt{t\left(1+\frac{t-1}{16}\right)}.$$

Since $t\ge2$,

$$(1+t)^2-4t\left(1+\frac{t-1}{16}\right)=\frac{(t-1)(3t-4)}{4}\ge\frac{(t-1)^2}{2}\ge\frac{(1+t)^2}{18},$$

so $m_N\alpha(\nu_N)/Q^N\le\frac12\sqrt{17/18}(1+t)$. By (12.16) and $(101/100)^2\cdot17/18<(99/100)^2$,

$$A(\omega)(1-\sigma)^{-N}\frac{m_N\alpha(\nu_N)}{Q^N}<(1+200\theta)\sqrt{\frac{17}{18}}(1+t)<\frac{99}{100}(1+t).$$

Since $(1-\sigma)^n<1$ and $\theta<\frac{1+t}{100}$, (12.15) implies

$$\frac{A\left((g^n)_*v\right)}{Q^n}<\frac{99}{100}(1+t)+\theta<1+t,$$

which contradicts (12.4).

Thus $n\gamma\le10^{12}$. Since $\gamma>0$, (1) holds.

Step 2. In this step we prove (3). Let b be an integer with $10^6\le b\le n/10^{20}$. By (1),

$$0<1-\left(\frac{e}{Q}\right)^b=1-(1-\gamma)^b\le b\gamma\le\frac{10^{12}b}{n}.\tag{12.18}$$

By Definition-Lemma 11.2, $(g^b)^*L=e^bL+J_b$, where $J_b\ge0$ and $\deg J_b=Q^b-e^b$. Since $v(L)=1$, $v\le t\,\mathrm{mult}_p$ by Lemma 11.6(1), and $\mathrm{mult}_p J_b\le\deg J_b$,

$$e^b\le w_b(L)=e^b+v(J_b)\le e^b+t\left(Q^b-e^b\right).$$

By (12.18),

$$1-\frac{10^{12}b}{n}\le\frac{w_b(L)}{Q^b}\le1+(t-1)\left(1-\left(\frac{e}{Q}\right)^b\right)\le1+\frac{10^{12}b}{n}(t-1).\tag{12.19}$$

Since $p\in L$, $w_b(L)\ge w_b(\mathfrak{m}_p)=m_b$. We show

$$\frac{w_b(L)-m_b}{Q^b}<\left(10^{12}+2\right)\frac{b}{n}(t-1).\tag{12.20}$$

If $w_b(L) = m_b$, then (12.20) holds since $t > 1$. Assume that $w_b(L) > m_b$. By Lemma 2.4 for g^b, w_b is a divisorial valuation, and the center of w_b is $g^b(p) = p$. Since $n - b \geq 1$, Lemma 12.2 for w_b and $j = n - b$ implies

$$A\left((g^n)_* v\right) \leq Q^{n-b} A(w_b) - \left(Q^{n-b} - e^{n-b}\right)\left(w_b(L) - m_b\right).$$

Since $b \leq n-1$, by Proposition 12.3, $A(w_b)/Q^b \leq 1 + t + \left(1 - (e/Q)^b\right)(t-1)$. Dividing by Q^n, (12.4) implies

$$1 + t < 1 + t + \left(1 - \left(\frac{e}{Q}\right)^b\right)(t-1) - \left(1 - \left(\frac{e}{Q}\right)^{n-b}\right)\frac{w_b(L) - m_b}{Q^b}.$$

Since $(e/Q)^{n-b} = (1-\gamma)^{n-b} \leq 1/(1 + (n-b)\gamma)$, $n - b \geq n/2$ and (12.18),

$$\frac{1 - (e/Q)^b}{1 - (e/Q)^{n-b}} \leq \frac{b\gamma\left(1 + (n-b)\gamma\right)}{(n-b)\gamma} = \frac{b}{n-b} + b\gamma \leq \left(2 + 10^{12}\right)\frac{b}{n}.$$

Thus (12.20) holds.

Assume that $\nu_b(L) > 1$. Since ν_b is divisorial, Theorem 2.9(4) and Theorem 11.1(4) imply $A(\nu_b) = A(w_b)/m_b < 11000$. By Lemma 8.4(3) for $\tau = \nu_b$ and the germ of L at p,

$$\alpha(\nu_b) - 1 \leq \left(\nu_b(L) - 1\right)\left(A(\nu_b) - \nu_b(L)\right) < 11000\left(\nu_b(L) - 1\right),$$

so $m_b\alpha(\nu_b) < m_b + 11000\left(w_b(L) - m_b\right)$. Since $\nu_b(L) > 1$ and $m_b \leq w_b(L)$, (12.19) and (12.20) imply

$$\begin{aligned}\frac{m_b^2\alpha(\nu_b)}{w_b(L)Q^b} < \frac{m_b\alpha(\nu_b)}{Q^b} &< \frac{w_b(L)}{Q^b} + 11000\frac{w_b(L) - m_b}{Q^b} \\ &< 1 + \left(10^{12} + 11000\left(10^{12} + 2\right)\right)\frac{b}{n}(t-1) < 1 + \frac{t-1}{4}.\end{aligned}$$

On the other hand, since $1 - (e/Q)^b \leq 1$ and $Q^{n-b} - 1 \geq Q - 1 \geq 4$, Proposition 12.3 implies

$$\frac{m_b^2\alpha(\nu_b)}{w_b(L)Q^b} > \frac{1+t}{2} - \frac{t-1}{2(Q-1)} \geq 1 + \frac{t-1}{2} - \frac{t-1}{8} > 1 + \frac{t-1}{4},$$

a contradiction. Thus $\nu_b(L) = 1$, that is, $w_b(L) = m_b$, and (12.19) implies

$$1 - \frac{10^{12}b}{n} \leq \frac{m_b}{Q^b} \leq 1 + \frac{10^{12}b}{n}(t-1).$$

This proves (3). □

Lemma 12.4. *Let X be a smooth surface, $p \in X$ a closed point, and μ_1, μ_2, μ_3 normalized valuations at p. If $\mu_1 \wedge \mu_3 \neq \mathrm{mult}_p$ and $\mu_2 \wedge \mu_3 = \mathrm{mult}_p$, then $\mu_1 \wedge \mu_2 = \mathrm{mult}_p$.*

Proof. By Theorem 2.9(1), $\mu_1 \wedge \mu_2, \mu_1 \wedge \mu_3 \in [\mathrm{mult}_p, \mu_1]$, and the segment $[\mathrm{mult}_p, \mu_1]$ is totally ordered. Let $\mu_0 := \min\{\mu_1 \wedge \mu_2, \mu_1 \wedge \mu_3\}$. Then $\mu_0 \leq \mu_2$ and $\mu_0 \leq \mu_3$, so $\mu_0 \leq \mu_2 \wedge \mu_3 = \mathrm{mult}_p$, that is, $\mu_0 = \mathrm{mult}_p$. Since $\mu_1 \wedge \mu_3 \neq \mathrm{mult}_p$, we have $\mu_0 = \mu_1 \wedge \mu_2$, so $\mu_1 \wedge \mu_2 = \mathrm{mult}_p$. □

Proof of Theorem 12.1(2). By Construction-Theorem 10.4, $t < 11$. We prove that ϕ has a non-zero term of degree k for some integer k with $2 \leq k < t$. In particular $t > 2$, and Theorem 12.1(2) follows.

Let $b := 10^6$ and $d_b := \max\left\{1 - \frac{m_b}{Q^b}, 0\right\}$. Since $n \geq 2^{200} > 10^{60}$, b is an integer with $10^6 \leq b \leq n/10^{20}$, so $10^{12}b/n \leq 10^{-8}$. Thus Theorem 12.1(3) implies

$$w_b(L) = m_b \quad \text{and} \quad d_b \leq \frac{10^{12}b}{n} \leq 10^{-8}. \tag{12.21}$$

By Theorem 11.1(1), $0 < b\sigma < \frac{b\log 32}{n} < \frac{4b}{n} < \frac{1}{4}$. Thus Bernoulli's inequality implies

$$\frac{3}{4} < 1 - b\sigma \le (1-\sigma)^b = \frac{\lambda^b}{Q^b} < 1. \tag{12.22}$$

Step 1. In this step we prove

$$1 + \frac{t-1}{3} < \frac{m_b\alpha(\nu_b)}{Q^b} \le t(1-\sigma)^b \quad \text{and} \quad 1-(1-\sigma)^b < t-1, \tag{12.23}$$

and we fix ω, θ_b, and ψ, which are used in **Step 2**.

Since $0 < e < Q$ by Construction-Theorem 10.1(1), $0 < 1 - e^b/Q^b < 1$. Since $b \le n-1$ and $Q = q^{17280} \ge 4$, $Q^{n-b} - 1 \ge Q - 1 \ge 3$. Thus Proposition 12.3 and $w_b(L) = m_b$ imply

$$\frac{m_b\alpha(\nu_b)}{Q^b} = \frac{m_b^2\alpha(\nu_b)}{w_b(L)Q^b} > \frac{1+t}{2} - \frac{t-1}{2(Q-1)} \ge 1 + \frac{t-1}{3},$$

which is the first inequality of (12.23).

Since $\lambda > \frac{9}{10}Q > 1$ by Construction-Theorem 10.1(1), Theorem 2.12(1) implies that there exists an eigenvaluation ω of g at p for which (a) or (b) of Theorem 2.12(1) holds. By Definition 2.11, $\omega(\mathfrak{m}_p) = 1$ and $g_*\omega = c(g,\omega)g_\bullet\omega = \lambda\omega$, so $(g^b)_*\omega = \lambda^b\omega$. By Theorem 11.1(1)(2),

$$1 < \omega(L) < 12 \quad \text{and} \quad 1 \le \alpha(\omega) < \omega(L) + 30\sigma < 13. \tag{12.24}$$

For any normalized valuation ν at p and any line M through p, Theorem 2.9(2) implies $\nu(M) = \alpha(\nu \wedge \nu_M)$, and Theorem 2.9(1) implies that $\alpha(\nu \wedge \nu_M) = 1$ if and only if $\nu \wedge \nu_M = \mathrm{mult}_p$. Since $\nu_b(L) = w_b(L)/m_b = 1$ and $\omega(L) > 1$,

$$\nu_b \wedge \nu_L = \mathrm{mult}_p \quad \text{and} \quad \omega \wedge \nu_L \ne \mathrm{mult}_p\,.$$

By Lemma 12.4 with $\mu_1 = \omega$, $\mu_2 = \nu_b$, and $\mu_3 = \nu_L$, $\nu_b \wedge \omega = \mathrm{mult}_p$.

Write $(g^b)_*\,\mathrm{mult}_p = a_b\theta_b$, where $a_b := \mathrm{mult}_p((g^b)^*\mathfrak{m}_p)$ and θ_b is a normalized valuation at p, and let $u_b := a_b/Q^b$. We show

$$u_b \le \frac{m_b}{Q^b} \quad \text{and} \quad u_b \le (1-\sigma)^b < 1. \tag{12.25}$$

By Lemma 2.10(1), $\mathrm{mult}_p \le v$, so $a_b \le v((g^b)^*\mathfrak{m}_p) = m_b$. Let M be a line through p with $M \ne L$. By Theorem 2.9(2), $\alpha(\nu_L \wedge \nu_M) = (L\cdot M)_p = 1$, so $\nu_L \wedge \nu_M = \mathrm{mult}_p$, and Lemma 12.4 with $\mu_1 = \omega$, $\mu_2 = \nu_M$, and $\mu_3 = \nu_L$ implies $\omega \wedge \nu_M = \mathrm{mult}_p$, that is, $\omega(M) = 1$. Since $\theta_b(M) \ge \theta_b(\mathfrak{m}_p) = 1$ and $\mathrm{mult}_p \le \omega$ by Lemma 2.10(1),

$$a_b \le a_b\theta_b(M) = \mathrm{mult}_p((g^b)^*M) \le \omega((g^b)^*M) = \lambda^b\omega(M) = \lambda^b.$$

Thus (12.25) follows from (12.22).

Since $t \in \mathbb{Q}$, v is divisorial. By Lemma 2.4, $w_b = (g^b)_*v$ is divisorial, hence so is ν_b. By [FJ04, Proposition 3.20(iii)(v)], the curve valuations ν_ψ of the irreducible formal curve germs ψ at p with $\nu_\psi > \nu_b$ define infinitely many tangent directions at ν_b, where the tangent directions at ν_b of normalized valuations $\mu, \mu' > \nu_b$ coincide if and only if $\mu \wedge \mu' > \nu_b$ (the tangent vectors at ν_b of [FJ04, Section 3.1.2]). Each of ω and θ_b defines at most one tangent direction at ν_b. Let ψ be an irreducible formal curve germ at p with $\nu_\psi > \nu_b$, and let $\mu \in \{\omega, \theta_b\}$. If $\mu \not\ge \nu_b$, then the elements $\nu_\psi \wedge \mu$ and ν_b of $[\mathrm{mult}_p, \nu_\psi]$ satisfy $\nu_\psi \wedge \mu < \nu_b$, since $\nu_\psi \wedge \mu \ge \nu_b$ would imply $\mu \ge \nu_b$. Thus $\nu_\psi \wedge \mu = \nu_b \wedge \mu$. If $\mu = \nu_b$, then $\nu_\psi \wedge \mu = \nu_b$. If $\mu > \nu_b$, then $\nu_\psi \wedge \mu = \nu_b$ as soon as the tangent directions of ν_ψ and μ at ν_b differ. Choosing ψ such that the tangent direction of ν_ψ at ν_b differs from that of μ for each $\mu \in \{\omega, \theta_b\}$ with $\mu > \nu_b$, and using $\nu_b \wedge \omega = \mathrm{mult}_p$, we obtain

$$\nu_\psi \wedge \omega = \nu_b \wedge \omega = \mathrm{mult}_p \quad \text{and} \quad \nu_\psi \wedge \theta_b = \nu_b \wedge \theta_b. \tag{12.26}$$

Since $\nu_b \wedge \nu_\psi = \nu_b$, Theorem 2.9(2) and (12.26) imply

$$\begin{aligned} v((g^b)^*\psi) &= m_b\nu_b(\psi) = m_b\alpha(\nu_b)\,\mathrm{mult}_p\,\psi, \\ \omega((g^b)^*\psi) &= \lambda^b\omega(\psi) = \lambda^b\,\mathrm{mult}_p\,\psi, \\ \mathrm{mult}_p((g^b)^*\psi) &= a_b\theta_b(\psi) = a_b\alpha(\theta_b \wedge \nu_b)\,\mathrm{mult}_p\,\psi. \end{aligned} \tag{12.27}$$

By Lemma 2.10(1) for v and for ω,

$$v \le t\,\mathrm{mult}_p \le t\omega. \tag{12.28}$$

Taking values of (12.28) at $(g^b)^*\psi$, (12.27) implies $m_b\alpha(\nu_b) \le t\lambda^b$, which is the second inequality of (12.23). Thus $1 + \frac{t-1}{3} < t(1-\sigma)^b$, so

$$1 - (1-\sigma)^b < \frac{2(t-1)}{3t} < t - 1.$$

Step 2. In this step we prove

$$d_b \le 1 - u_b < 71(t-1). \tag{12.29}$$

The first inequality follows from (12.25). Since $\theta_b(L) \ge \theta_b(\mathfrak{m}_p) = 1$, one of the following cases holds.

Case 1: $\theta_b(L) > 1$. Then $\theta_b \wedge \nu_L \ne \mathrm{mult}_p$, and Lemma 12.4 with $\mu_1 = \theta_b$, $\mu_2 = \nu_b$, and $\mu_3 = \nu_L$ implies $\theta_b \wedge \nu_b = \mathrm{mult}_p$. By (12.27), $\mathrm{mult}_p((g^b)^*\psi) = a_b\,\mathrm{mult}_p\,\psi$. Taking values of the first inequality of (12.28) at $(g^b)^*\psi$, (12.27) implies $m_b\alpha(\nu_b) \le ta_b$. Thus (12.23) implies $1 + \frac{t-1}{3} < tu_b$, so $1 - u_b < \frac{2(t-1)}{3t} < t - 1$.

Case 2: $\theta_b(L) = 1$. Then $\theta_b \wedge \nu_L = \mathrm{mult}_p$, and Lemma 12.4 with $\mu_1 = \omega$, $\mu_2 = \theta_b$, and $\mu_3 = \nu_L$ implies $\theta_b \wedge \omega = \mathrm{mult}_p$. Since $(g^b)_*\,\mathrm{mult}_p = a_b\theta_b$, $(g^b)_*\omega = \lambda^b\omega$, $\alpha(\mathrm{mult}_p \wedge \omega) = \alpha(\mathrm{mult}_p) = 1$, and $\alpha(\theta_b \wedge \omega) = 1$, Lemma 8.3 for the germ of g^b at p and the valuations mult_p and ω implies that $\alpha(\theta_b) < +\infty$ and that the matrix

$$\begin{pmatrix} \deg_p g^b - a_b^2\alpha(\theta_b) & \deg_p g^b - a_b\lambda^b \\ \deg_p g^b - a_b\lambda^b & \left(\deg_p g^b - \lambda^{2b}\right)\alpha(\omega) \end{pmatrix}$$

is positive semidefinite. Since $\alpha(\omega) > 0$, the lower right entry implies $\lambda^{2b} \le \deg_p g^b$. Since the topological degree of g^b is Q^{2b}, $\deg_p g^b \le Q^{2b}$. Since $\alpha(\theta_b) \ge 1$ and the lower right entry is non-negative, the determinant of the matrix implies

$$\left(\deg_p g^b - a_b\lambda^b\right)^2 \le \left(\deg_p g^b - a_b^2\right)\left(\deg_p g^b - \lambda^{2b}\right)\alpha(\omega). \tag{12.30}$$

Let $\zeta_1 := \deg_p g^b - \lambda^{2b}$ and $\zeta_2 := \lambda^b(\lambda^b - a_b)$. Then $\zeta_1 \ge 0$ since $\lambda^{2b} \le \deg_p g^b$, $\zeta_2 \ge 0$ by (12.25), and

$$\deg_p g^b - a_b\lambda^b = \zeta_1 + \zeta_2 \quad \text{and} \quad \deg_p g^b - a_b^2 = \zeta_1 + (\lambda^b - a_b)(\lambda^b + a_b) \le \zeta_1 + 2\zeta_2.$$

Thus (12.30) implies

$$(\zeta_1 + \zeta_2)^2 \le \alpha(\omega)\zeta_1(\zeta_1 + 2\zeta_2).$$

If $\zeta_1 = 0$, then $\zeta_2 = 0$. If $\zeta_1 > 0$, then $\xi := \zeta_2/\zeta_1$ satisfies $\xi^2 - 2(\alpha(\omega) - 1)\xi - (\alpha(\omega) - 1) \le 0$, so

$$\xi \le \alpha(\omega) - 1 + \sqrt{\alpha(\omega)(\alpha(\omega) - 1)} < 2\alpha(\omega).$$

Hence, for $\zeta_1 = 0$ and for $\zeta_1 > 0$,

$$\zeta_2 \le 2\alpha(\omega)\zeta_1 \le 2\alpha(\omega)(Q^{2b} - \lambda^{2b}).$$

Dividing by Q^{2b}, we obtain

$$(1-\sigma)^b\left((1-\sigma)^b - u_b\right) \le 2\alpha(\omega)\left(1 - (1-\sigma)^{2b}\right) \le 4\alpha(\omega)\left(1 - (1-\sigma)^b\right).$$

By (12.22) and (12.24),

$$(1-\sigma)^b - u_b < \frac{16\cdot 13}{3}\left(1-(1-\sigma)^b\right) < 70\left(1-(1-\sigma)^b\right).$$

Thus (12.23) implies

$$1-u_b < 71\left(1-(1-\sigma)^b\right) < 71(t-1).$$

Thus (12.29) holds in Case 1 and in Case 2. By (12.21) and (12.29),

$$d_b^2 < 71\cdot 10^{-8}(t-1). \tag{12.31}$$

Step 3. In this step we prove that there do not exist affine linear coordinates x', y' centered at p with $v = \mathrm{mon}_{x',y'}(1,t)$. Assume that such x', y' exist, and let $P := \{y' = 0\}$. Since $v(L) = 1 < t = v(y')$, $P \neq L$.

Let M be a line through p. By Lemma 8.6 for g^b, the coordinates x', y', and $\beta = t$,

$$w_b(M) = v((g^b)^*M) \le Q^b + (t-1)\,\mathrm{mult}_P(g^b)^*M.$$

Since $P \neq L$ and $\deg(g^{b-1})^*M = Q^{b-1}$, Construction-Theorem 10.1(2) implies

$$\mathrm{mult}_P(g^b)^*M = e_P(g)\,\mathrm{mult}_{g(P)}(g^{b-1})^*M \le \frac{Q^b}{100},$$

so

$$w_b(M) \le Q^b\left(1+\frac{t-1}{100}\right) \quad \text{for any line } M \text{ through } p. \tag{12.32}$$

Let

$$\beta_b := \max\{\nu_b(M) \mid M \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

Since $\nu_b(M) \ge \nu_b(\mathfrak{m}_p) = 1$, $\beta_b \ge 1$. Since ν_b is divisorial, $A(\nu_b) = A(w_b)/m_b < +\infty$ by Theorem 2.9(4). By Lemma 8.5(2) for $\tau = \nu_b$ and since $\beta_b \ge 1$,

$$\alpha(\nu_b) \le \beta_b + (\beta_b - 1)\left(A(\nu_b) - 1 - \beta_b\right) \le \beta_b + (\beta_b - 1)\left(A(\nu_b) - 2\right).$$

Multiplying by m_b/Q^b, we obtain

$$\frac{m_b\alpha(\nu_b)}{Q^b} \le \frac{\beta_b m_b}{Q^b} + (\beta_b - 1)\frac{A(w_b) - 2m_b}{Q^b}. \tag{12.33}$$

We bound the three quantities on the right side of (12.33). By (12.32), $\beta_b m_b/Q^b \le 1 + \frac{t-1}{100}$, so (12.21) implies

$$0 \le \beta_b - 1 \le \frac{1 + \frac{t-1}{100} - \frac{m_b}{Q^b}}{m_b/Q^b} \le \frac{d_b + \frac{t-1}{100}}{1-10^{-8}}.$$

By Theorem 2.9(3), $A(\nu_b) \ge 1 + \alpha(\nu_b) \ge 2$, so $A(w_b) - 2m_b \ge 0$. By Proposition 12.3 and $0 < e < Q$, $A(w_b)/Q^b \le 1 + t + (t-1) = 2t$, so

$$0 \le \frac{A(w_b) - 2m_b}{Q^b} \le 2t - \frac{2m_b}{Q^b} \le 2(t-1) + 2d_b.$$

By (12.21), (12.31), and $t - 1 < 10$,

$$\begin{aligned}
&\left(d_b + \frac{t-1}{100}\right)\left(2(t-1) + 2d_b\right) \\
&\quad = 2d_b^2 + \frac{101}{50}(t-1)d_b + \frac{(t-1)^2}{50} \\
&\quad < (t-1)\left(142\cdot 10^{-8} + 3\cdot 10^{-8} + \frac{1}{5}\right) < (t-1)\left(10^{-5} + \frac{1}{5}\right).
\end{aligned}$$

Thus (12.33) implies

$$\frac{m_b\alpha(\nu_b)}{Q^b} < 1 + \frac{t-1}{100} + \frac{(t-1)(10^{-5}+1/5)}{1-10^{-8}} < 1 + \frac{t-1}{100} + \frac{21(t-1)}{100} < 1 + \frac{t-1}{4},$$

which contradicts (12.23).

Step 4. In this step we prove that ϕ has a non-zero term of degree k for some integer k with $2 \le k < t$. Assume that ϕ has no such term. Since $\phi(0) = 0$ and $\deg\phi \le t$, there exist $c_1, c_2 \in \mathbb{C}$ with $\phi(x) = c_1x + c_2x^{\lfloor t\rfloor}$, where $c_2 = 0$ if $t \notin \mathbb{Z}$. Let $y' := y - c_1x$ and $z := y' - c_2x^{\lfloor t\rfloor} = y - \phi(x)$. Then x, y' are affine linear coordinates centered at p, and $v = \mathrm{mon}_{x,z}(1,t)$. Since $c_2x^{\lfloor t\rfloor}$ is 0 or a non-zero multiple of x^t, the automorphism $(x,z) \to (x, z + c_2x^{\lfloor t\rfloor})$ maps each monomial $x^{i_1}z^{i_2}$ to a sum of monomials $x^{j_1}z^{j_2}$ with $j_1 + tj_2 = i_1 + ti_2$, so

$$v = \mathrm{mon}_{x,z}(1,t) = \mathrm{mon}_{x,y'}(1,t),$$

which contradicts **Step 3**. □

13. Proof of the main theorem

Proof of Theorem 1.5. Let $q \ge 2$ be an integer, let f be an endomorphism of $\mathbb{P}^2$ with $f^*\mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^2}(q)$, and let B_n, C, g, and Q be as in Set-up 5.1. Let $n \ge 2^{200}$ be an integer, and assume that $(\mathbb{P}^2, B_{17280n})$ is not lc. We derive a contradiction.

Step 1. Since $n \ge 2^{200} > 10^{60} > 10^{16}$, we may let p, x, y, t, ϕ, v, L, and e, and w_b, m_b, ν_b for any integer $b \ge 0$, be as in Construction-Theorem 10.4. By Construction-Theorem 10.4(2) and Construction-Theorem 10.1, $g(p) = p$ and L is a line through p with $g(L) = L$. In the proof of Construction-Theorem 10.1, Theorem 9.1 implies $p \in C$, Proposition 6.1(3) implies that $g(p) = p$ and that L is a line, and Theorem 7.1 implies $\frac{9}{10}Q < c_\infty(g,p) < Q$. By Theorem 12.1(1)(2), whose proof uses Theorem 11.1,

$$0 < 1 - \frac{e}{Q} \le \frac{10^{12}}{n} \quad \text{and} \quad 2 < t < 11.$$

Thus for any positive integer j,

$$0 \le 1 - \frac{e^j}{Q^j} \le j\left(1 - \frac{e}{Q}\right) \le \frac{10^{12}j}{n}. \tag{13.1}$$

In particular, since $n > 10^{60}$,

$$1 - \frac{e^4}{Q^4} < 10^{-8}. \tag{13.2}$$

Let

$$I := \{10^6 + 4i \mid i \in \mathbb{Z},\ 0 \le i \le 600\}.$$

Any $b \in I$ satisfies $10^6 \le b < 10^7 < n/10^{20}$. Thus $b \le n-1$ and $\frac{10^{12}b}{n} < 10^{-41}$ for any $b \in I$.

Let $b \in I$. By Theorem 12.1(3), $w_b(L) = m_b$, that is, $\nu_b(L) = 1$, and since $t - 1 < 10$,

$$1 - 10^{-8} \le 1 - \frac{10^{12}b}{n} \le \frac{m_b}{Q^b} \le 1 + \frac{10^{12}b}{n}(t-1) \le 1 + 10^{-7}. \tag{13.3}$$

By (13.1), $1 - \frac{e^b}{Q^b} < 10^{-8}$. Since $w_b(L) = m_b$, $2 < t$, $t - 1 < 10$, and $Q^{n-b} - 1 \ge 1$, Proposition 12.3 implies

$$\frac{A(w_b)}{Q^b} \le 1 + t + 10^{-7} < 12 + 10^{-7} \quad \text{and} \quad \frac{m_b}{Q^b}\alpha(\nu_b) > \frac{1+t}{2} - \frac{10^{-7}}{2} > \frac{3}{2} - 10^{-7}.$$

Since $A(w_b) = m_bA(\nu_b)$, (13.3) implies

$$A(\nu_b) < \frac{12 + 10^{-7}}{1 - 10^{-8}} < 13 \quad \text{and} \quad \alpha(\nu_b) > \frac{3/2 - 10^{-7}}{1 + 10^{-7}} > \frac{13}{10}.$$

By Theorem 2.9(3), $\alpha(\nu_b) \le A(\nu_b) - 1 < 12$. Therefore, for any $b \in I$,

$$\frac{13}{10} < \alpha(\nu_b) < 12 \quad \text{and} \quad A(\nu_b) < 13. \tag{13.4}$$

Step 2. In this step, for any $b \in I$, we construct a divisorial monomial valuation $\mu_b \le \nu_b$ and bound $\alpha(\mu_b)$ away from 1.

Let $b \in I$ and

$$\beta_b := \max\{\nu_b(H) \mid H \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

By (13.4), $A(\nu_b) < +\infty$. For any line H through p, Theorem 2.9(1)(2) implies

$$\nu_b(H) = \alpha(\nu_b \wedge \nu_H) \le \alpha(\nu_b),$$

so $\beta_b \le \alpha(\nu_b)$. By Lemma 8.5(2),

$$\frac{3}{10} < \alpha(\nu_b) - 1 \le (\beta_b - 1)\,(A(\nu_b) - 1)\,,$$

where $0 < A(\nu_b) - 1 < 12$ by Theorem 2.9(3) and (13.4). Therefore

$$\frac{1}{40} < \beta_b - 1 \quad \text{and} \quad \beta_b \le \alpha(\nu_b) < 12. \tag{13.5}$$

Let M_b be a line through p with $\nu_b(M_b) = \beta_b$. Since $\nu_b(L) = 1 < \beta_b$, $M_b \ne L$, so there exist affine linear coordinates x_b, y_b centered at p with $L = \{x_b = 0\}$ and $M_b = \{y_b = 0\}$. Let $\mu_b := \nu_b \wedge \nu_{M_b}$. By Lemma 8.5(4),

$$\mu_b = \mathrm{mon}_{x_b,y_b}(1, \beta_b), \quad \alpha(\mu_b) = \beta_b, \quad \text{and} \quad A(\mu_b) = 1 + \beta_b. \tag{13.6}$$

In particular, $\mu_b(\mathfrak{m}_p) = \mu_b(L) = 1$. Since $t \in \mathbb{Q}$, v is divisorial, so $w_b = (g^b)_* v$ is divisorial by Lemma 2.4, and

$$\beta_b = \nu_b(y_b) = \frac{w_b(y_b)}{w_b(\mathfrak{m}_p)} \in \mathbb{Q}.$$

Thus μ_b, a monomial valuation with rational weights, is divisorial. Since $\mu_b \le \nu_b$, we have $\nu_b \wedge \mu_b = \mu_b$, so by Definition 8.1,

$$\|\delta_{\nu_b} - \delta_{\mu_b}\|^2 = \alpha(\nu_b) - 2\alpha(\mu_b) + \alpha(\mu_b) = \alpha(\nu_b) - \beta_b. \tag{13.7}$$

Step 3. In this step we fix an integer $0 \le i \le 599$, let $b := 10^6 + 4i$, and compare $(g^4)_*\nu_b$ with $(g^4)_*\mu_b$ by Lemma 8.3. Since $0 \le i \le 599$, the integers b and $b+4$ belong to I.

Since $(g^4)_* w_b = w_{b+4}$, we have $(g^4)_*\nu_b = \frac{m_{b+4}}{m_b}\nu_{b+4}$. Let

$$a := \frac{m_{b+4}}{Q^4 m_b}, \quad a' := \frac{((g^4)_*\mu_b)(\mathfrak{m}_p)}{Q^4}, \quad \mu_b' := \frac{(g^4)_*\mu_b}{Q^4 a'}, \quad \text{and} \quad u := \frac{((g^4)_*\mu_b)(L)}{Q^4}.$$

Since $g(p) = p$, $a' > 0$ and μ_b' is a normalized valuation at p, and μ_b' is divisorial by Lemma 2.4. Moreover,

$$(g^4)_*\nu_b = Q^4 a \nu_{b+4} \quad \text{and} \quad (g^4)_*\mu_b = Q^4 a' \mu_b'.$$

Since $\mu_b \le \nu_b$ and $\nu_{b+4}(L) = 1$,

$$Q^4 u = \mu_b\left((g^4)^* L\right) \le \nu_b\left((g^4)^* L\right) = ((g^4)_*\nu_b)(L) = Q^4 a \nu_{b+4}(L) = Q^4 a.$$

Write $(g^4)^* L = e^4 L + J_4$ as in Definition-Lemma 11.2. Since $J_4 \ge 0$ and $\mu_b(L) = 1$, $Q^4 u \ge e^4$. Thus by (13.2),

$$1 - 10^{-8} < \frac{e^4}{Q^4} \le u \le a. \tag{13.8}$$

Since g^4 is an endomorphism of $\mathbb{P}^2$ of degree Q^4, $\deg_p g^4 \le Q^8$. By (13.4) and (13.6), ν_b and μ_b have finite skewness. Lemma 8.3 for $h = g^4$ and the valuations ν_b, μ_b implies that ν_{b+4} and μ_b' have finite skewness, and evaluating the positive semidefinite form at $(1, -1)$ gives

$$\deg_p g^4 \cdot \|\delta_{\nu_b} - \delta_{\mu_b}\|^2 - \left\|Q^4 a \delta_{\nu_{b+4}} - Q^4 a' \delta_{\mu_b'}\right\|^2 \ge 0.$$

Since $\deg_p g^4 \le Q^8$, (13.7) implies

$$a^2\alpha(\nu_{b+4}) - 2aa'\alpha(\nu_{b+4} \wedge \mu'_b) + a'^2\alpha(\mu'_b) \le \alpha(\nu_b) - \beta_b. \tag{13.9}$$

Step 4. In this step we bound the values of $(g^4)_*\mu_b$ on the lines $H \ne L$ through p and $A((g^4)_*\mu_b)$, and we bound $\alpha(\mu'_b)$ when $\mu'_b(L) = 1$.

Let $H \ne L$ be a line through p. By Definition-Lemma 11.2 for $j = 4$ and the irreducible curve M_b, $\operatorname{mult}_{M_b}(g^4)^*H \le 10^{-8}Q^4$. By Lemma 8.6 for the endomorphism g^4 of degree Q^4, the coordinates x_b, y_b, and $\beta_b \ge 1$, where $M_b = \{y_b = 0\}$, and by (13.5),

$$\begin{aligned}((g^4)_*\mu_b)(H) = \mu_b\left((g^4)^*H\right) &\le Q^4 + (\beta_b - 1)\operatorname{mult}_{M_b}(g^4)^*H \\ &\le Q^4\left(1 + 10^{-8}(\beta_b - 1)\right) < Q^4(1 + 11 \cdot 10^{-8}).\end{aligned} \tag{13.10}$$

By (13.6), $\mu_b(\mathfrak{m}_p) = \mu_b(L) = 1$ and $A(\mu_b) = 1 + \beta_b$. Thus Lemma 10.3(3) for $w = \mu_b$ and $j = 4$, where $m = \mu_b(\mathfrak{m}_p) = 1$ and $\zeta = \alpha(\mu_b)/\mu_b(L) = \beta_b$, implies

$$A((g^4)_*\mu_b) \le e^4 + (2Q^4 - e^4)\beta_b + A(\mu_b) - 1 - \beta_b = e^4 + (2Q^4 - e^4)\beta_b.$$

Thus by (13.2) and (13.5),

$$\frac{A((g^4)_*\mu_b)}{Q^4} \le 1 + \beta_b + \left(1 - \frac{e^4}{Q^4}\right)(\beta_b - 1) < 13 + 11 \cdot 10^{-8}. \tag{13.11}$$

Assume that $\mu'_b(L) = 1$. Then $a' = a'\mu'_b(L) = u$. Since the equation of any line through p belongs to $\mathfrak{m}_p$, $a' \le ((g^4)_*\mu_b)(H)/Q^4$ for any line $H \ne L$ through p. Thus by (13.8) and (13.10),

$$1 - 10^{-8} < a' < 1 + 11 \cdot 10^{-8}.$$

Let

$$\beta'_b := \max\{\mu'_b(H) \mid H \subset \mathbb{P}^2 \text{ is a line through } p\}.$$

By (13.10), for any line $H \ne L$ through p,

$$\mu'_b(H) = \frac{((g^4)_*\mu_b)(H)}{Q^4a'} < \frac{1 + 11 \cdot 10^{-8}}{1 - 10^{-8}}.$$

Since $\mu'_b(L) = 1$,

$$1 \le \beta'_b < \frac{1 + 11 \cdot 10^{-8}}{1 - 10^{-8}} < 1 + 13 \cdot 10^{-8}.$$

By (13.11),

$$A(\mu'_b) = \frac{A((g^4)_*\mu_b)}{Q^4a'} < \frac{13 + 11 \cdot 10^{-8}}{1 - 10^{-8}} < 14.$$

Lemma 8.5(2) for $\tau = \mu'_b$ implies

$$\alpha(\mu'_b) - 1 \le (\beta'_b - 1)(A(\mu'_b) - 1) < 13 \cdot 10^{-8} \cdot 13.$$

Therefore, if $\mu'_b(L) = 1$, then

$$\alpha(\mu'_b) < 1 + 169 \cdot 10^{-8} < 1 + 2 \cdot 10^{-6}. \tag{13.12}$$

Step 5. In this step we prove

$$\alpha(\nu_{b+4}) < \alpha(\nu_b) - \frac{1}{50}. \tag{13.13}$$

Since $\mu'_b(L) \ge \mu'_b(\mathfrak{m}_p) = 1$, either $\mu'_b(L) > 1$ or $\mu'_b(L) = 1$.

Case 1: $\mu'_b(L) > 1$. By Theorem 2.9(2),

$$\alpha(\mu'_b \wedge \nu_L) = \mu'_b(L) > 1 \quad \text{and} \quad \alpha(\nu_{b+4} \wedge \nu_L) = \nu_{b+4}(L) = 1,$$

so $\mu'_b \wedge \nu_L \ne \operatorname{mult}_p$ and $\nu_{b+4} \wedge \nu_L = \operatorname{mult}_p$ by Theorem 2.9(1). Thus Lemma 12.4 with $\mu_1 = \mu'_b$, $\mu_2 = \nu_{b+4}$, and $\mu_3 = \nu_L$ implies $\nu_{b+4} \wedge \mu'_b = \operatorname{mult}_p$, that is,

$$\alpha(\nu_{b+4} \wedge \mu'_b) = 1.$$

Moreover, $\alpha(\mu'_b) \geq \alpha(\mu'_b \wedge \nu_L) = \mu'_b(L) = u/a'$, and $a' \leq u$ since the equation of L belongs to $\mathfrak{m}_p$. Thus by (13.9),

$$a^2\alpha(\nu_{b+4}) \leq \alpha(\nu_b) - \beta_b + 2aa' - a'^2 \cdot \frac{u}{a'} = \alpha(\nu_b) - \beta_b + a'(2a-u).$$

By (13.8), $2a - u \geq a > 0$, so $a'(2a-u) \leq u(2a-u) = a^2 - (a-u)^2 \leq a^2$. Thus

$$\alpha(\nu_{b+4}) \leq 1 + \frac{\alpha(\nu_b) - \beta_b}{a^2}.$$

Case 2: $\mu'_b(L) = 1$. Since $\nu_{b+4} \wedge \mu'_b \leq \mu'_b$, Theorem 2.9(1) implies $\alpha(\nu_{b+4} \wedge \mu'_b) \leq \alpha(\mu'_b)$. Since $a, a' > 0$, (13.9) implies

$$\begin{aligned}\alpha(\nu_b) - \beta_b &\geq a^2\alpha(\nu_{b+4}) + a'^2\alpha(\mu'_b) - 2aa'\alpha(\mu'_b)\\ &= a^2\left(\alpha(\nu_{b+4}) - \alpha(\mu'_b)\right) + (a-a')^2\alpha(\mu'_b)\\ &\geq a^2\left(\alpha(\nu_{b+4}) - \alpha(\mu'_b)\right).\end{aligned}$$

Thus by (13.12),

$$\alpha(\nu_{b+4}) < 1 + 2\cdot 10^{-6} + \frac{\alpha(\nu_b) - \beta_b}{a^2}.$$

In Case 1 and in Case 2,

$$\alpha(\nu_{b+4}) < 1 + 2\cdot 10^{-6} + (\alpha(\nu_b) - \beta_b)a^{-2}.$$

By (13.5), $0 \leq \alpha(\nu_b) - \beta_b < 12$ and $\beta_b - 1 > \frac{1}{40}$, and by (13.3),

$$a^{-2} = \left(\frac{m_b/Q^b}{m_{b+4}/Q^{b+4}}\right)^2 \leq \left(\frac{1+10^{-7}}{1-10^{-8}}\right)^2 < 1 + 3\cdot 10^{-6}.$$

Thus

$$\begin{aligned}\alpha(\nu_{b+4}) &< 1 + 2\cdot 10^{-6} + (\alpha(\nu_b) - \beta_b)(1 + 3\cdot 10^{-6})\\ &< \alpha(\nu_b) - (\beta_b - 1) + 38\cdot 10^{-6}\\ &< \alpha(\nu_b) - \frac{1}{40} + 38\cdot 10^{-6} < \alpha(\nu_b) - \frac{1}{50}.\end{aligned}$$

Step 6. In this step we conclude the proof.

By (13.13) for $b = 10^6 + 4i$ with $0 \leq i \leq 599$, and by (13.4) for $b = 10^6$,

$$\alpha(\nu_{10^6+2400}) < \alpha(\nu_{10^6}) - \frac{600}{50} < 12 - 12 = 0.$$

This contradicts $\alpha(\nu_{10^6+2400}) \geq 1$, which holds by Theorem 2.9(1). Therefore $(\mathbb{P}^2, B_{17280n})$ is lc for any integer $n \geq 2^{200}$. For $n = 2^{200}$, $17280n = N$, so $\left(\mathbb{P}^2, \frac{1}{q^N-1}R(f^N)\right)$ is lc. □

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